

DRAWING

TITLE=K22

ABBREV=DRAWING

LAST_MODIFIED=

DRAWING TITLE

SCH, D8, MLB

DRAWING NUMBER

051-9504

SIZE

D

REVISION

5.3.0

BRANCH

NOTICE OF PROPRIETARY PROPERTY:

THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:

I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE

II NOT TO REPRODUCE OR COPY IT

III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART

IV ALL RIGHTS RESERVED

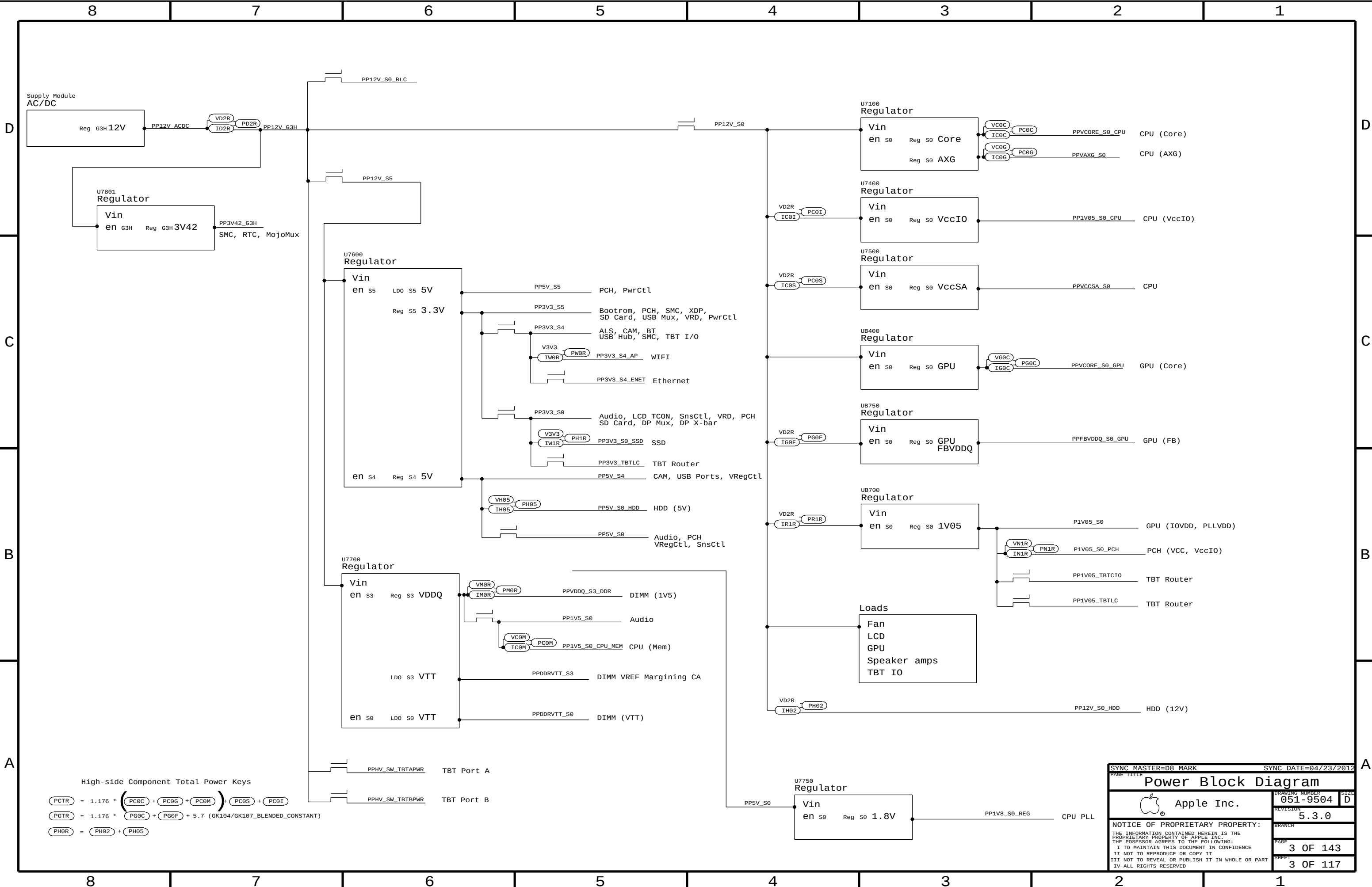
PAGE

1 OF 143

SHEET

1 OF 117

8	7	6	5	4	3	2	1																								
System Block diagram can be found on Kismet PATH: Kismet > K70/72 > Block Diagrams > K70 Block Diagram																															
D8_MLB01/05/2012 <table><tr><td colspan="2">PAGE TITLE</td><td colspan="2">System Block Diagram</td></tr><tr><td colspan="2"> Apple Inc.</td><td>DRAWING NUMBER</td><td>051-9504</td></tr><tr><td colspan="2"></td><td>REVISION</td><td>5.3.0</td></tr><tr><td colspan="2">NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED</td><td>BRANCH</td><td>PAGE</td></tr><tr><td colspan="2"></td><td></td><td>2 OF 143</td></tr><tr><td colspan="2"></td><td>SHEET</td><td>2 OF 117</td></tr></table>								PAGE TITLE		System Block Diagram		 Apple Inc.		DRAWING NUMBER	051-9504			REVISION	5.3.0	NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		BRANCH	PAGE				2 OF 143			SHEET	2 OF 117
PAGE TITLE		System Block Diagram																													
 Apple Inc.		DRAWING NUMBER	051-9504																												
		REVISION	5.3.0																												
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		BRANCH	PAGE																												
			2 OF 143																												
		SHEET	2 OF 117																												
8	7	6	5	4	3	2	1																								



SYNC MASTER=D8 MARK

SYNC DATE=04/23/2012

Power Block Diagram

Apple Inc.

THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
IV ALL RIGHTS RESERVED

DRAWING NUMBER

051-9504

REVISION

5.3.0

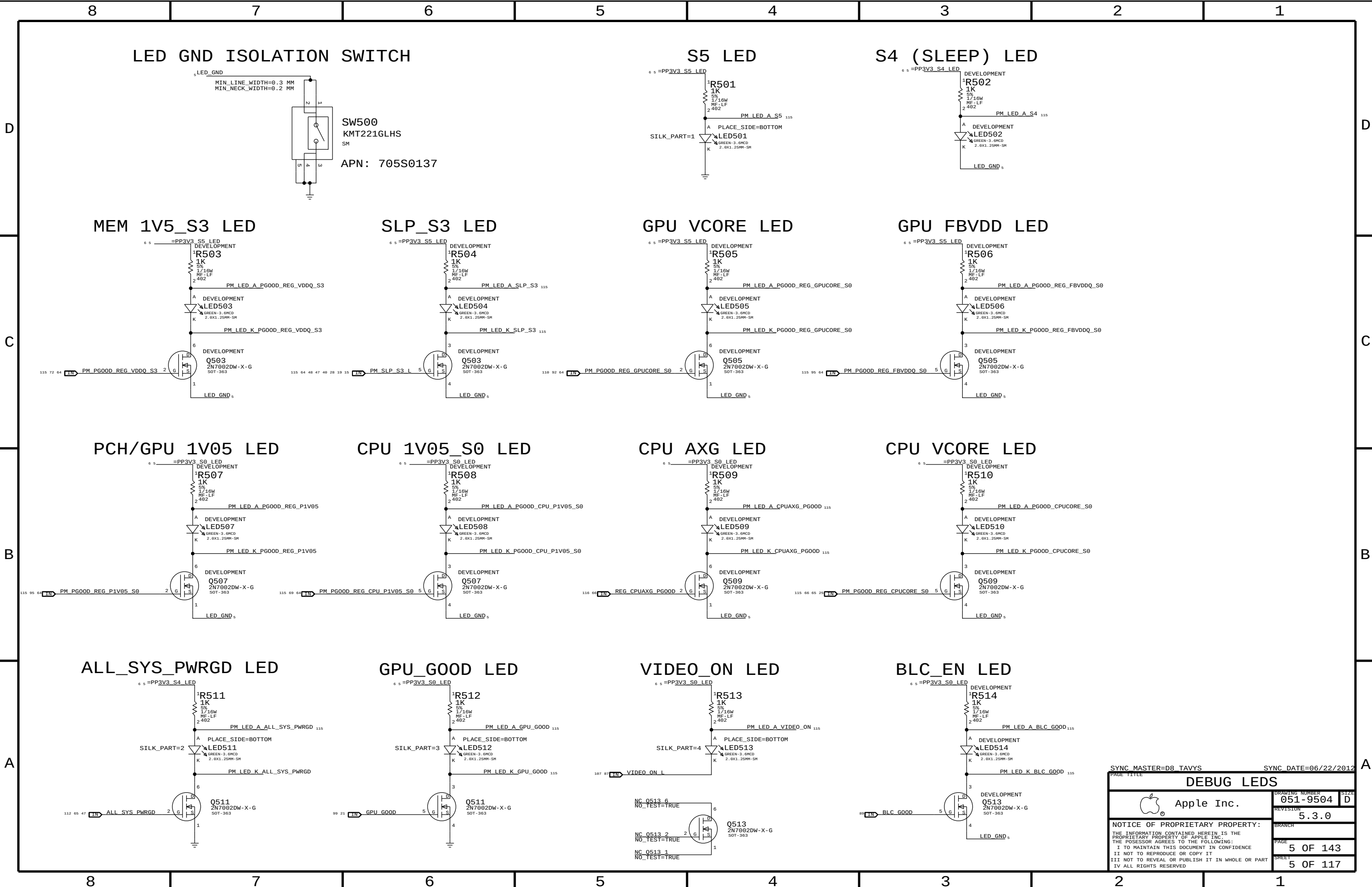
PAGE

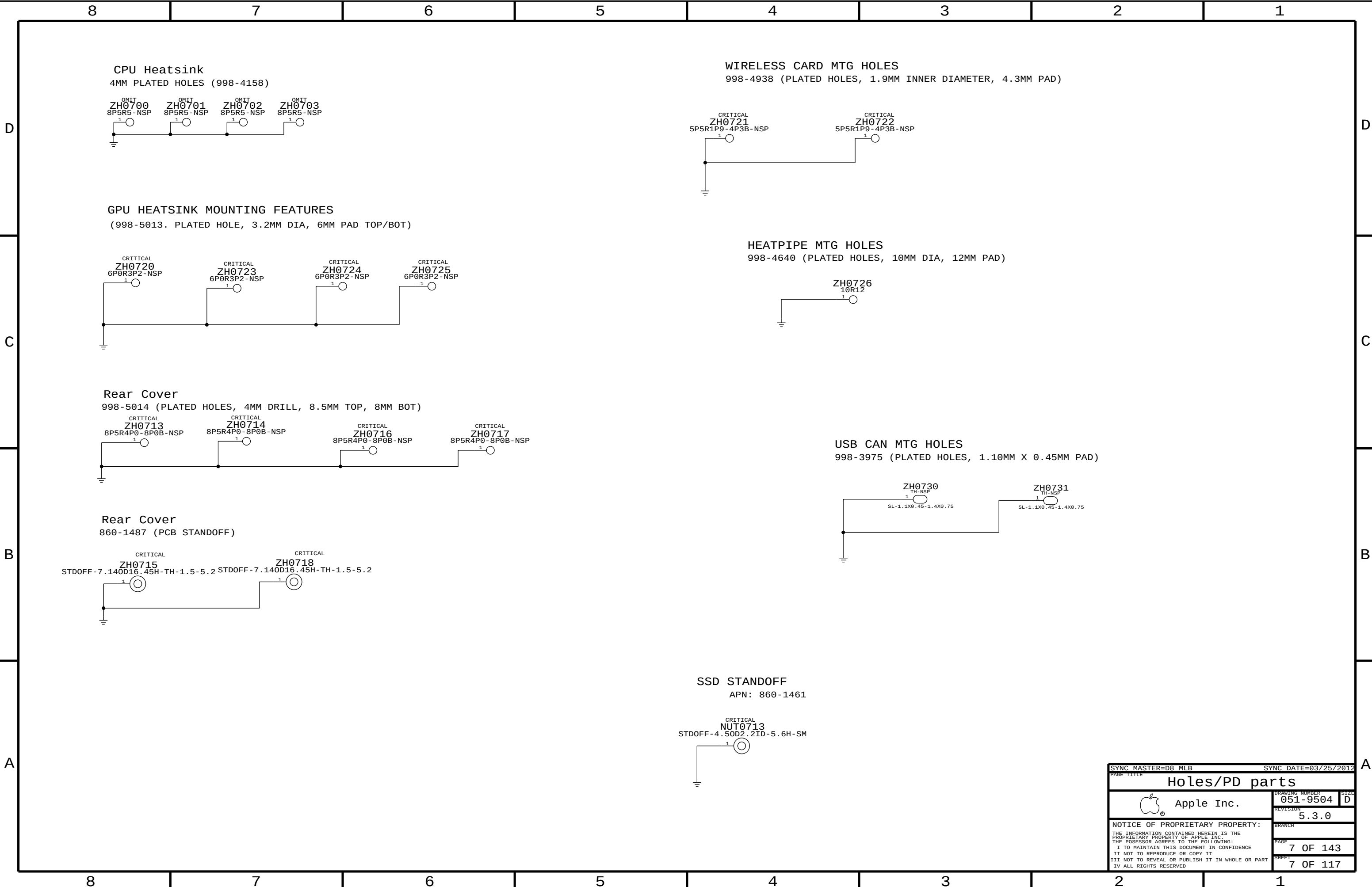
3 OF 143

SHEET

3 OF 117

8		7		6		5		4		3		2		1	
BOM Variants															
BOM NUMBER		BOM NAME				BOM OPTIONS									
639-3816		PCBA, MLB, 2.9G, 4C, GK107, SAM, D8				D8_COMMON, D8, CPU:4C_2P9GHZ, GPU:107EGE, FB:BOTH_SAMSUNG, EEEE:F2FR									
639-3952		PCBA, MLB, 3.1G, 4C, GK107, SAM, D8				D8_COMMON, D8, CPU:4C_3P1GHZ, GPU:107EGE, FB:BOTH_SAMSUNG, EEEE:F49T									
639-4092		PCBA, MLB, 2.9G, 4C, GK107, HYN.D8				D8_COMMON, D8, CPU:4C_2P9GHZ, GPU:107EGE, FB:BOTH_HYNIX, EEEE:F653									
639-4093		PCBA, MLB, 3.1G, 4C, GK107, HYN.D8				D8_COMMON, D8, CPU:4C_3P1GHZ, GPU:107EGE, FB:BOTH_HYNIX, EEEE:F654									
085-4433		PCBA, MLB, DEV, D8				DEVELOPMENT, D8_DEVEL									





SYNC MASTER=D8_MLB		SYNC DATE=03/25/2012	
PAGE TITLE			
Holes/PD parts			
 Apple Inc.		DRAWING NUMBER	051-9504
		REVISION	5.3.0
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		BRANCH	
		PAGE	7 OF 143
		SHEET	7 OF 117

8	7	6	5	4	3	2	1		
CPU Reserved		PCH PCIe		PCH Unused Display		PCH SATA		PCH Test Points	
10 TP_CPU_RSVD<16..1> == NC_CPU_RSVD<16..1> MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_PCIE1_D2RN == NC_PCIE1_D2RN MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_CRT_IG_RED == NC_CRT_IG_RED MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_SATA_C_R2D_CN == NC_SATA_C_R2D_CN MAKE_BASE=TRUE NO_TEST=TRUE		21 TP_PCH_TP1 == NC_PCH_TP1 MAKE_BASE=TRUE NO_TEST=TRUE	
10 TP_CPU_RSVD<46..19> == NC_CPU_RSVD<46..19> MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_PCIE1_D2RP == NC_PCIE1_D2RP MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_CRT_IG_GREEN == NC_CRT_IG_GREEN MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_SATA_C_R2D_CP == NC_SATA_C_R2D_CP MAKE_BASE=TRUE NO_TEST=TRUE		21 TP_PCH_TP2 == NC_PCH_TP2 MAKE_BASE=TRUE NO_TEST=TRUE	
10 CPU_CFG<15..12> == TP_CPU_CFG<15..12> MAKE_BASE=TRUE		10 TP_PCIE1_R2D_CN == NC_PCIE1_R2D_CN MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_CRT_IG_BLUE == NC_CRT_IG_BLUE MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_SATA_C_D2RN == NC_SATA_C_D2RN MAKE_BASE=TRUE NO_TEST=TRUE		21 TP_PCH_TP3 == NC_PCH_TP3 MAKE_BASE=TRUE NO_TEST=TRUE	
CPU Memory		10 TP_PCIE1_R2D_CP == NC_PCIE1_R2D_CP MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_CRT_IG_HSYNC == NC_CRT_IG_HSYNC MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_SATA_C_D2RP == NC_SATA_C_D2RP MAKE_BASE=TRUE NO_TEST=TRUE		21 TP_PCH_TP4 == NC_PCH_TP4 MAKE_BASE=TRUE NO_TEST=TRUE	
12 TP_MEM_A_DQ_CB<7..0> == NC_MEM_A_DQ_CB<7..0> MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_PCIE2_D2RN == NC_PCIE2_D2RN MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_CRT_IG_VSYNC == NC_CRT_IG_VSYNC MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_SATA_D_R2D_CN == NC_SATA_D_R2D_CN MAKE_BASE=TRUE NO_TEST=TRUE		21 TP_PCH_TP5 == NC_PCH_TP5 MAKE_BASE=TRUE NO_TEST=TRUE	
12 TP_MEM_A_DQS_N<8> == NC_MEM_A_DQS_N<8> MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_PCIE2_D2RP == NC_PCIE2_D2RP MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_CRT_IG_DDC_CLK == NC_CRT_IG_DDC_CLK MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_SATA_D_R2D_CP == NC_SATA_D_R2D_CP MAKE_BASE=TRUE NO_TEST=TRUE		21 TP_PCH_TP6 == NC_PCH_TP6 MAKE_BASE=TRUE NO_TEST=TRUE	
12 TP_MEM_A_DQS_P<8> == NC_MEM_A_DQSPX<8> MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_PCIE2_R2D_CN == NC_PCIE2_R2D_CN MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_CRT_IG_DDC_DATA == NC_CRT_IG_DDC_DATA MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_SATA_D_D2RN == NC_SATA_D_D2RN MAKE_BASE=TRUE NO_TEST=TRUE		21 TP_PCH_TP7 == NC_PCH_TP7 MAKE_BASE=TRUE NO_TEST=TRUE	
12 TP_MEM_B_DQ_CB<7..0> == NC_MEM_B_DQ_CB<7..0> MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_PCIE2_R2D_CP == NC_PCIE2_R2D_PNX MAKE_BASE=TRUE NO_TEST=TRUE		10 DP_IG_B_MLN<3..0> == NC_DP_IG_B_MLN<3..0> MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_SATA_D_D2RP == NC_SATA_D_D2RPX MAKE_BASE=TRUE NO_TEST=TRUE		21 TP_PCH_TP8 == NC_PCH_TP8 MAKE_BASE=TRUE NO_TEST=TRUE	
12 TP_MEM_B_DQS_N<8> == NC_MEM_B_DQSNX<8> MAKE_BASE=TRUE NO_TEST=TRUE		10 DMI_MIDBUS_CLK100M_N == NC_DMI_MIDBUS_CLK100NX MAKE_BASE=TRUE NO_TEST=TRUE		10 DP_IG_B_MLP<3..0> == NC_DP_IG_B_MLPX<3..0> MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_SATA_E_R2D_CN == NC_SATA_E_R2D_CN MAKE_BASE=TRUE NO_TEST=TRUE		21 TP_PCH_TP9 == NC_PCH_TP9 MAKE_BASE=TRUE NO_TEST=TRUE	
12 TP_MEM_B_DQS_P<8> == NC_MEM_B_DQSPX<8> MAKE_BASE=TRUE NO_TEST=TRUE		10 DMI_MIDBUS_CLK100M_P == NC_DMI_MIDBUS_CLK100PX MAKE_BASE=TRUE NO_TEST=TRUE		10 DP_IG_B_AUX_N == NC_DP_IG_B_AUXNX MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_SATA_E_R2D_CP == NC_SATA_E_R2D_CPX MAKE_BASE=TRUE NO_TEST=TRUE		21 TP_PCH_TP10 == NC_PCH_TP10 MAKE_BASE=TRUE NO_TEST=TRUE	
		10 TP_PCIE_CLK100M_PE0N == NC_PCIE_CLK100M_PE0NX MAKE_BASE=TRUE NO_TEST=TRUE		10 DP_IG_B_AUX_P == NC_DP_IG_B_AUXPX MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_SATA_E_D2RN == NC_SATA_E_D2RNX MAKE_BASE=TRUE NO_TEST=TRUE		21 TP_PCH_TP11 == NC_PCH_TP11 MAKE_BASE=TRUE NO_TEST=TRUE	
		10 TP_PCIE_CLK100M_PE0P == NC_PCIE_CLK100M_PE0PX MAKE_BASE=TRUE NO_TEST=TRUE		10 DP_IG_B_HPD == NC_DP_IG_B_HPDX MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_SATA_E_D2RP == NC_SATA_E_D2RPX MAKE_BASE=TRUE NO_TEST=TRUE		21 TP_PCH_TP12 == NC_PCH_TP12 MAKE_BASE=TRUE NO_TEST=TRUE	
		10 TP_PCIE_CLK100M_PE4N == NC_PCIE_CLK100M_PE4NX MAKE_BASE=TRUE NO_TEST=TRUE		10 DP_IG_B_DDC_CLK == NC_DP_IG_B_DDC_CLKX MAKE_BASE=TRUE NO_TEST=TRUE				21 TP_PCH_TP13 == NC_PCH_TP13 MAKE_BASE=TRUE NO_TEST=TRUE	
		10 TP_PCIE_CLK100M_PE4P == NC_PCIE_CLK100M_PE4PX MAKE_BASE=TRUE NO_TEST=TRUE		10 DP_IG_B_DDC_DATA == NC_DP_IG_B_DDC_DATAX MAKE_BASE=TRUE NO_TEST=TRUE				21 TP_PCH_TP14 == NC_PCH_TP14 MAKE_BASE=TRUE NO_TEST=TRUE	
		10 TP_PCIE_CLK100M_PE5N == NC_PCIE_CLK100M_PESNX MAKE_BASE=TRUE NO_TEST=TRUE		10 DP_IG_C_MLN<3..0> == NC_DP_IG_C_MLN<3..0> MAKE_BASE=TRUE NO_TEST=TRUE				21 TP_PCH_TP15 == NC_PCH_TP15 MAKE_BASE=TRUE NO_TEST=TRUE	
		10 TP_PCIE_CLK100M_PE5P == NC_PCIE_CLK100M_PESPX MAKE_BASE=TRUE NO_TEST=TRUE		10 DP_IG_C_MLP<3..0> == NC_DP_IG_C_MLPX<3..0> MAKE_BASE=TRUE NO_TEST=TRUE				21 TP_PCH_TP16 == NC_PCH_TP16 MAKE_BASE=TRUE NO_TEST=TRUE	
		21 TP_PCIE_CLK100M_PE6N == NC_PCIE_CLK100M_PEGNX MAKE_BASE=TRUE NO_TEST=TRUE		10 DP_IG_C_AUX_N == NC_DP_IG_C_AUXNX MAKE_BASE=TRUE NO_TEST=TRUE				21 TP_PCH_TP17 == NC_PCH_TP17 MAKE_BASE=TRUE NO_TEST=TRUE	
		21 TP_PCIE_CLK100M_PE6P == NC_PCIE_CLK100M_PEGPX MAKE_BASE=TRUE NO_TEST=TRUE		10 DP_IG_C_AUX_P == NC_DP_IG_C_AUXPX MAKE_BASE=TRUE NO_TEST=TRUE				21 TP_PCH_TP18 == NC_PCH_TP18 MAKE_BASE=TRUE NO_TEST=TRUE	
		21 TP_PCIE_CLK100M_PE7N == NC_PCIE_CLK100M_PETNX MAKE_BASE=TRUE NO_TEST=TRUE		10 DP_IG_C_HPD == NC_DP_IG_C_HPDX MAKE_BASE=TRUE NO_TEST=TRUE				21 TP_PCH_TP19 == NC_PCH_TP19 MAKE_BASE=TRUE NO_TEST=TRUE	
		21 TP_PCIE_CLK100M_PE7P == NC_PCIE_CLK100M_PETPX MAKE_BASE=TRUE NO_TEST=TRUE		10 DP_IG_C_CTRL_CLK == NC_DP_IG_C_CTRL_CLKX MAKE_BASE=TRUE NO_TEST=TRUE				21 TP_PCH_TP20 == NC_PCH_TP20 MAKE_BASE=TRUE NO_TEST=TRUE	
				10 DP_IG_C_CTRL_DATA == NC_DP_IG_C_CTRL_DATAX MAKE_BASE=TRUE NO_TEST=TRUE					
		10 TP_PE_TX_N<3..0> == NC_PE_TNX<3..0> MAKE_BASE=TRUE NO_TEST=TRUE		10 DP_IG_D_MLN<3..0> == NC_DP_IG_D_MLN<3..0> MAKE_BASE=TRUE NO_TEST=TRUE		PCH Reserved		PCH PCI	
		10 TP_PE_TX_P<3..0> == NC_PE_TPX<3..0> MAKE_BASE=TRUE NO_TEST=TRUE		10 DP_IG_D_MLP<3..0> == NC_DP_IG_D_MLPX<3..0> MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_PCH_RESERVE_0 == NC_PCH_RESERVE_0 MAKE_BASE=TRUE NO_TEST=TRUE		20 TP_PCI_AD<31..0> == NC_PCI_AD<31..0> MAKE_BASE=TRUE NO_TEST=TRUE	
		10 TP_PE_RX_N<3..0> == NC_PE_RXN<3..0> MAKE_BASE=TRUE NO_TEST=TRUE		10 DP_IG_D_AUXN == NC_DP_IG_D_AUXNX MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_PCH_RESERVE_1 == NC_PCH_RESERVE_1 MAKE_BASE=TRUE NO_TEST=TRUE		20 TP_PCI_C_BE_L<3..0> == NC_PCI_C_BE_L<3..0> MAKE_BASE=TRUE NO_TEST=TRUE	
		10 TP_PE_RX_P<3..0> == NC_PE_RPX<3..0> MAKE_BASE=TRUE NO_TEST=TRUE		10 DP_IG_D_AUXP == NC_DP_IG_D_AUXPX MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_PCH_RESERVE_2 == NC_PCH_RESERVE_2 MAKE_BASE=TRUE NO_TEST=TRUE		20 TP_PCI_PAR == NC_PCI_PAR MAKE_BASE=TRUE NO_TEST=TRUE	
				10 DP_IG_D_HPD == NC_DP_IG_D_HPDX MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_PCH_RESERVE_3 == NC_PCH_RESERVE_3 MAKE_BASE=TRUE NO_TEST=TRUE		20 TP_PCI_RESET_L == NC_PCI_RESET_L MAKE_BASE=TRUE NO_TEST=TRUE	
				10 DP_IG_D_CTRL_CLK == NC_DP_IG_D_CTRL_CLKX MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_PCH_RESERVE_4 == NC_PCH_RESERVE_4 MAKE_BASE=TRUE NO_TEST=TRUE		20 TP_PCH_PCI_GNT0_L == NC_PCH_PCI_GNT0_L MAKE_BASE=TRUE NO_TEST=TRUE	
				10 DP_IG_D_CTRL_DATA == NC_DP_IG_D_CTRL_DATAX MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_PCH_RESERVE_5 == NC_PCH_RESERVE_5 MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_PCH_INIT3V3_L == NC_PCH_INIT3V3_L MAKE_BASE=TRUE NO_TEST=TRUE	
		PCH USB				10 TP_PCH_RESERVE_6 == NC_PCH_RESERVE_6 MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_LPC_DREQ0_L == NC_LPC_DREQ0_L MAKE_BASE=TRUE NO_TEST=TRUE	
		20 USB_PCH_4_N == NC_USB_PCH_4NX MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_SDVO_TVCLKINN == NC_SDVO_TVCLKINNXX MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_PCH_RESERVE_7 == NC_PCH_RESERVE_7 MAKE_BASE=TRUE NO_TEST=TRUE		PCH Miscellaneous	
		20 USB_PCH_4_P == NC_USB_PCH_4PX MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_SDVO_TVCLKINP == NC_SDVO_TVCLKINPX MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_PCH_RESERVE_8 == NC_PCH_RESERVE_8 MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_HDA_SDIN1 == NC_HDA_SDIN1 MAKE_BASE=TRUE NO_TEST=TRUE	
		20 USB_PCH_5_N == NC_USB_PCH_5NX MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_SDVO_STALLN == NC_SDVO_STALLNX MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_PCH_RESERVE_9 == NC_PCH_RESERVE_9 MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_HDA_SDIN2 == NC_HDA_SDIN2 MAKE_BASE=TRUE NO_TEST=TRUE	
		20 USB_PCH_5_P == NC_USB_PCH_5PX MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_SDVO_STALLP == NC_SDVO_STALLPX MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_PCH_RESERVE_10 == NC_PCH_RESERVE_10 MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_HDA_SDIN3 == NC_HDA_SDIN3 MAKE_BASE=TRUE NO_TEST=TRUE	
		20 USB_PCH_6_N == NC_USB_PCH_6NX MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_SDVO_INTN == NC_SDVO_INTNX MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_PCH_RESERVE_11 == NC_PCH_RESERVE_11 MAKE_BASE=TRUE NO_TEST=TRUE		21 TP_PCH_PWM0 == NC_PCH_PWM0 MAKE_BASE=TRUE NO_TEST=TRUE	
		20 USB_PCH_6_P == NC_USB_PCH_6PX MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_SDVO_INTP == NC_SDVO_INTPX MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_PCH_RESERVE_12 == NC_PCH_RESERVE_12 MAKE_BASE=TRUE NO_TEST=TRUE		21 TP_PCH_PWM1 == NC_PCH_PWM1 MAKE_BASE=TRUE NO_TEST=TRUE	
		20 USB_PCH_11_N == NC_USB_PCH_11NX MAKE_BASE=TRUE NO_TEST=TRUE				10 TP_PCH_RESERVE_13 == NC_PCH_RESERVE_13 MAKE_BASE=TRUE NO_TEST=TRUE		21 TP_PCH_PWM2 == NC_PCH_PWM2 MAKE_BASE=TRUE NO_TEST=TRUE	
		20 USB_PCH_11_P == NC_USB_PCH_11PX MAKE_BASE=TRUE NO_TEST=TRUE				10 TP_PCH_RESERVE_14 == NC_PCH_RESERVE_14 MAKE_BASE=TRUE NO_TEST=TRUE		21 TP_PCH_PWM3 == NC_PCH_PWM3 MAKE_BASE=TRUE NO_TEST=TRUE	
		20 USB_PCH_12_N == NC_USB_PCH_12NX MAKE_BASE=TRUE NO_TEST=TRUE				10 TP_PCH_RESERVE_15 == NC_PCH_RESERVE_15 MAKE_BASE=TRUE NO_TEST=TRUE		21 TP_PCH_SST == NC_PCH_SST MAKE_BASE=TRUE NO_TEST=TRUE	
		20 USB_PCH_12_P == NC_USB_PCH_12PX MAKE_BASE=TRUE NO_TEST=TRUE				10 TP_PCH_RESERVE_16 == NC_PCH_RESERVE_16 MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_PCH_CL_CLK1 == NC_PCH_CL_CLK1 MAKE_BASE=TRUE NO_TEST=TRUE	
		20 USB_PCH_13_N == NC_USB_PCH_13NX MAKE_BASE=TRUE NO_TEST=TRUE				10 TP_PCH_RESERVE_17 == NC_PCH_RESERVE_17 MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_PCH_CL_DATA1 == NC_PCH_CL_DATA1 MAKE_BASE=TRUE NO_TEST=TRUE	
		20 USB_PCH_13_P == NC_USB_PCH_13PX MAKE_BASE=TRUE NO_TEST=TRUE				10 TP_PCH_RESERVE_18 == NC_PCH_RESERVE_18 MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_PCH_CL_RST1 == NC_PCH_CL_RST1 MAKE_BASE=TRUE NO_TEST=TRUE	
		PCH Clocks				10 TP_PCH_RESERVE_19 == NC_PCH_RESERVE_19 MAKE_BASE=TRUE NO_TEST=TRUE		20 TP_PCI_CLK33M_OUT2 == NC_PCI_CLK33M_OUT2 MAKE_BASE=TRUE NO_TEST=TRUE	
		10 TP_PCH_CLKOUT_DPN == NC_PCH_CLKOUT_DPNX MAKE_BASE=TRUE NO_TEST=TRUE				10 TP_PCH_RESERVE_20 == NC_PCH_RESERVE_20 MAKE_BASE=TRUE NO_TEST=TRUE		20 TP_PCI_CLK33M_OUT3 == NC_PCI_CLK33M_OUT3 MAKE_BASE=TRUE NO_TEST=TRUE	
		10 TP_PCH_CLKOUT_DPP == NC_PCH_CLKOUT_DPPX MAKE_BASE=TRUE NO_TEST=TRUE				10 TP_PCH_RESERVE_21 == NC_PCH_RESERVE_21 MAKE_BASE=TRUE NO_TEST=TRUE			
		10 PCH_CLK25M_XTALOUT == NC_PCH_CLK25M_XTALOUT MAKE_BASE=TRUE NO_TEST=TRUE				10 TP_PCH_RESERVE_22 == NC_PCH_RESERVE_22 MAKE_BASE=TRUE NO_TEST=TRUE			
		10 TP_PCH_GPI064_CLKOUTFLEX0 == NC_PCH_GPI064_CLKOUTFLEX0 MAKE_BASE=TRUE NO_TEST=TRUE				10 TP_PCH_RESERVE_23 == NC_PCH_RESERVE_23 MAKE_BASE=TRUE NO_TEST=TRUE			
		10 TP_PCH_GPI065_CLKOUTFLEX1 == NC_PCH_GPI065_CLKOUTFLEX1 MAKE_BASE=TRUE NO_TEST=TRUE				10 TP_PCH_RESERVE_24 == NC_PCH_RESERVE_24 MAKE_BASE=TRUE NO_TEST=TRUE			
		10 TP_PCH_GPI066_CLKOUTFLEX2 == NC_PCH_GPI066_CLKOUTFLEX2 MAKE_BASE=TRUE NO_TEST=TRUE				10 TP_PCH_RESERVE_25 == NC_PCH_RESERVE_25 MAKE_BASE=TRUE NO_TEST=TRUE			
		10 TP_PCH_GPI067_CLKOUTFLEX3 == NC_PCH_GPI067_CLKOUTFLEX3 MAKE_BASE=TRUE NO_TEST=TRUE				10 TP_PCH_RESERVE_26 == NC_PCH_RESERVE_26 MAKE_BASE=TRUE NO_TEST=TRUE			
						10 TP_PCH_RESERVE_27 == NC_PCH_RESERVE_27 MAKE_BASE=TRUE NO_TEST=TRUE			
						10 TP_PCH_RESERVE_28 == NC_PCH_RESERVE_28 MAKE_BASE=TRUE NO_TEST=TRUE			
						PCH and CPU FDI			
						10 TP_CPU_FDI_TX_N<7..0> == NC_CPU_FDI_TNX<7..0> MAKE_BASE=TRUE NO_TEST=TRUE			
						10 TP_CPU_FDI_TX_P<7..0> == NC_CPU_FDI_TPX<7..0> MAKE_BASE=TRUE NO_TEST=TRUE			
						10 PCH_FDI_RX_N<7..0> == NC_PCH_FDI_RNX<7..0> MAKE_BASE=TRUE NO_TEST=TRUE			
						10 PCH_FDI_RX_P<7..0> == NC_PCH_FDI_RPX<7..0> MAKE_BASE=TRUE NO_TEST=TRUE			
						10 TP_CPU_FDI_FSYNC<1..0> == NC_CPU_FDI_FSYNC<1..0> MAKE_BASE=TRUE NO_TEST=TRUE			
						10 TP_CPU_FDI_LSYNC<1..0> == NC_CPU_FDI_LSYNC<1..0> MAKE_BASE=TRUE NO_TEST=TRUE			
						10 TP_CPU_FDI_INT == NC_CPU_FDI_INT MAKE_BASE=TRUE NO_TEST=TRUE			
						10 PCH_FDI_FSYNC<1..0> == NC_PCH_FDI_FSYNC<1..0> MAKE_BASE=TRUE NO_TEST=TRUE			
						10 PCH_FDI_LSYNC<1..0> == NC_PCH_FDI_LSYNC<1..0> MAKE_BASE=TRUE NO_TEST=TRUE			
						10 PCH_FDI_INT == NC_PCH_FDI_INT MAKE_BASE=TRUE NO_TEST=TRUE			
8	7	6	5	4	3	2	1		

SYNC MASTER=D8.MLB.ULTIMATE

SYNC DATE=04/02/2012

Page Title

Unused Signal Aliases

Apple Inc.

DRAWING NUMBER

051-9504

SIZE

D

REVISION

5.3.0

NOTICE OF PROPRIETARY PROPERTY:
THE INFORMATION CONTAINED HEREIN IS THE
PROPRIETARY PROPERTY OF APPLE INC.
THE POSSESSOR AGREES TO THE FOLLOWING:
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
I NOT TO REPRODUCE OR COPY IT
I NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
I ALL RIGHTS RESERVED

PAGE

8 OF 143

SHEET

8 OF 117

SYNC MASTER=D8_MLB_ULTIMATE

SYNC DATE=04/02/2012

PAGE TITLE

Unused Signal Aliases

 Apple Inc.

DRAWING NUMBER
051-9504

SIZE
D

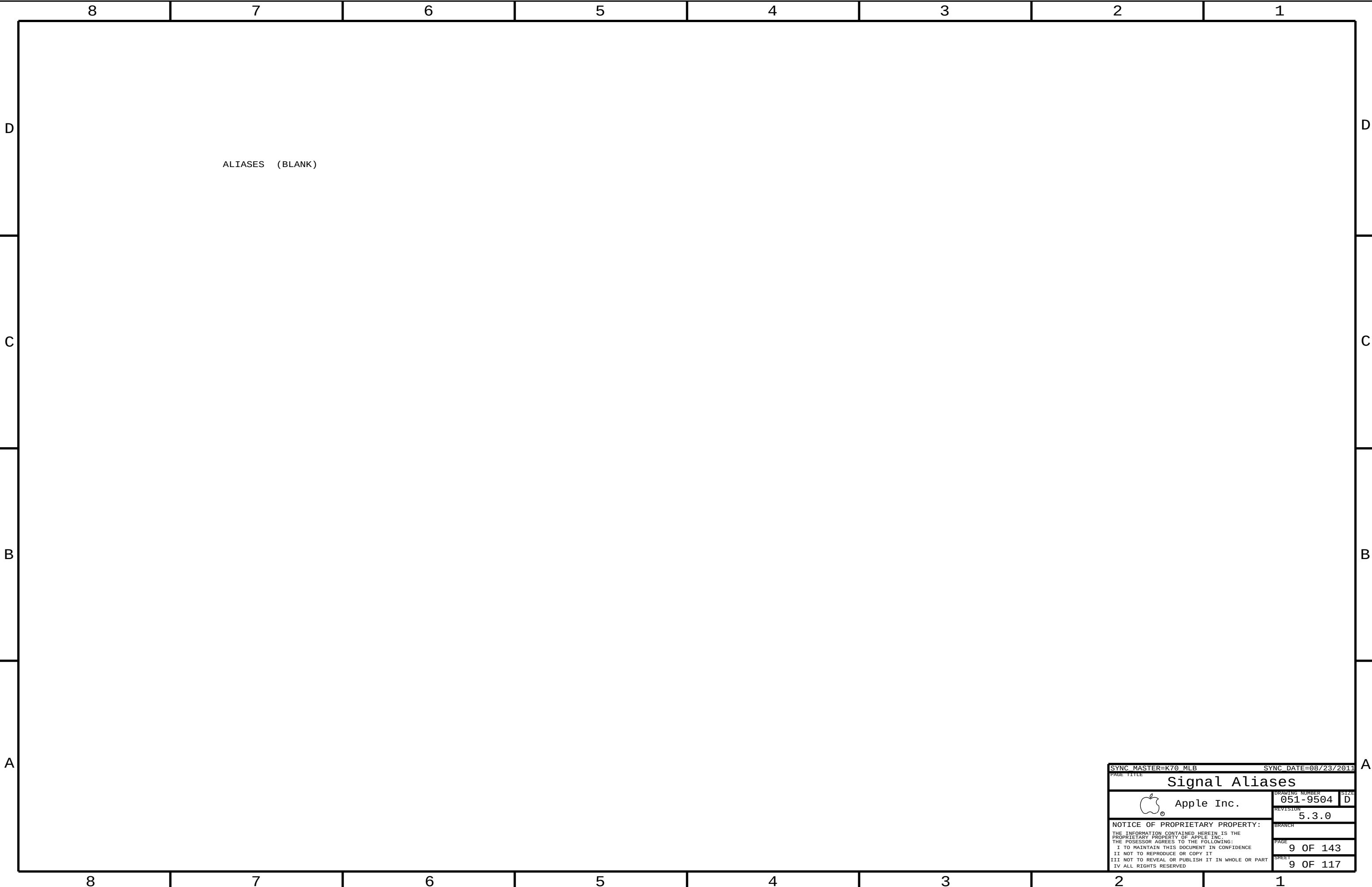
REVISION
5.3.0

BRANCH

NOTICE OF PROPRIETARY PROPERTY:
THE INFORMATION CONTAINED HEREIN IS THE
PROPRIETARY PROPERTY OF APPLE INC.
THE POSSESSOR AGREES TO THE FOLLOWING:
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
IV ALL RIGHTS RESERVED

PAGE
8 OF 143

SHEET
8 OF 117



SYNC MASTER=K70 MLB

SYNC DATE=08/23/2011

PAGE TITLE

Signal Aliases

 Apple Inc.

NOTICE OF PROPRIETARY PROPERTY:
THE INFORMATION CONTAINED HEREIN IS THE
PROPRIETARY PROPERTY OF APPLE INC.
THE POSSESSOR AGREES TO THE FOLLOWING:
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
IV ALL RIGHTS RESERVED

DRAWING NUMBER

051-9504

REVISION

5.3.0

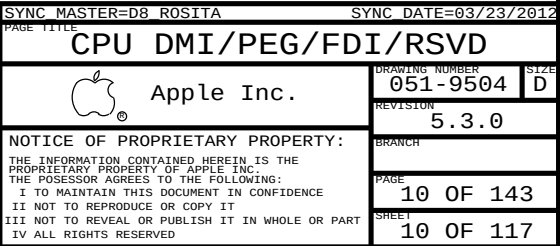
BRANCH

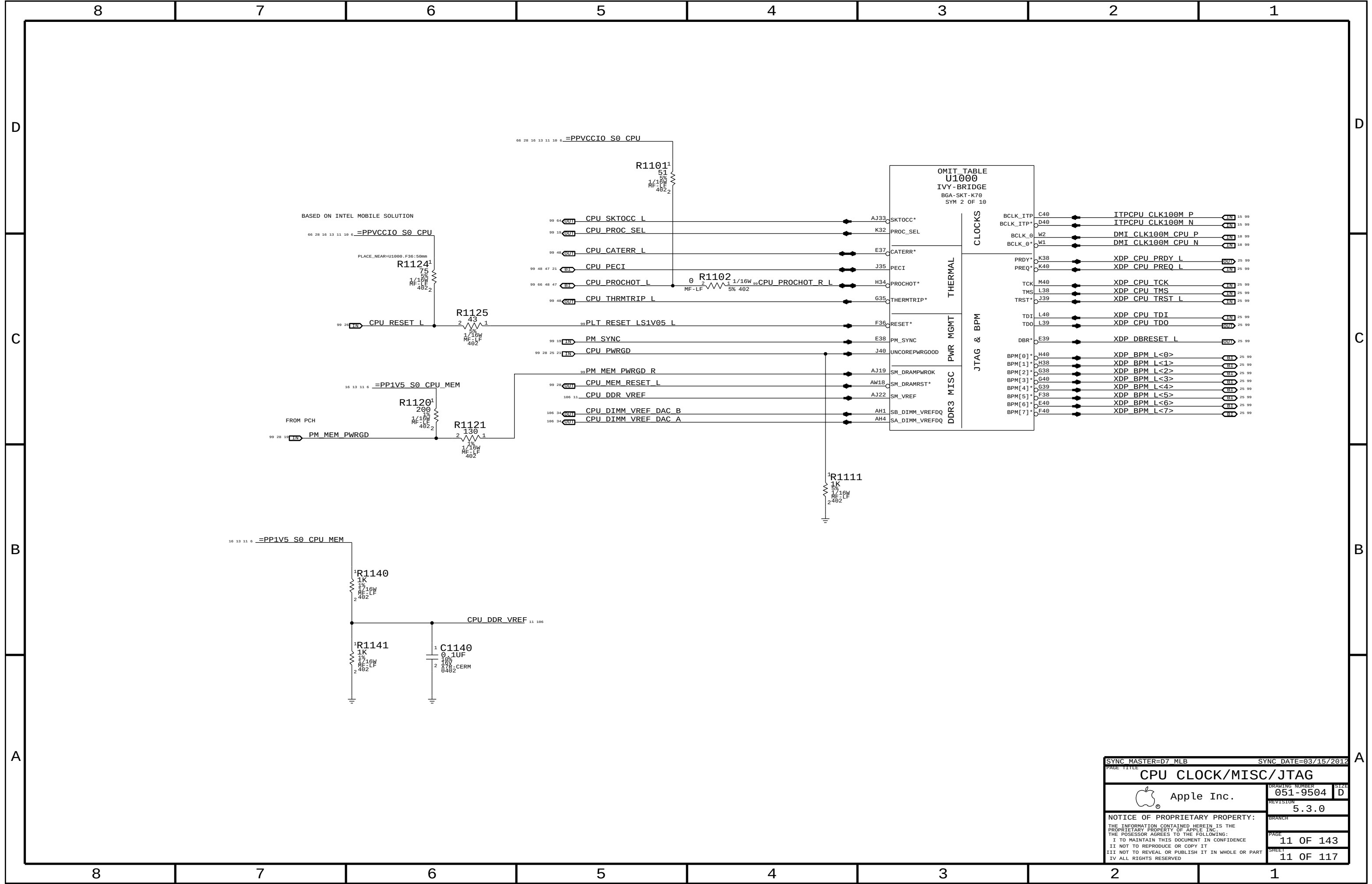
PAGE

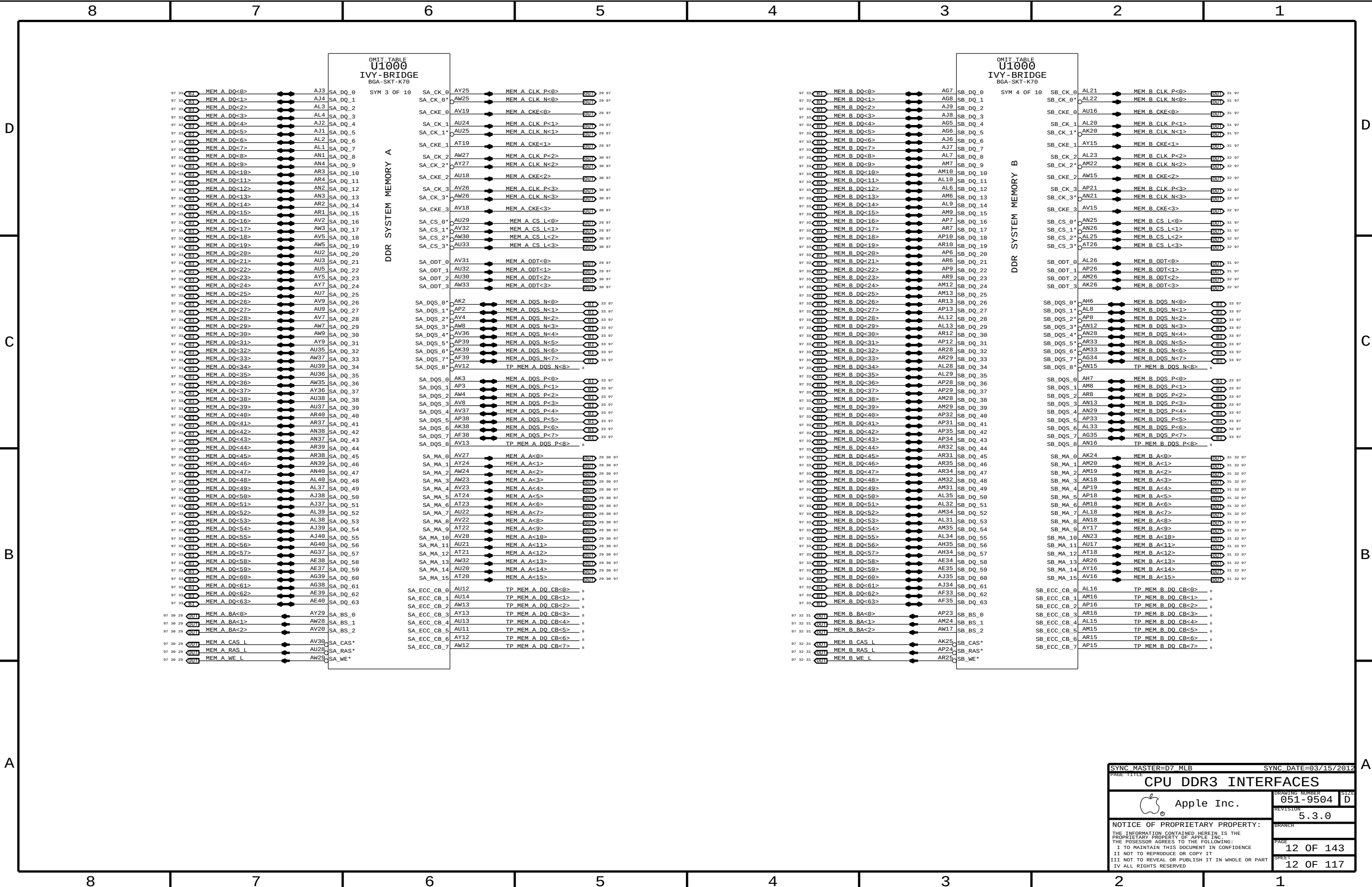
9 OF 143

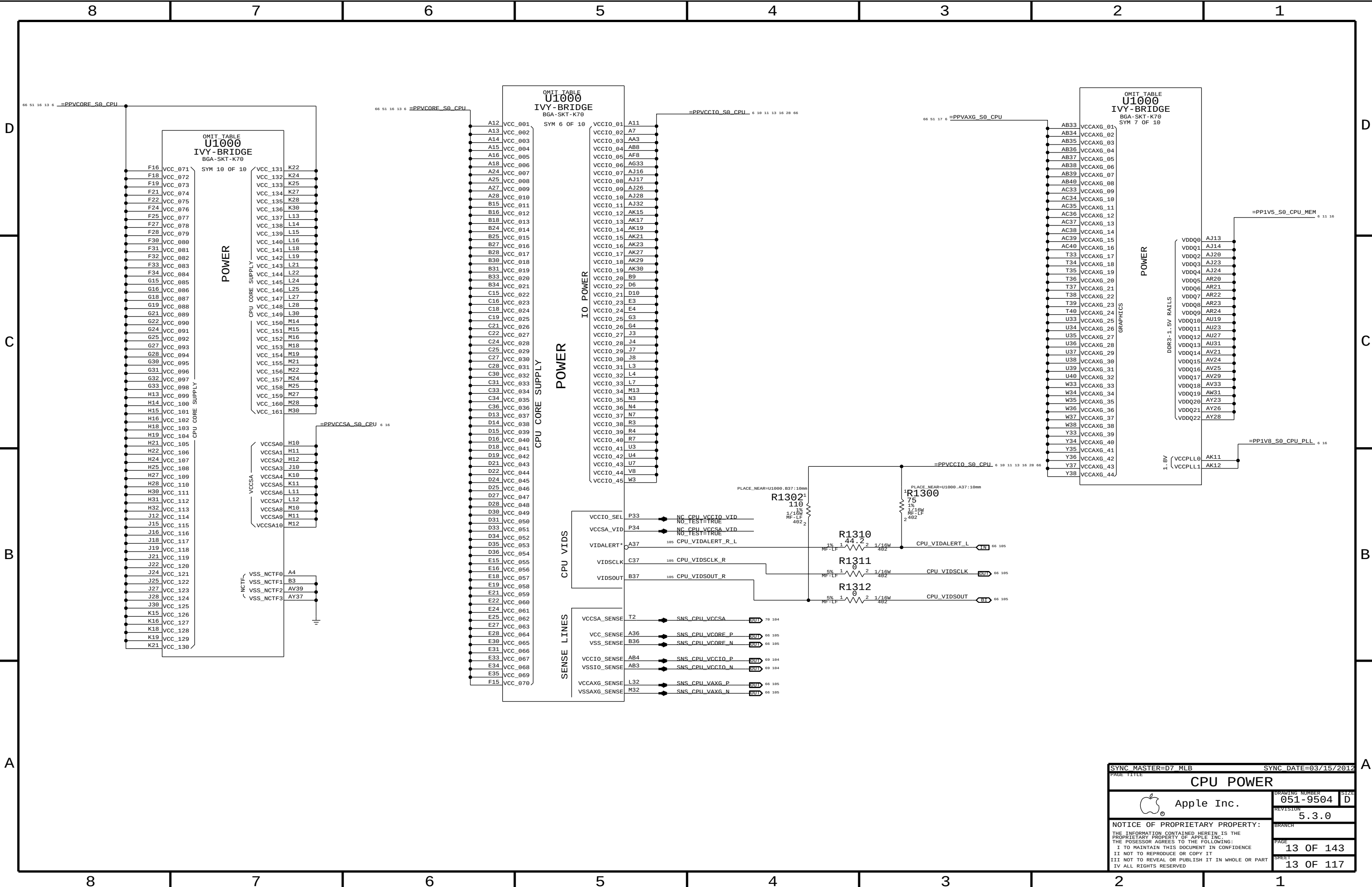
SHEET

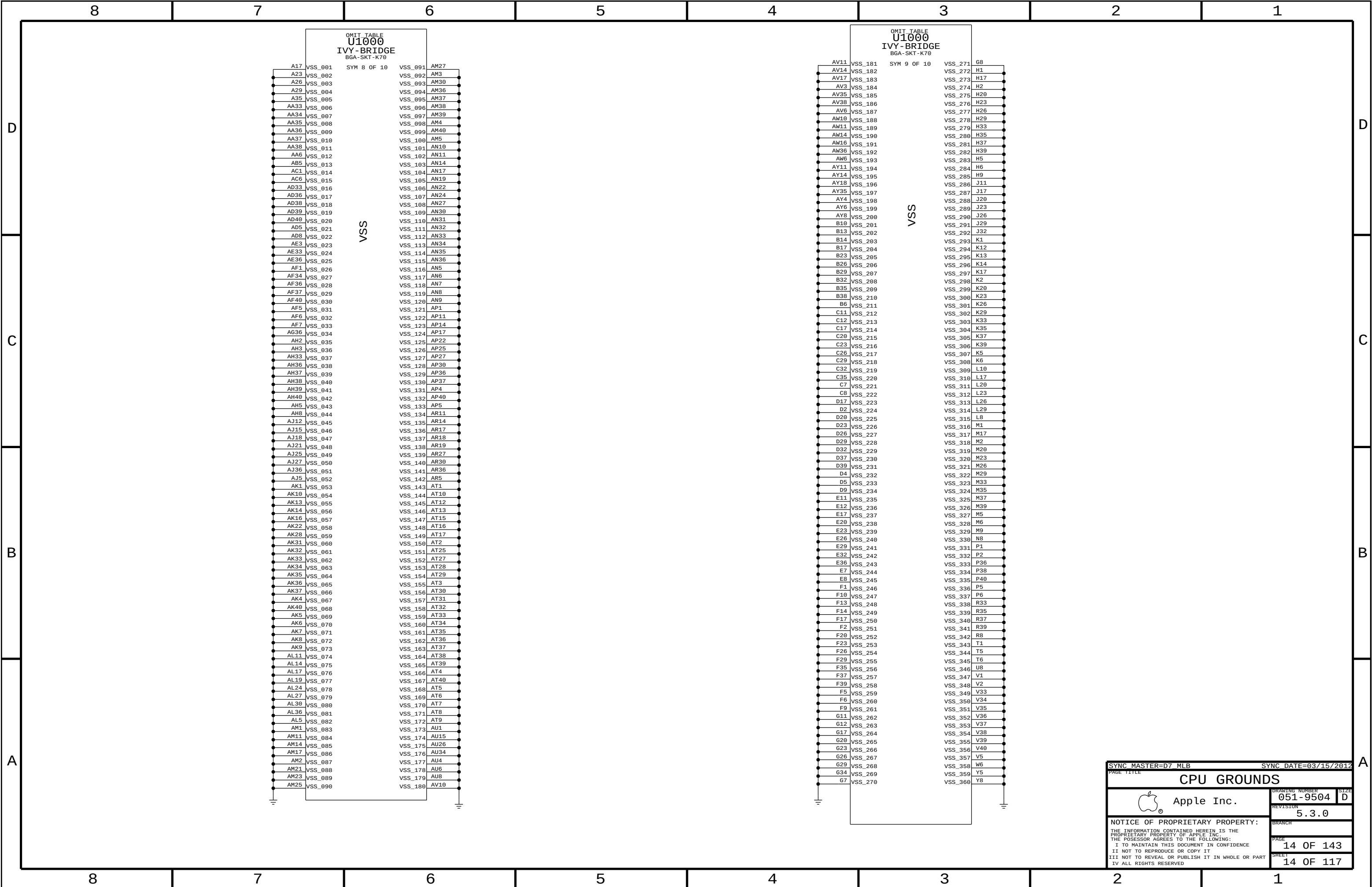
9 OF 117







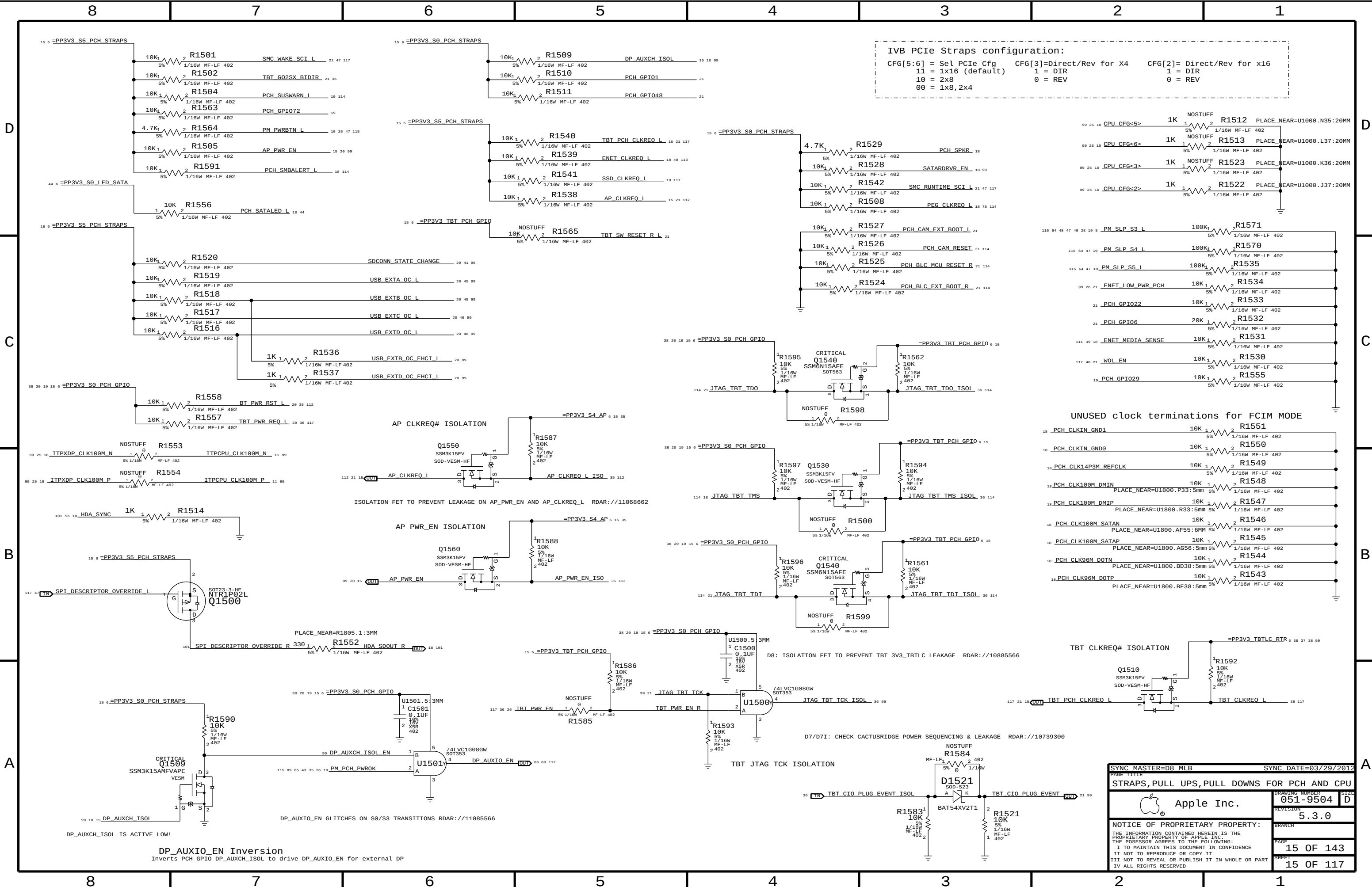




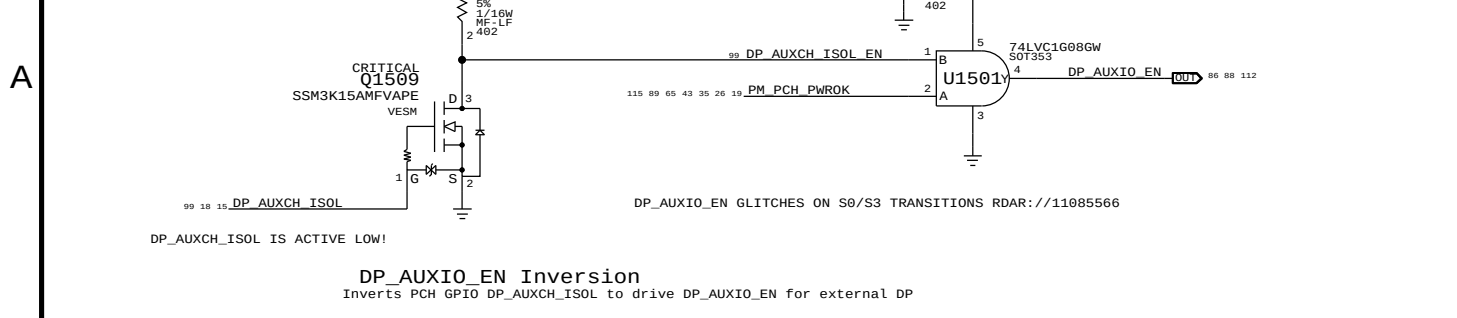
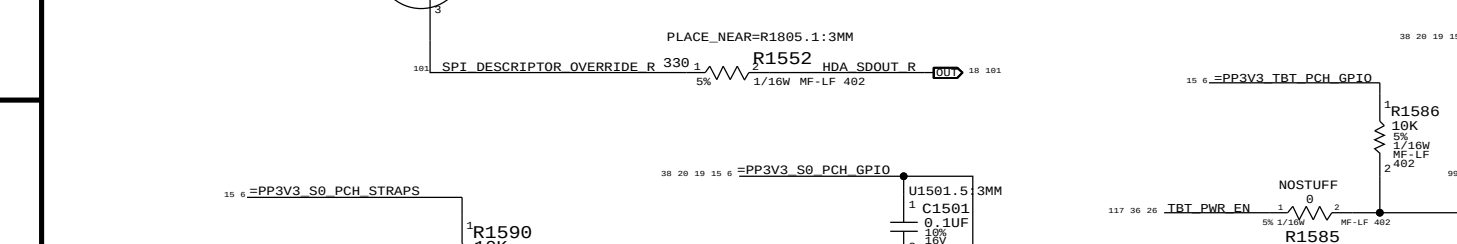
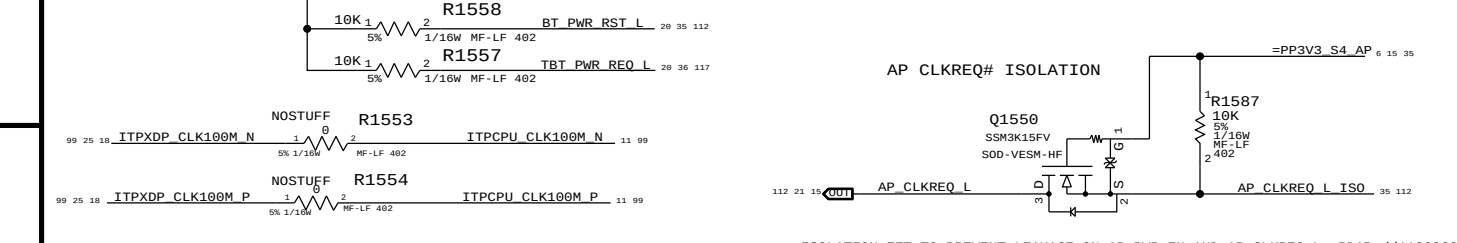
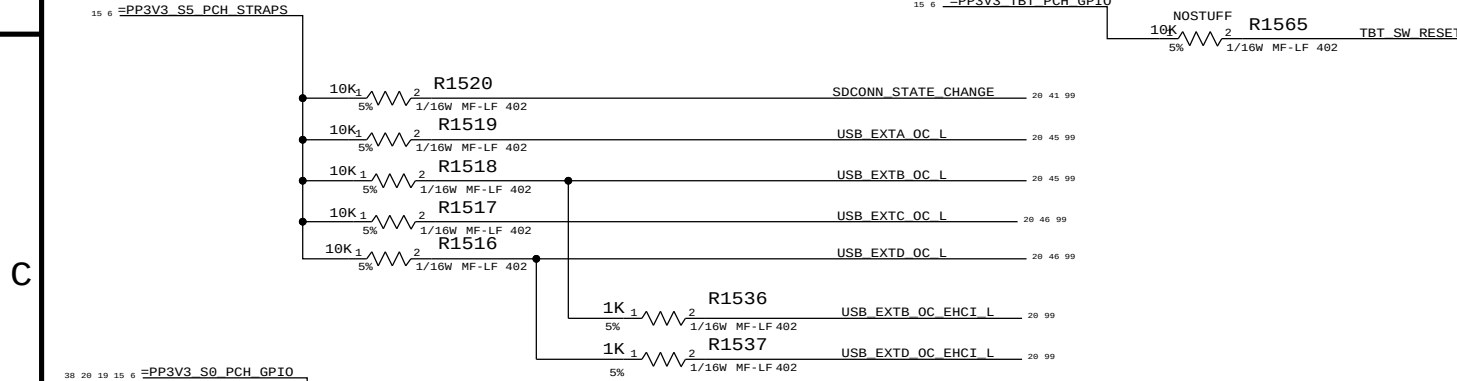
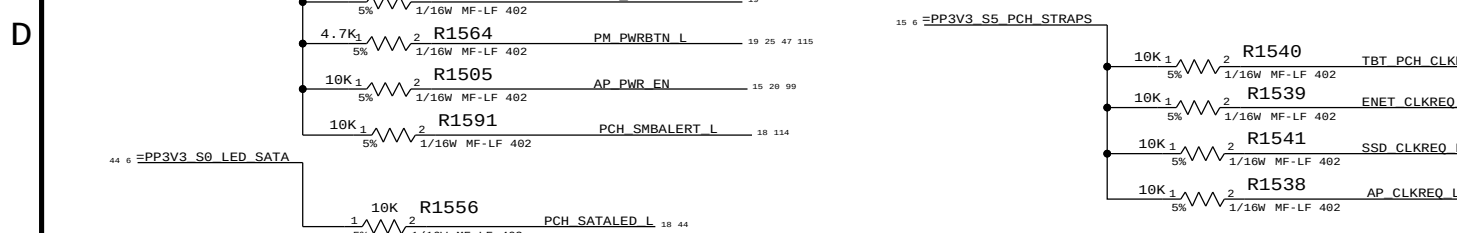
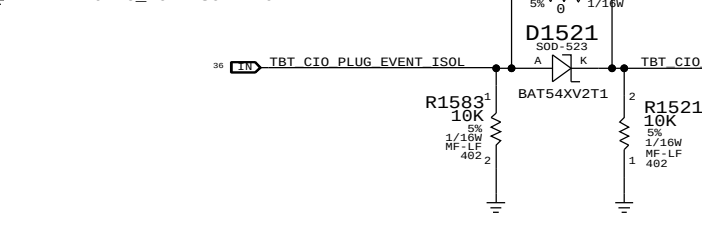
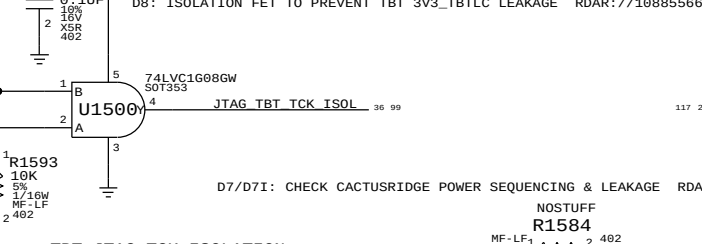
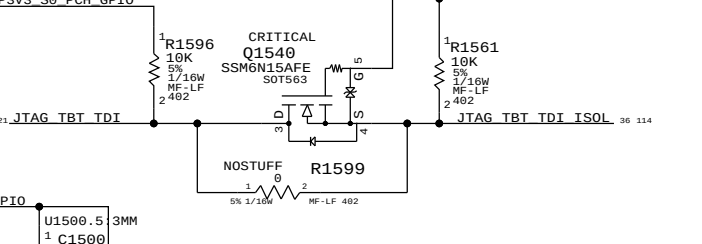
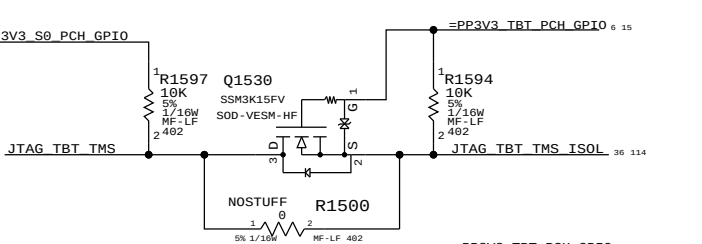
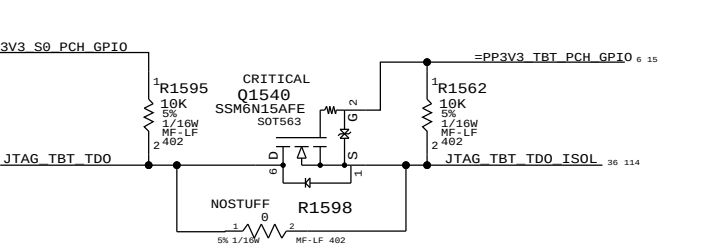
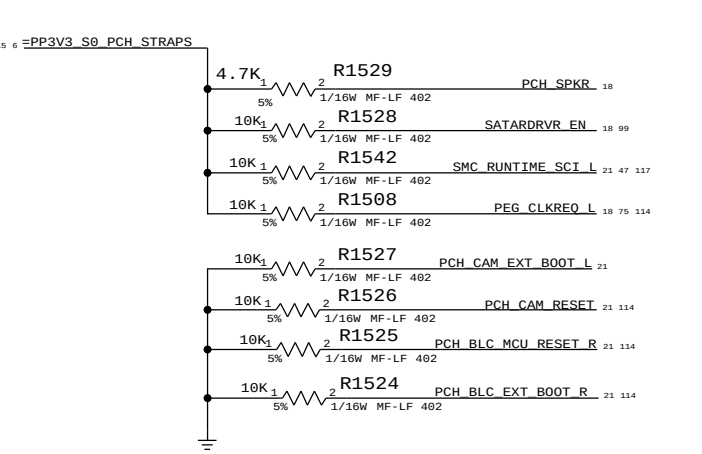
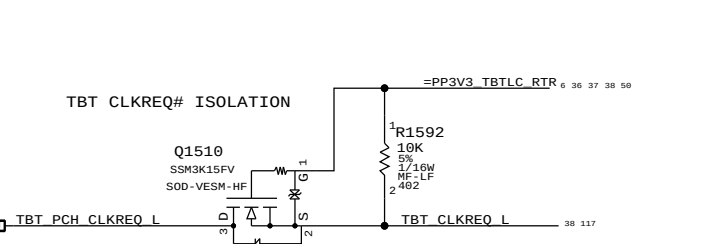
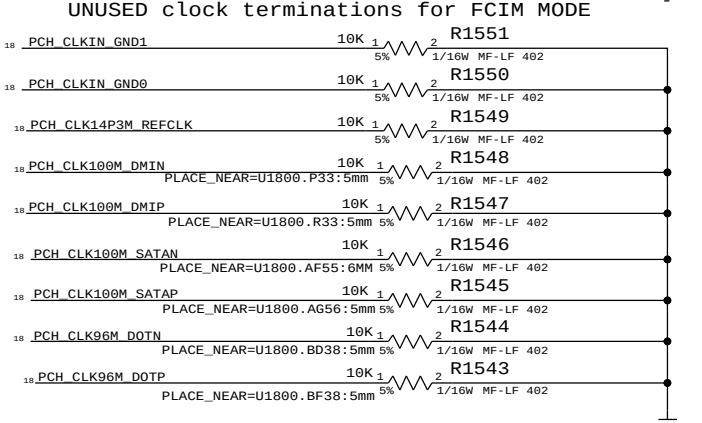
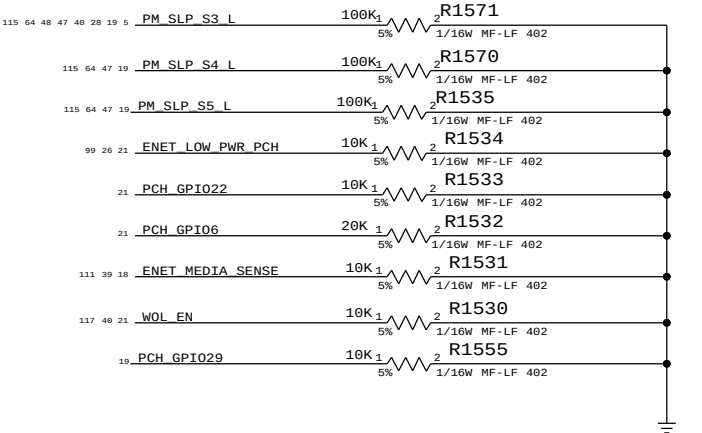
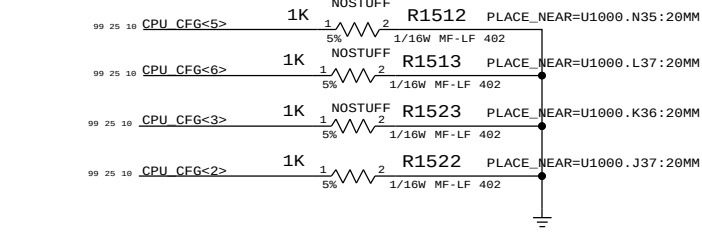
SYNC MASTER=D7_MLB

SYNC DATE=03/15/2012

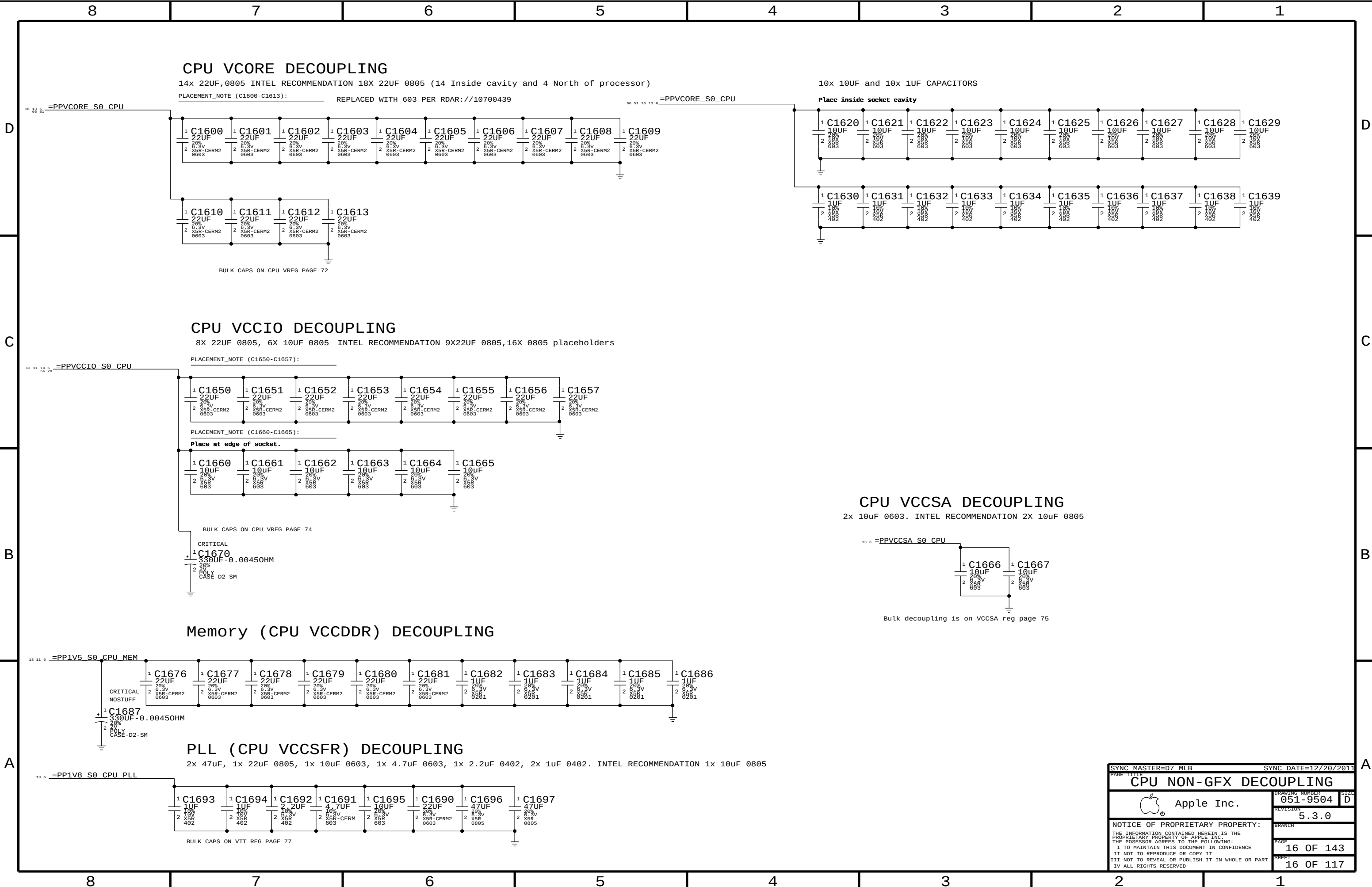
PAGE TITLE		CPU GROUNDS	
 Apple Inc.	DRAWING NUMBER	051-9504	SIZE
	REVISION	5.3.0	
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		BRANCH	
		PAGE	14 OF 143
		SHEET	14 OF 117



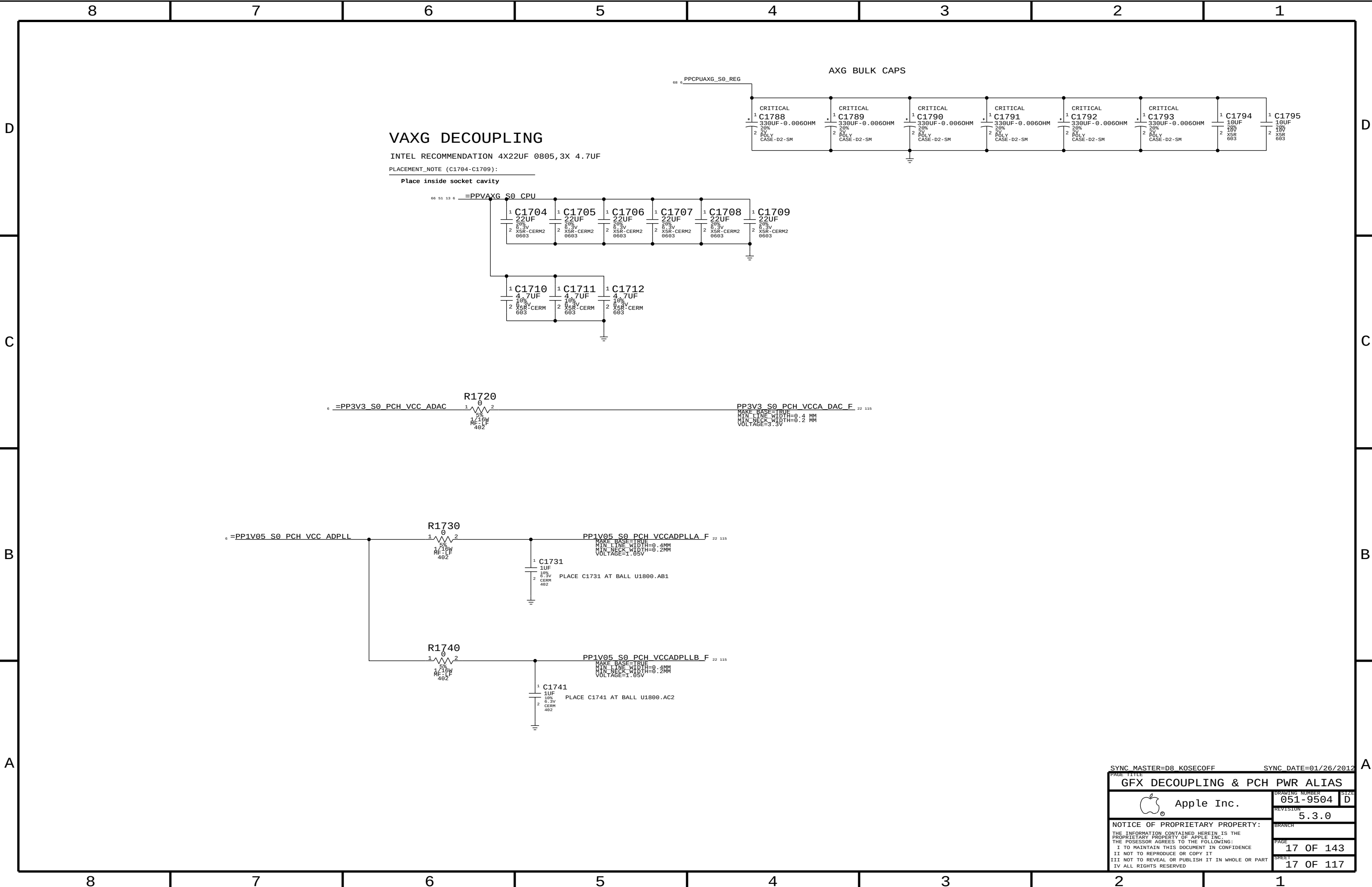
IVB PCIe Straps configuration:
CFG[5:6] = Sel PCIe Cfg CFG[3]=Direct/Rev for X4 CFG[2]= Direct/Rev for x16
11 = 1x16 (default) 1 = DIR 1 = DIR
10 = 2x8 0 = REV 0 = REV
00 = 1x8,2x4



SYNC MASTER=D8_MLB		SYNC DATE=03/29/2012	
PAGE TITLE		STRAPS,PULL UPS,PULL DOWNS FOR PCH AND CPU	
		DRAWING NUMBER	051-9504
		REVISION	5.3.0
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:		PAGE	15 OF 143
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		SHEET	15 OF 117
II NOT TO REPRODUCE OR COPY IT			
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IV ALL RIGHTS RESERVED			



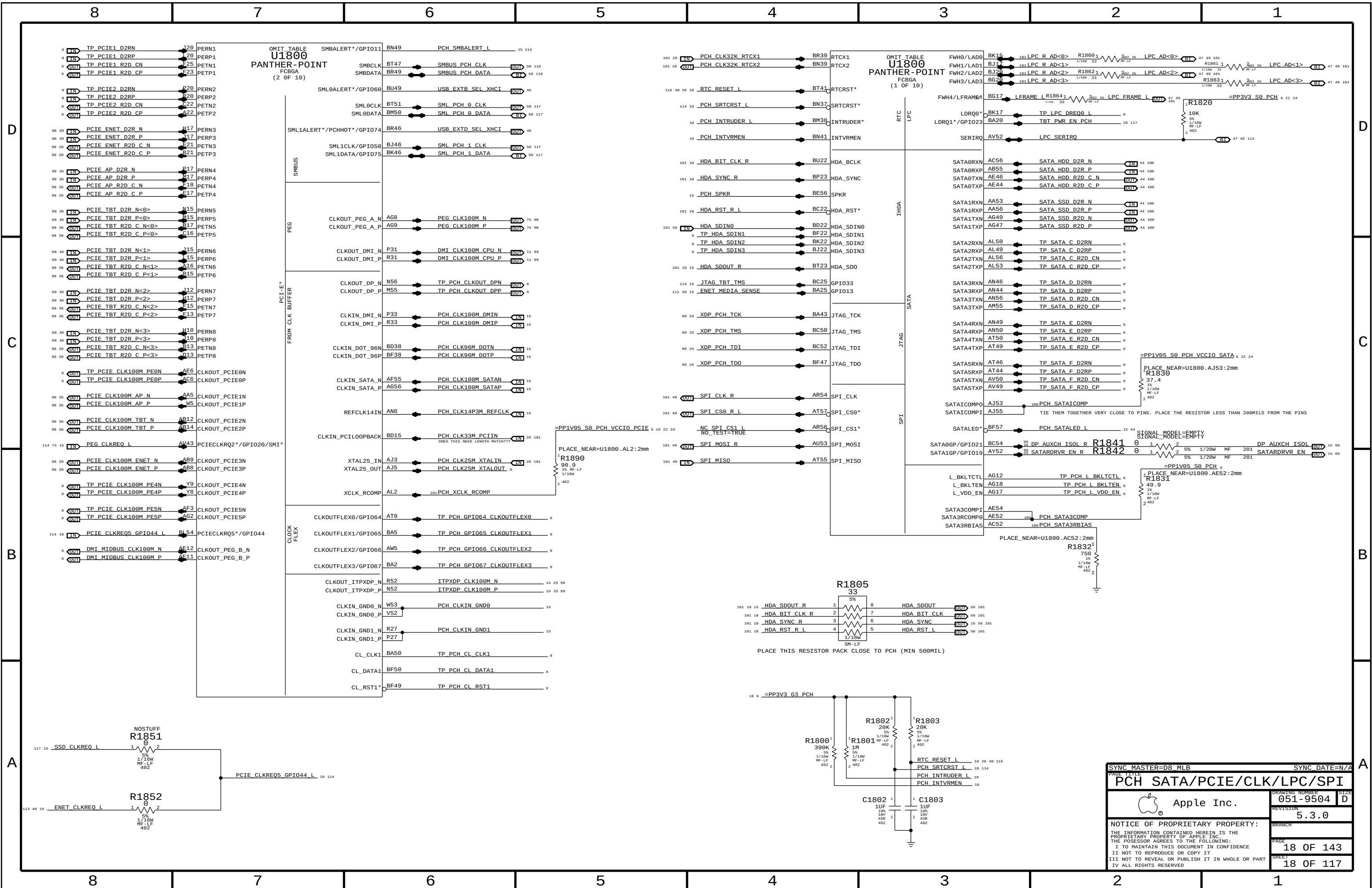
SYNC MASTER=D7_MLB		SYNC DATE=12/20/2011	
PAGE TITLE			
CPU NON-GFX DECOUPLING			
 Apple Inc.	DRAWING NUMBER		SIZE
	051-9504		D
	REVISION		
		5.3.0	
NOTICE OF PROPRIETARY PROPERTY:			
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC.			
THE POSSESSOR AGREES TO THE FOLLOWING:			
I I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE			
II NOT TO REPRODUCE OR COPY IT			
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IV ALL RIGHTS RESERVED			
		BRANCH	
		PAGE	16 OF 143
		SHEET	16 OF 117

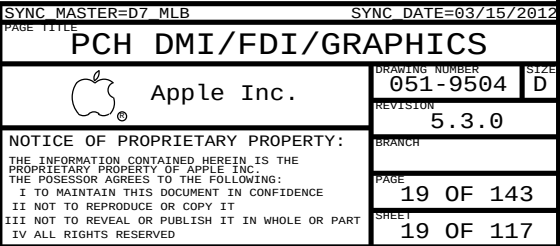


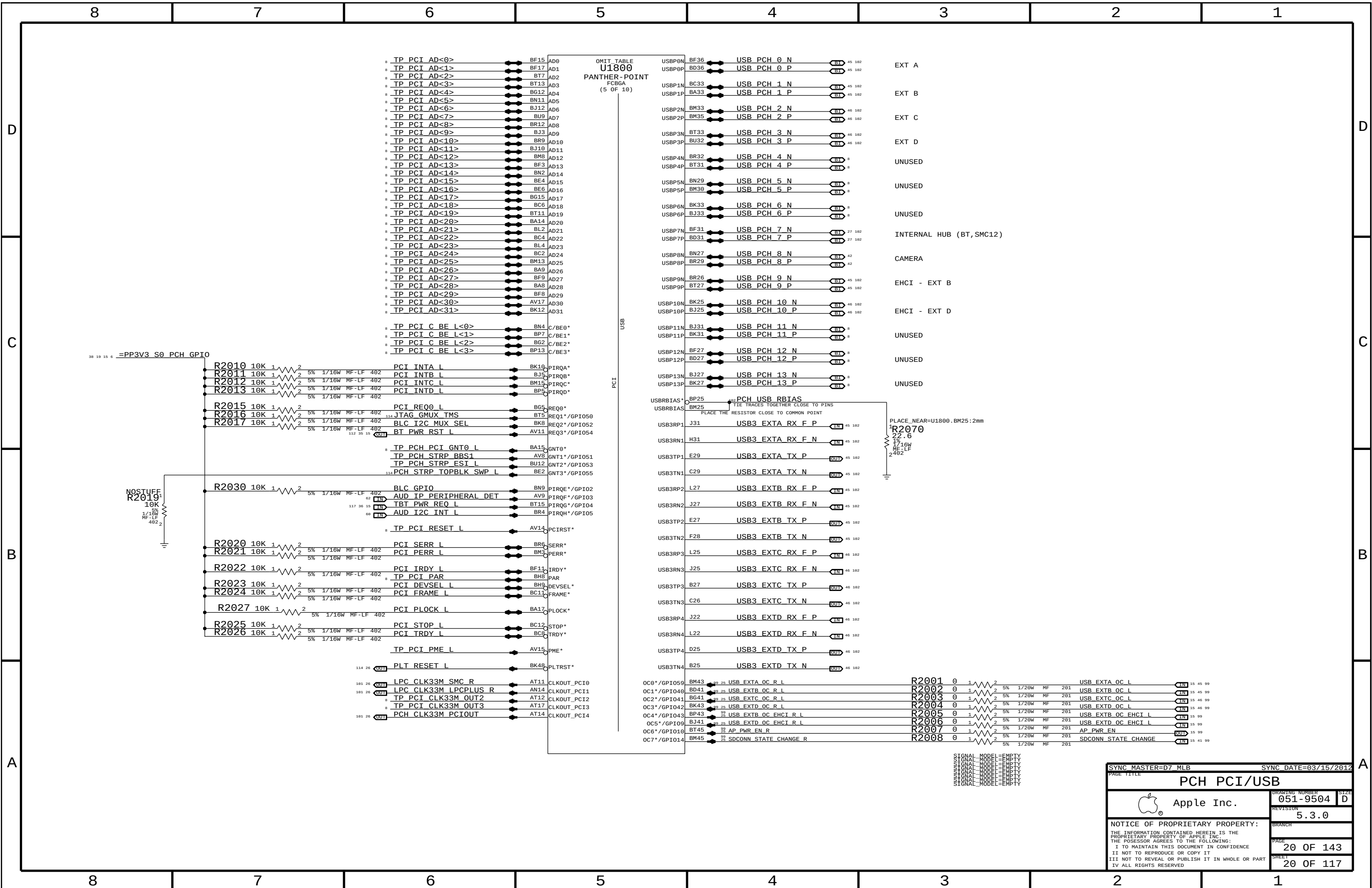
SYNC MASTER=D8 KOSECOFF

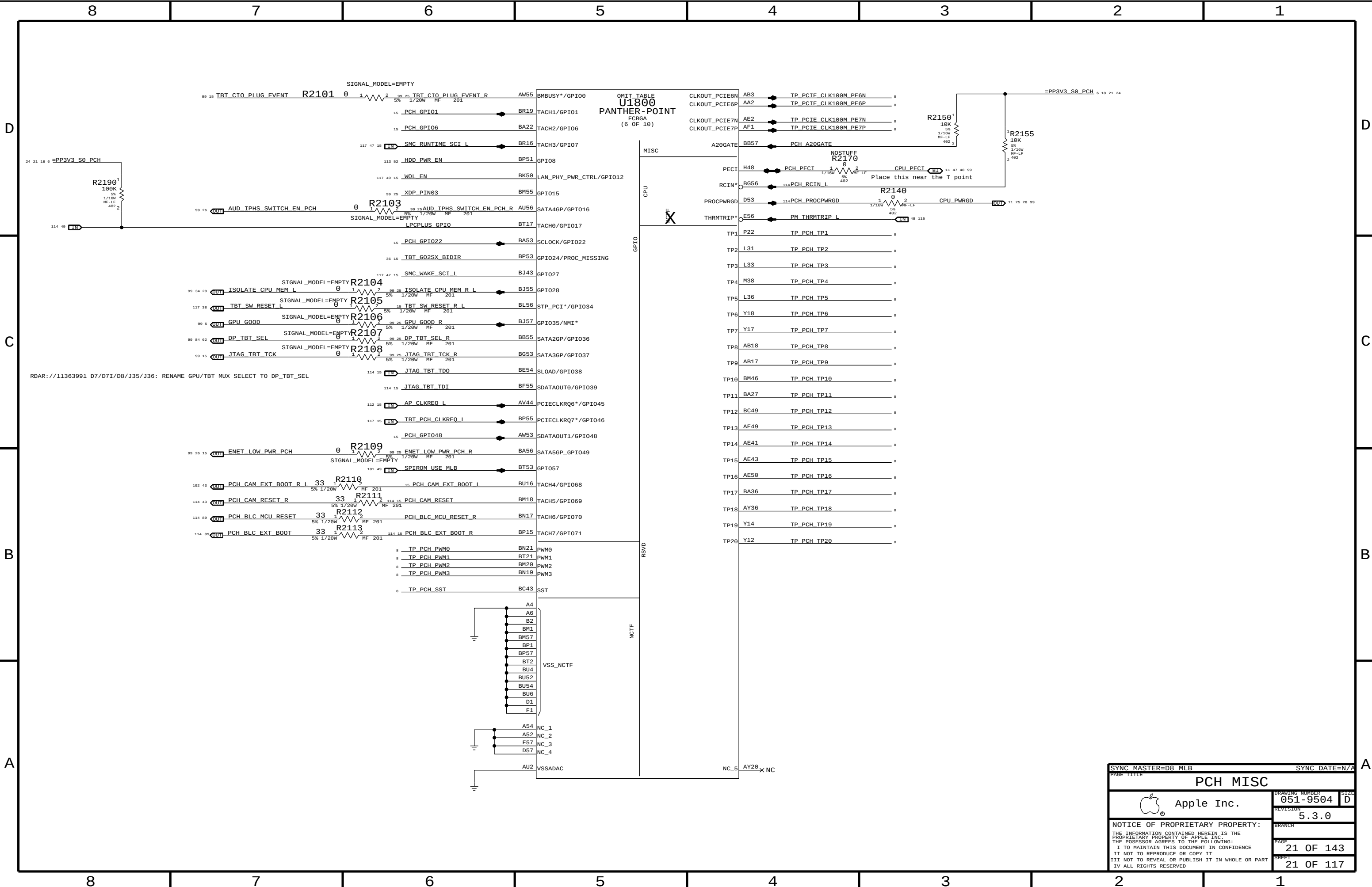
SYNC DATE=01/26/2012

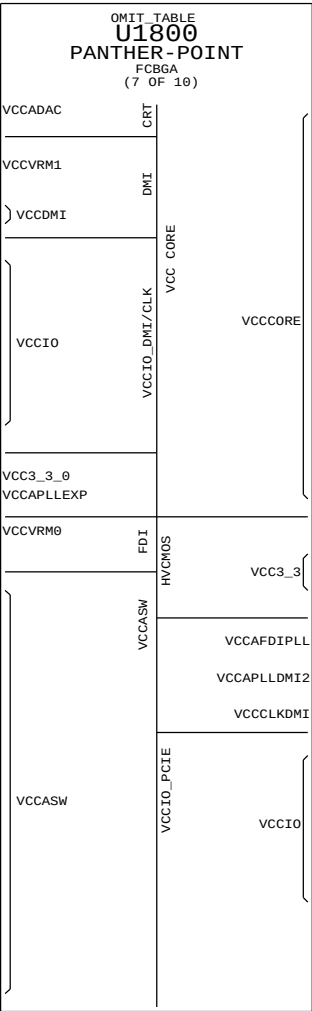
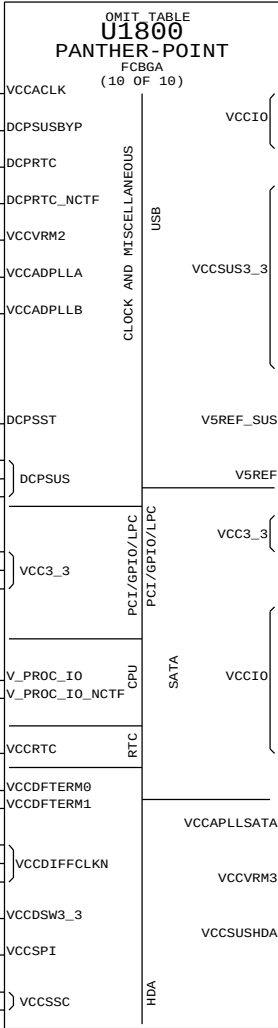
PAGE TITLE		
GFX DECOUPLING & PCH PWR ALIAS		
 Apple Inc.	DRAWING NUMBER	051-9504
	REVISION	5.3.0
	BRANCH	
NOTICE OF PROPRIETARY PROPERTY:		PAGE
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:		17 OF 143
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		SHEET
II NOT TO REPRODUCE OR COPY IT		
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART		17 OF 117
IV ALL RIGHTS RESERVED		

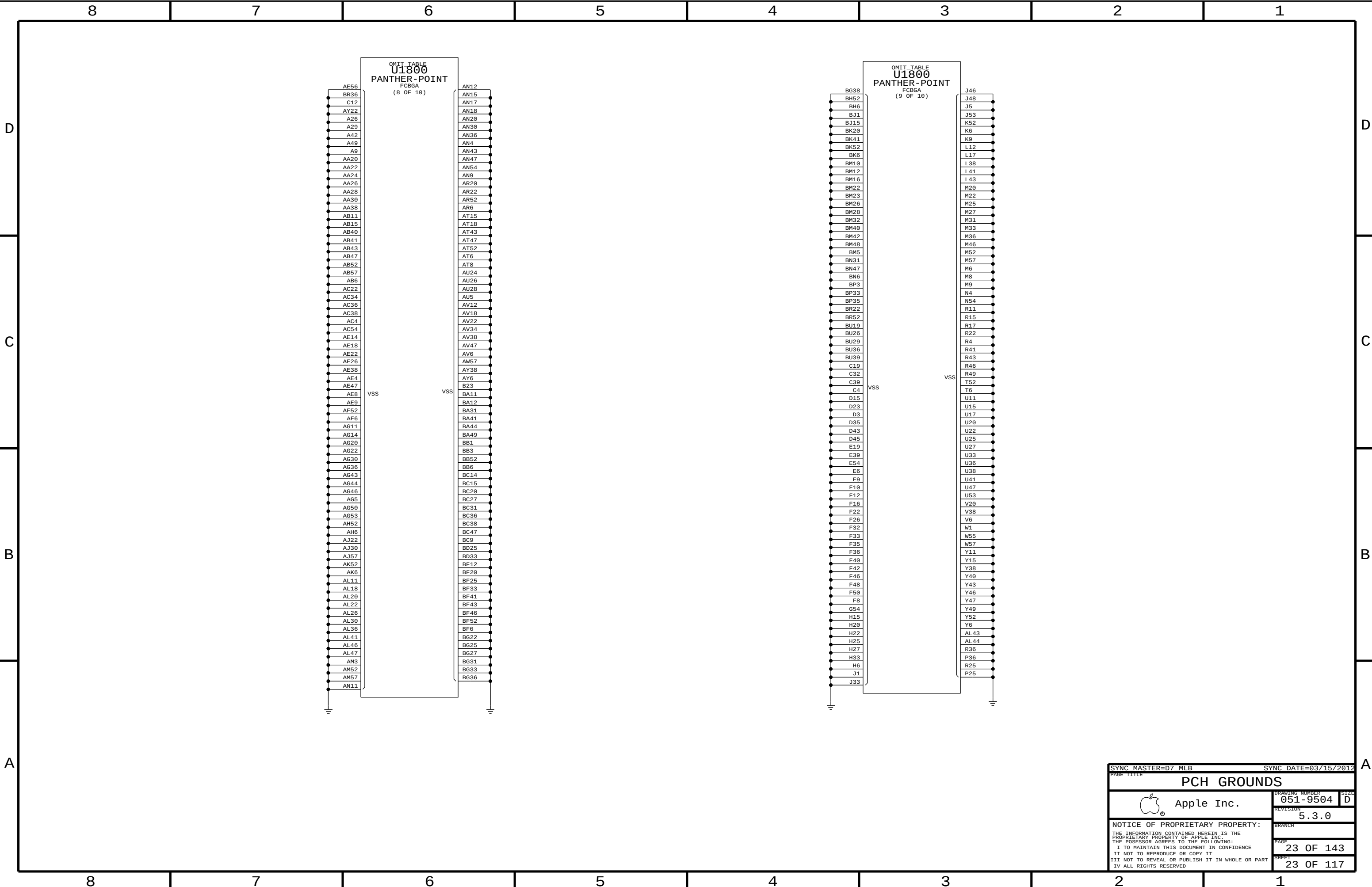












SYNC MASTER=D7_MLB

SYNC DATE=03/15/2012

PAGE TITLE

PCH GROUNDS

 Apple Inc.

DRAWING NUMBER

051-9504

SIZE

D

NOTICE OF PROPRIETARY PROPERTY:
THE INFORMATION CONTAINED HEREIN IS THE
PROPRIETARY PROPERTY OF APPLE INC.
THE POSSESSOR AGREES TO THE FOLLOWING:
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
IV ALL RIGHTS RESERVED

REVISION

5.3.0

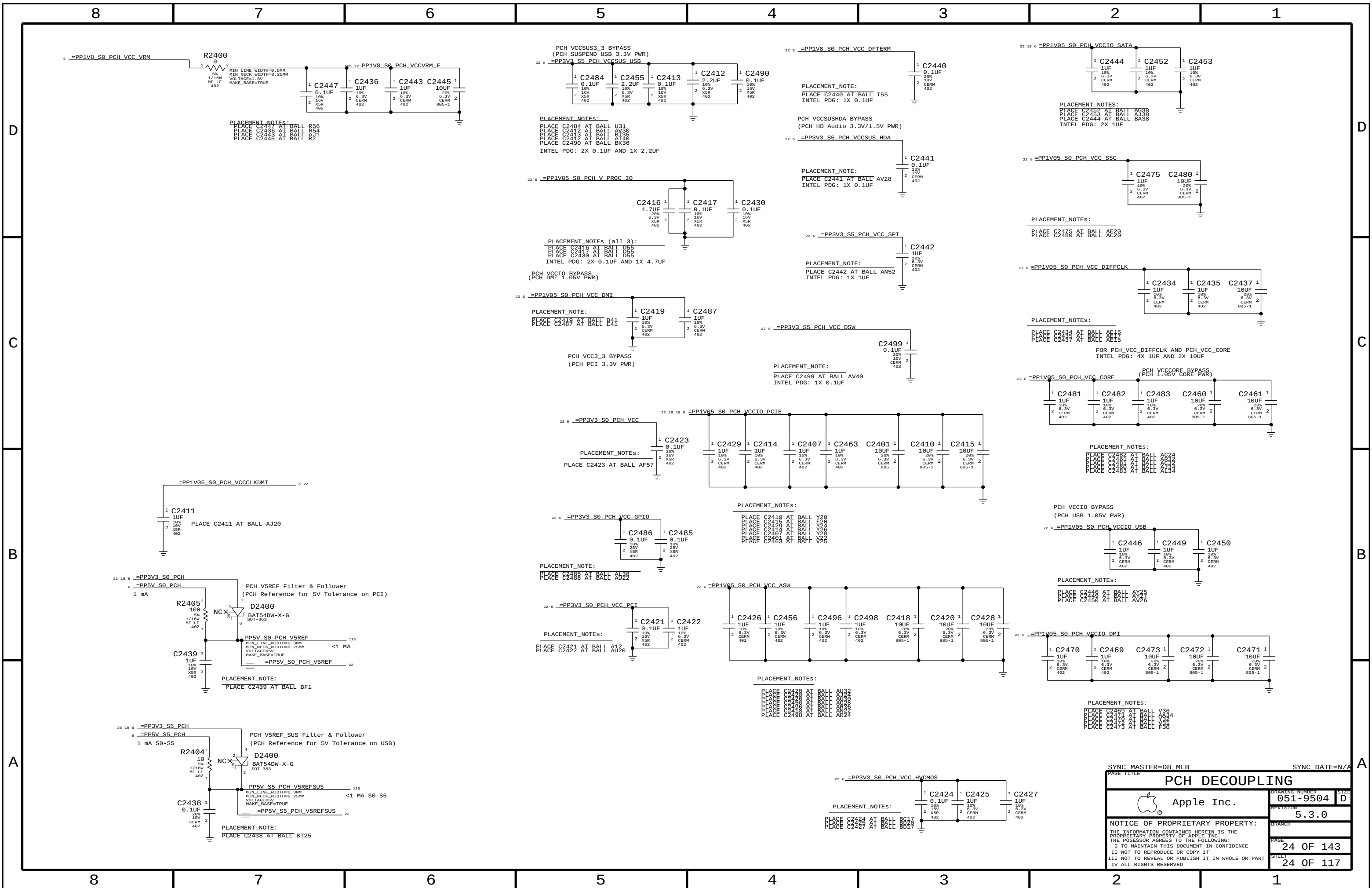
BRANCH

PAGE

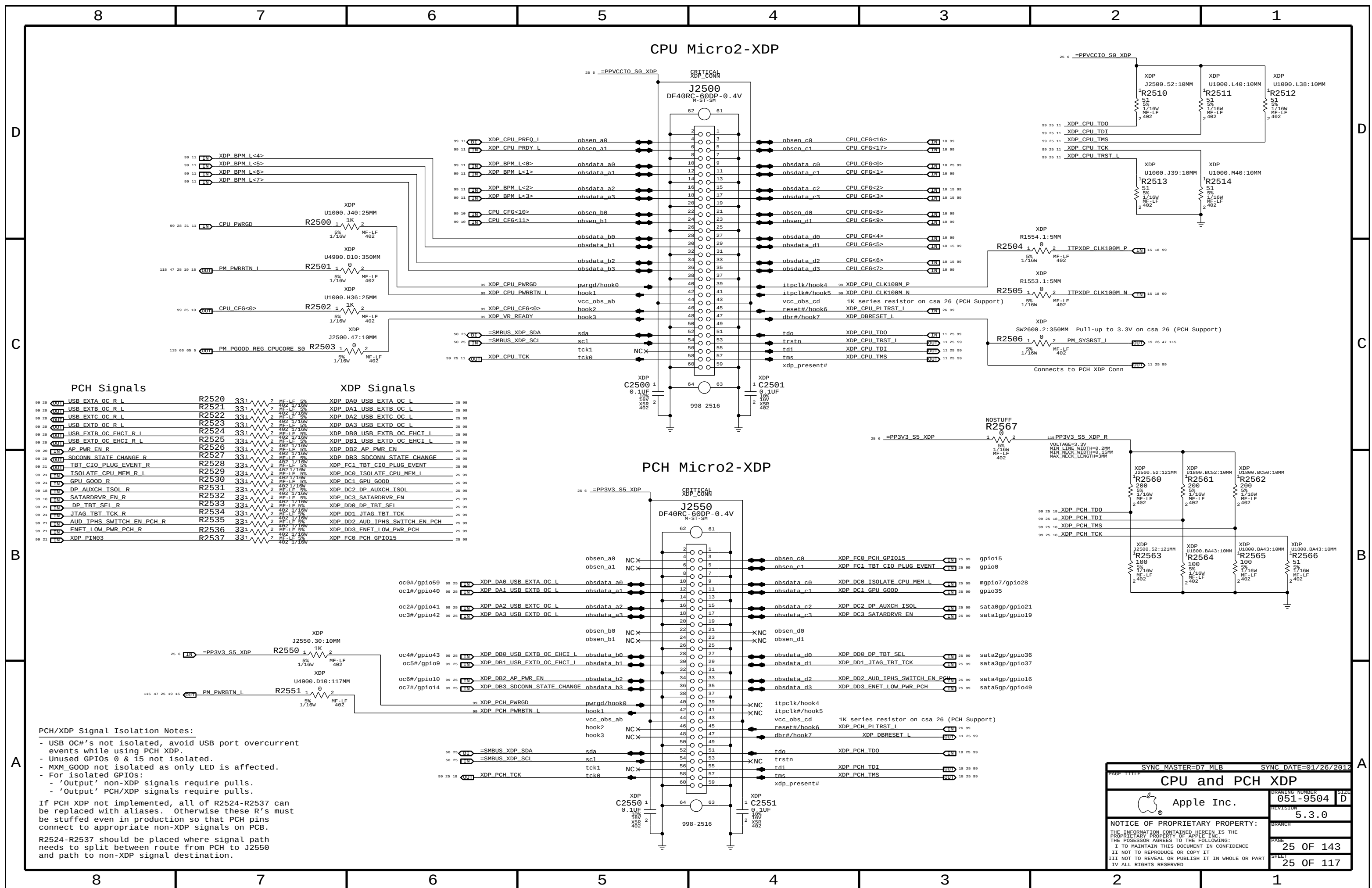
23 OF 143

SHEET

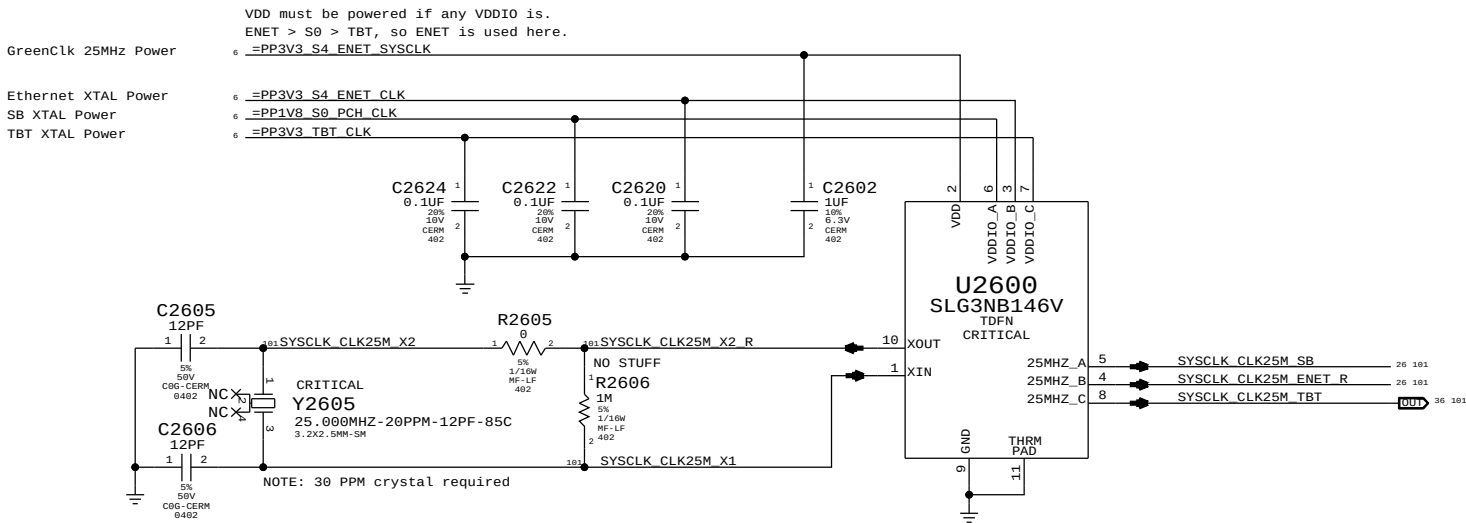
23 OF 117



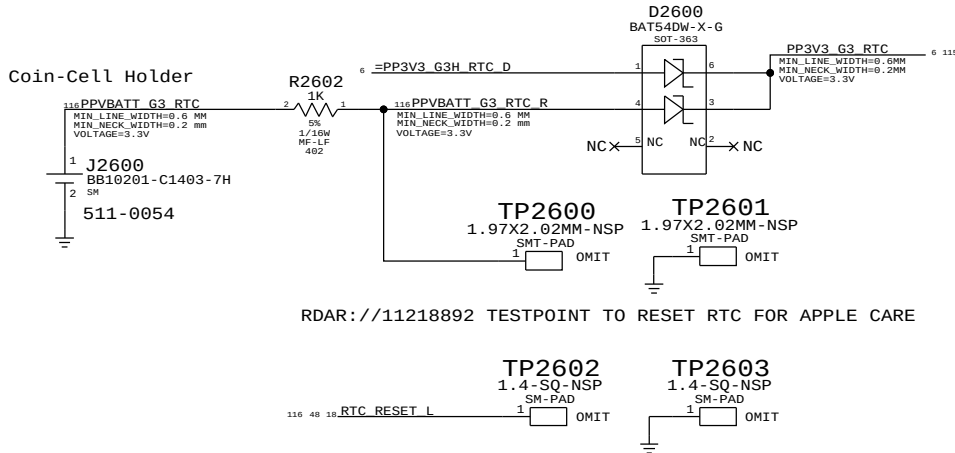
PAGE TITLE		SYNC MASTER=D8_MLB		SYNC DATE=N/A	
PCH DECOUPLING		DRAWING NUMBER		SIZE	
Apple Inc.		051-9504		D	
NOTICE OF PROPRIETARY PROPERTY:		REVISION		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:		5.3.0		PAGE	
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		II NOT TO REPRODUCE OR COPY IT		24 OF 143	
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART		IV ALL RIGHTS RESERVED		SHEET	
				24 OF 117	



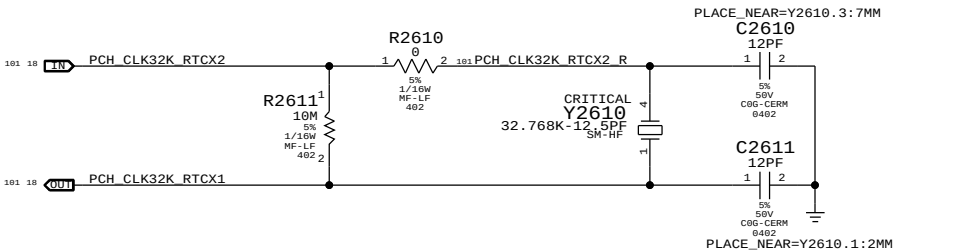
System 25MHz Clock Generator



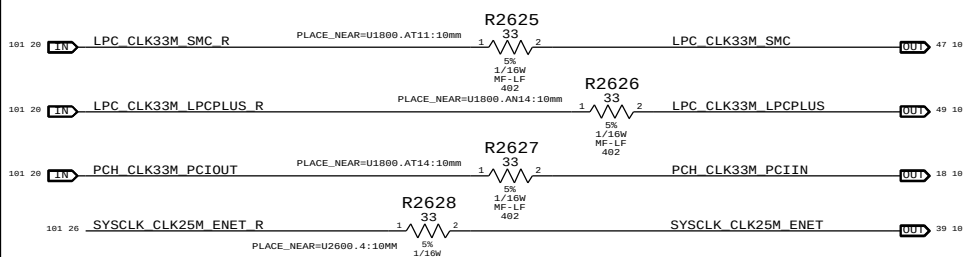
RTC Power Sources



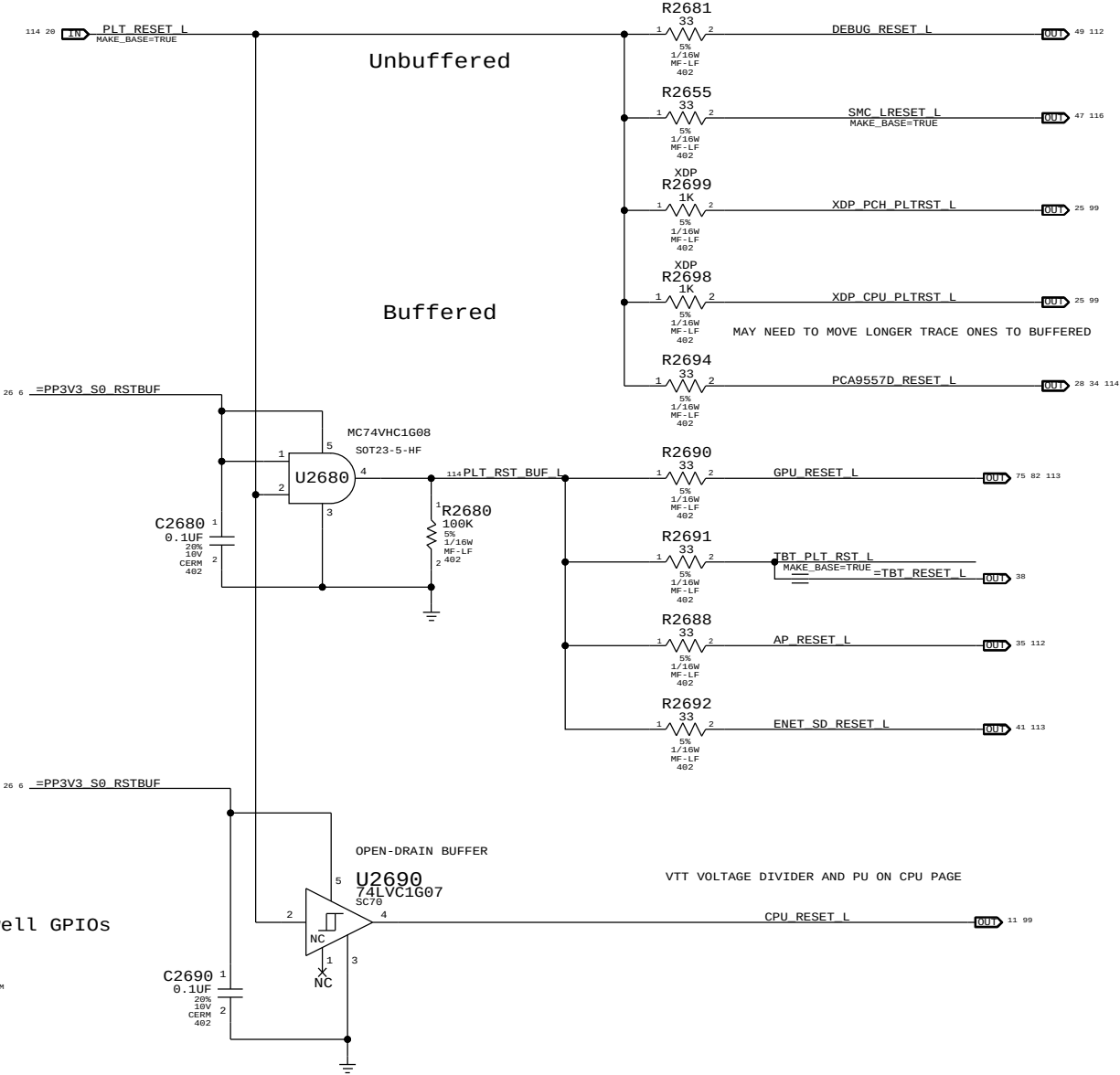
PCH RTC Crystal



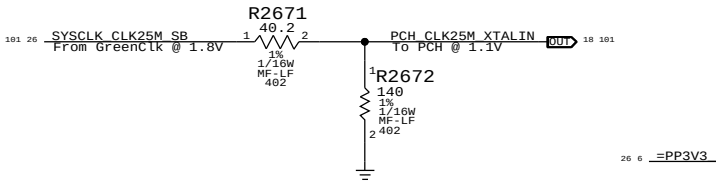
Clock series termination



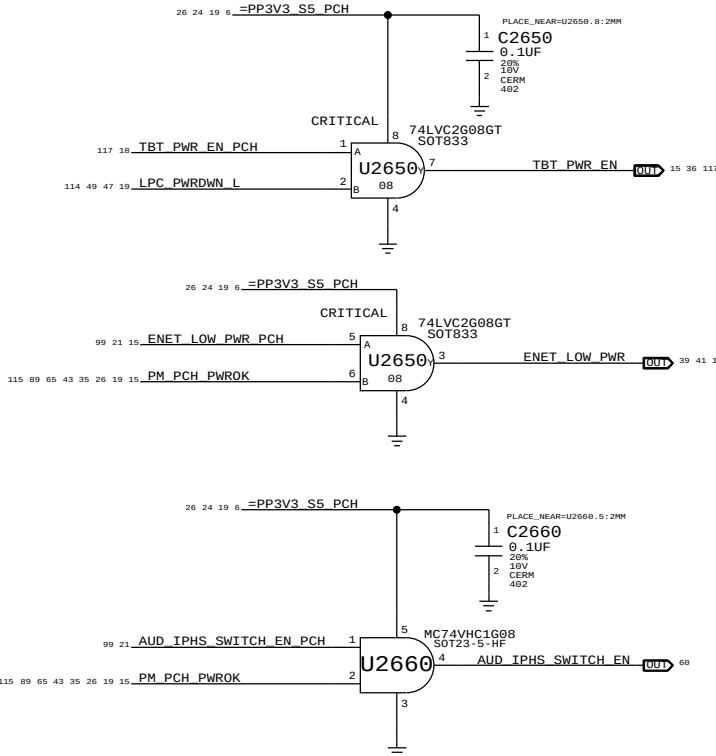
Platform Reset Connections



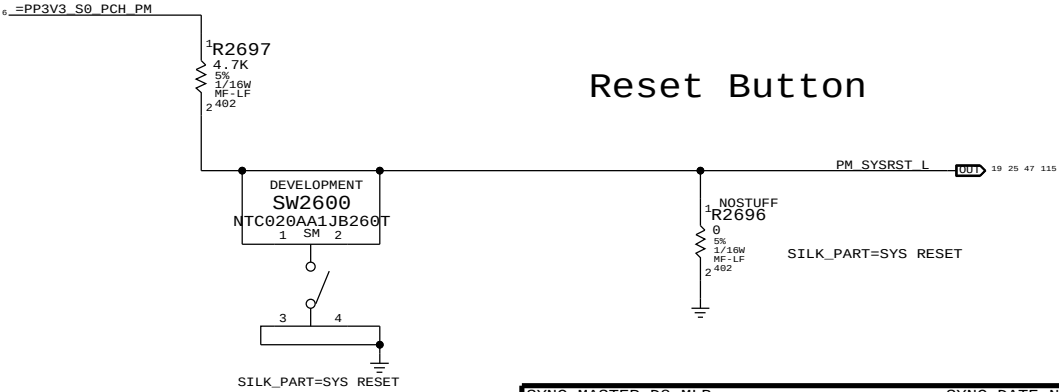
PCH 25MHZ CLOCK



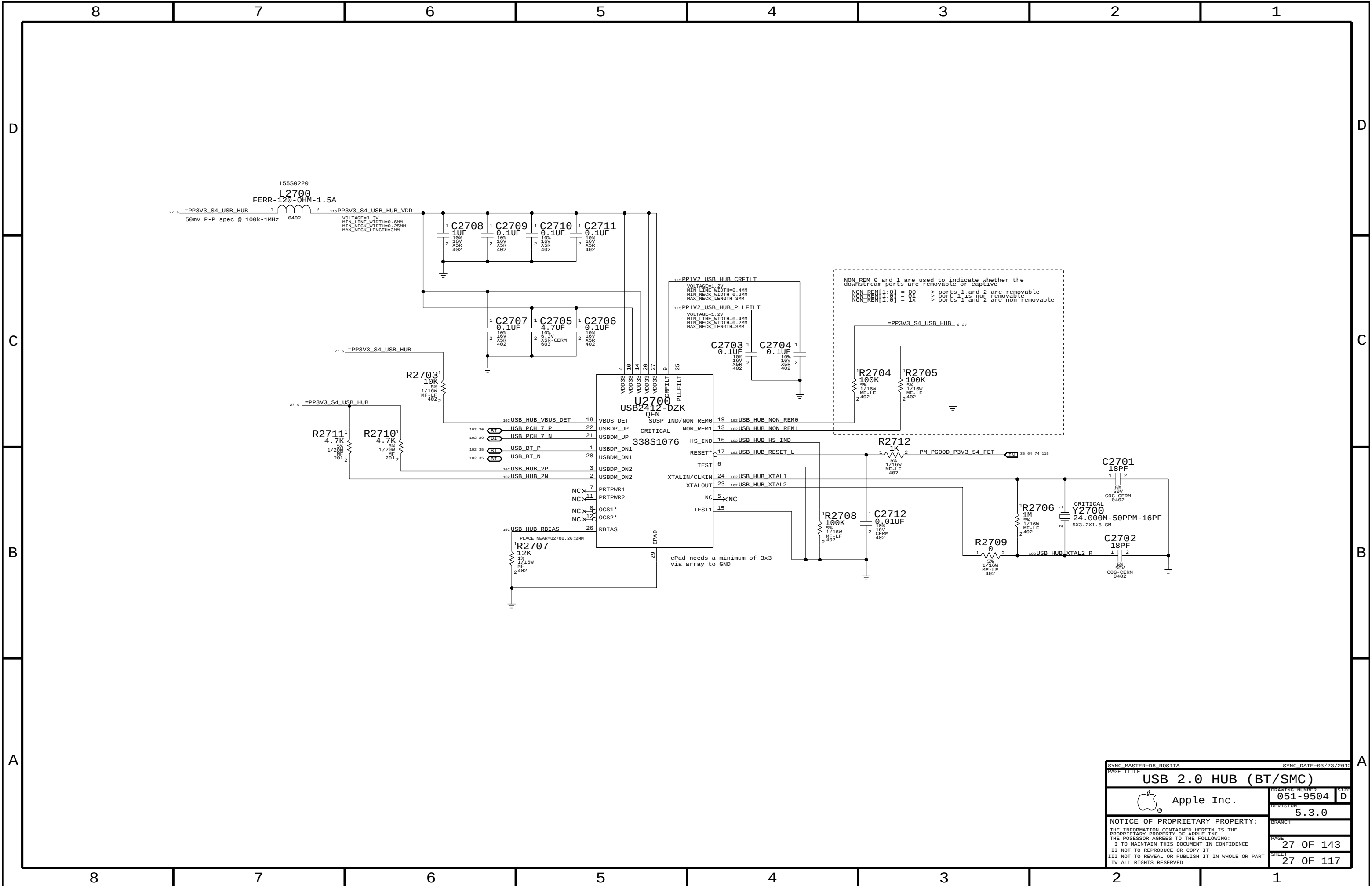
GPIO Isolation to prevent glitches on critical core well GPIOs



Reset Button



SYNC MASTER=D8_MLB		SYNC DATE=N/A	
PAGE TITLE			
CHIPSET SUPPORT			
 Apple Inc.	DRAWING NUMBER	051-9504	SIZE D
	REVISION	5.3.0	
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		BRANCH	
		PAGE	26 OF 143
		SHEET	26 OF 117



SYNC MASTER=D8 ROSITA		SYNC DATE=03/23/2012	
PAGE TITLE			
USB 2.0 HUB (BT/SMC)			
 Apple Inc.		DRAWING NUMBER	SIZE
		051-9504	D
		REVISION	
		5.3.0	
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC.		PAGE	
THE POSSESSOR AGREES TO THE FOLLOWING:		27 OF 143	
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE			
II NOT TO REPRODUCE OR COPY IT			
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART		SHEET	
IV ALL RIGHTS RESERVED		27 OF 117	

MEM_RESET_L Generator

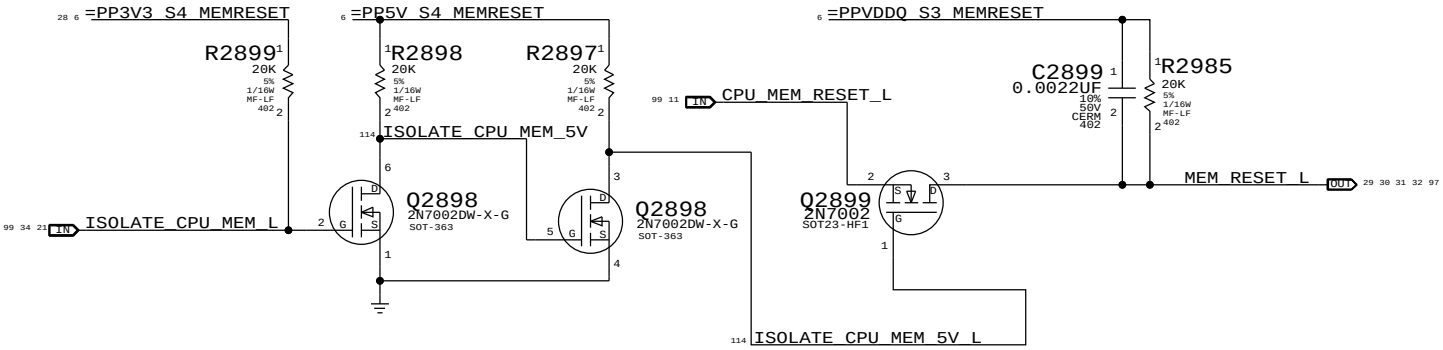
The circuits below handle MEMVTT power during S0->S3->S0 transitions, as well as isolating the CPU's SM_DRAMRST# output from the S0-DIMMs when necessary.

ISOLATE_CPU_MEM_L GPIO state during S3<->S0 transitions determines behaviour of signals.

WHEN HIGH: MEM_RESET_L NOT ISOLATED.

WHEN LOW: MEM_RESET_L IS ISOLATED.

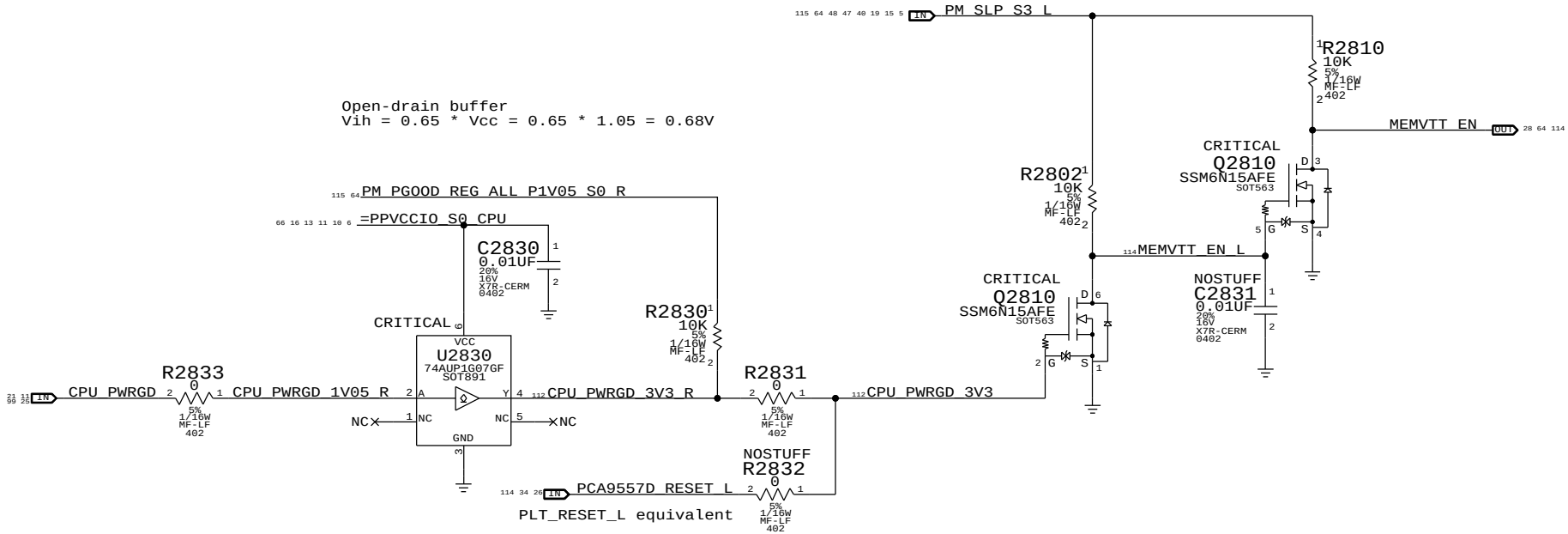
MEM_RESET_L = !ISOLATE_CPU_MEM_L + CPU_MEM_RESET_L (Block CPU from driving MEM_RESET_L in S3)



MEMVTT_EN Generator

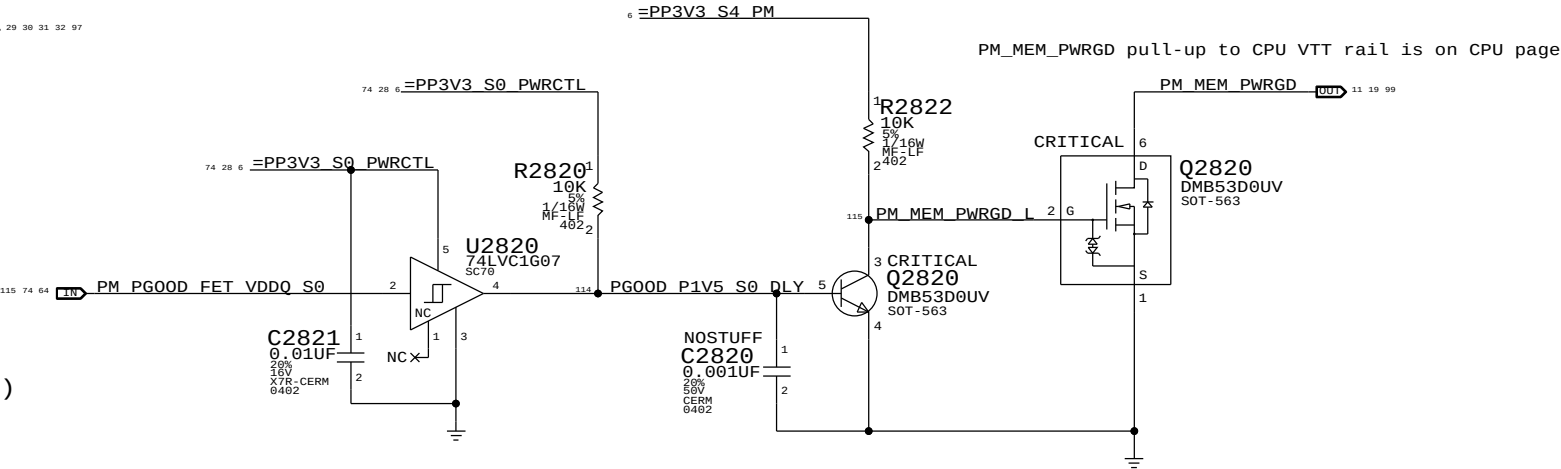
rdar://11117167 Enables MEMVTT when PCH drives CPU PWRGD.

CPU does not drive MEM_CKE until VCCORE activated but CPU 1V5 (VDDQ) leaks into it. Clamping MEMVTT will keep the MEM_CKE low until CPU actively controls it. MEMVTT Clamp actively holds MEMVTT rail low until MEMVTT is enabled. MEMVTT_EN = CPU_PWRGD * PM_SLP_S3_L (VTT is enabled when PCH tells CPU to enable VCCORE)



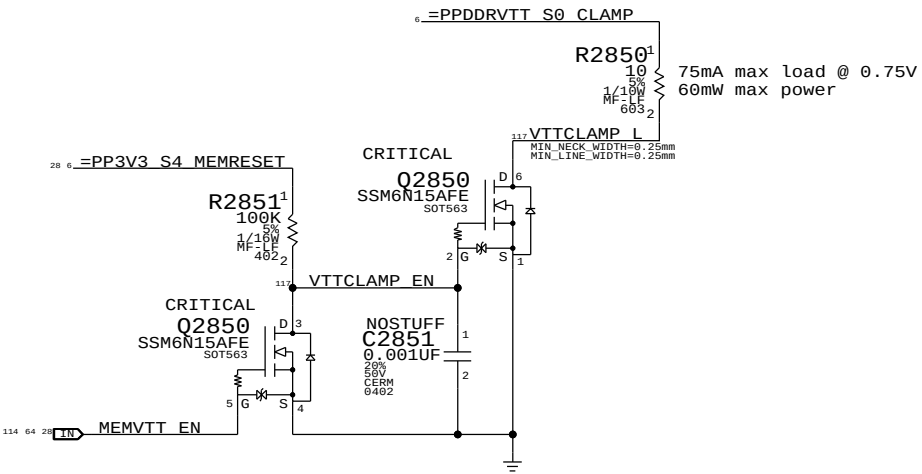
1V5 S0 "PGOOD" for CPU

With optional delay from 1V5 S0 PG00D



MEMVTT Clamp

Ensures CKE signals are held low in S3 and in S0 before CPU PWRGD



Step	ISOLATE_CPU_MEM_L	PM_SLP_S3_L	CPU_PWRGD	CPU_MEM_RESET_L	MEM_RESET_L	MEMVTT_EN
S0	0	1	1	CPU_MEM_RESET_L	1	1
1	0	1	1	1	1	1
2	0	1	1	1	1	1
3	0	0	0	X	1	0
4	0	0	0	X	1	0
5	0	1	1	0 (*)	1	1
6	0	1	1	1	1	1
S0	7	1	1	CPU_MEM_RESET_L	1	1

(*) CPU_MEM_RESET_L asserts due to loss of PM_MEM_PWRGD, must wait for software to clear before deasserting ISOLATE_CPU_MEM_L GPIO.

NOTE: On a S5->S0 transition, ISOLATE_CPU_MEM_L will default low. Rails will power-up as if from S3, but MEM_RESET_L now needs to be asserted in S0. Software must de-assert ISOLATE_CPU_MEM_L and then generate a valid reset cycle on CPU_MEM_RESET_L.

SYNC MASTER=D8 MARK

SYNC DATE=04/23/2012

CPU Memory S3 Support

Apple Inc.

051-9504

5.3.0

NOTICE OF PROPRIETARY PROPERTY:

THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:

I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE

II NOT TO REPRODUCE OR COPY IT

III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART

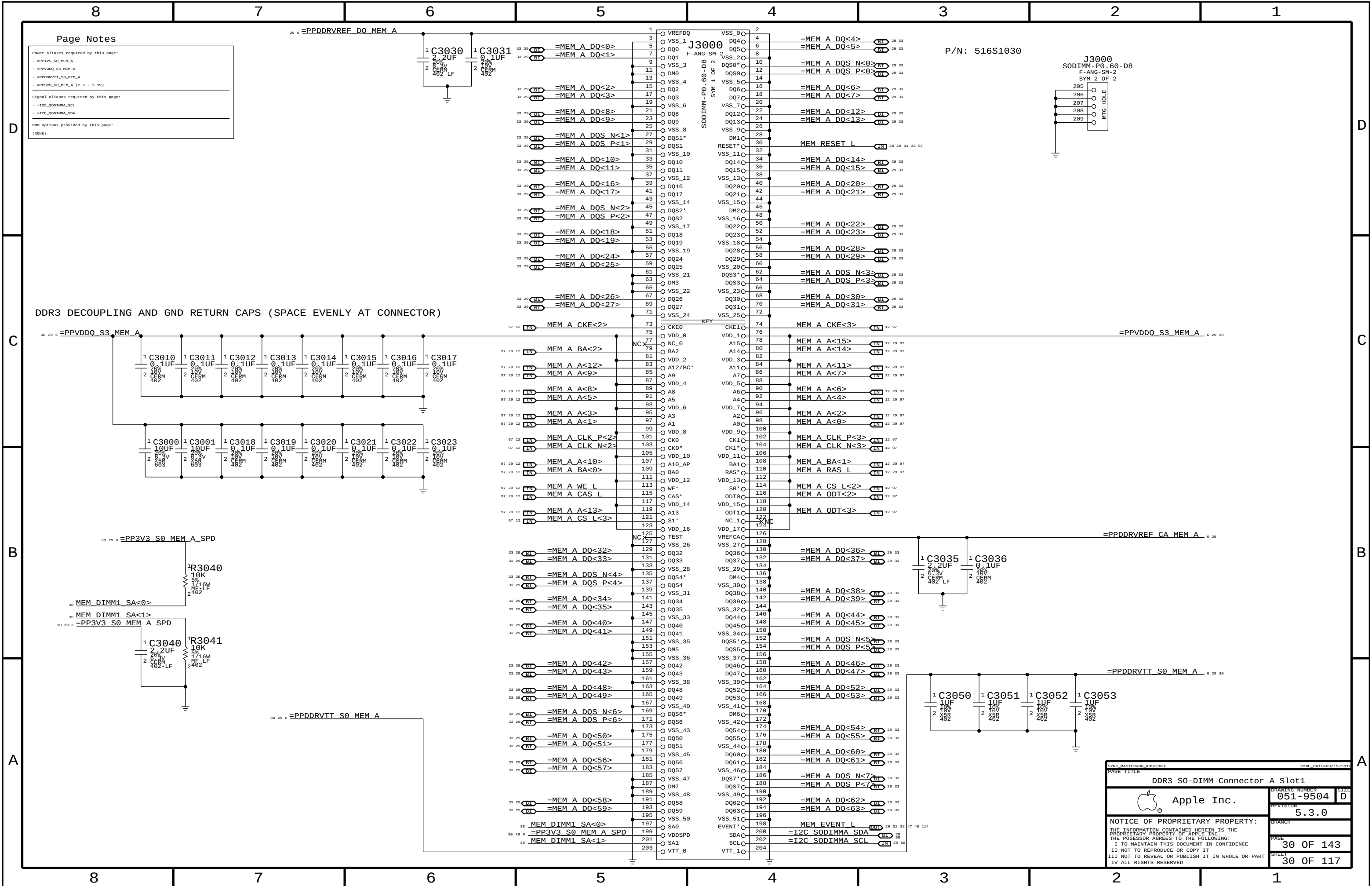
IV ALL RIGHTS RESERVED

PAGE

28 OF 143

SHEET

28 OF 117



Page Notes

Power aliases required by this page:

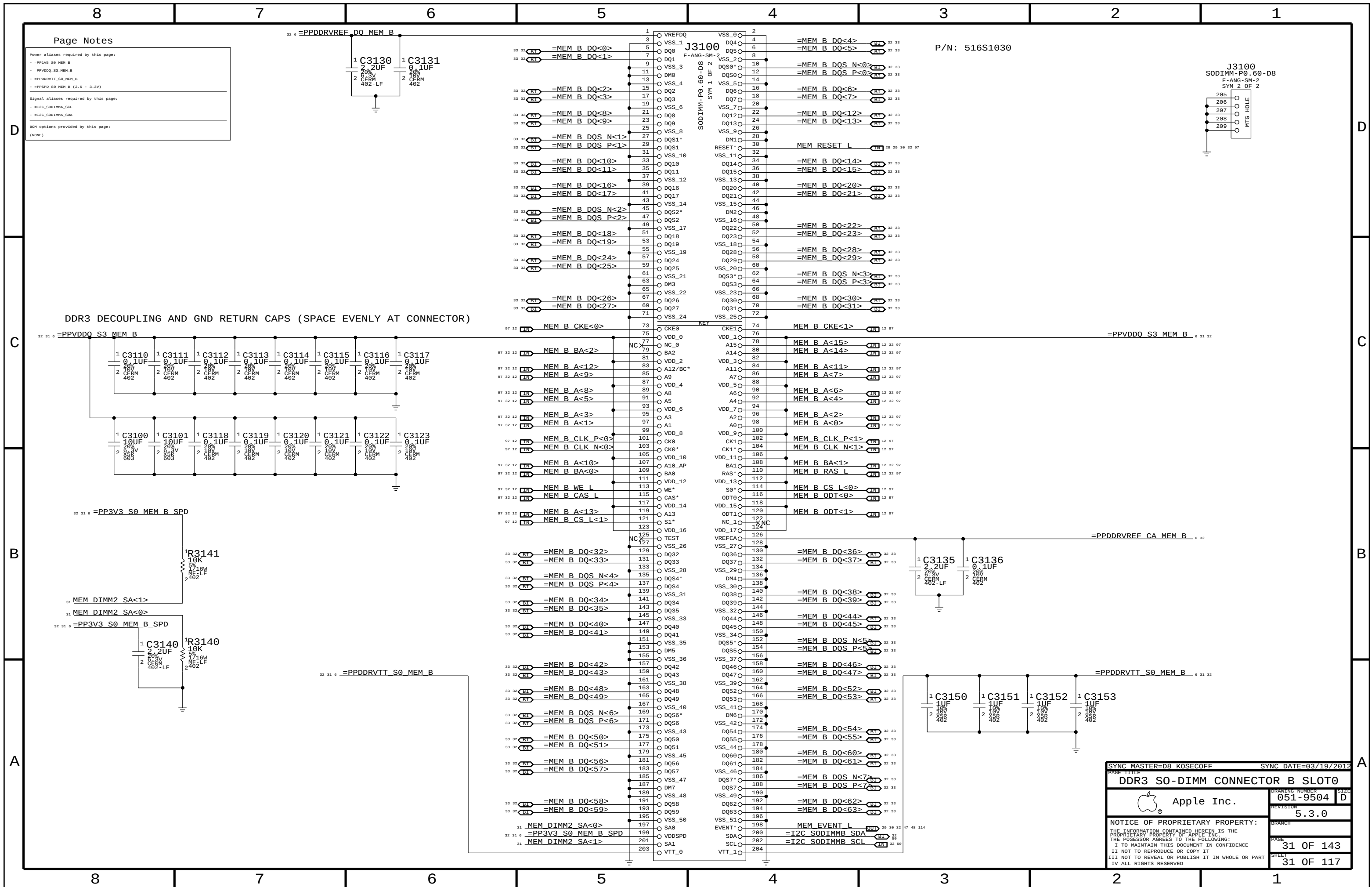
- =PP3V3_S0_MEM_A
- =PPVDDQ_S3_MEM_A
- =PPDDRVTT_S0_MEM_A
- =PPSPD_S0_MEM_A (2.5 - 3.3V)

Signal aliases required by this page:

- =I2C_SODIMMA_SCL
- =I2C_SODIMMA_SDA

BOM options provided by this page:

(NONE)



8		7		6		5		4		3		2		1	
D	97 12	MEM A DQS N<0>	==	=MEM A DQS N<0>	29 30	97 12	MEM B DQS N<0>	==	=MEM B DQS N<0>	31 32	D				
	97 12	MEM A DQS P<0>	==	=MEM A DQS P<0>	29 30	97 12	MEM B DQS P<0>	==	=MEM B DQS P<0>	31 32					
	97 12	MEM A DQ<7>	==	=MEM A DQ<7>	29 30	97 12	MEM B DQ<7>	==	=MEM B DQ<3>	31 32					
	97 12	MEM A DQ<6>	==	=MEM A DQ<6>	29 30	97 12	MEM B DQ<6>	==	=MEM B DQ<6>	31 32					
	97 12	MEM A DQ<5>	==	=MEM A DQ<4>	29 30	97 12	MEM B DQ<5>	==	=MEM B DQ<5>	31 32					
	97 12	MEM A DQ<4>	==	=MEM A DQ<1>	29 30	97 12	MEM B DQ<4>	==	=MEM B DQ<4>	31 32					
	97 12	MEM A DQ<3>	==	=MEM A DQ<3>	29 30	97 12	MEM B DQ<3>	==	=MEM B DQ<7>	31 32					
	97 12	MEM A DQ<2>	==	=MEM A DQ<2>	29 30	97 12	MEM B DQ<2>	==	=MEM B DQ<2>	31 32					
	97 12	MEM A DQ<1>	==	=MEM A DQ<5>	29 30	97 12	MEM B DQ<1>	==	=MEM B DQ<1>	31 32					
	97 12	MEM A DQ<0>	==	=MEM A DQ<0>	29 30	97 12	MEM B DQ<0>	==	=MEM B DQ<0>	31 32					
C	97 12	MEM A DQS N<1>	==	=MEM A DQS N<1>	29 30	97 12	MEM B DQS N<1>	==	=MEM B DQS N<1>	31 32	C				
	97 12	MEM A DQS P<1>	==	=MEM A DQS P<1>	29 30	97 12	MEM B DQS P<1>	==	=MEM B DQS P<1>	31 32					
	97 12	MEM A DQ<15>	==	=MEM A DQ<15>	29 30	97 12	MEM B DQ<15>	==	=MEM B DQ<11>	31 32					
	97 12	MEM A DQ<14>	==	=MEM A DQ<14>	29 30	97 12	MEM B DQ<14>	==	=MEM B DQ<10>	31 32					
	97 12	MEM A DQ<13>	==	=MEM A DQ<13>	29 30	97 12	MEM B DQ<13>	==	=MEM B DQ<13>	31 32					
	97 12	MEM A DQ<12>	==	=MEM A DQ<12>	29 30	97 12	MEM B DQ<12>	==	=MEM B DQ<12>	31 32					
	97 12	MEM A DQ<11>	==	=MEM A DQ<11>	29 30	97 12	MEM B DQ<11>	==	=MEM B DQ<15>	31 32					
	97 12	MEM A DQ<10>	==	=MEM A DQ<10>	29 30	97 12	MEM B DQ<10>	==	=MEM B DQ<14>	31 32					
	97 12	MEM A DQ<9>	==	=MEM A DQ<9>	29 30	97 12	MEM B DQ<9>	==	=MEM B DQ<8>	31 32					
	97 12	MEM A DQ<8>	==	=MEM A DQ<8>	29 30	97 12	MEM B DQ<8>	==	=MEM B DQ<9>	31 32					
B	97 12	MEM A DQS N<2>	==	=MEM A DQS N<2>	29 30	97 12	MEM B DQS N<2>	==	=MEM B DQS N<2>	31 32	B				
	97 12	MEM A DQS P<2>	==	=MEM A DQS P<2>	29 30	97 12	MEM B DQS P<2>	==	=MEM B DQS P<2>	31 32					
	97 12	MEM A DQ<23>	==	=MEM A DQ<23>	29 30	97 12	MEM B DQ<23>	==	=MEM B DQ<22>	31 32					
	97 12	MEM A DQ<22>	==	=MEM A DQ<22>	29 30	97 12	MEM B DQ<22>	==	=MEM B DQ<18>	31 32					
	97 12	MEM A DQ<21>	==	=MEM A DQ<21>	29 30	97 12	MEM B DQ<21>	==	=MEM B DQ<21>	31 32					
	97 12	MEM A DQ<20>	==	=MEM A DQ<20>	29 30	97 12	MEM B DQ<20>	==	=MEM B DQ<20>	31 32					
	97 12	MEM A DQ<19>	==	=MEM A DQ<18>	29 30	97 12	MEM B DQ<19>	==	=MEM B DQ<23>	31 32					
	97 12	MEM A DQ<18>	==	=MEM A DQ<19>	29 30	97 12	MEM B DQ<18>	==	=MEM B DQ<19>	31 32					
	97 12	MEM A DQ<17>	==	=MEM A DQ<17>	29 30	97 12	MEM B DQ<17>	==	=MEM B DQ<17>	31 32					
	97 12	MEM A DQ<16>	==	=MEM A DQ<16>	29 30	97 12	MEM B DQ<16>	==	=MEM B DQ<16>	31 32					
A	97 12	MEM A DQS N<3>	==	=MEM A DQS N<3>	29 30	97 12	MEM B DQS N<3>	==	=MEM B DQS N<3>	31 32	A				
	97 12	MEM A DQS P<3>	==	=MEM A DQS P<3>	29 30	97 12	MEM B DQS P<3>	==	=MEM B DQS P<3>	31 32					
	97 12	MEM A DQ<31>	==	=MEM A DQ<31>	29 30	97 12	MEM B DQ<31>	==	=MEM B DQ<31>	31 32					
	97 12	MEM A DQ<30>	==	=MEM A DQ<30>	29 30	97 12	MEM B DQ<30>	==	=MEM B DQ<30>	31 32					
	97 12	MEM A DQ<29>	==	=MEM A DQ<29>	29 30	97 12	MEM B DQ<29>	==	=MEM B DQ<29>	31 32					
	97 12	MEM A DQ<28>	==	=MEM A DQ<28>	29 30	97 12	MEM B DQ<28>	==	=MEM B DQ<28>	31 32					
	97 12	MEM A DQ<27>	==	=MEM A DQ<27>	29 30	97 12	MEM B DQ<27>	==	=MEM B DQ<27>	31 32					
	97 12	MEM A DQ<26>	==	=MEM A DQ<26>	29 30	97 12	MEM B DQ<26>	==	=MEM B DQ<26>	31 32					
	97 12	MEM A DQ<25>	==	=MEM A DQ<25>	29 30	97 12	MEM B DQ<25>	==	=MEM B DQ<25>	31 32					
	97 12	MEM A DQ<24>	==	=MEM A DQ<24>	29 30	97 12	MEM B DQ<24>	==	=MEM B DQ<24>	31 32					

SYNC MASTER=D8 KOSECOFF

SYNC DATE=03/19/2012

PAGE TITLE

DDR3 ALIASES AND BITSWAPS

 Apple Inc.

DRAWING NUMBER
051-9504

SIZE
D

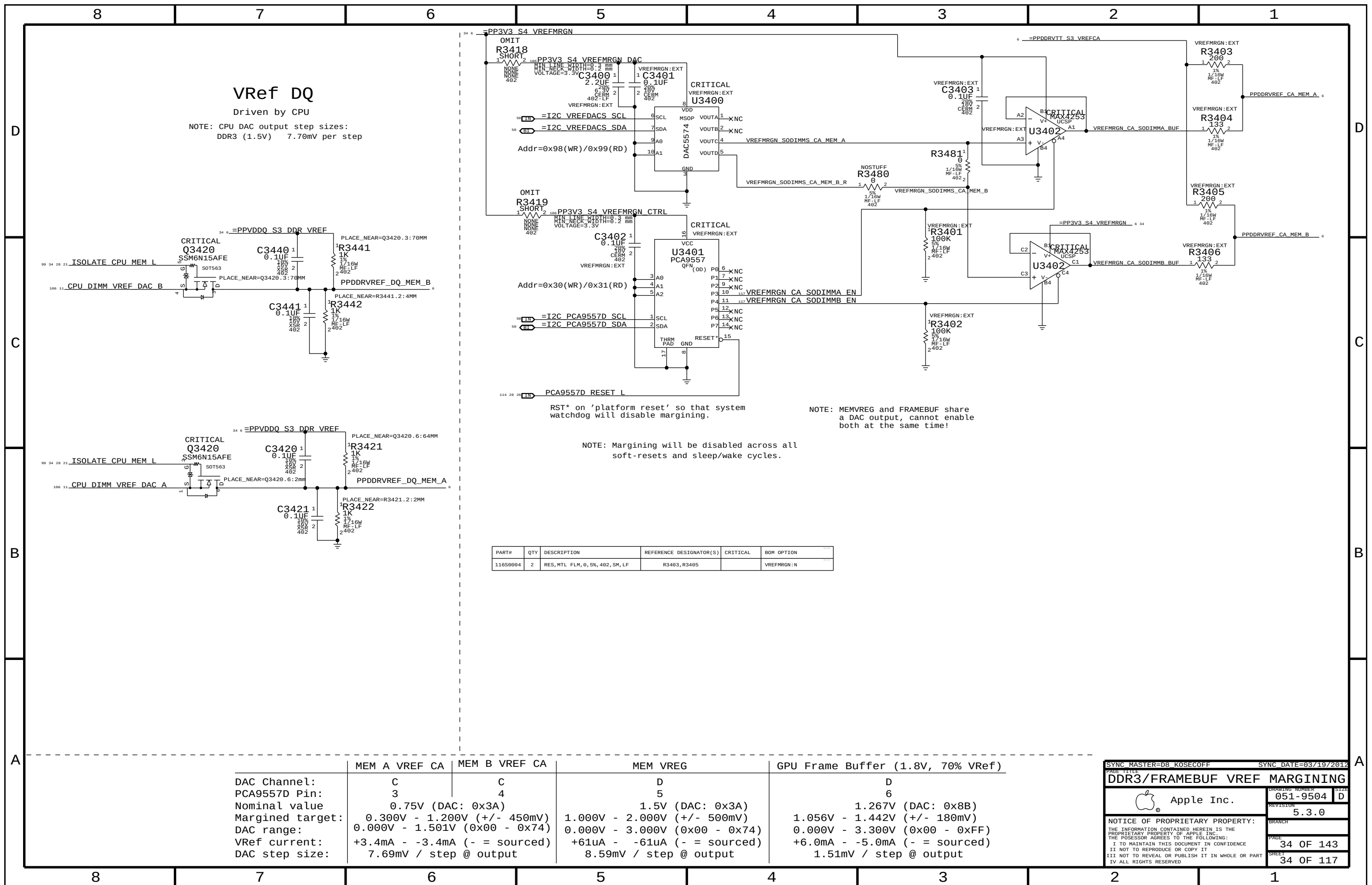
REVISION
5.3.0

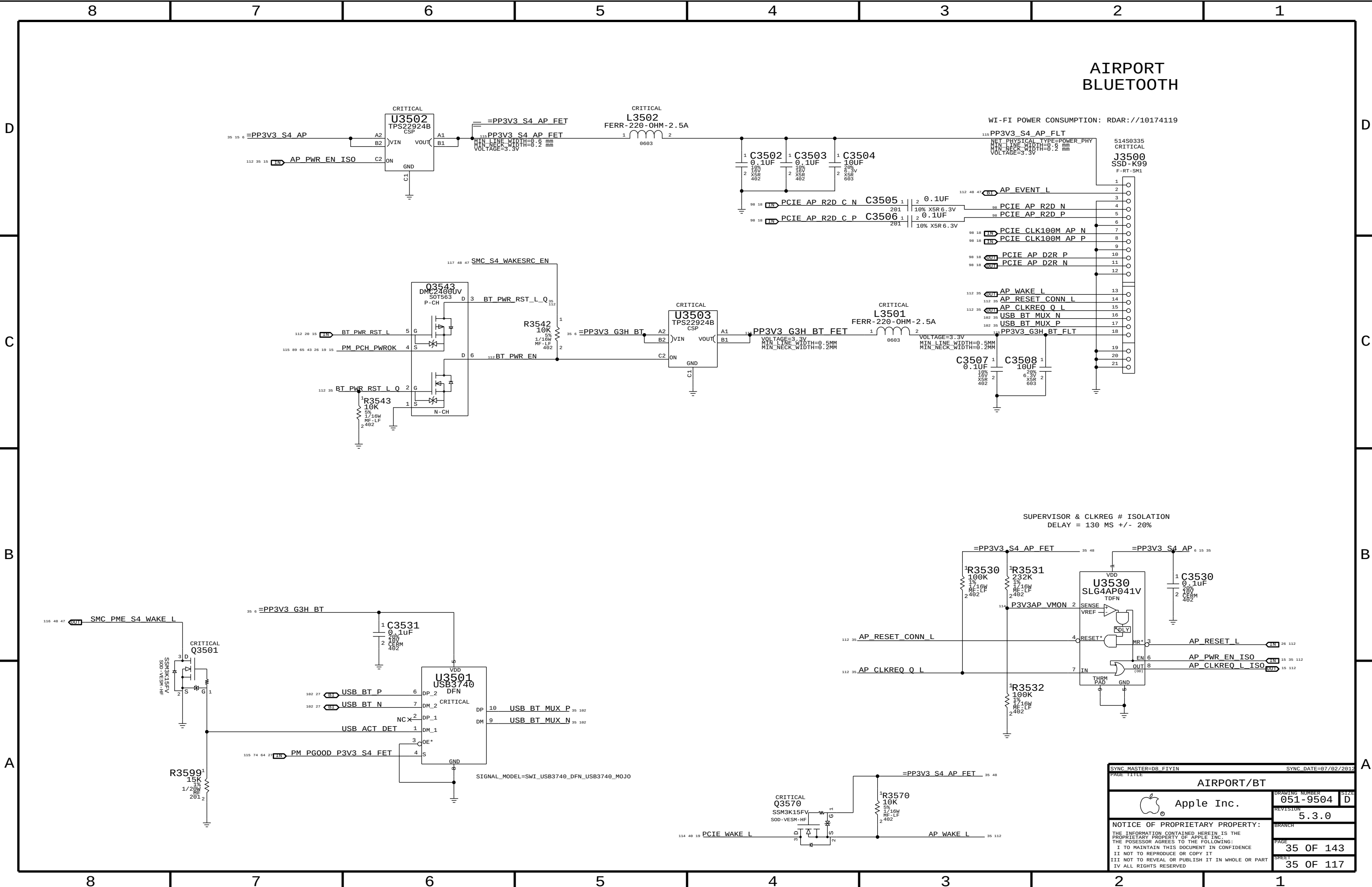
BRANCH

NOTICE OF PROPRIETARY PROPERTY:
THE INFORMATION CONTAINED HEREIN IS THE
PROPRIETARY PROPERTY OF APPLE INC.
THE POSSESSOR AGREES TO THE FOLLOWING:
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
IV ALL RIGHTS RESERVED

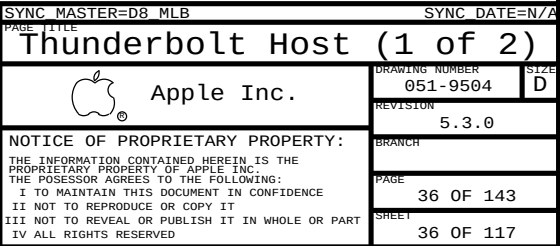
PAGE
33 OF 143

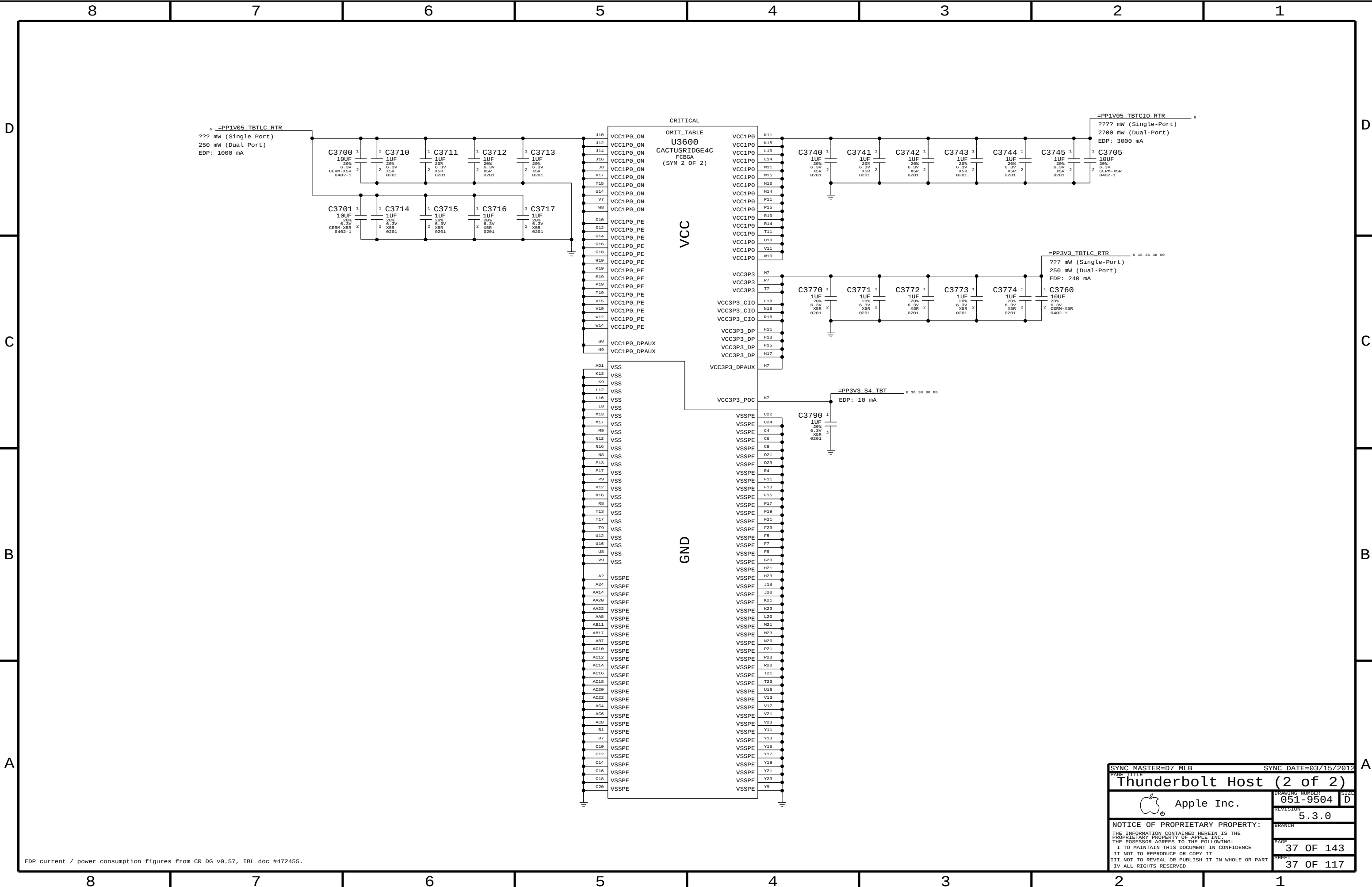
SHEET
33 OF 117





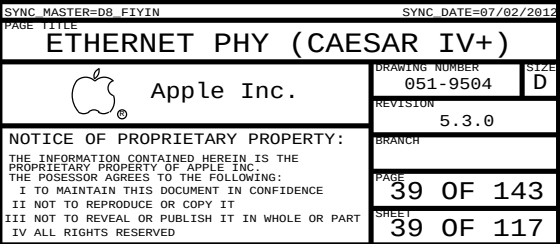
SYNC MASTER=D8 FIYIN		SYNC DATE=07/02/2012	
PAGE TITLE		AIRPORT/BT	
Apple Inc.		DRAWING NUMBER	051-9504
		REVISION	5.3.0
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:		PAGE	35 OF 143
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		SHEET	35 OF 117
II NOT TO REPRODUCE OR COPY IT			
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IV ALL RIGHTS RESERVED			

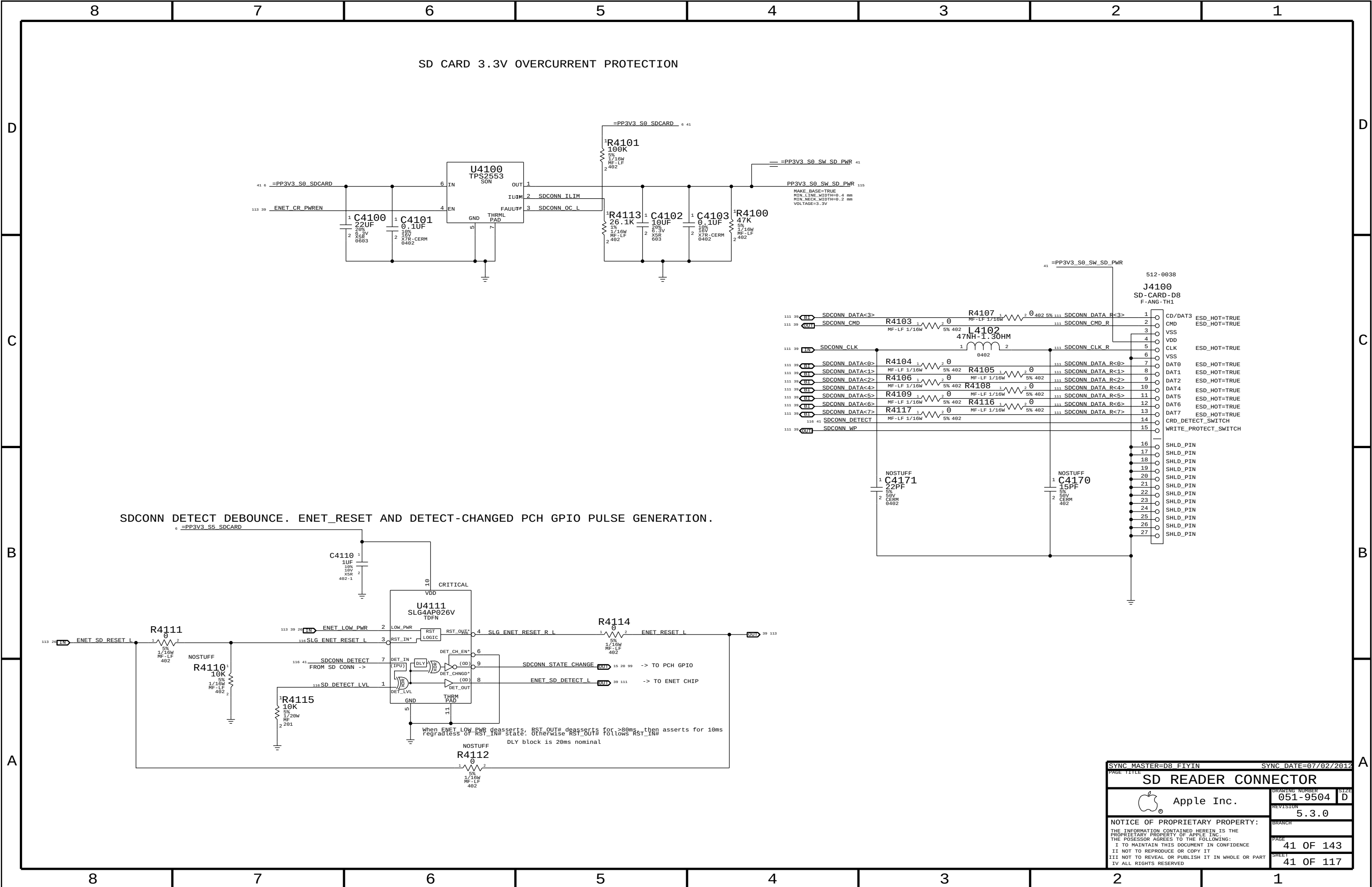




EDP current / power consumption figures from CR DG v0.57, IBL doc #472455.

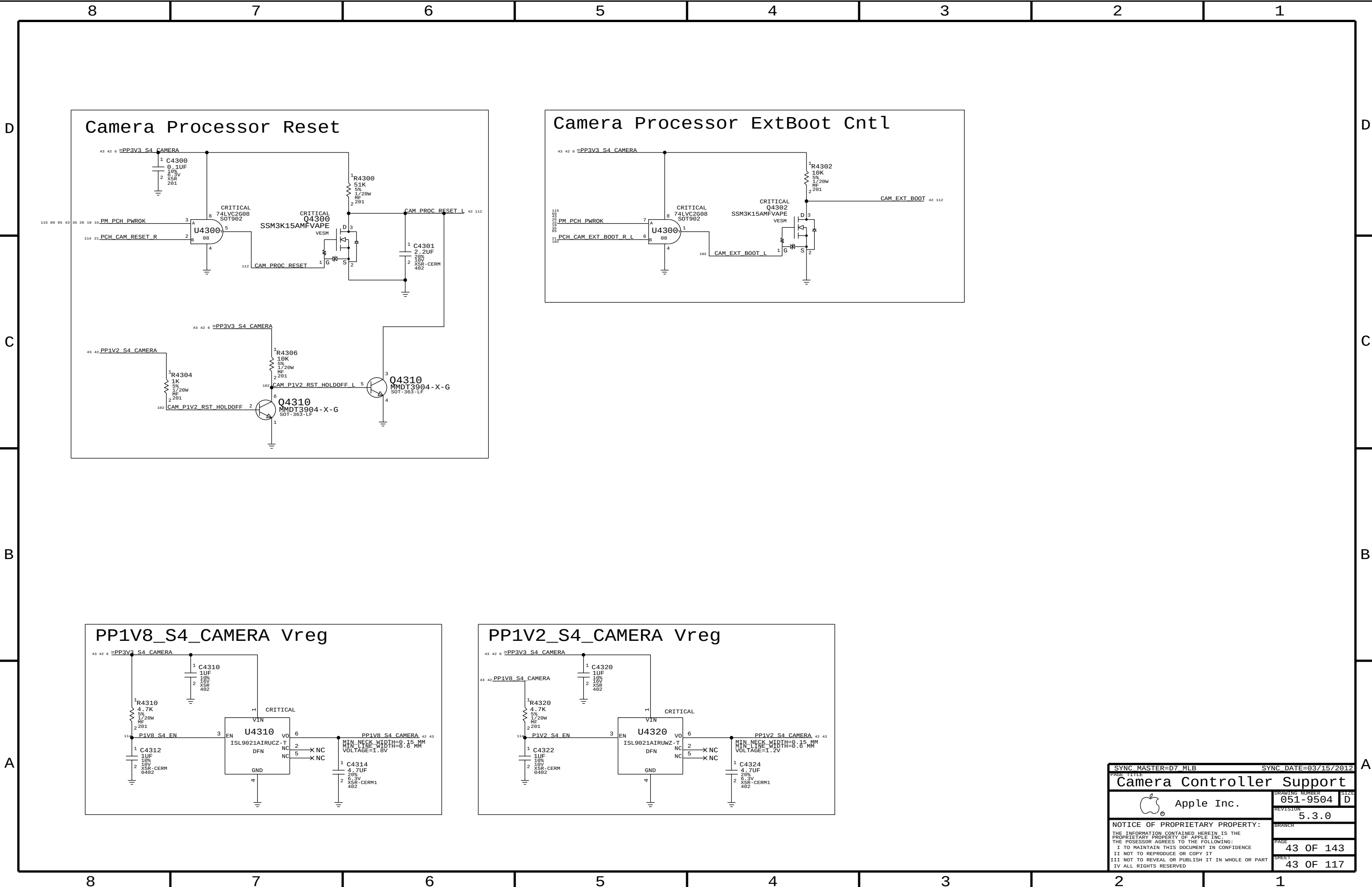
PAGE 1/11		SYNC DATE=03/15/2012	
Thunderbolt Host (2 of 2)			
Apple Inc.		DRAWING NUMBER	051-9504
		REVISION	5.3.0
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:		PAGE	
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		37 OF 143	
II NOT TO REPRODUCE OR COPY IT		SHEET	
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART		37 OF 117	
IV ALL RIGHTS RESERVED			





D

A



D

C

B

A

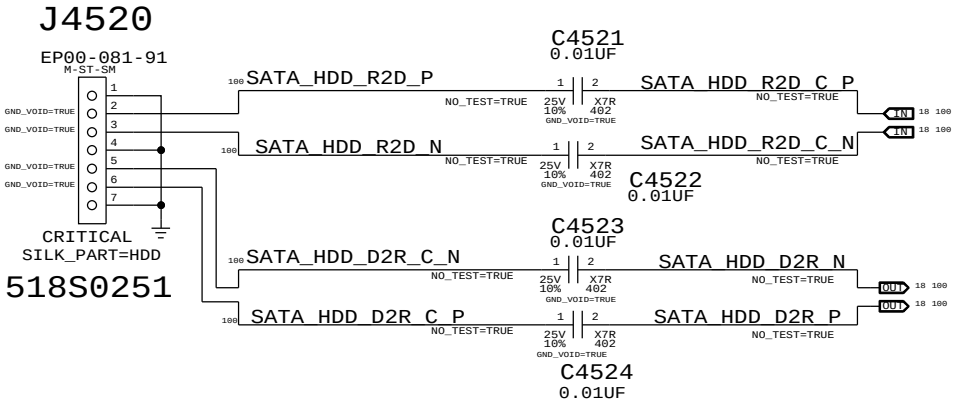
D

C

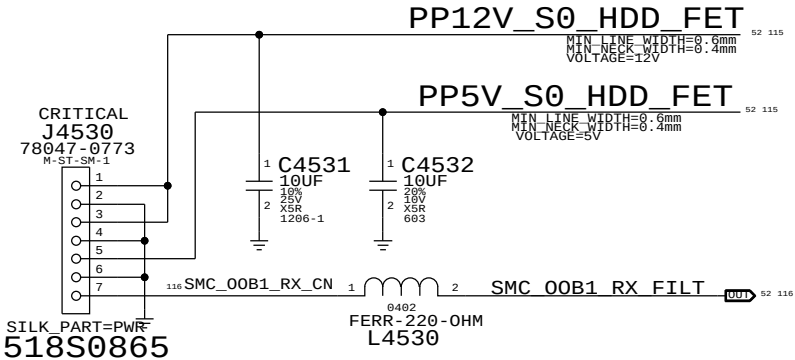
B

A

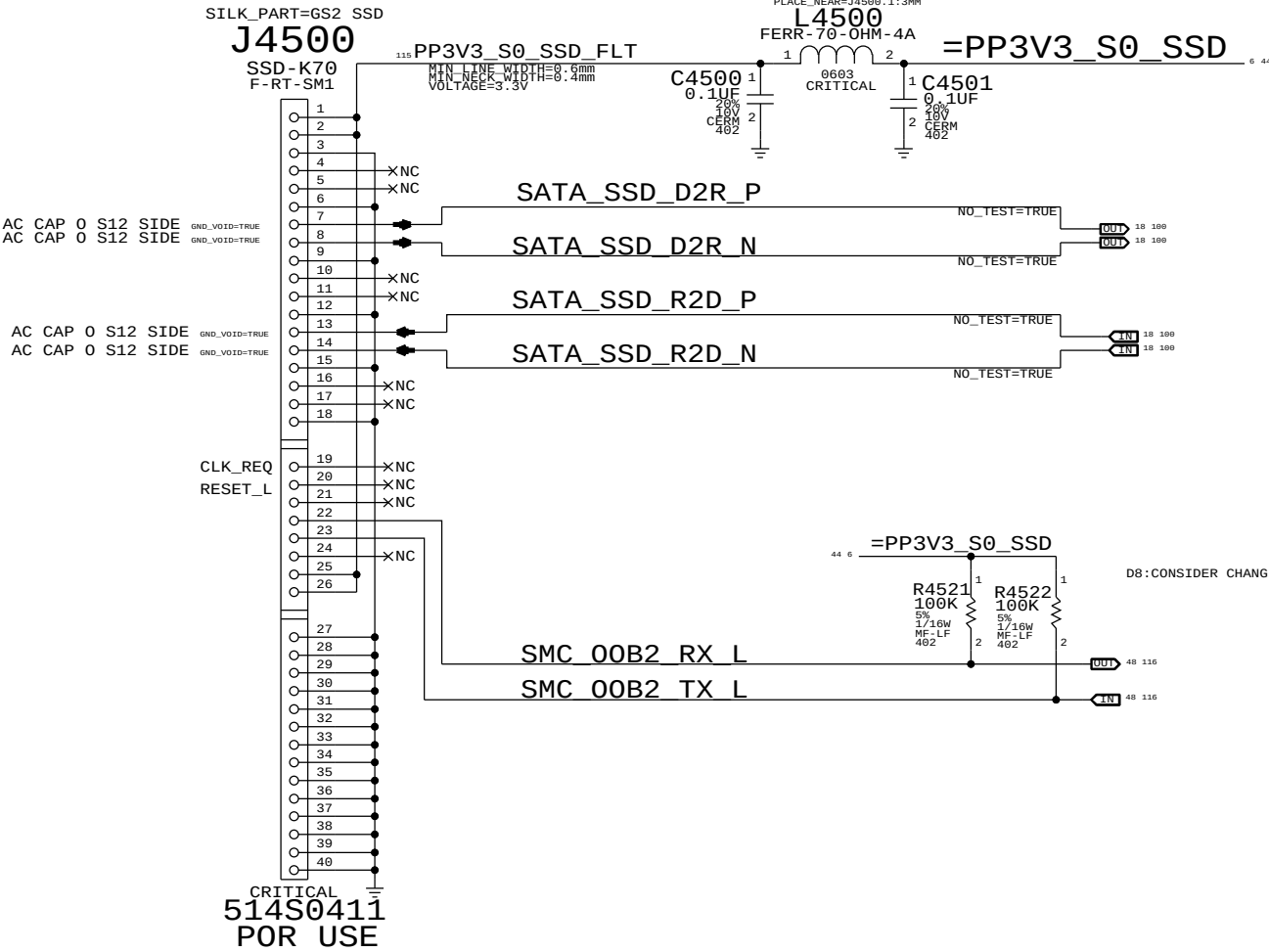
HDD DATA



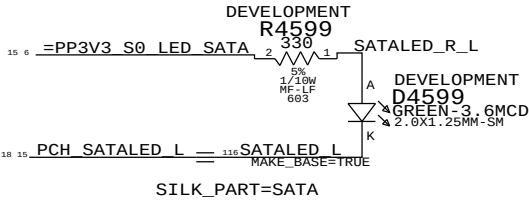
HDD POWER



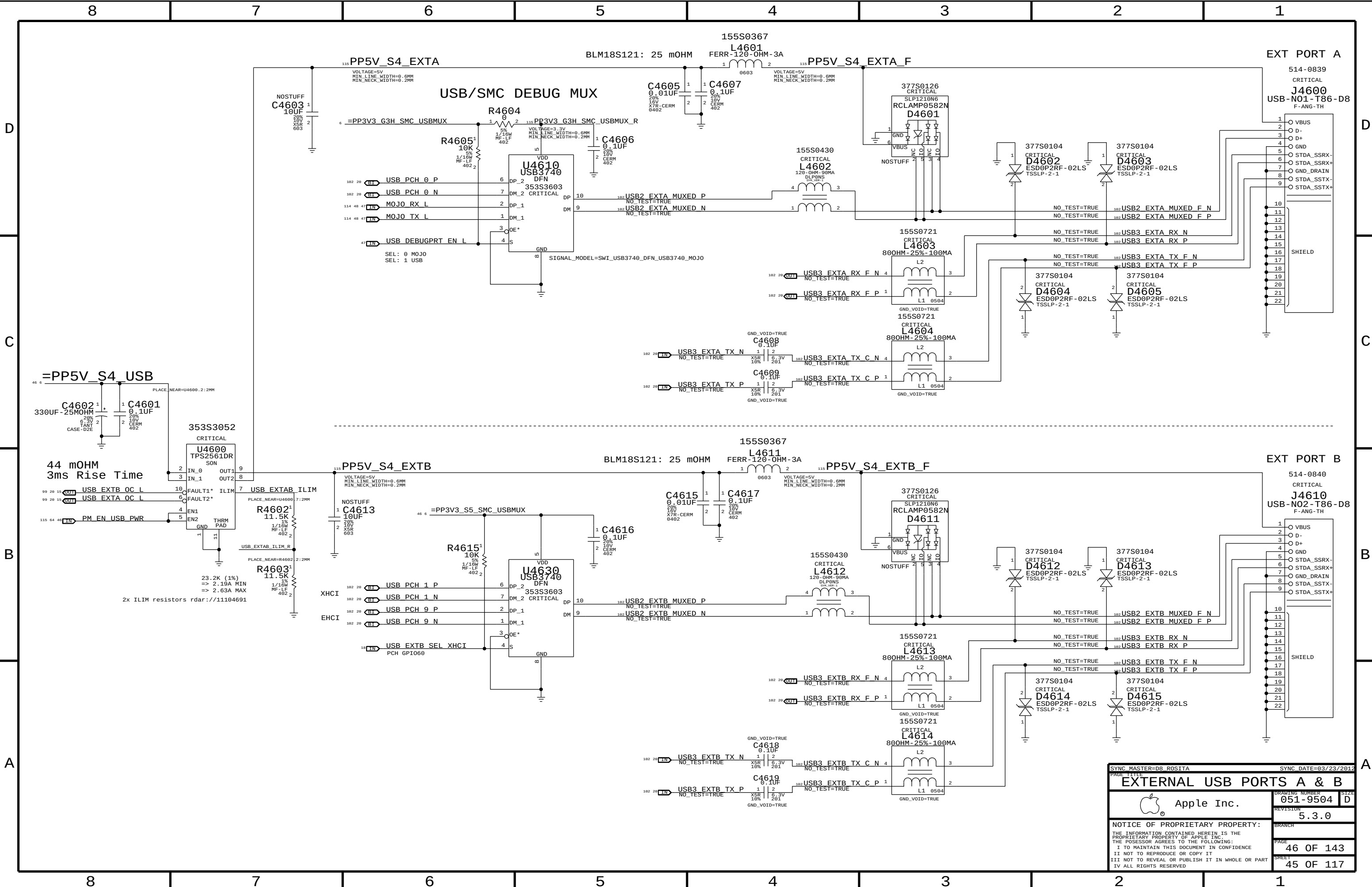
GUMSTICK2



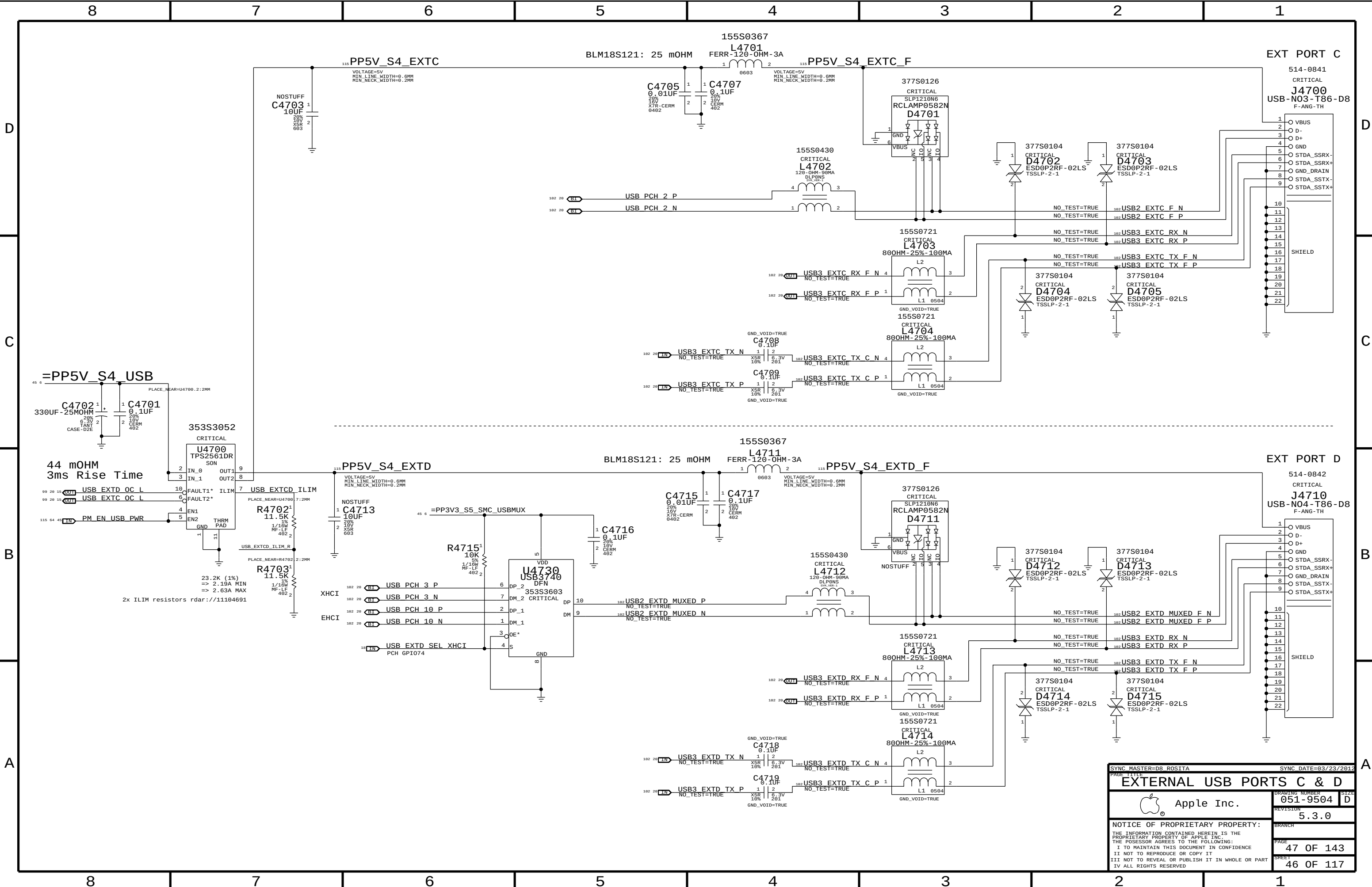
SATA Activity LED



SYNC MASTER=D8_JERRY		SYNC DATE=01/31/2012	
PAGE TITLE		SATA Connectors	
 Apple Inc.		DRAWING NUMBER	051-9504
		SIZE	D
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		REVISION	5.3.0
		BRANCH	
		PAGE	45 OF 143
		SHEET	44 OF 117



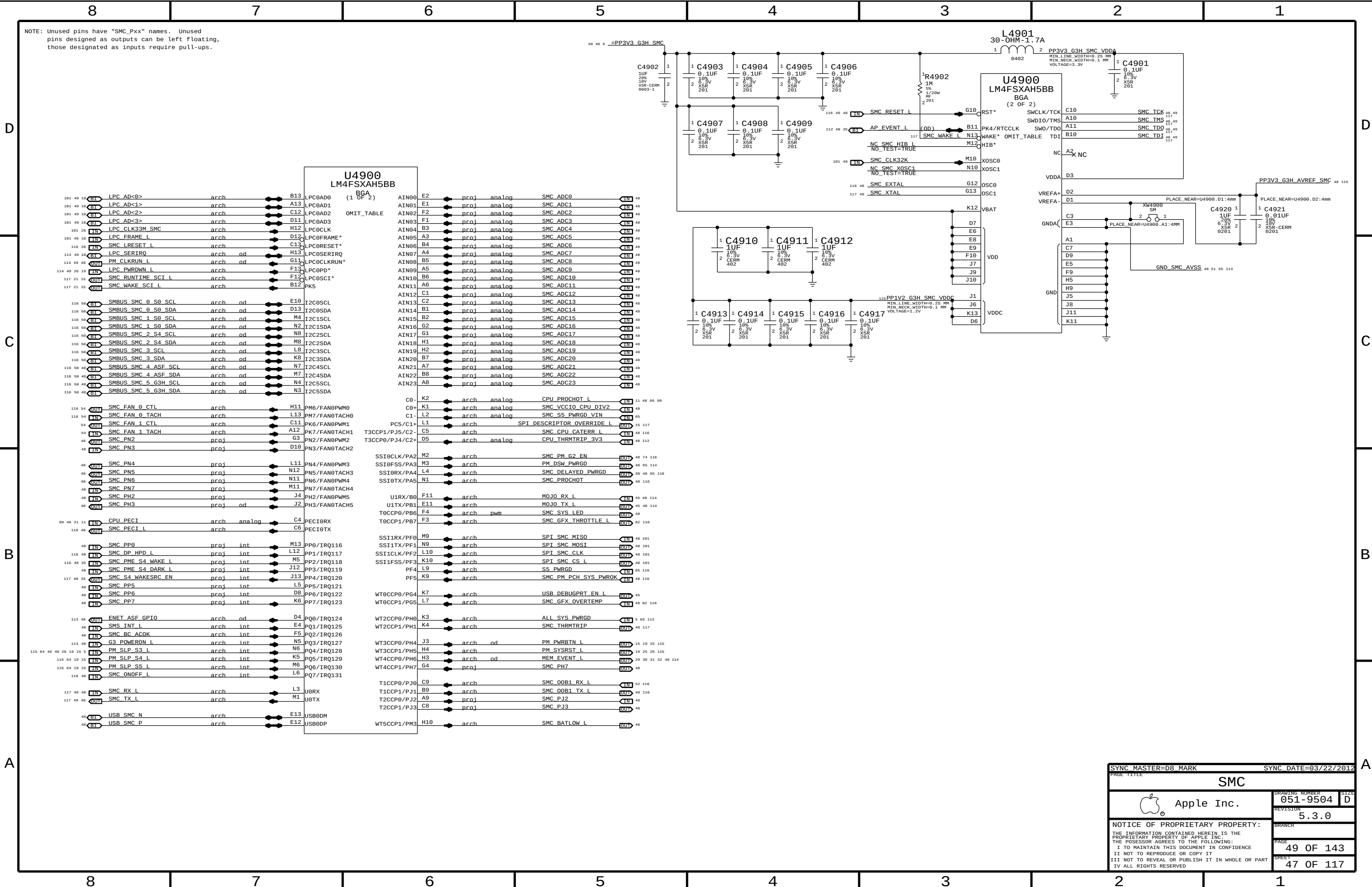
SYNC MASTER=D8 ROSITA		SYNC DATE=03/23/2012	
EXTERNAL USB PORTS A & B		DRAWING NUMBER	
Apple Inc.		051-9504	
NOTICE OF PROPRIETARY PROPERTY:		REVISION	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:		5.3.0	
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		BRANCH	
II NOT TO REPRODUCE OR COPY IT		PAGE	
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART		46 OF 143	
IV ALL RIGHTS RESERVED		SHEET	
		45 OF 117	

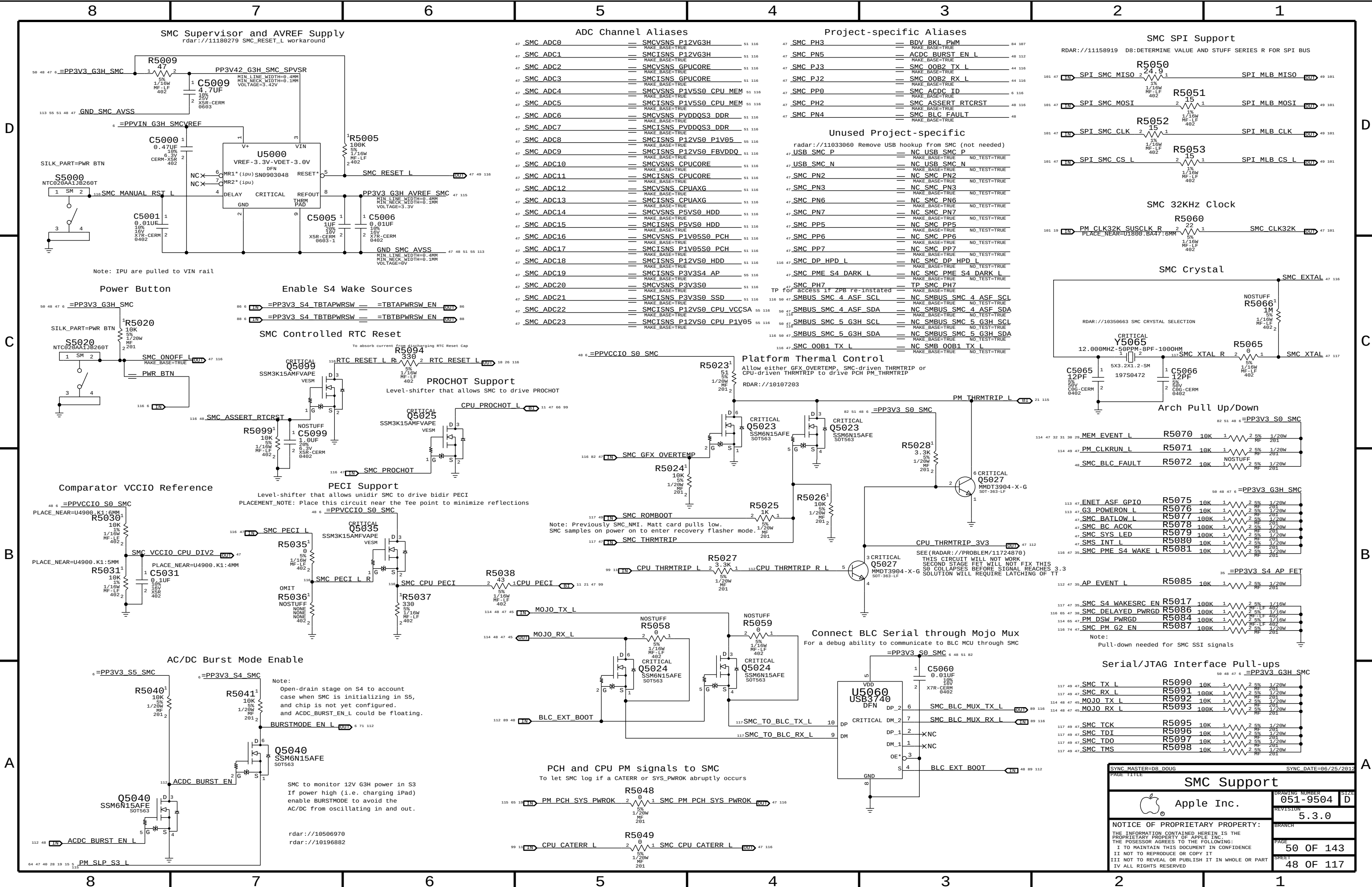


SYNC MASTER=D8 ROSITA

SYNC DATE=03/23/2012

EXTERNAL USB PORTS C & D		DRAWING NUMBER	051-9504	SIZE	D
Apple Inc.		REVISION	5.3.0		
NOTICE OF PROPRIETARY PROPERTY:		BRANCH			
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:		PAGE	47 OF 143		
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		SHEET	46 OF 117		
II NOT TO REPRODUCE OR COPY IT					
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART					
IV ALL RIGHTS RESERVED					





SMC Supervisor and AVREF Supply

rdar://11180279 SMC_RESET_L workaround

ADC Channel Aliases

Project-specific Aliases

SMC SPI Support

RDAR://11158919 D8:DETERMINE VALUE AND STUFF SERIES R FOR SPI BUS

Unused Project-specific

radar://11033060 Remove USB hookup from SMC (not needed)

SMC 32KHz Clock

SMC Crystal

RDAR://10350663 SMC CRYSTAL SELECTION

Platform Thermal Control

Allow either GFX OVERTEMP, SMC-driven THRMTrip or CPU-driven THRMTrip to drive PCH PM_THRMTrip

RDAR://10107203

Connect BLC Serial through Mojo Mux

For a debug ability to communicate to BLC MCU through SMC

Serial/JTAG Interface Pull-ups

PCH and CPU PM signals to SMC

To let SMC log if a CATERR or SYS_PWROK abruptly occurs

SYNC MASTER=D8 DOUG

SYNC DATE=06/25/2012

PAGE TITLE

SMC Support

Apple Inc.

NOTICE OF PROPRIETARY PROPERTY:

THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
IV ALL RIGHTS RESERVED

DRAWING NUMBER

051-9504

REVISION

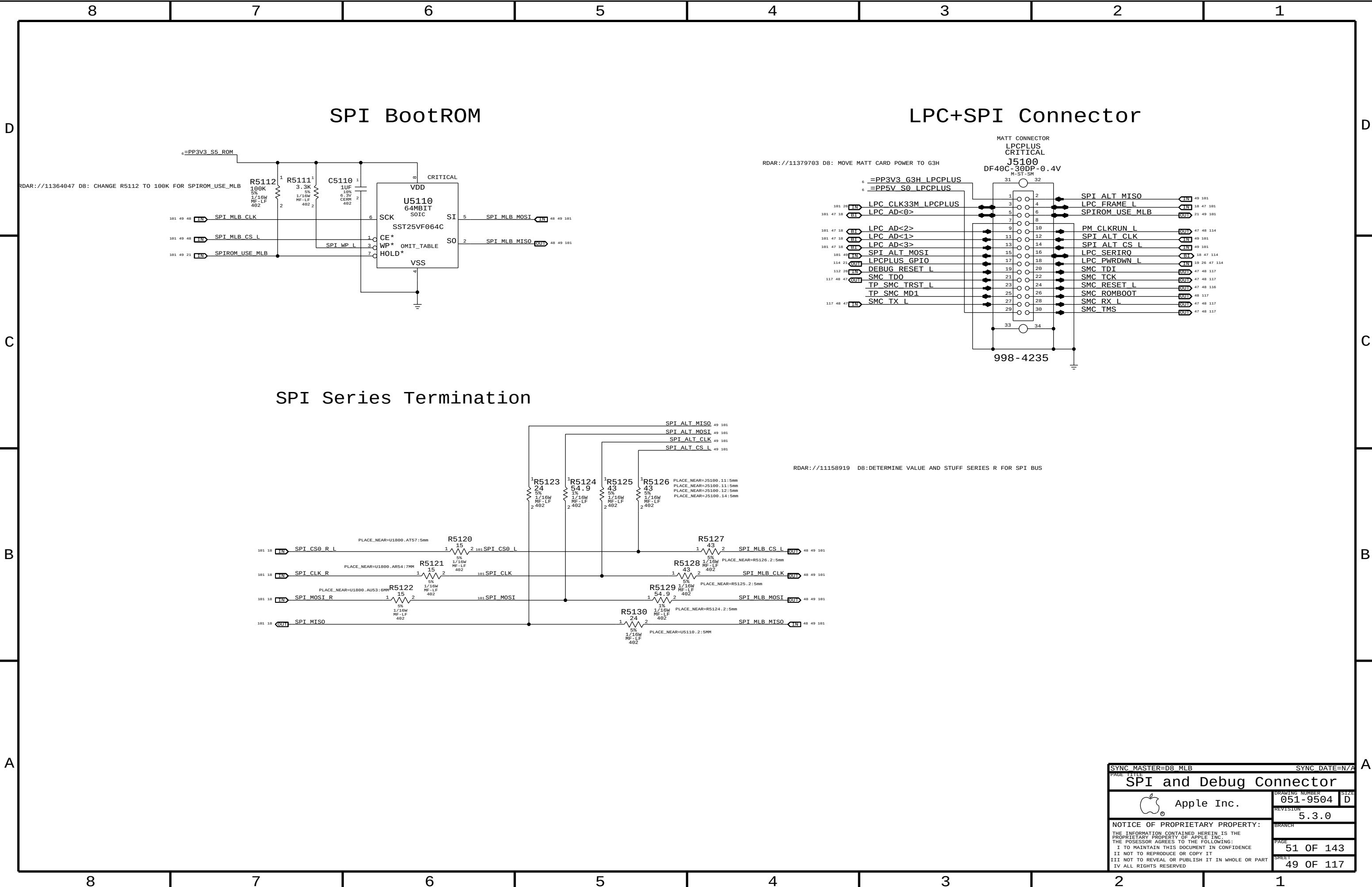
5.3.0

PAGE

50 OF 143

SHEET

48 OF 117

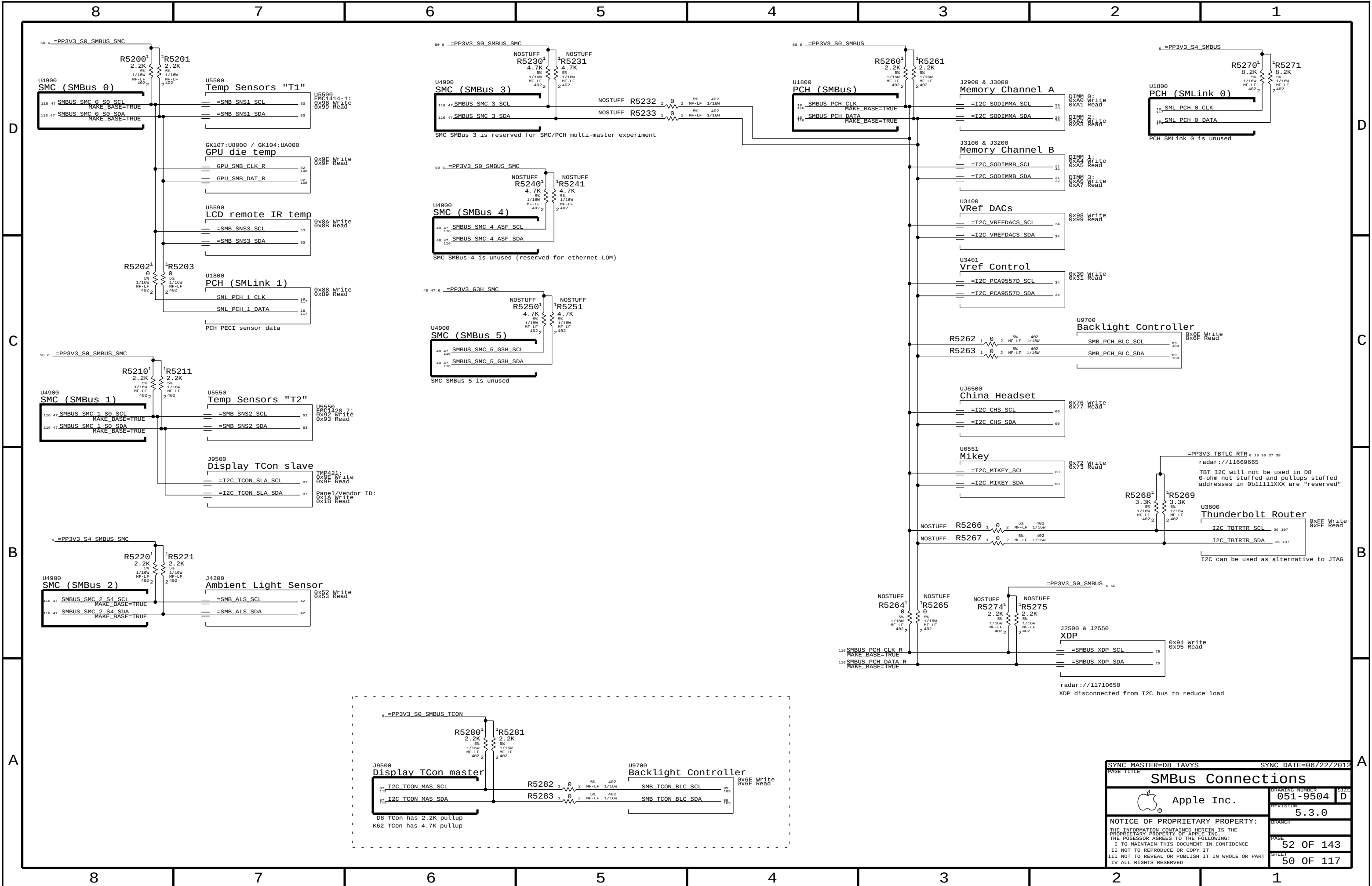


SPI BootROM

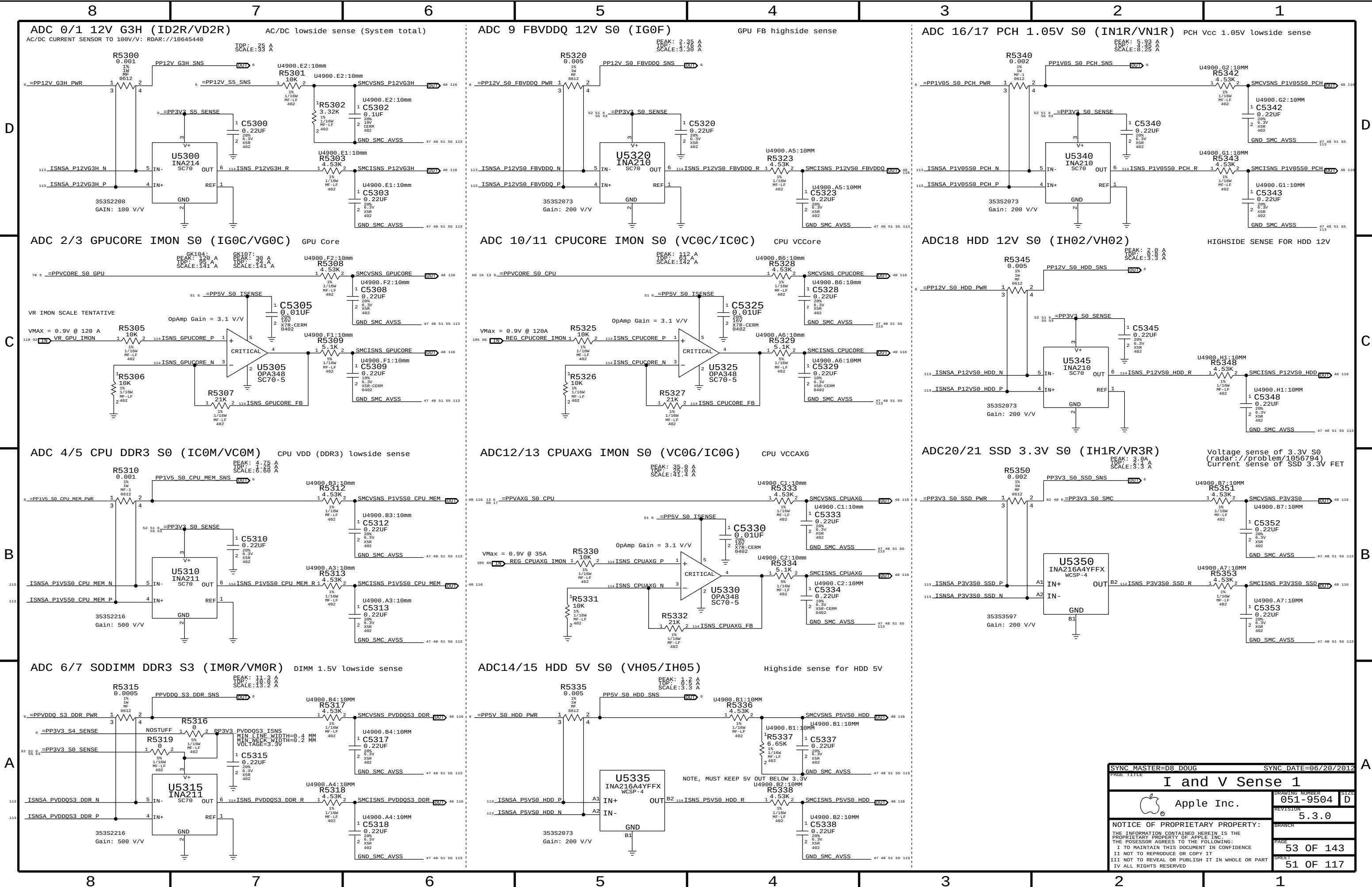
LPC+SPI Connector

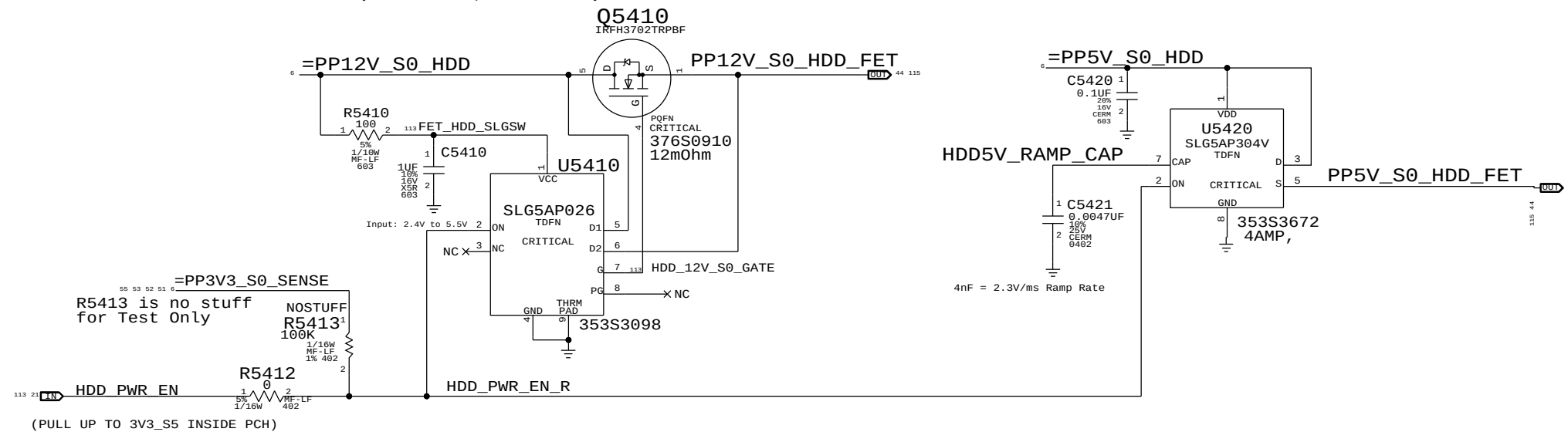
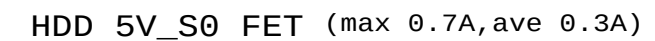
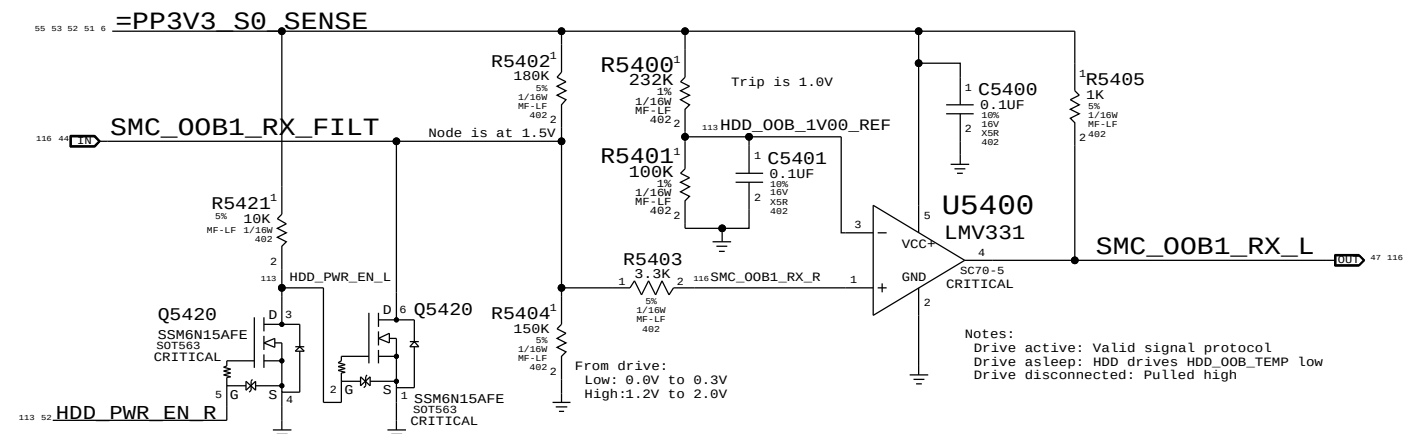
SPI Series Termination

SYNC MASTER=DB_MLB		SYNC DATE=N/A	
PAGE TITLE			
SPI and Debug Connector			
 Apple Inc.		DRAWING NUMBER	051-9504
		SIZE	D
		REVISION	5.3.0
		BRANCH	
NOTICE OF PROPRIETARY PROPERTY:			
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:			
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE			
II NOT TO REPRODUCE OR COPY IT			
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IV ALL RIGHTS RESERVED			
		PAGE	51 OF 143
		SHEET	49 OF 117



SYNC MASTER=D8 TAVYS		SYNC DATE=06/22/2012	
PAGE TITLE		SMBus Connections	
Apple Inc.		DRAWING NUMBER	051-9504
		REVISION	5.3.0
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:		PAGE	52 OF 143
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		SHEET	50 OF 117
II NOT TO REPRODUCE OR COPY IT			
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IV ALL RIGHTS RESERVED			





REMOVE R5413 AND SHORT R5412 AFTER HDD_PWR_EN WORKS

SYNCH-MASTER-DB- JERRY- - - - -		SYNCH-DATE=02/25/2012	
PAGE TITLE			
HDD/SSD Temp Sense			
 Apple Inc.		DRAWING NUMBER 051-9504	
		SIZE D	
		REVISION 5.3.0	
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE I NOT TO REPRODUCE OR COPY IT I NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART I ALL RIGHTS RESERVED		PAGE 54 OF 143	
		SHEET 52 OF 117	

D

C

B

A

```

47 [IN] SMC_FAN_1_CTL == NC_SMC_FAN_1_CTL
                        MAKE_BASE=TRUE NO_TEST=TRUE

47 [OUT] SMC_FAN_1_TACH == NC_SMC_FAN_1_TACH
                         MAKE_BASE=TRUE NO_TEST=TRUE

```

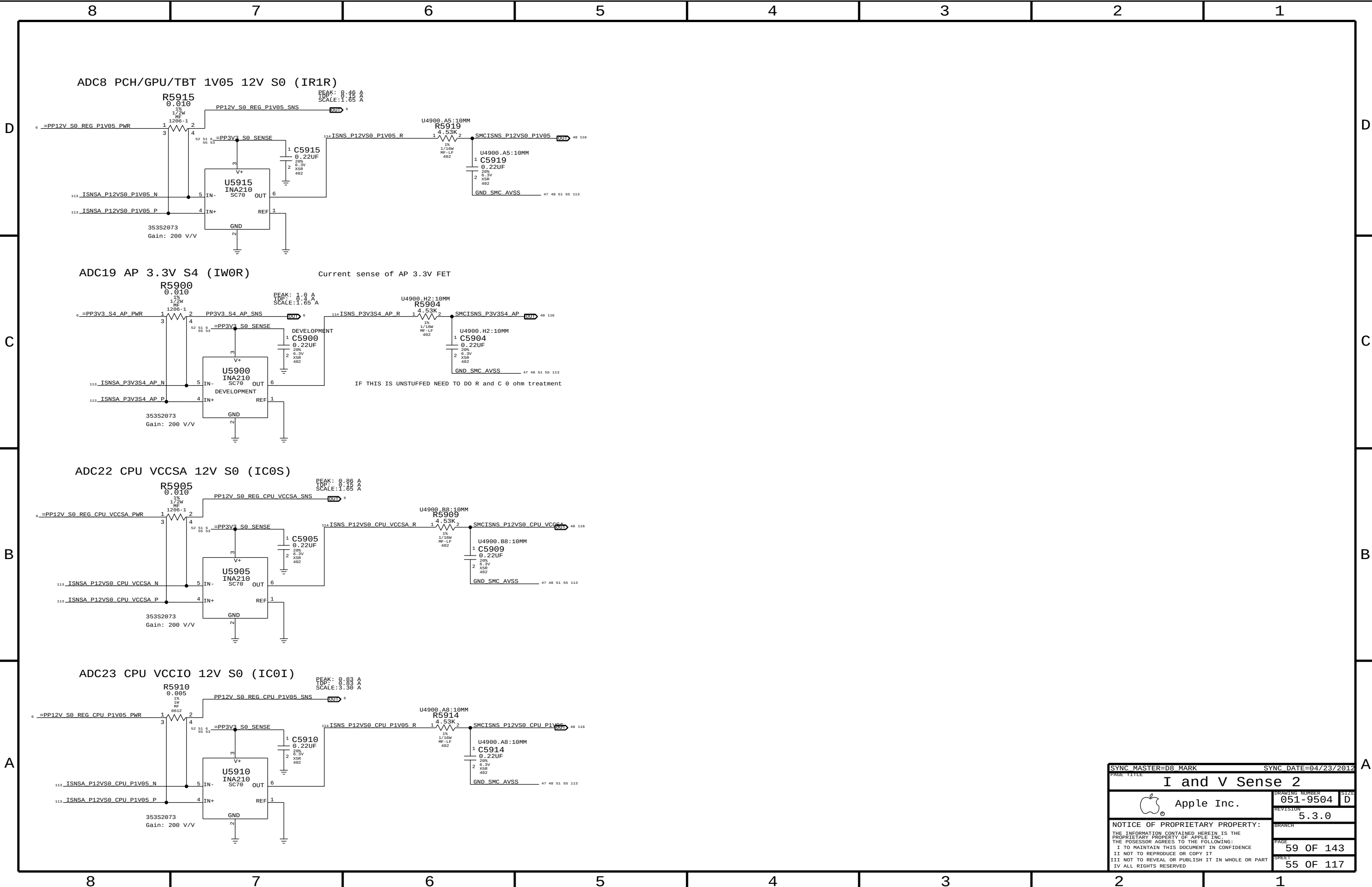


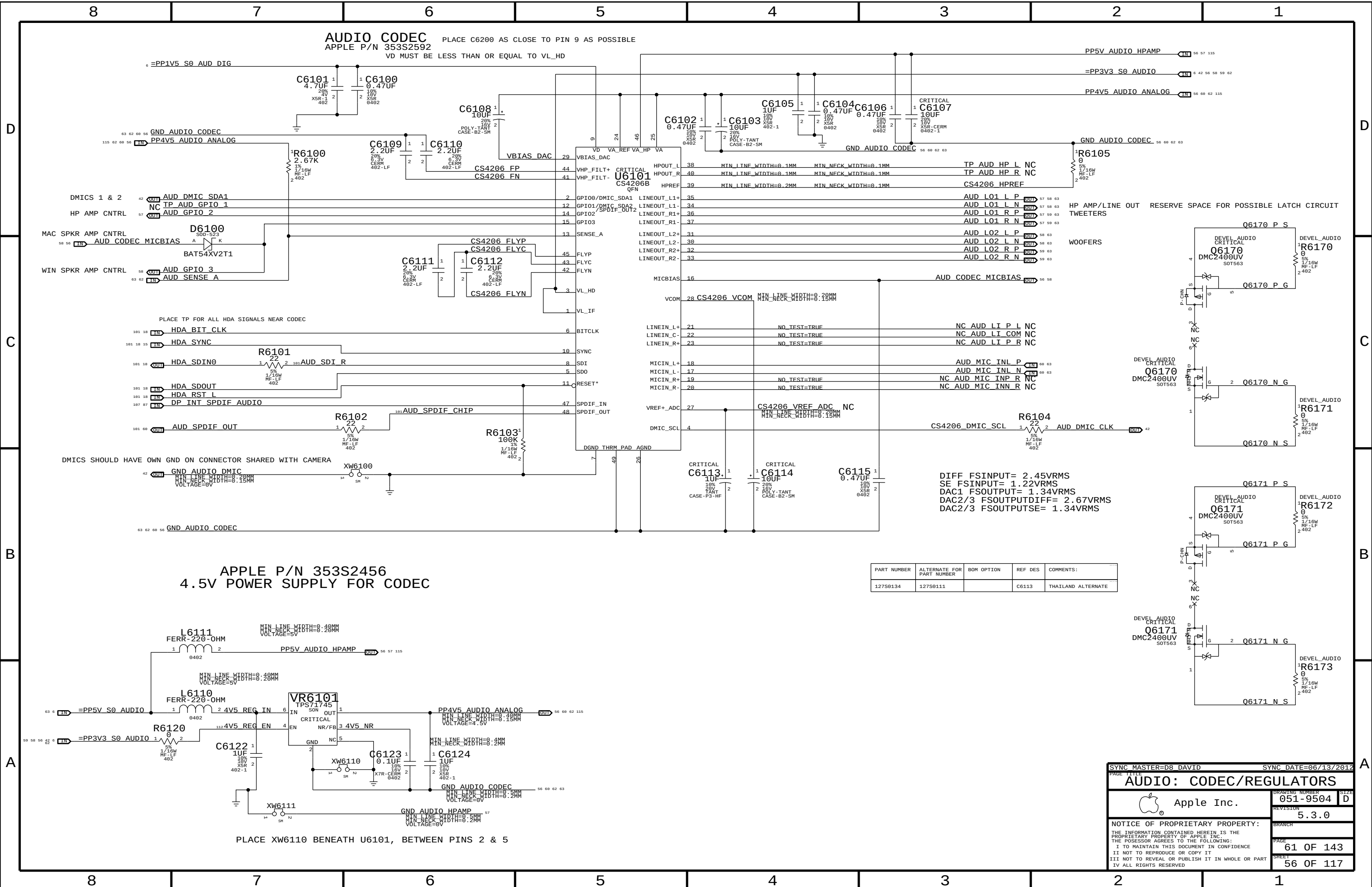
C

B

A

SYNC MASTER=D7 MLB		SYNC DATE=05/24/2012	
PAGE TITLE			
System Fan			
 Apple Inc.	DRAWING NUMBER		SIZE
	051-9504		D
		REVISION	
		5.3.0	
NOTICE OF PROPRIETARY PROPERTY:			
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:			
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE			
II NOT TO REPRODUCE OR COPY IT			
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IV ALL RIGHTS RESERVED			
		BRANCH	
		PAGE	
		56 OF 143	
		SHEET	
		54 OF 117	





PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
127S0134	127S0111		C6113	THAILAND ALTERNATE

SYNC MASTER=D8 DAVID

SYNC DATE=06/13/2012

AUDIO: CODEC/REGULATORS

Apple Inc.

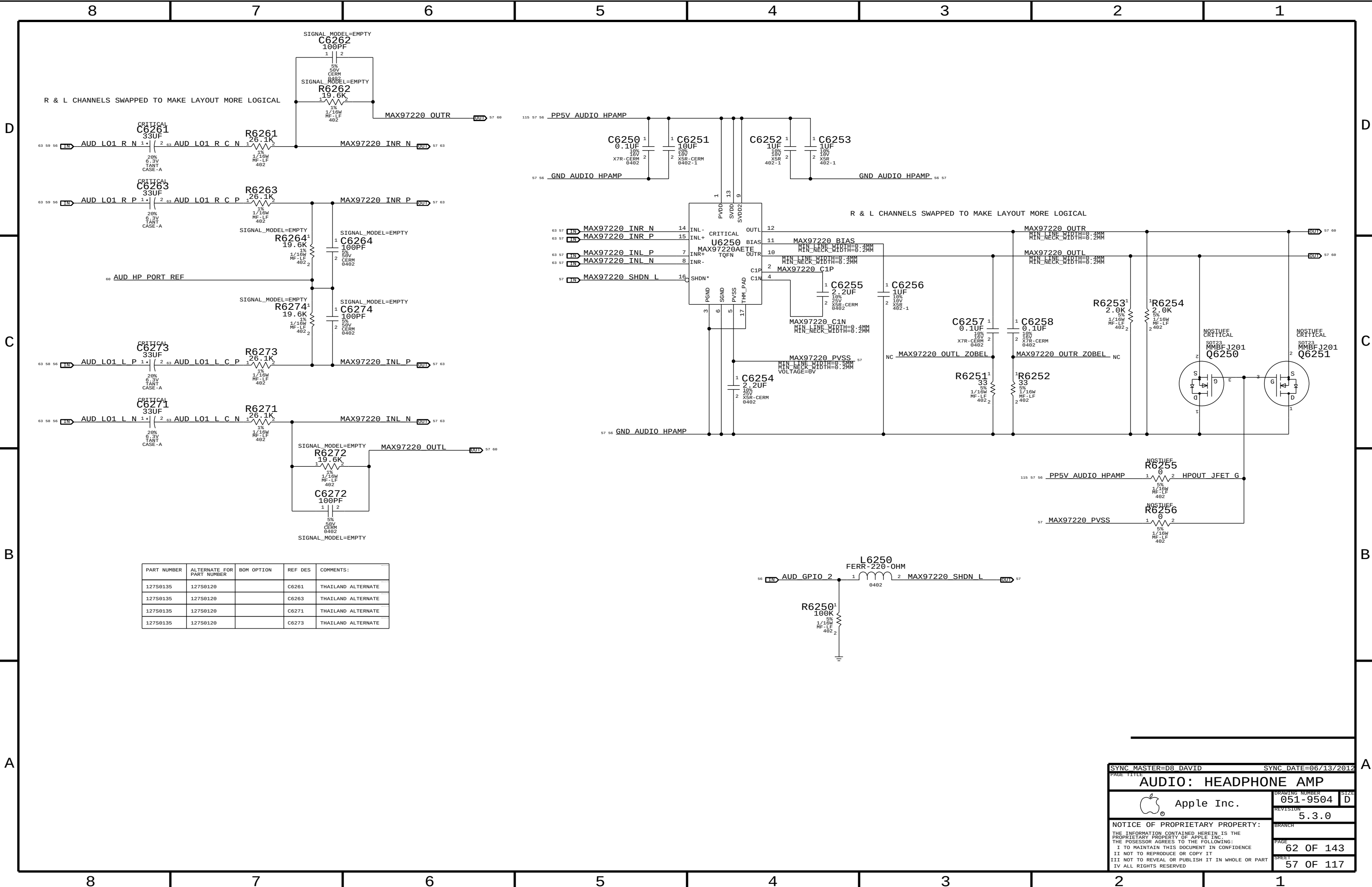
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
IV ALL RIGHTS RESERVED

DRAWING NUMBER
051-9504

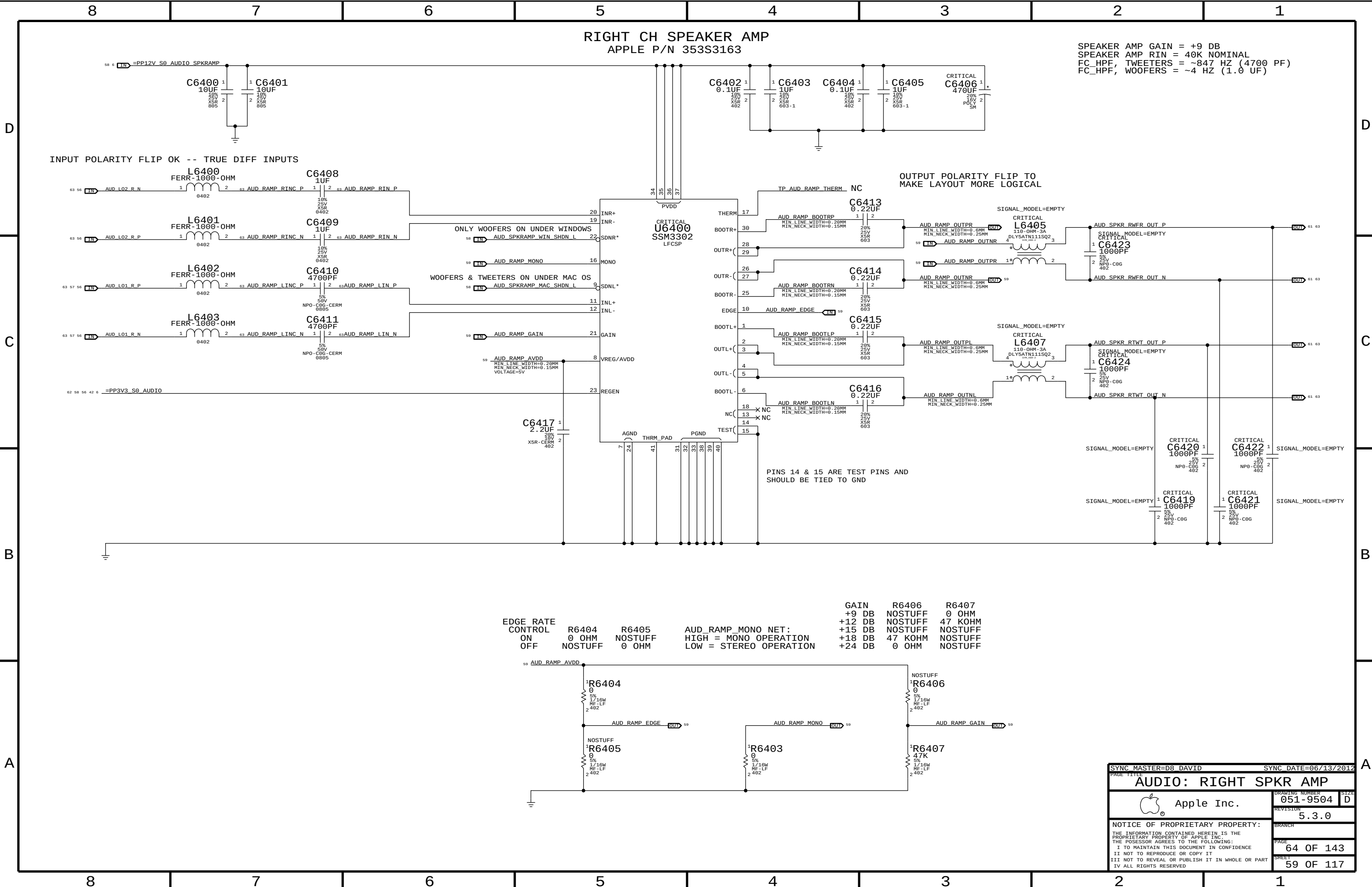
REVISION
5.3.0

PAGE
61 OF 143

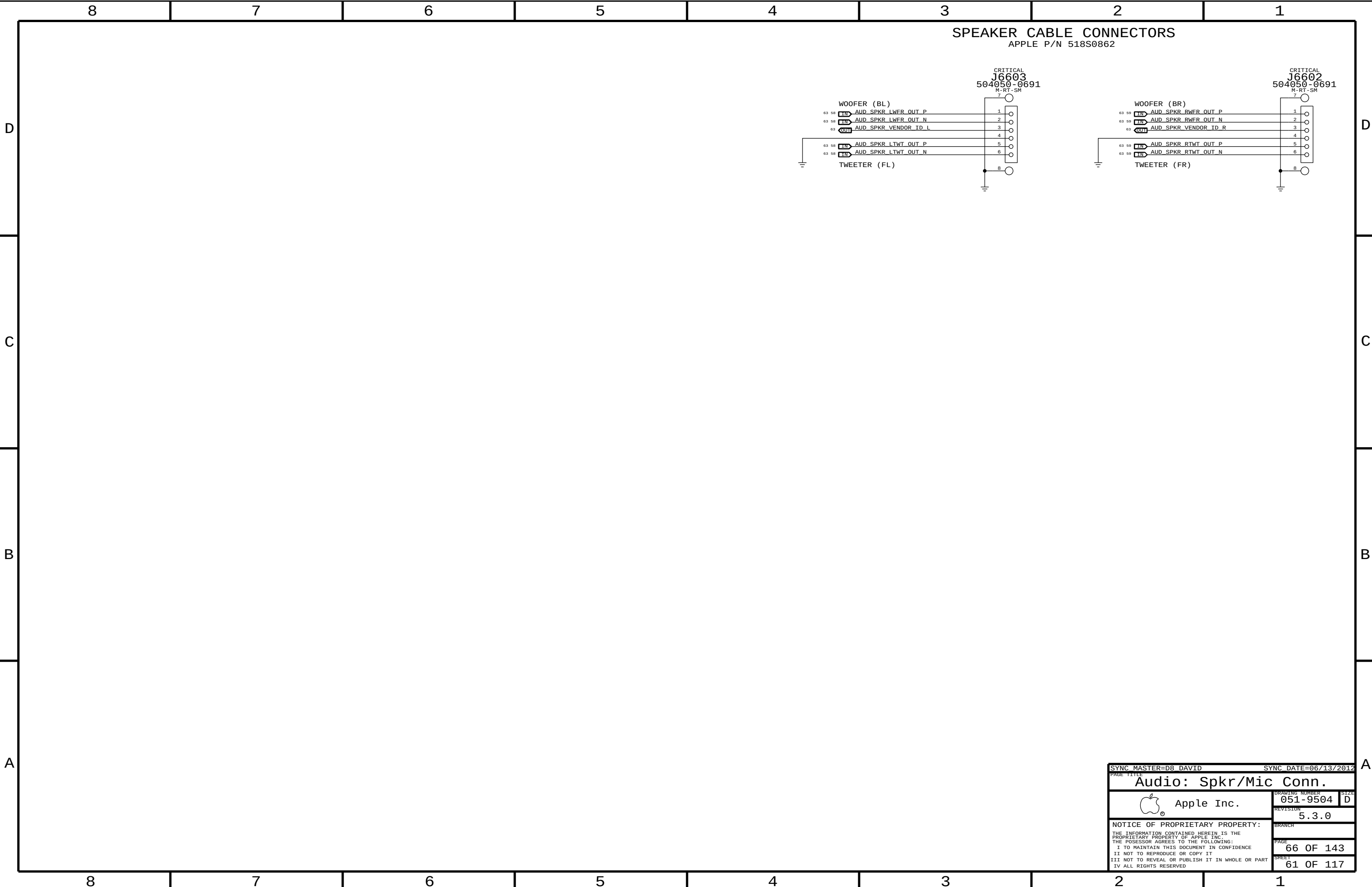
SHEET
56 OF 117



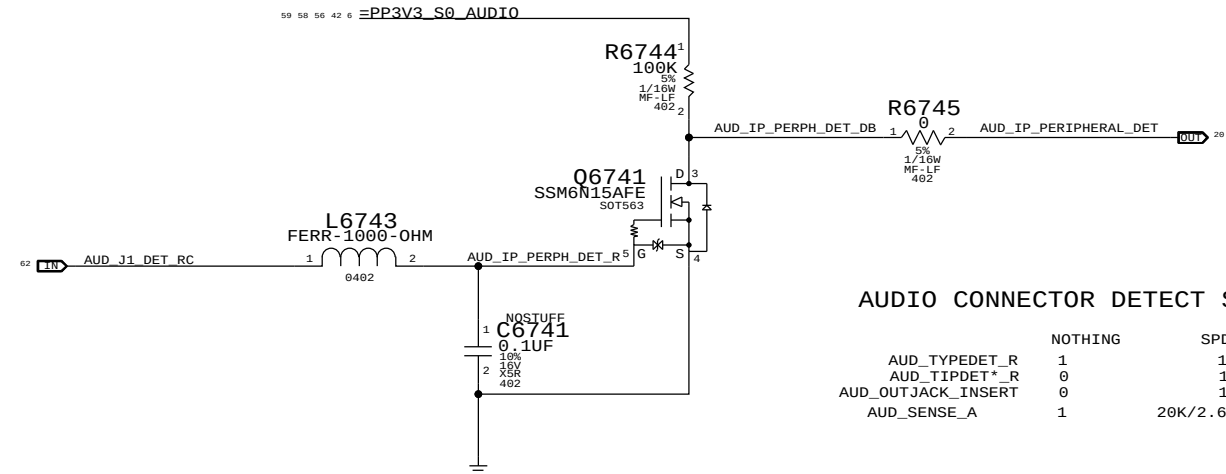
PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
127S0135	127S0120		C6261	THAILAND ALTERNATE
127S0135	127S0120		C6263	THAILAND ALTERNATE
127S0135	127S0120		C6271	THAILAND ALTERNATE
127S0135	127S0120		C6273	THAILAND ALTERNATE



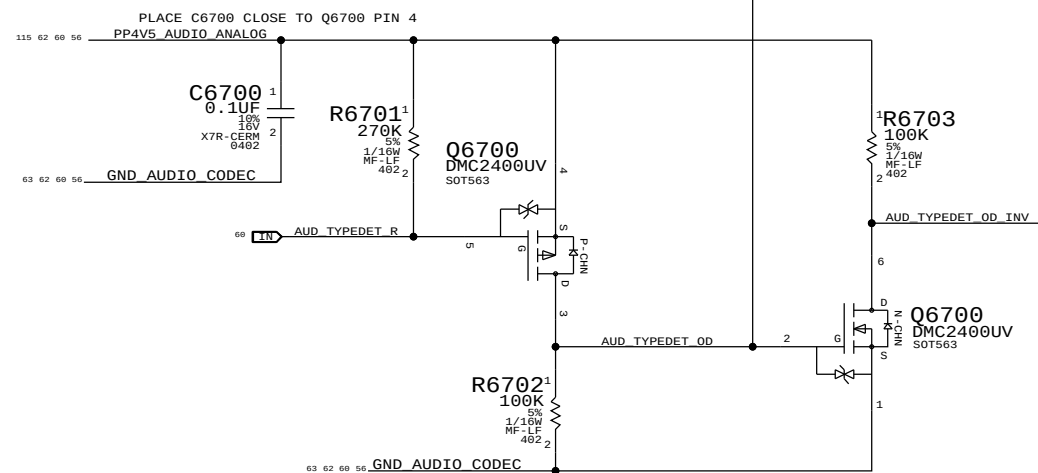
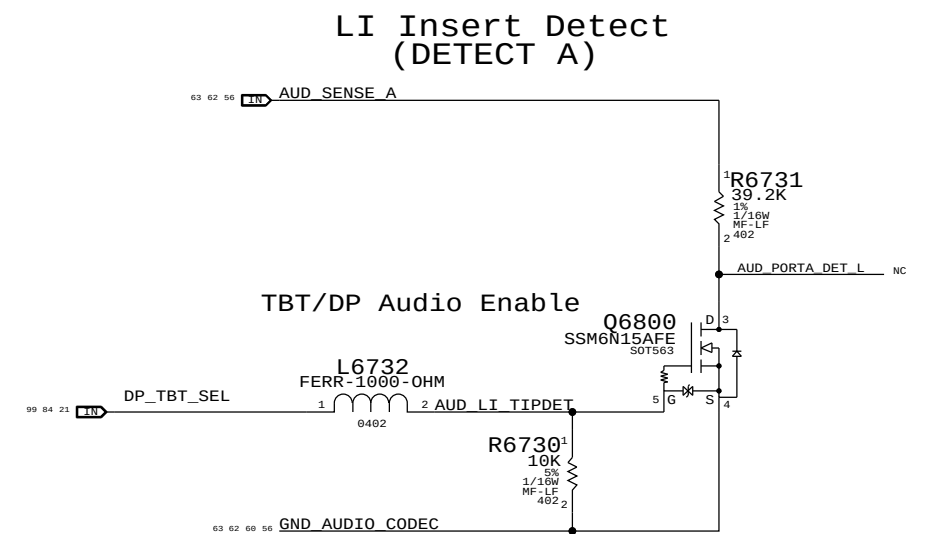
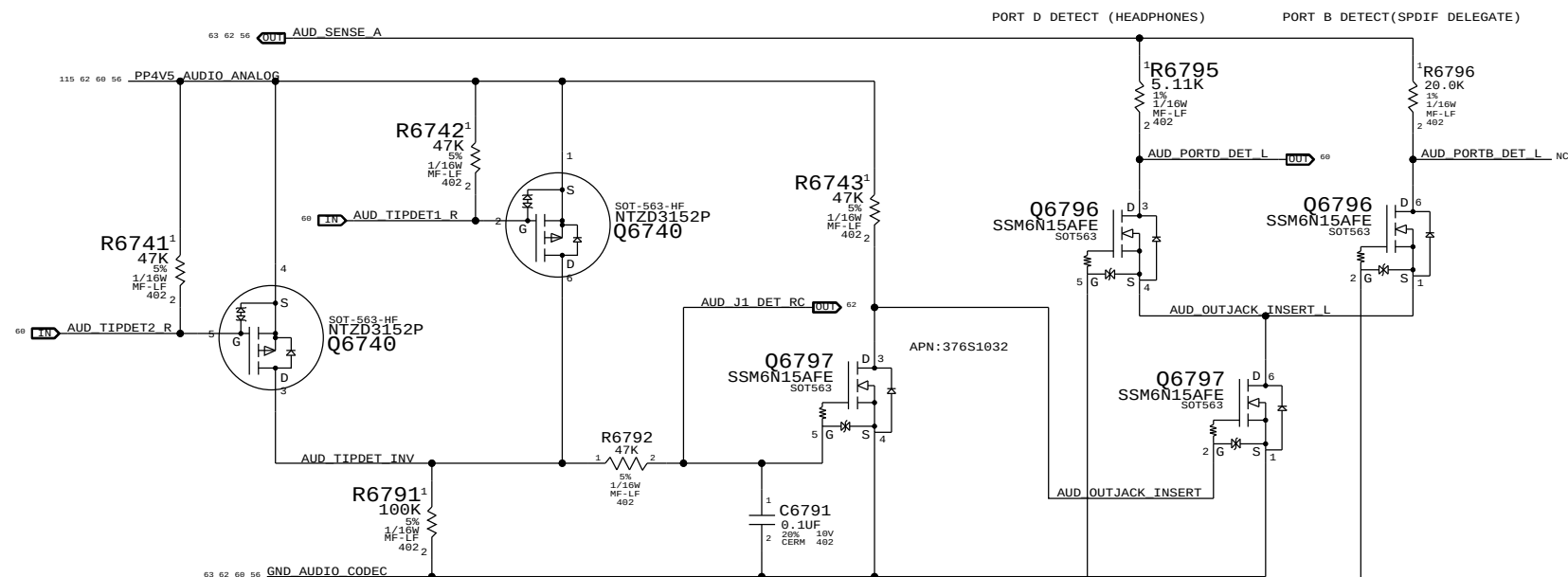
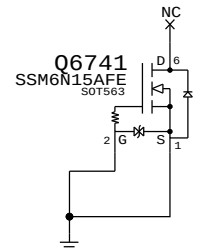
PAGE TITLE		PAGE TITLE	
AUDIO: RIGHT SPKR AMP		051-9504	
Apple Inc.		5.3.0	
NOTICE OF PROPRIETARY PROPERTY:		64 OF 143	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:		59 OF 117	
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE			
II NOT TO REPRODUCE OR COPY IT			
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IV ALL RIGHTS RESERVED			



IPHS HS Detect Debounce CKT



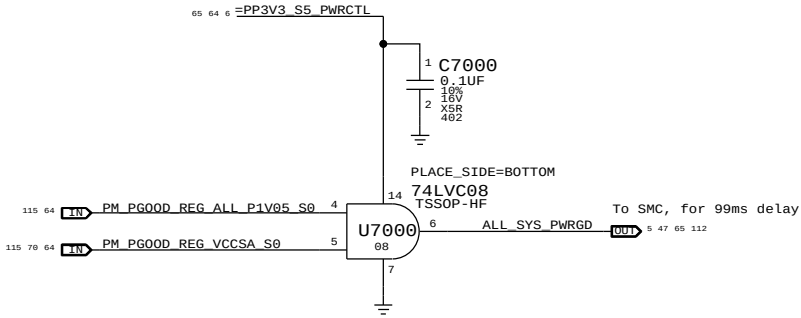
AUDIO CONNECTOR DETECT STATES			
	NOTHING	SPDIF	HEADPHONE
AUD_TYPEDET_R	1	1	0
AUD_TIPDET*_R	0	1	1
AUD_OUTJACK_INSERT	0	1	1
AUD_SENSE_INA	1	20K/2.67K RDIV	5.11K/2.67K RDIV



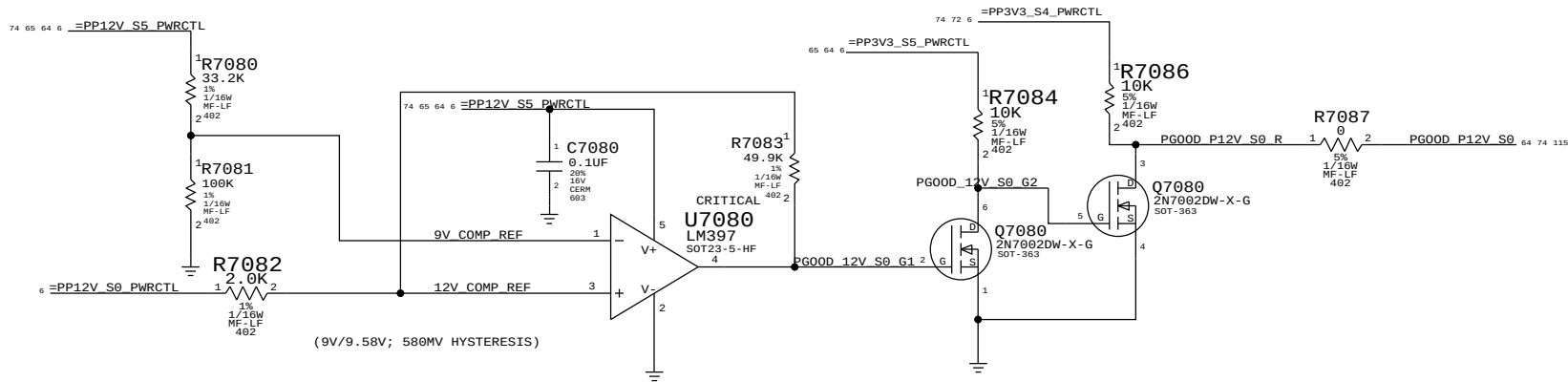
SYNC MASTER=D8 DAVID		SYNC DATE=06/13/2012	
PAGE 1111			
AUDIO: Detects/Grounding			
	DRAWING NUMBER		SIZE
	051-9504		D
Apple Inc.		REVISION	
		5.3.0	
NOTICE OF PROPRIETARY PROPERTY:			
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:			
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE			
I NOT TO REPRODUCE OR COPY IT			
I NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IN ALL RIGHTS RESERVED			

Platform Power Good derive SMC ALL_SYS_PWRGD

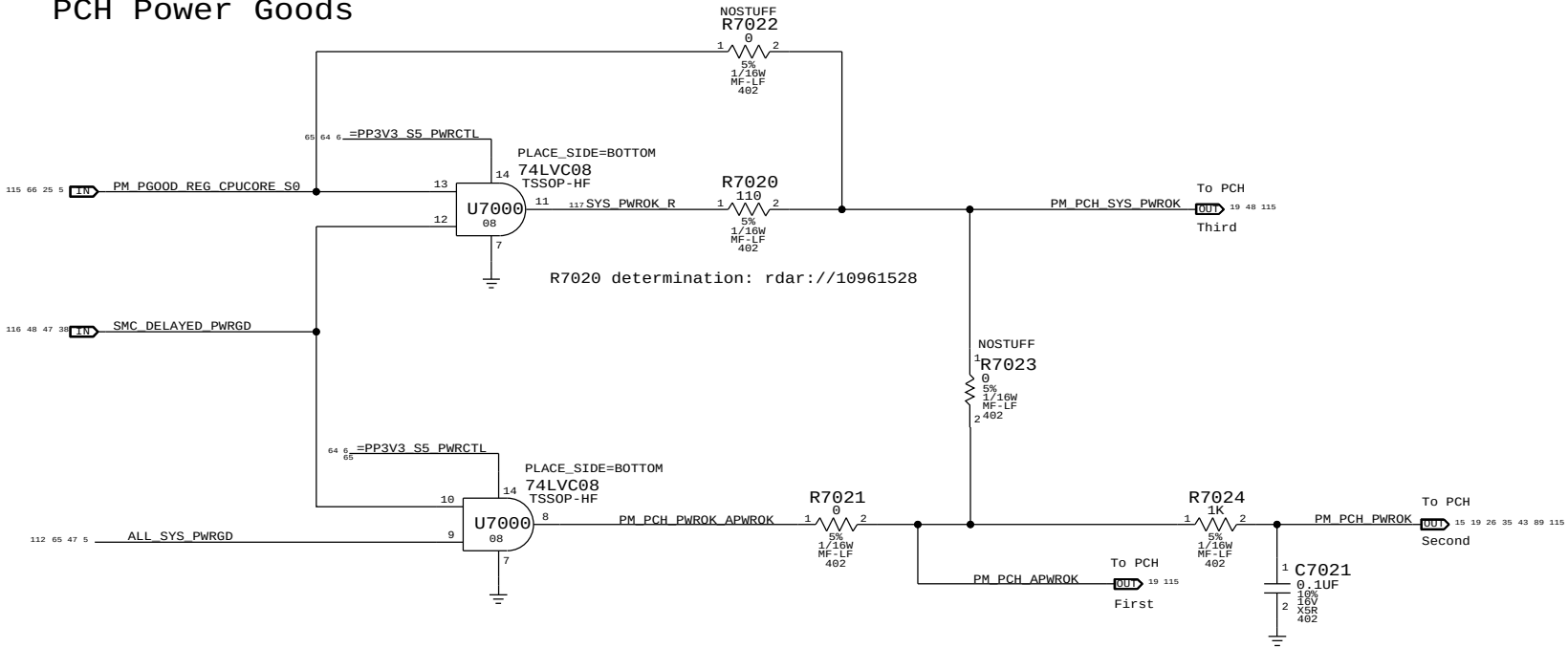
The end of the power sequence for S0 rails except CPU CORE.



PGOOD comparators for PP12V_S0



PCH Power Goods



radar://11043352 Need AND Gate to deassert PM_PCH_PWROK to PCH when unexpected power loss happens

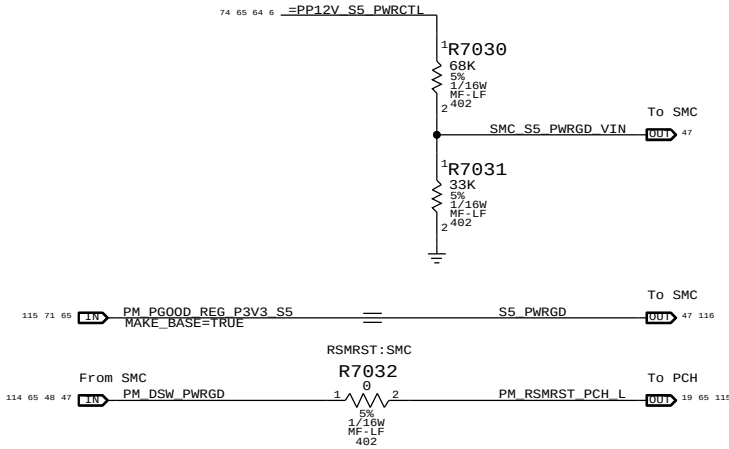
Resume Reset

Intel Doc# 29517 Maho Bay PDG, Section 22.13
Intel Doc# 29562 Panther Point EDS, Section 8.7 and 8.8

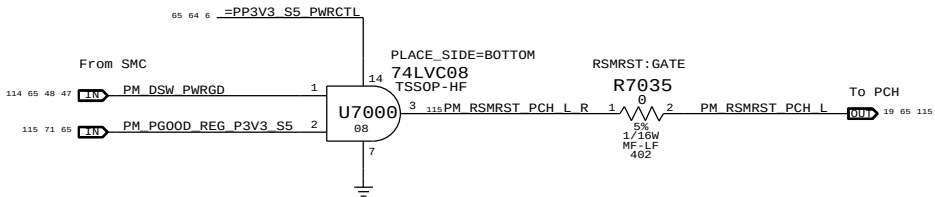
Note:
The iMac K70K72 designs does not support Deep Sx modes so both DPWROK and RSMRST# signals are shorted together

Requirements:
Power on:
Asserted at least 10 ms after all suspend well power is valid
Power off or loss of AC:
Transition to 0.8V or less before VccSUS3_3 drops to 2.90 V
to allow PCH to switch suspend well to battery without excessive loading

Primary method:
The SMC guarantees proper assertion and de-assertion of RSMRST# for normal operation.
SMC de-asserts RSMRST# (PM_DSW_PWRGD) when S5_PWRGD input is asserted and SMC_S5_PWRGD_VIN input is above comparator input level of 1.5 V.
SMC asserts RSMRST# (PM_DSW_PWRGD) when SMC_S5_PWRGD_VIN input drops from 1.8 V to 1.5 V (as implemented) when 12 V S5 rail drops to 10 V.



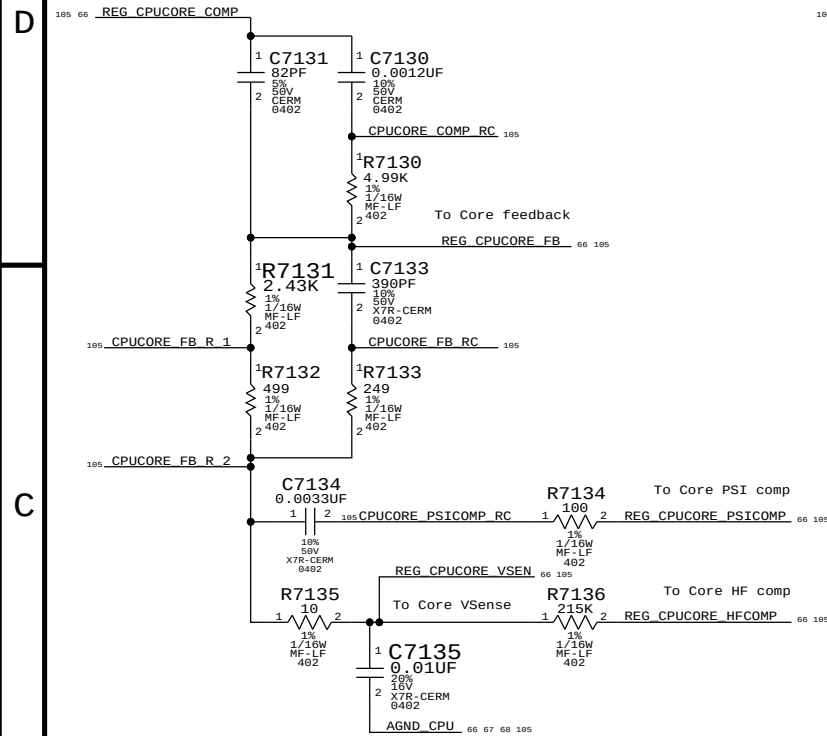
Secondary method:
The SMC guarantees proper assertion and de-assertion of RSMRST# for normal operation via PM_DSW_PWRGD.
RSMRST# is asserted when power good from regulator is de-asserted in the event AC is lost. Power good de-assertion should happen quickly enough to meet Intel spec.



SYNC MASTER=D8 MARK		SYNC DATE=04/23/2012	
PAGE TITLE		PM Power Good	
Apple Inc.		DRAWING NUMBER	051-9504
		REVISION	5.3.0
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:		PAGE	70 OF 143
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		SHEET	65 OF 117
II NOT TO REPRODUCE OR COPY IT			
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IV ALL RIGHTS RESERVED			

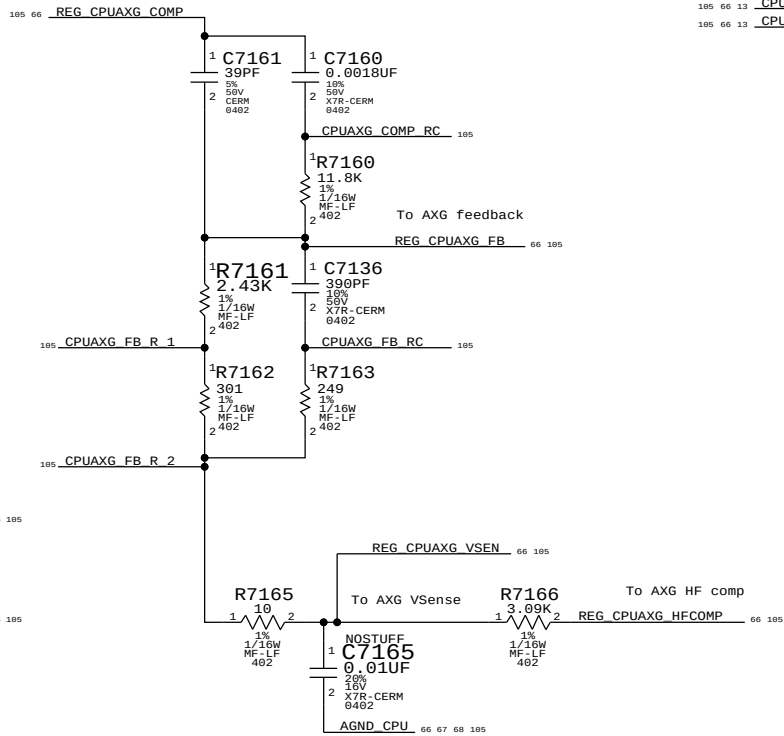
```
Max avg current: 63 A (BUDGET)
Max peak current: 110 A (BUDGET)
OC trip point: ? A (nom)/? A (min)
Switching freq: 290 kHz
```

Core compensation and feedback



```
Max avg current: 12.7 A (BUDGET)
Max peak current: 30.0 A (BUDGET)
OC trip point: ? A (nom)/? A (min)
Switching freq: 290 kHz
```

AXG compensation and feedback



Pull-ups 1

28 16 13 11 10 9 8 7 6 5 4 3 2 1

C7117
0.1uF
18V
X7R-CERM
0402

R7117
54.9
1k/16W
NF-1F
402

R7118
90.9
1k/16W
NF-1F
402

R7119
110
1k/16W
NF-1F
402

NOSTUFF

AGND_CPU

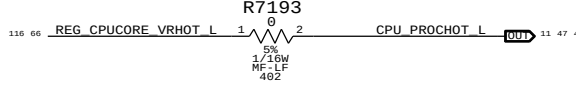
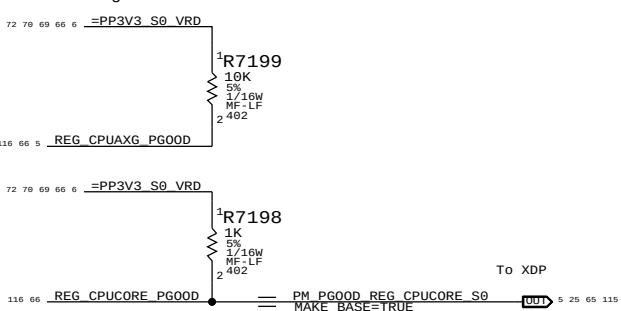
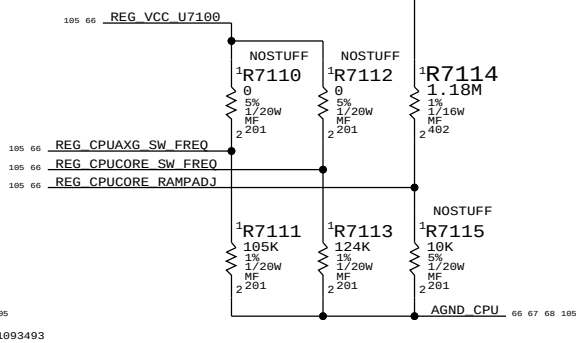
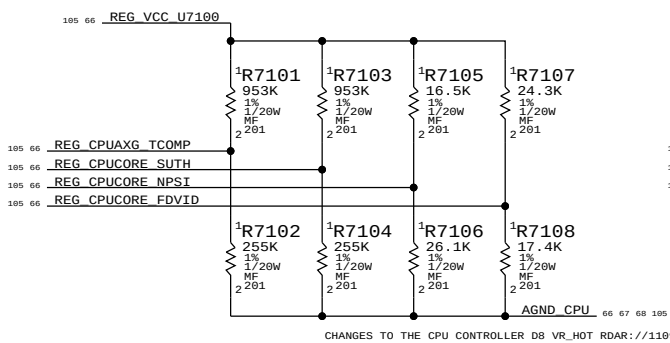
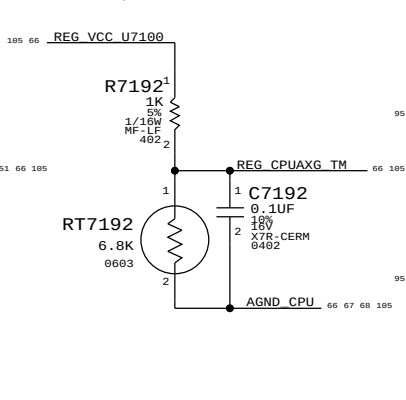
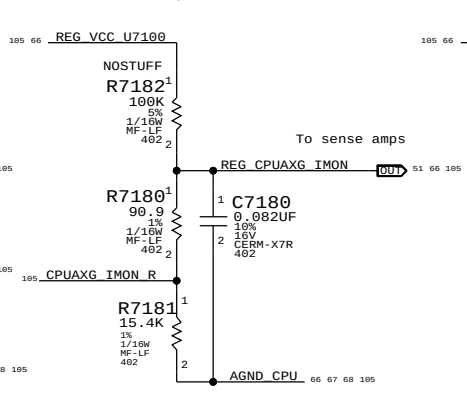
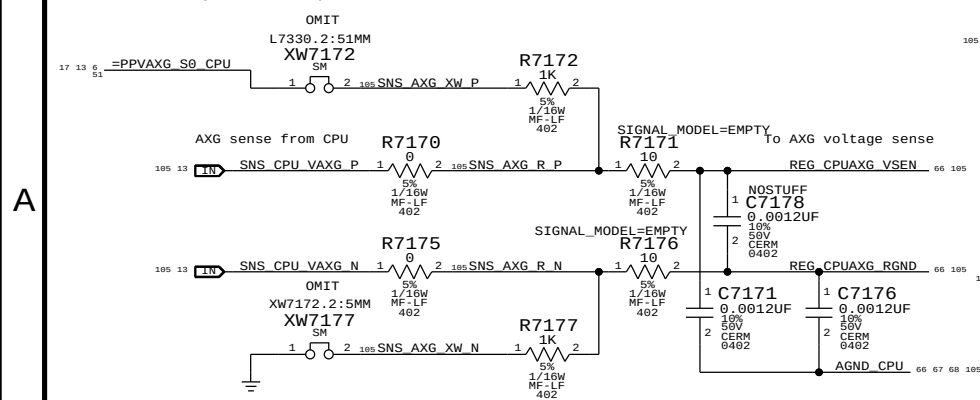
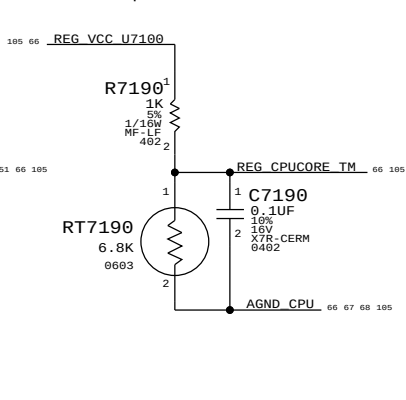
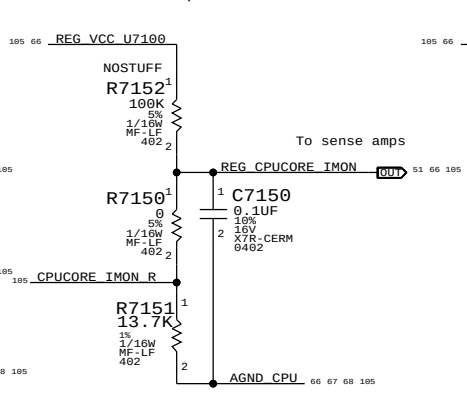
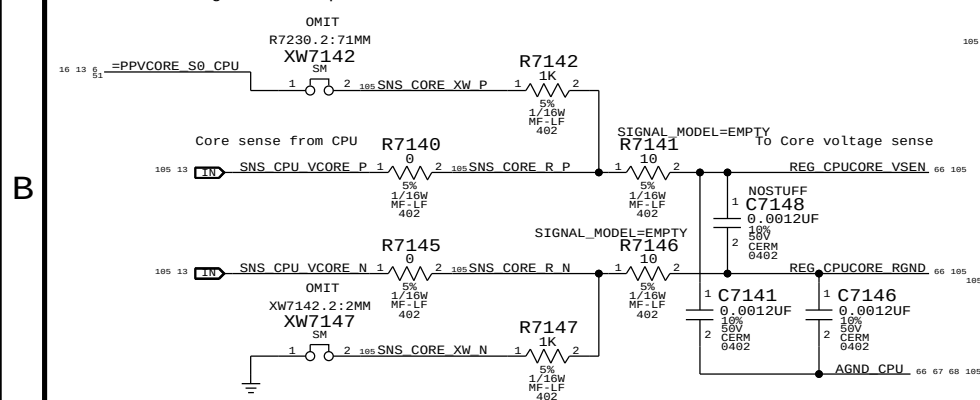
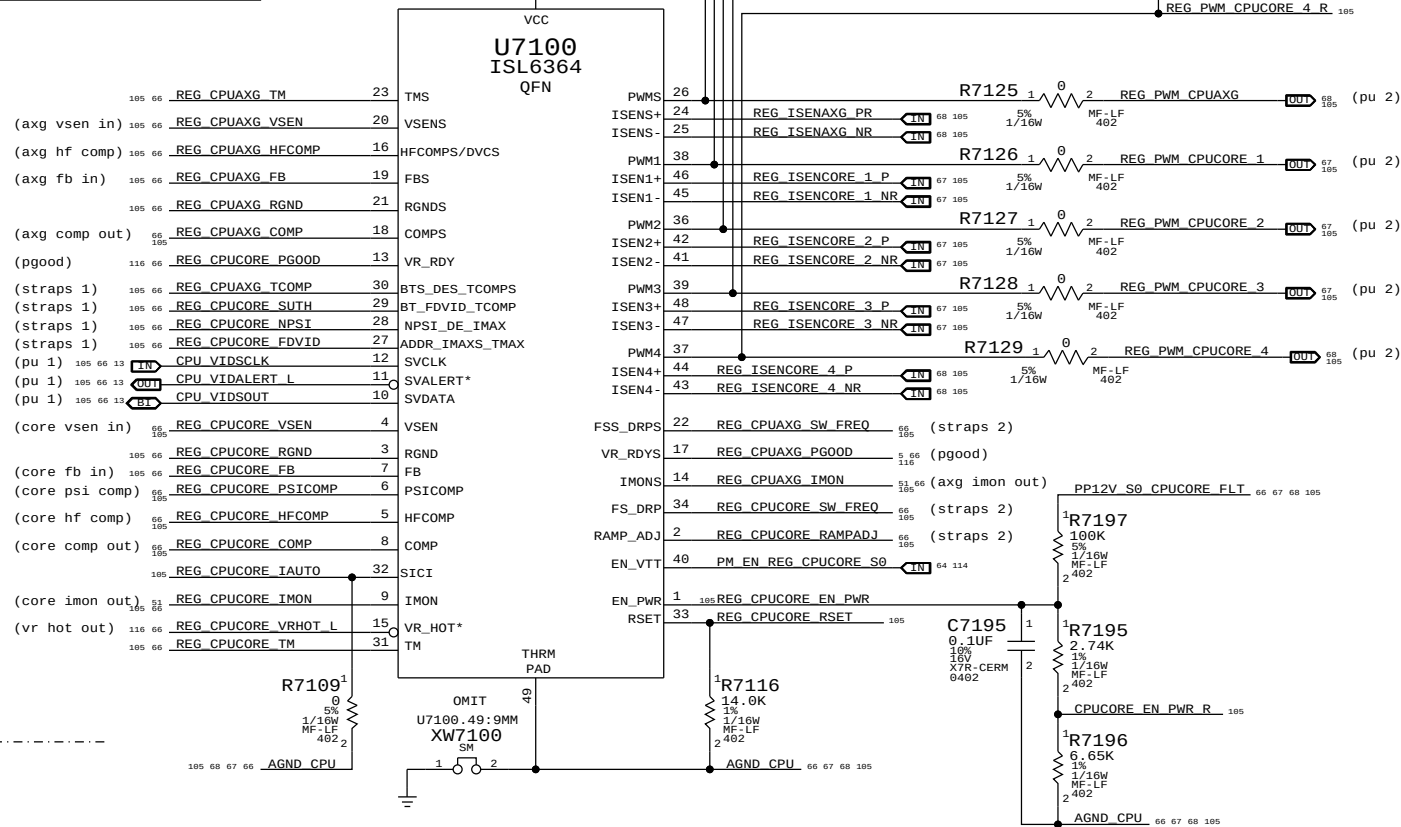
CPU_VIDSClk

CPU_VIDALERT_L

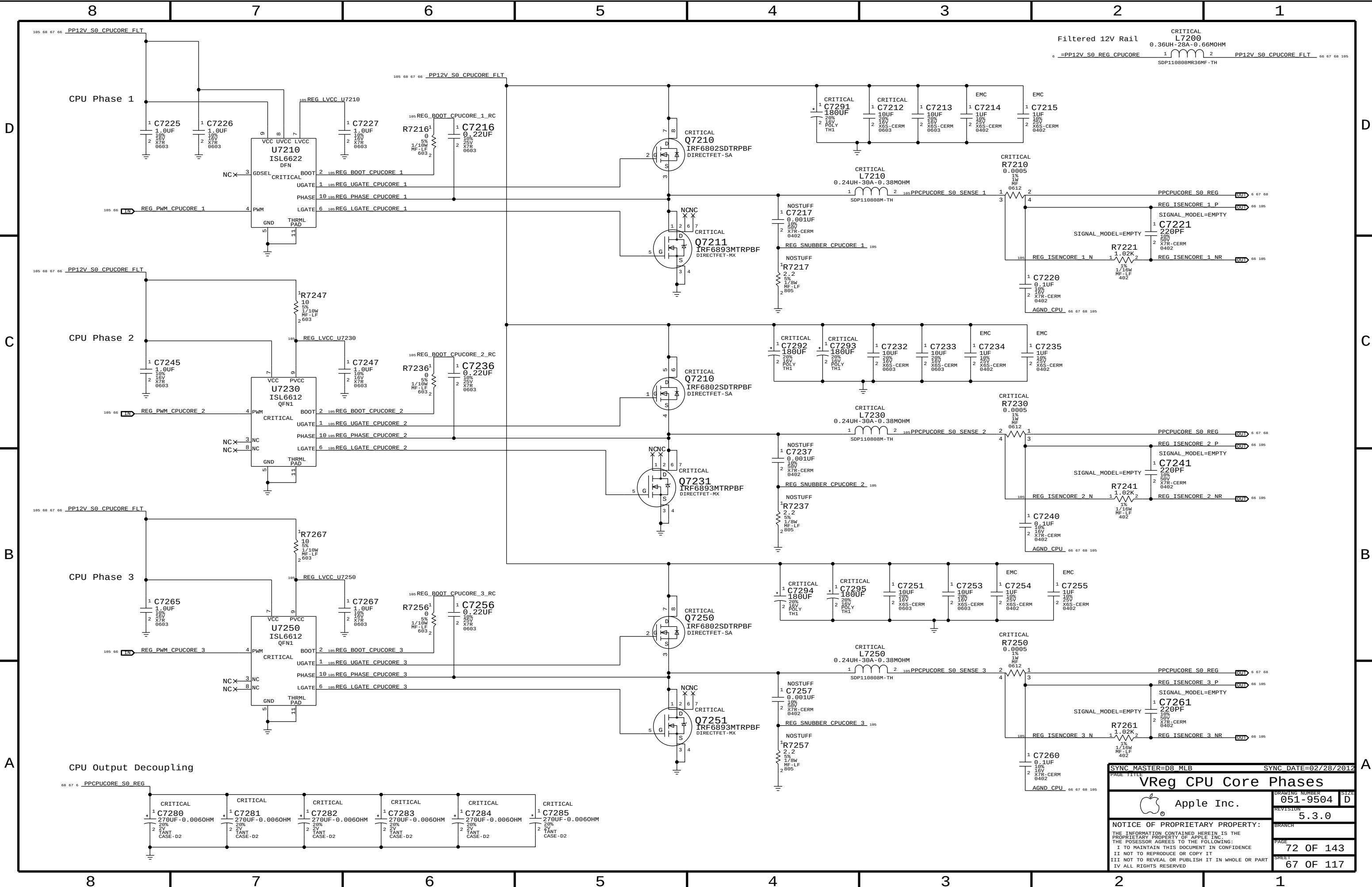
CPU_VIDSOUT

7

Pull-ups 2



SYNC MASTER=D8 MLB		SYNC DATE=02/28/201	
PAGE TITLE			
VReg CPU Core/AXG Cntl			
 Apple Inc.		DRAWING NUMBER 051-9504	
		REVISION 5.3.0	
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE I, NOT TO REPRODUCE OR COPY IT I, NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART I, ALL RIGHTS RESERVED		PAGE 71 OF 143	
		SHEET 66 OF 117	



SYNC MASTER=D8_MLB

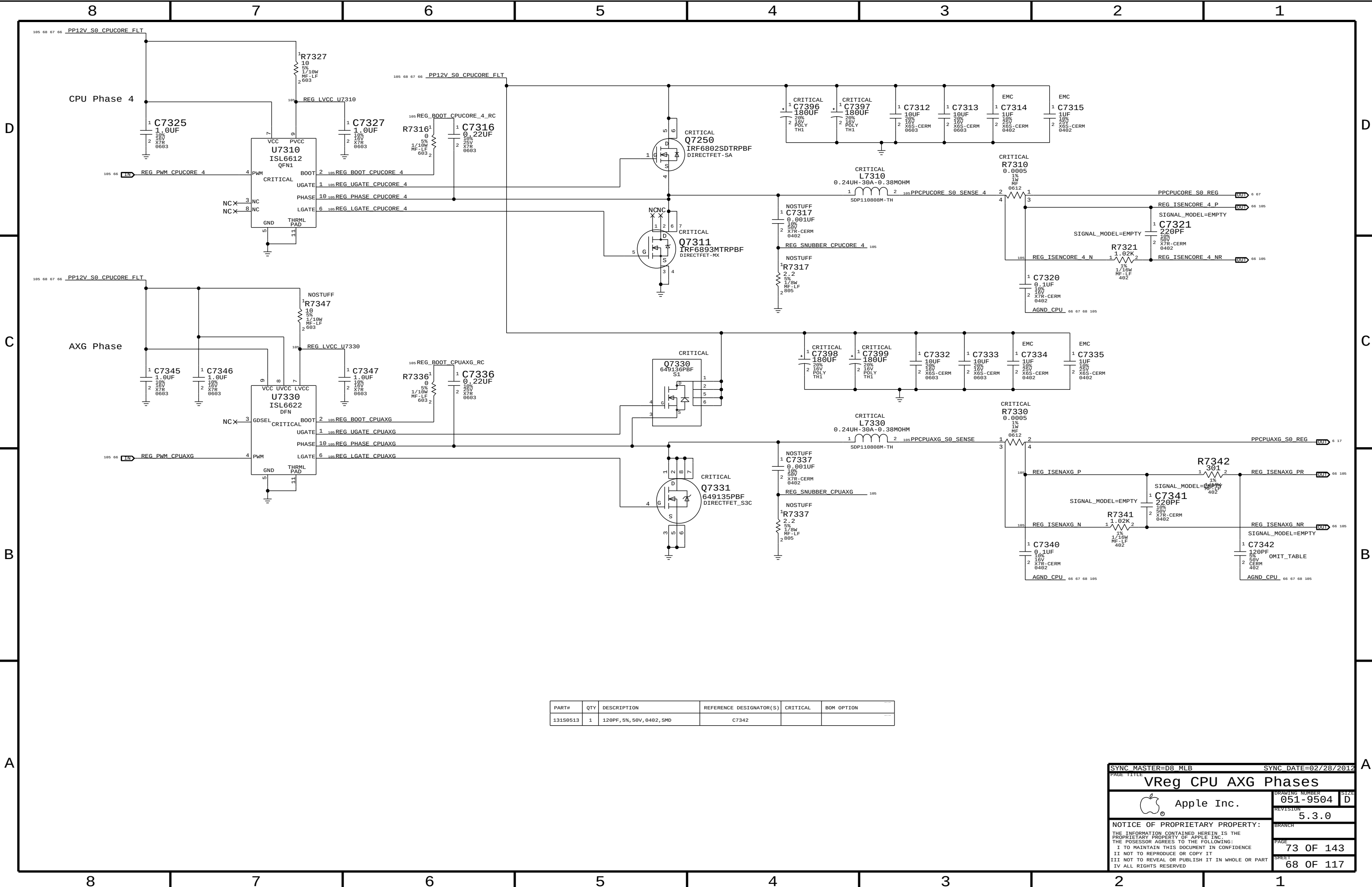
SYNC DATE=02/28/2012

VReg CPU Core Phases

Apple Inc.

THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
IV ALL RIGHTS RESERVED

DRAWING NUMBER	051-9504	SIZE	D
REVISION	5.3.0	BRANCH	
PAGE	72 OF 143	SHEET	67 OF 117



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
131S0513	1	120PF, 5%, 50V, 0402, SMD	C7342		

SYNC MASTER=D8_MLB

SYNC DATE=02/28/2012

VReg CPU AXG Phases

Apple Inc.

051-9504

5.3.0

73 OF 143

68 OF 117

NOTICE OF PROPRIETARY PROPERTY:

THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
IV ALL RIGHTS RESERVED

D

C



A

D

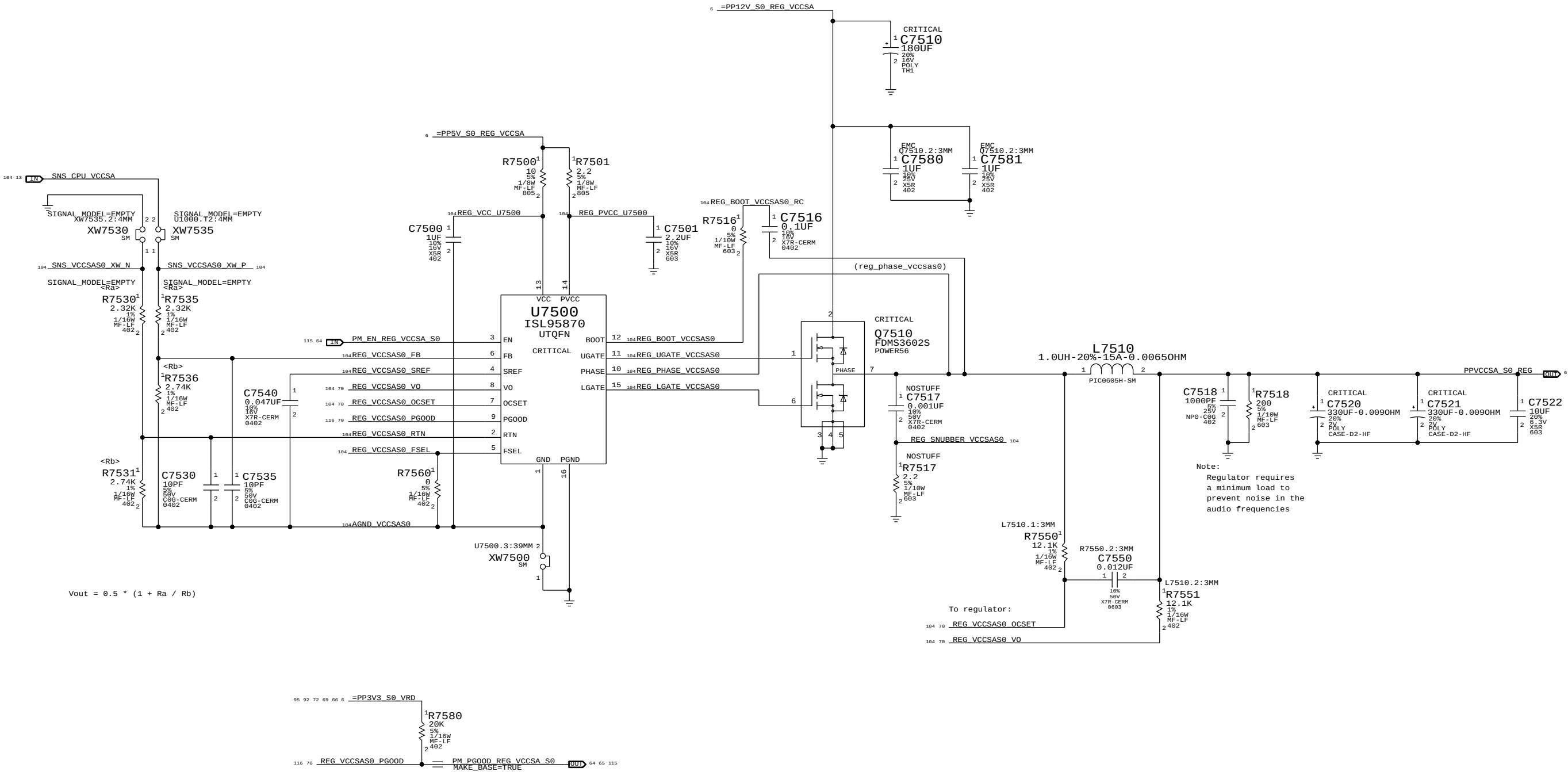
C

B

A

CPU VccSA (0.925V) S0 Regulator

Max avg current: 12.07 A (BUDGET)
Max peak current: 30 A (BUDGET)
OC trip point: ? A (min)/? A (max)
Switching freq: 500 KHz



$$V_{out} = 0.5 * (1 + R_a / R_b)$$

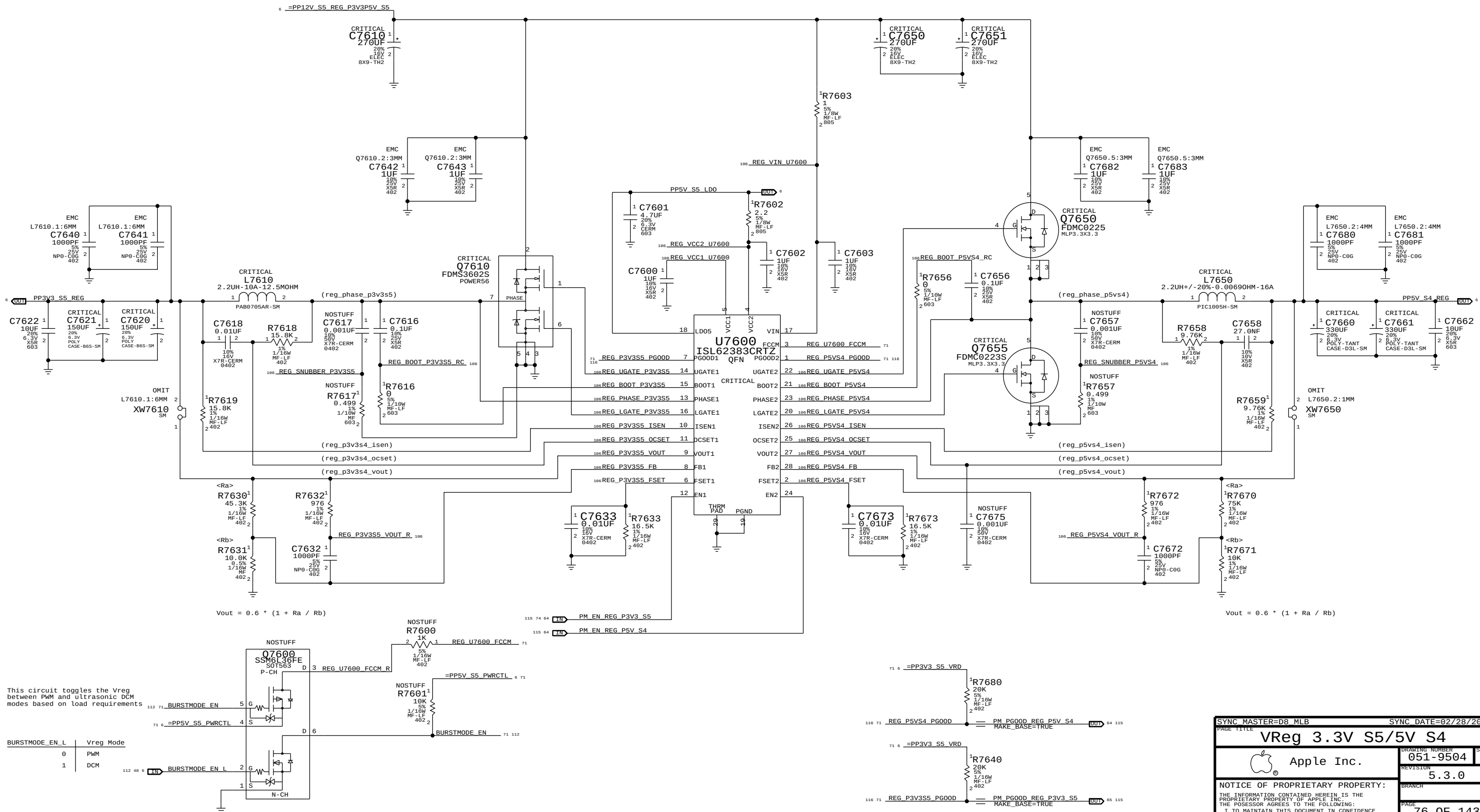
SYNC MASTER=D8 KOSECOFF		SYNC DATE=02/25/2012	
PAGE TITLE		VReg CPU VccSA S0	
Apple Inc.		DRAWING NUMBER	051-9504
		REVISION	5.3.0
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:		PAGE	75 OF 143
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		SHEET	70 OF 117
II NOT TO REPRODUCE OR COPY IT			
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IV ALL RIGHTS RESERVED			

3.3V S5 Regulator

Max avg current: 6 A (design)/ 4.85 A (budget)
Max peak current: ? A (design)/ 6.6 A (budget)
OC trip point: ? A (nom)/? A (min)
Switching freq: 350 kHz

5V S4 Regulator

Max avg current: 10 A (design)/ 6.08 A (budget)
Max peak current: ? A (design)/ 6.9 A (budget)
OC trip point: ? A (nom)/? A (min)
Switching freq: 350 kHz



SYNC MASTER=D8 MLB		SYNC DATE=02/28/2012	
PAGE TITLE		VReg 3.3V S5/5V S4	
Apple Inc.		DRAWING NUMBER	051-9504
		REVISION	5.3.0
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:		PAGE	76 OF 143
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		SHEET	71 OF 117
II NOT TO REPRODUCE OR COPY IT			
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IV ALL RIGHTS RESERVED			

D

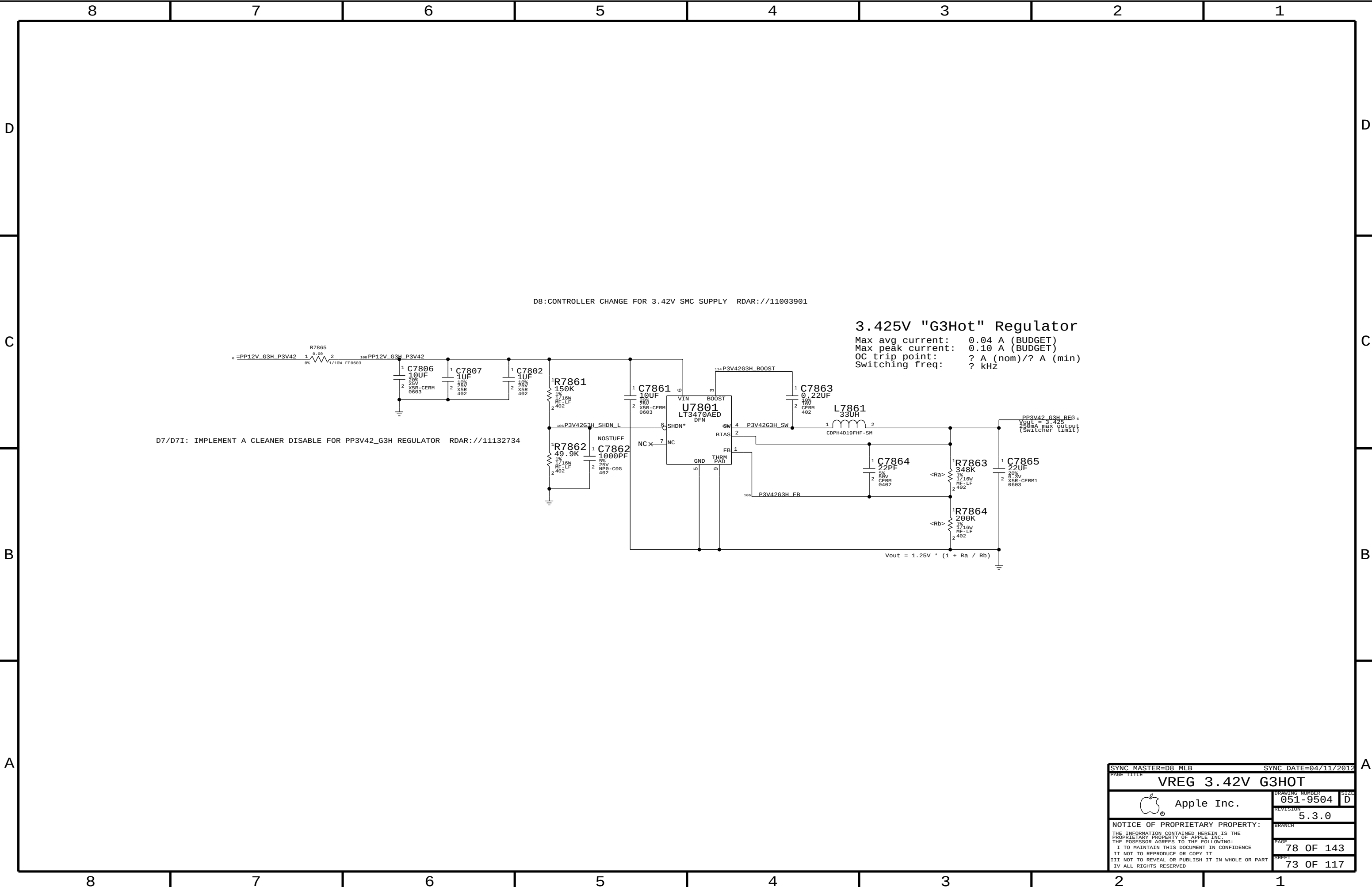
C

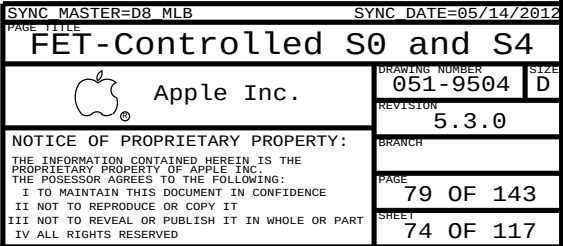


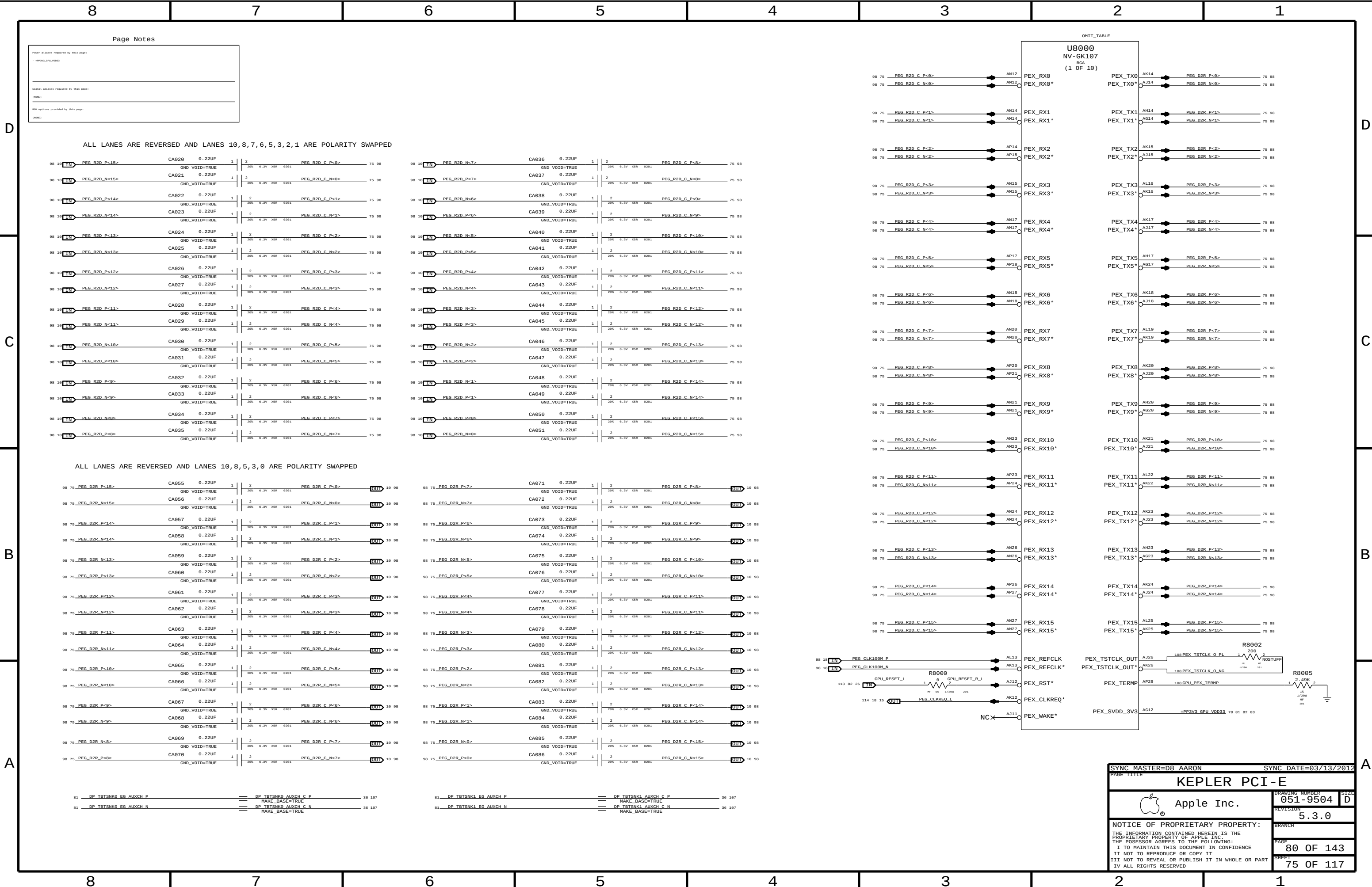
A

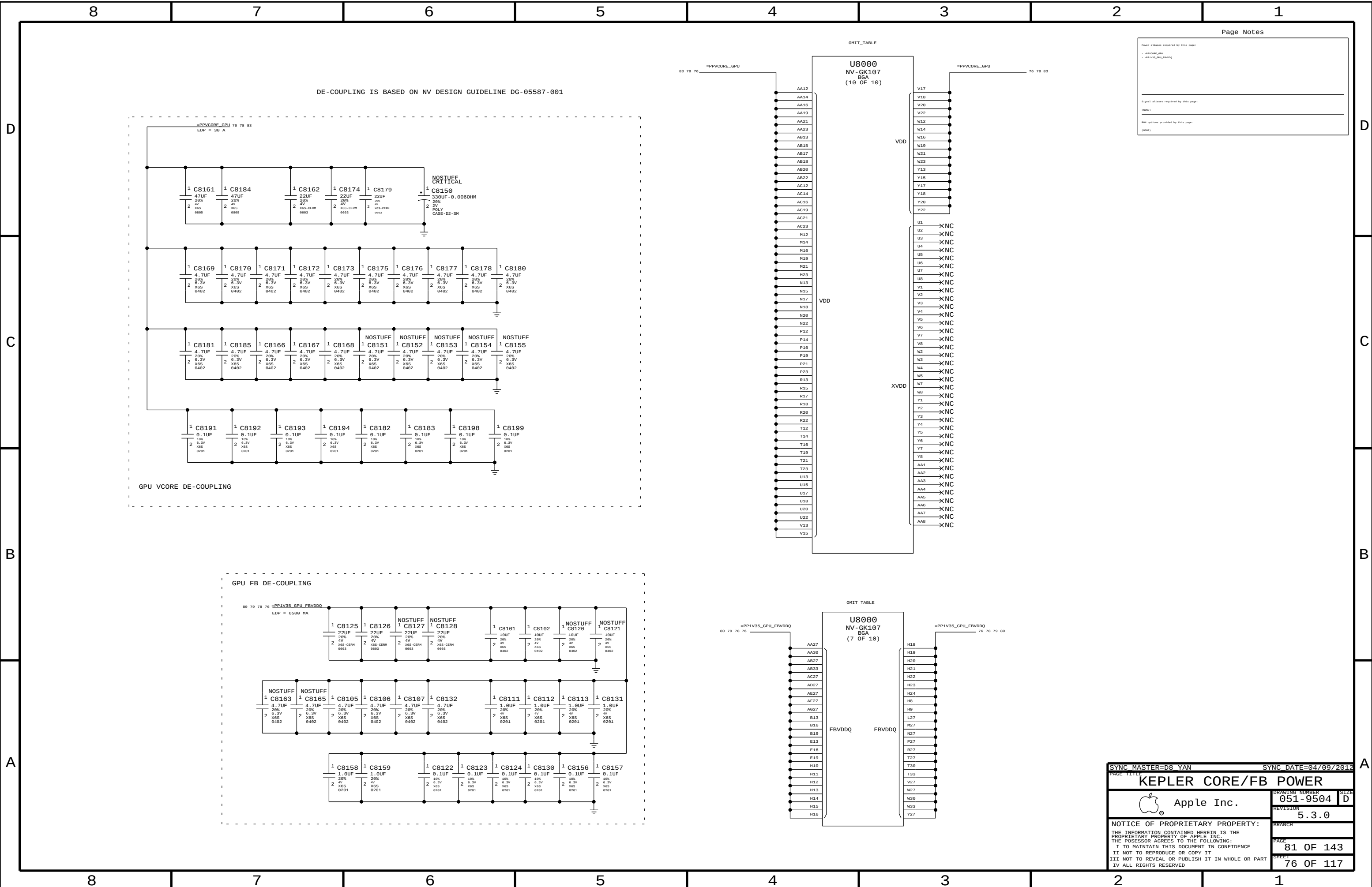
D

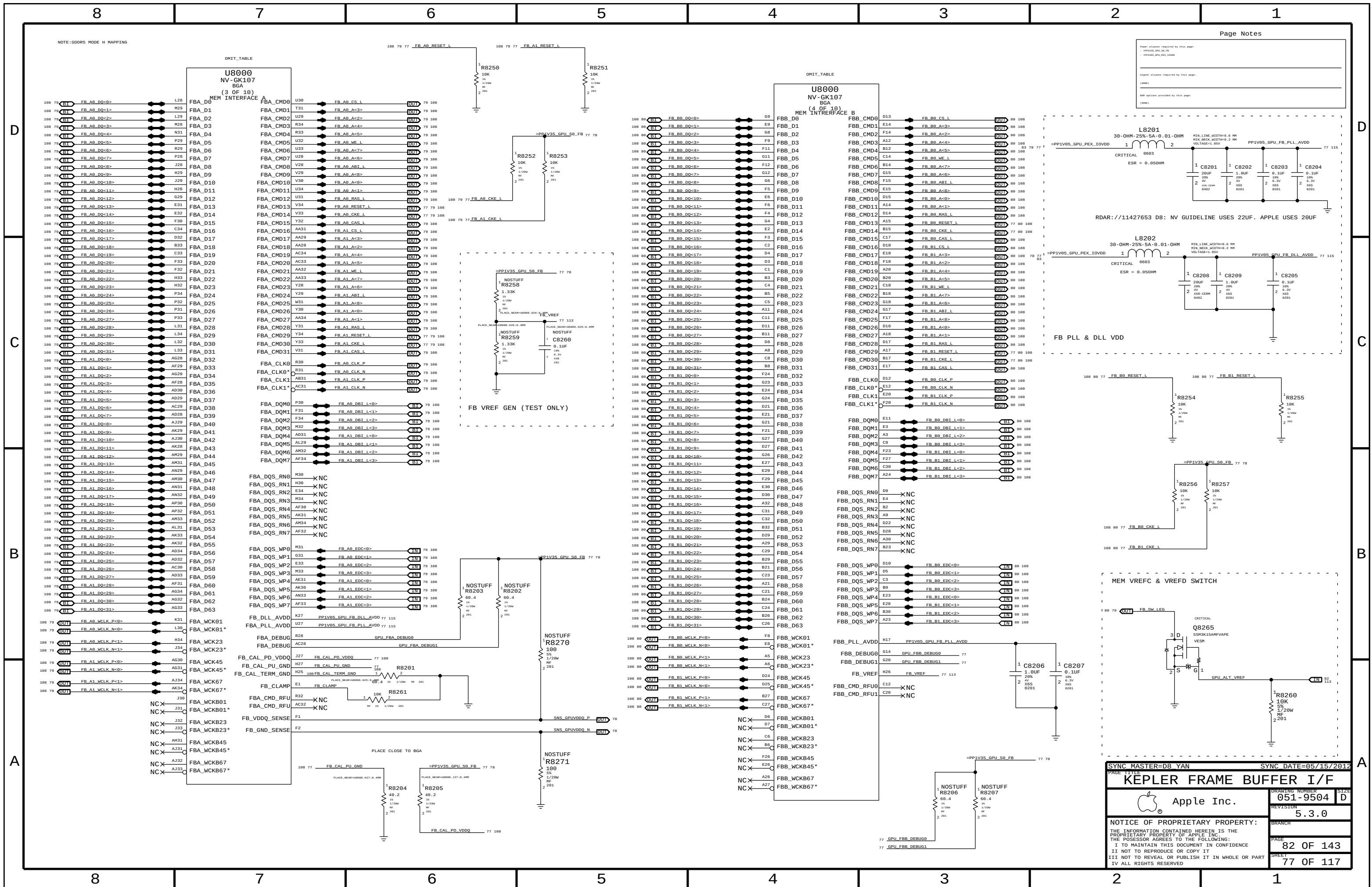
A





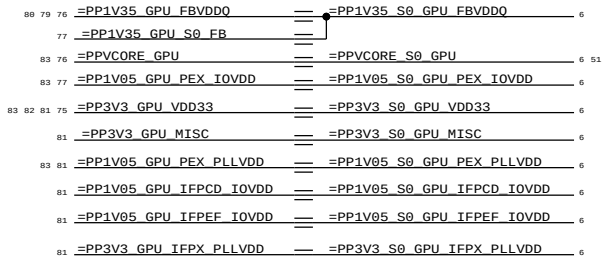




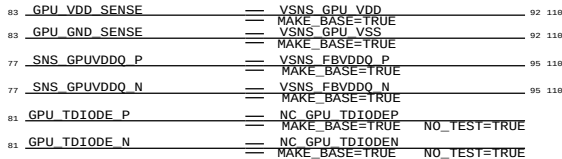


D

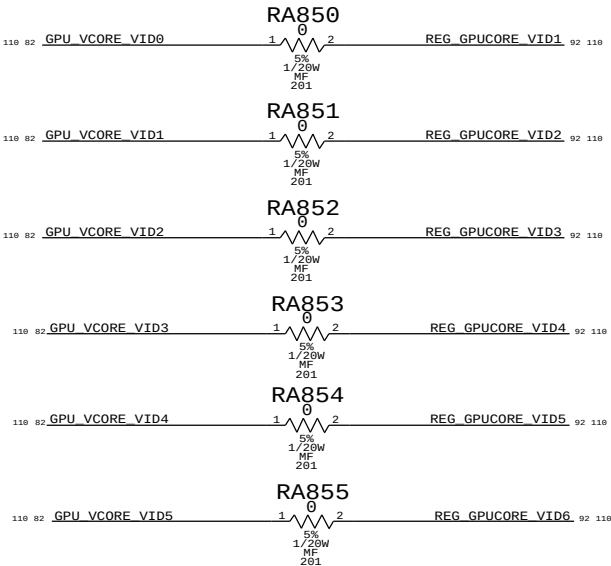
GPU POWER ALIAS



GPU SIGNAL & SENSE ALIAS



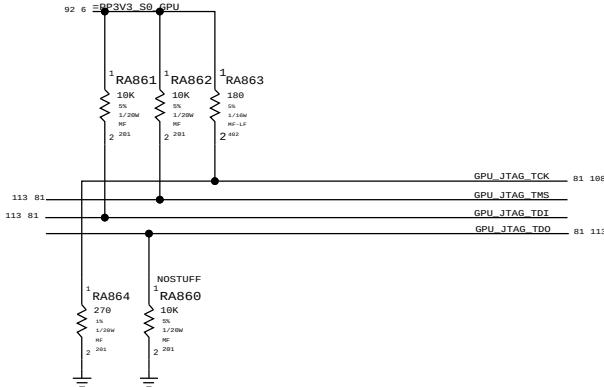
GPU VIDS ALIAS (VR VID0 IS TIED LOW)



THESE POWER ALIASES ARE CRAETED TO MATCH D8 GK104

B

PU/PD IS BASED ON RECOMMENDATION FROM NV FOR NVIDIA GPU JTAG DEBUGGER



A

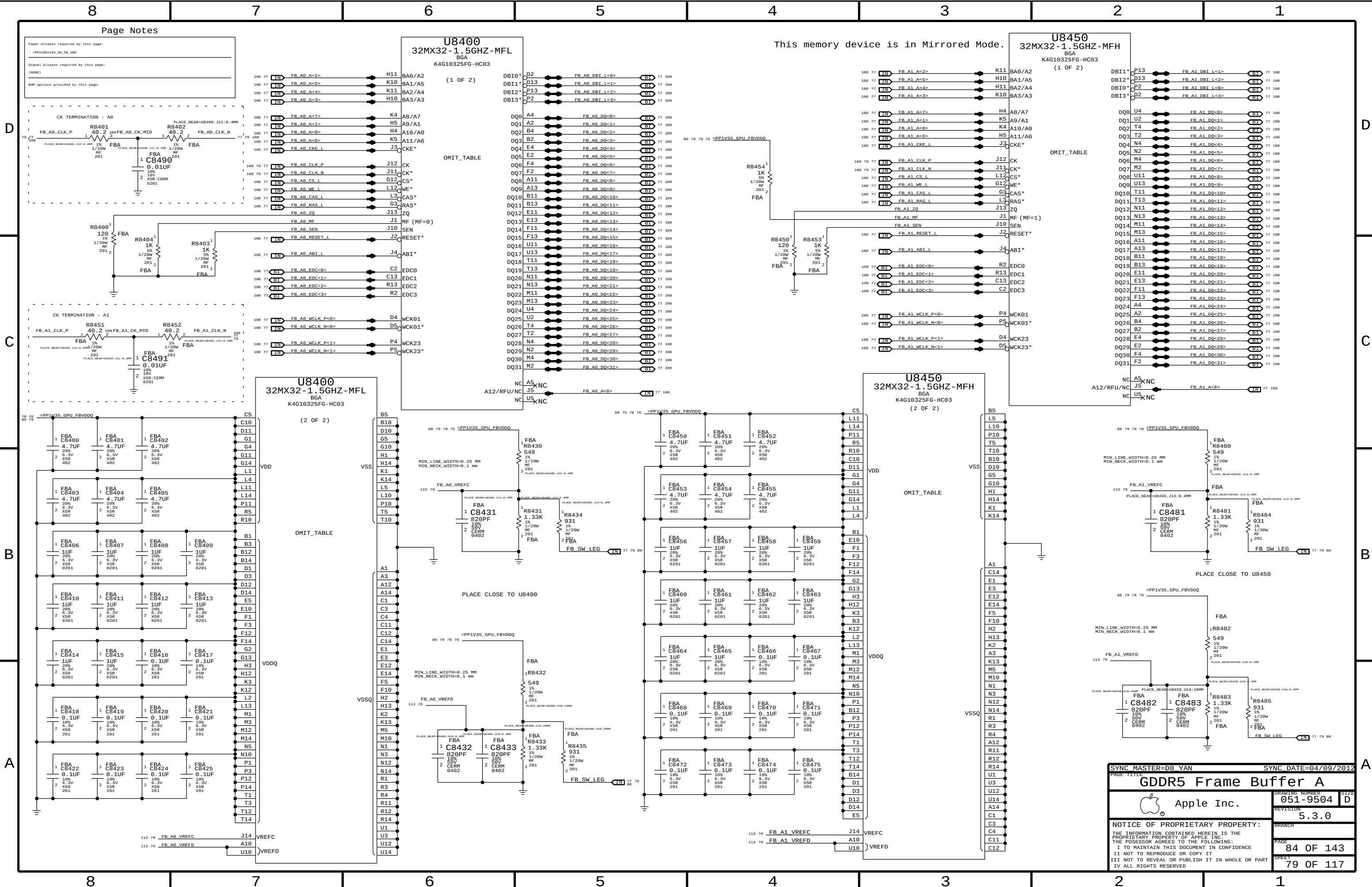
D

C

B

A

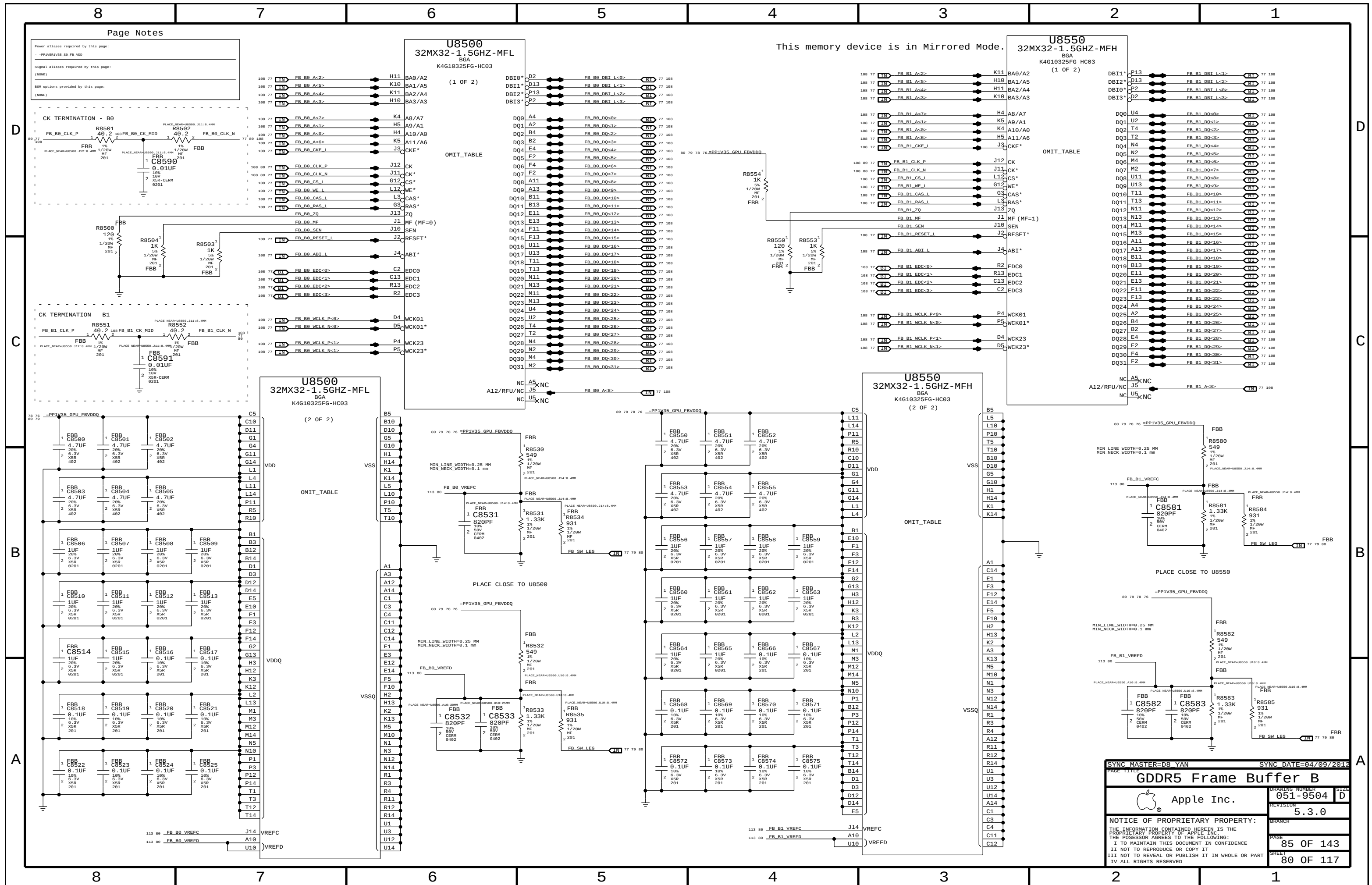
SYNC MASTER=D8_YAN		SYNC DATE=04/09/2012	
PAGE TITLE		GPU SIGNAL & POWER ALIASES	
 Apple Inc.		DRAWING NUMBER	051-9504
		REVISION	5.3.0
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		BRANCH	
		PAGE	83 OF 143
		SHEET	78 OF 117

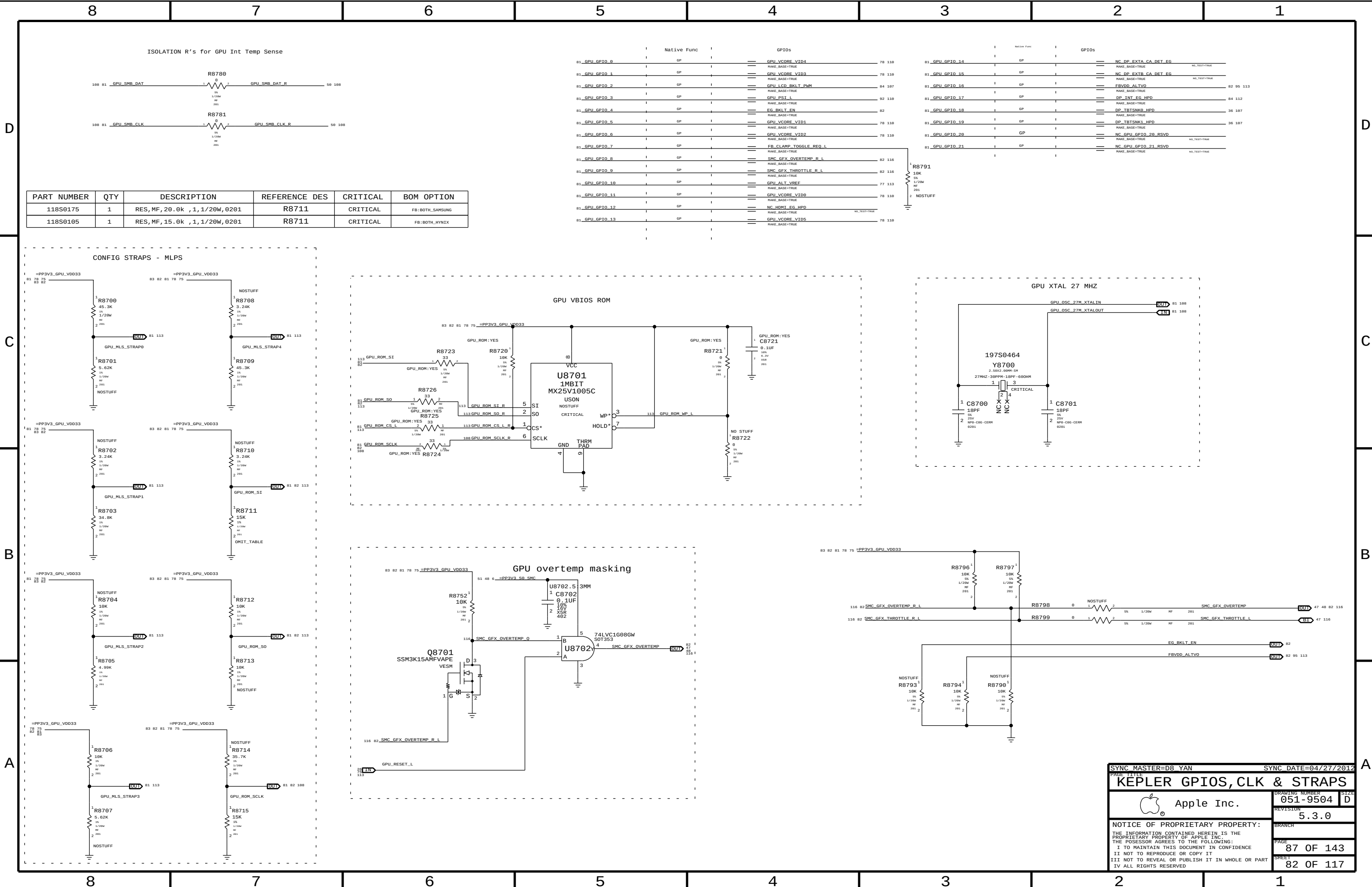


SYNC MASTER=D8_YAN

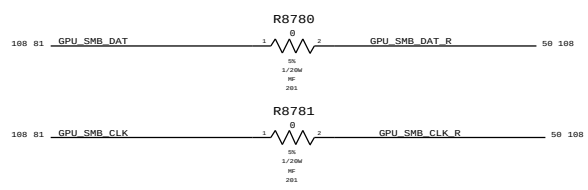
SYNC DATE=04/09/2012

PAGE TITLE		GDDR5 Frame Buffer A	
Apple Inc.		DRAWING NUMBER	051-9504
REVISION		5.3.0	SIZE D
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:		PAGE	
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		84 OF 143	
II NOT TO REPRODUCE OR COPY IT		SHEET	
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART		79 OF 117	
IV ALL RIGHTS RESERVED			





ISOLATION R's for GPU Int Temp Sense

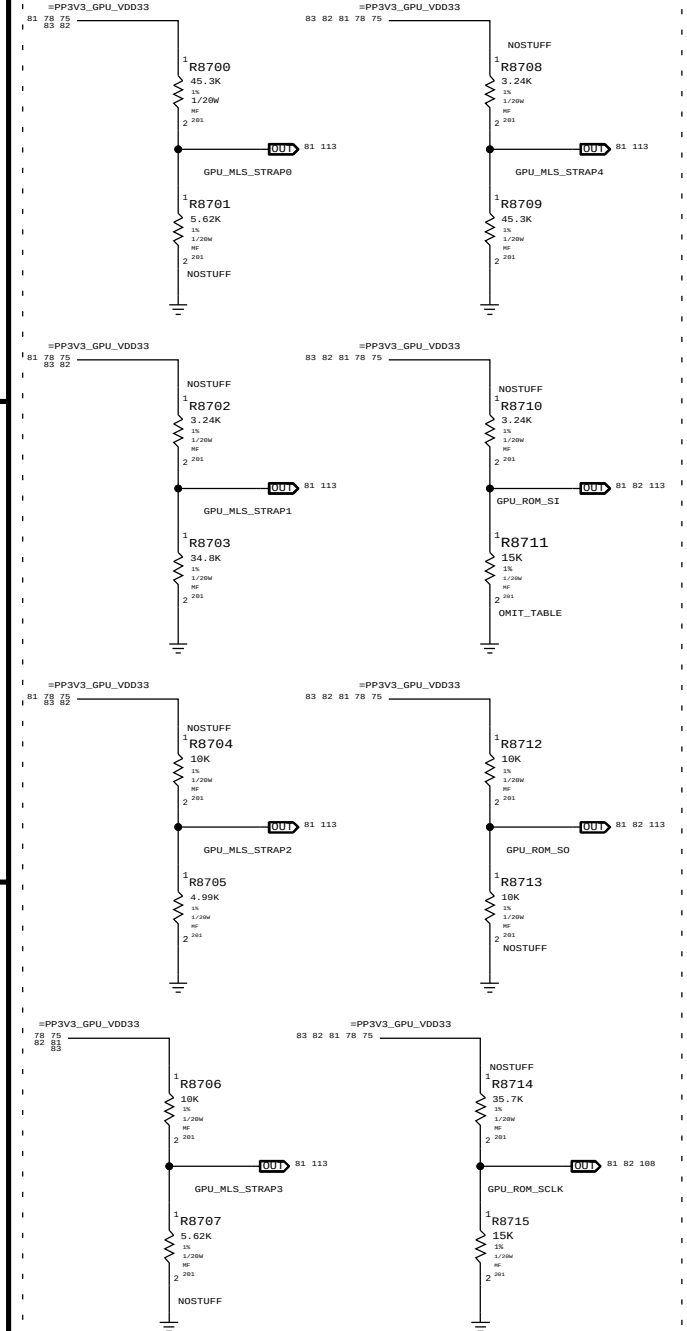


	Native Func		GPIOs	
81_GPU_GPIO_0	GP		GPU_VCORE_VID4	78 110
81_GPU_GPIO_1	GP		GPU_VCORE_VID3	78 110
81_GPU_GPIO_2	GP		GPU_LCD_BKLT_PWM	84 107
81_GPU_GPIO_3	GP		GPU_PSI_L	92 110
81_GPU_GPIO_4	GP		EG_BKLT_EN	82
81_GPU_GPIO_5	GP		GPU_VCORE_VID1	78 110
81_GPU_GPIO_6	GP		GPU_VCORE_VID2	78 110
81_GPU_GPIO_7	GP		FB_CLAMP_TOGGLE_REQ_L	82 116
81_GPU_GPIO_8	GP		SMC_GFX_OVERTEMP_R_L	82 116
81_GPU_GPIO_9	GP		SMC_GFX_THROTTLE_R_L	82 116
81_GPU_GPIO_10	GP		GPU_ALT_VREF	77 113
81_GPU_GPIO_11	GP		GPU_VCORE_VID0	78 110
81_GPU_GPIO_12	GP		NC_HDMI_EG_HPD	NO_TEST+TRUE
81_GPU_GPIO_13	GP		GPU_VCORE_VID5	78 110

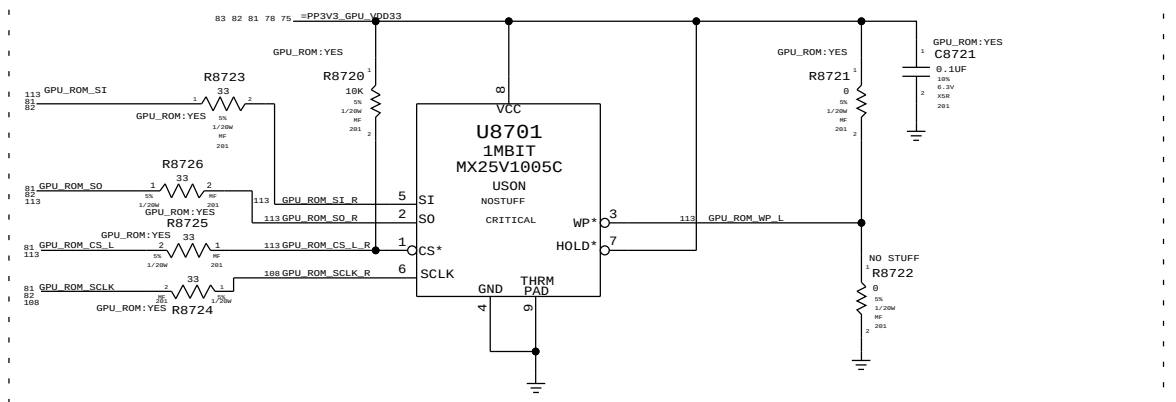
	Native Func		GPIOs	
81_GPU_GPIO_14	GP		NC_DP_EXTR_CA_DET_EG	NO_TEST+TRUE
81_GPU_GPIO_15	GP		NC_DP_EXTR_CA_DET_EG	NO_TEST+TRUE
81_GPU_GPIO_16	GP		FBVDD_ALTV0	82 95 113
81_GPU_GPIO_17	GP		DP_INT_EG_HPD	84 112
81_GPU_GPIO_18	GP		DP_TBTSNK0_HPD	36 107
81_GPU_GPIO_19	GP		DP_TBTSNK1_HPD	36 107
81_GPU_GPIO_20	GP		NC_GPU_GPIO_20_RSVD	NO_TEST+TRUE
81_GPU_GPIO_21	GP		NC_GPU_GPIO_21_RSVD	NO_TEST+TRUE

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
118S0175	1	RES,MF,20.0k ,1,1/20W,0201	R8711	CRITICAL	FB:BOTH_SAMSUNG
118S0105	1	RES,MF,15.0k ,1,1/20W,0201	R8711	CRITICAL	FB:BOTH_HYWIX

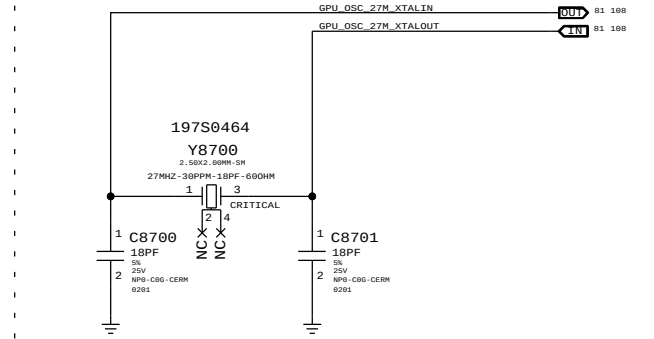
CONFIG STRAPS - MLPS



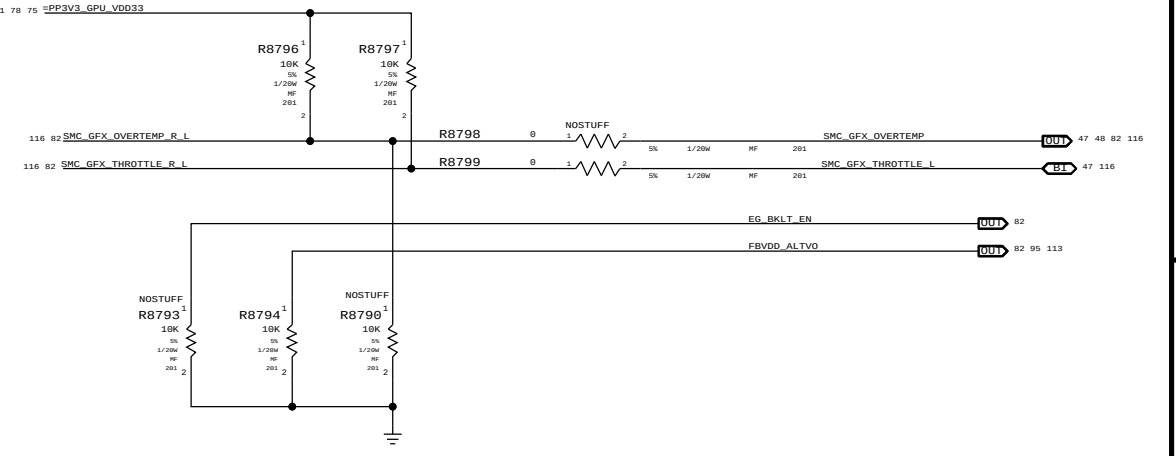
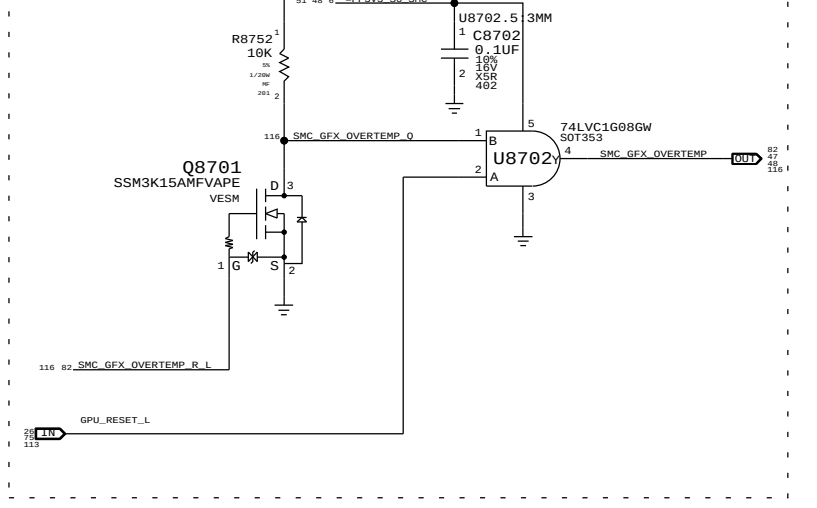
GPU VBIOS ROM



GPU XTAL 27 MHZ



GPU overtemp masking



SYNC MASTER=D8_YAN

SYNC DATE=04/27/2012

KEPLER GPIOs,CLK & STRAPS

Apple Inc.

NOTICE OF PROPRIETARY PROPERTY:
THE INFORMATION CONTAINED HEREIN IS THE
PROPRIETARY PROPERTY OF APPLE INC.
THE POSSESSOR AGREES TO THE FOLLOWING:
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
IV ALL RIGHTS RESERVED

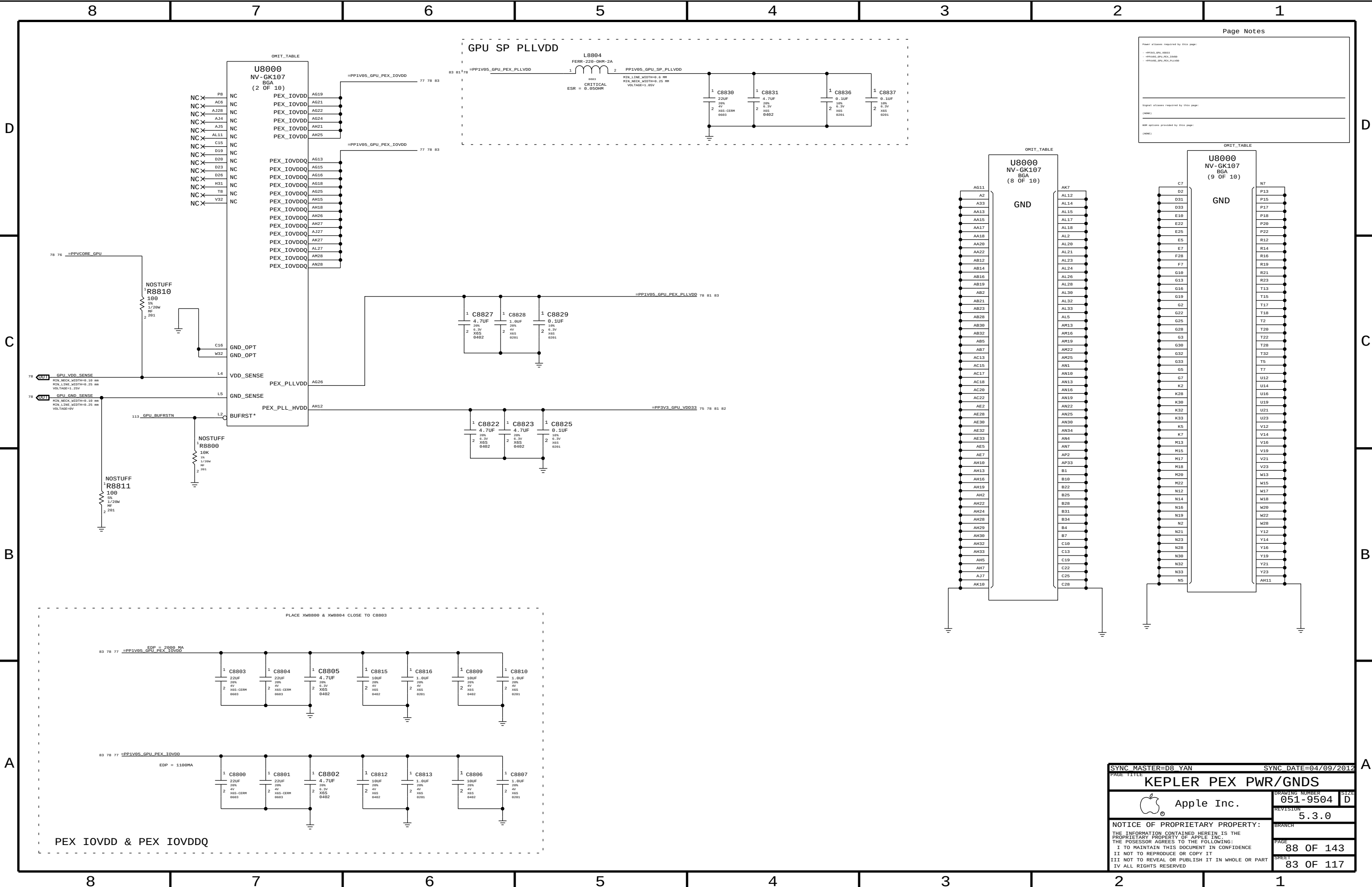
DRAWING NUMBER
051-9504

REVISION
5.3.0

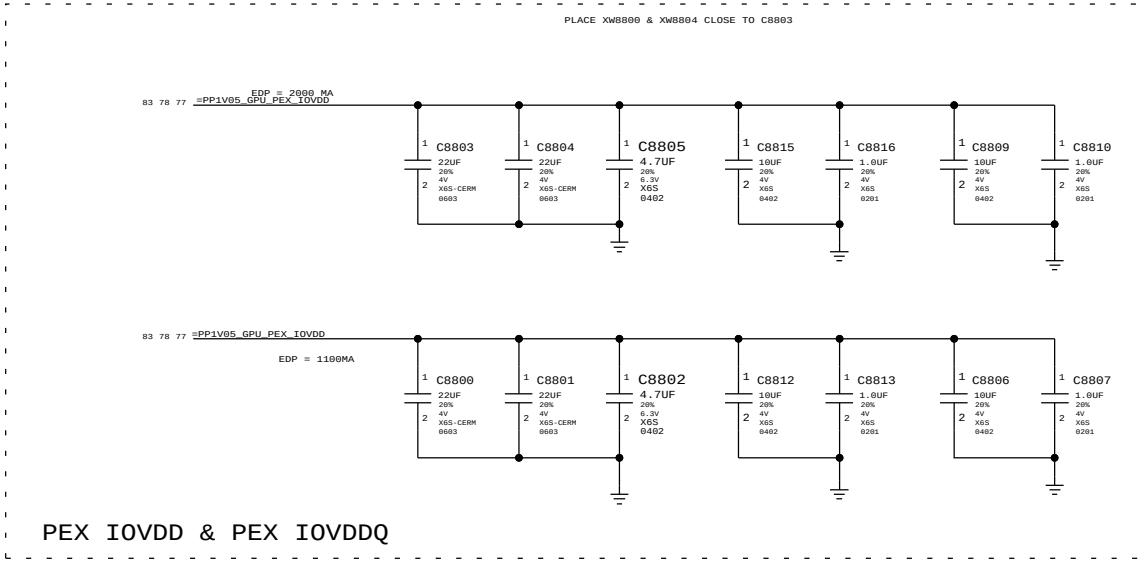
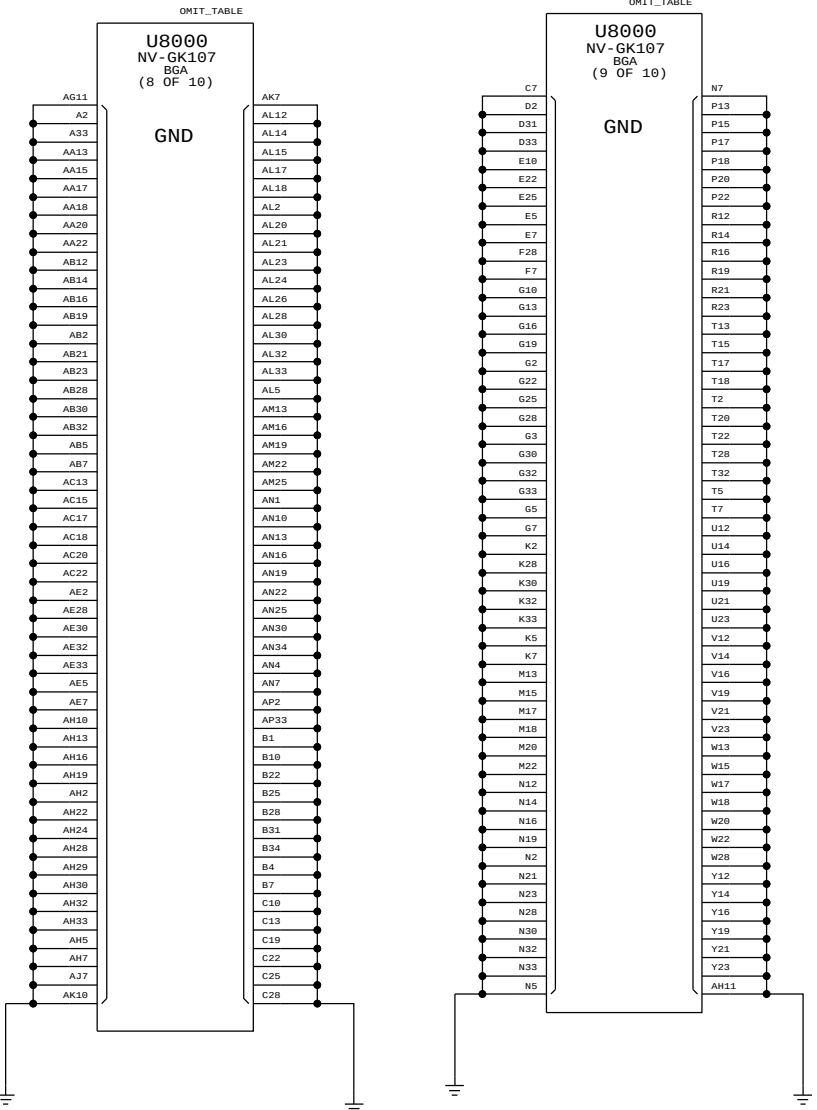
BRANCH

PAGE
87 OF 143

SHEET
82 OF 117



Page Notes	
Power aliases required by this page:	
- PP1V05_GPU_PEX_PLLVDD	
- PP1V05_GPU_PEX_IOVDD	
Signal aliases required by this page:	
(NONE)	
SOP options provided by this page:	
(NONE)	



SYNC MASTER=D8_YAN

SYNC DATE=04/09/2012

KEPLER PEX PWR/GNDS

Apple Inc.

THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:

I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE

II NOT TO REPRODUCE OR COPY IT

III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART

IV ALL RIGHTS RESERVED

DRAWING NUMBER

051-9504

REVISION

5.3.0

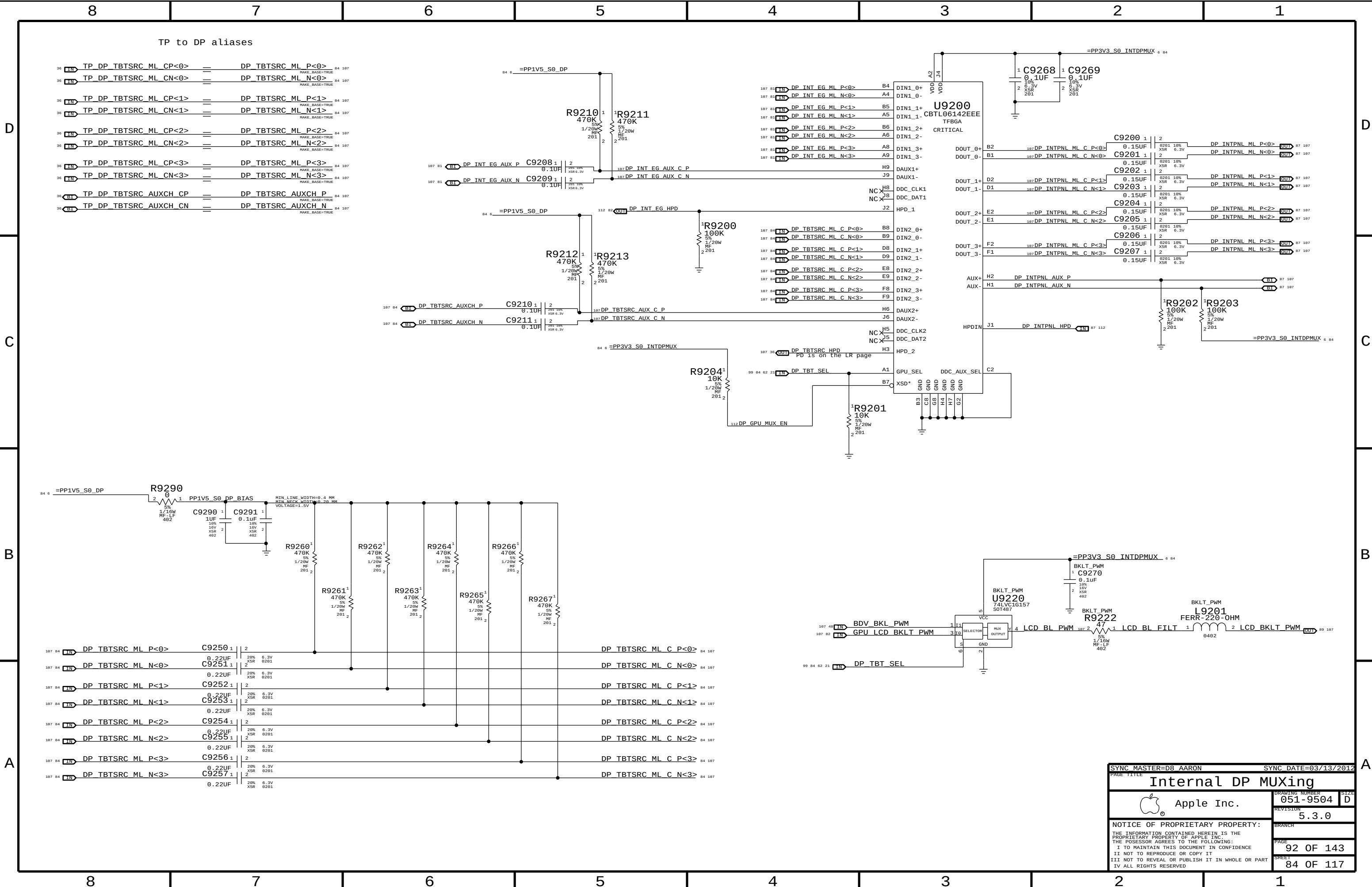
BRANCH

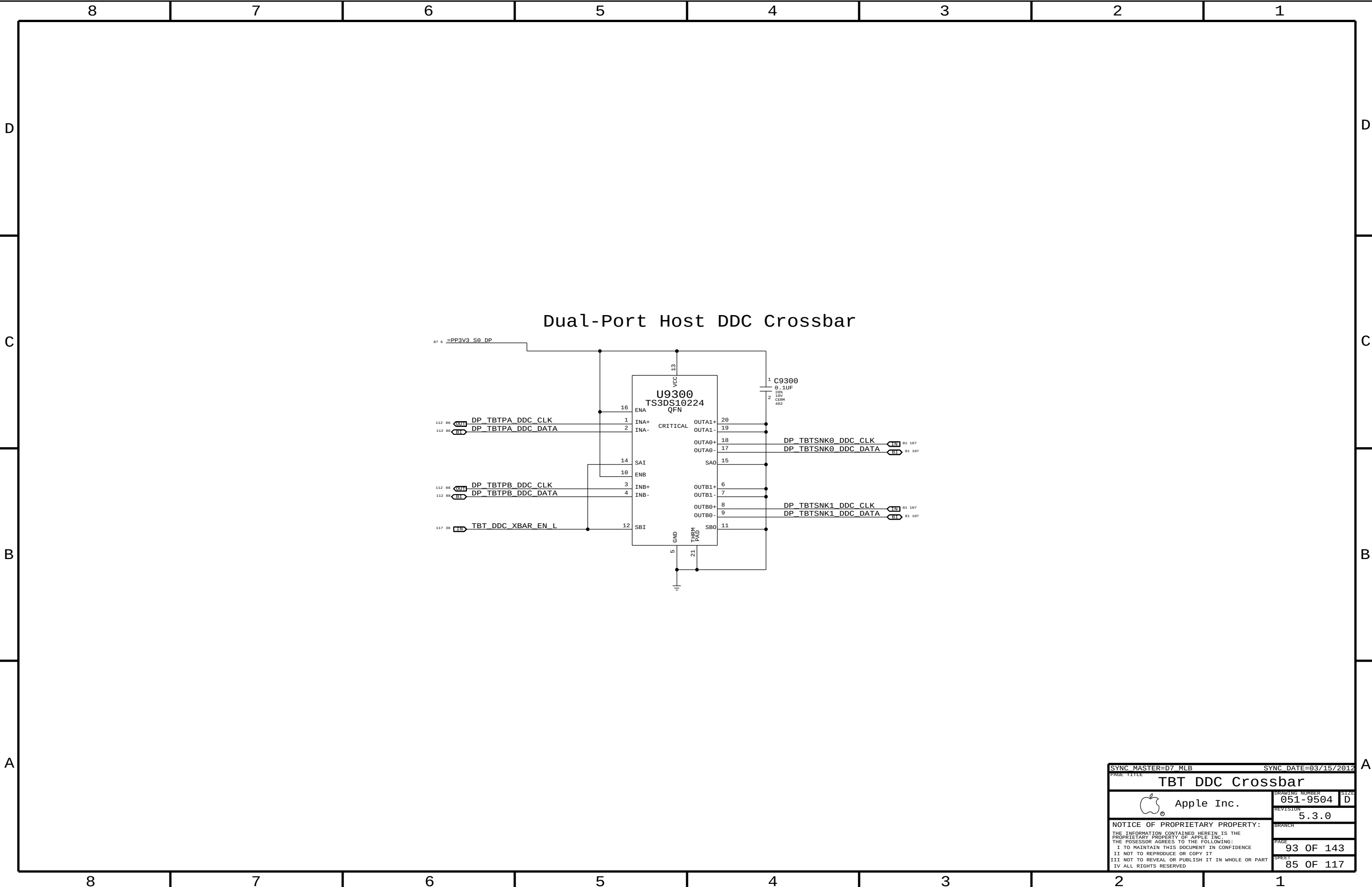
PAGE

88 OF 143

SHEET

83 OF 117





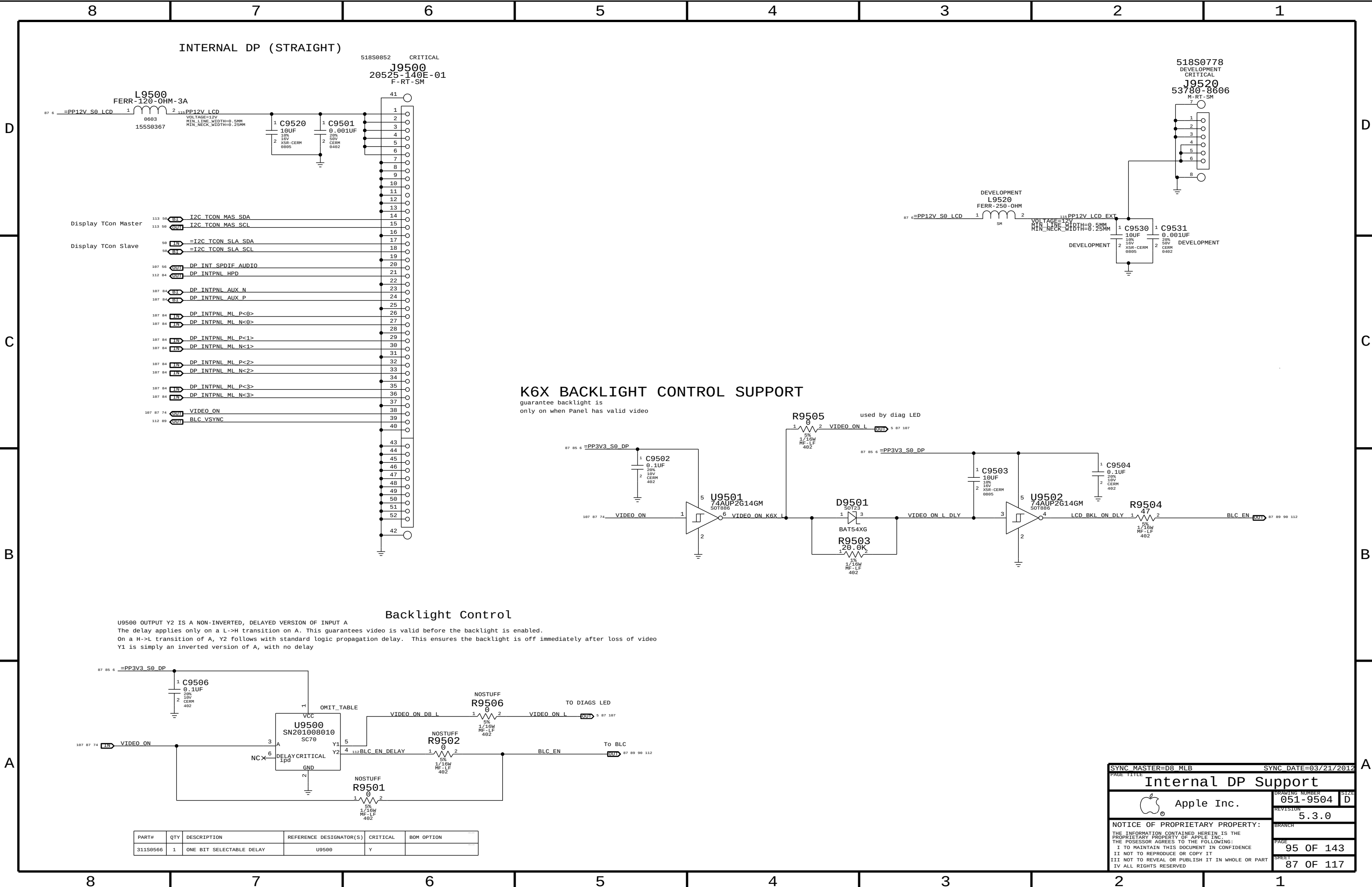
8	7	6	5	4	3	2	1
---	---	---	---	---	---	---	---



D

CB

B



INTERNAL DP (STRAIGHT)

518S0852 CRITICAL
J9500
20525-140E-01
F-RT-SM

518S0778 DEVELOPMENT
CRITICAL
J9520
53780-8606
M-RT-SM

K6X BACKLIGHT CONTROL SUPPORT

guarantee backlight is
only on when Panel has valid video

Backlight Control

U9500 OUTPUT Y2 IS A NON-INVERTED, DELAYED VERSION OF INPUT A
The delay applies only on a L->H transition on A. This guarantees video is valid before the backlight is enabled.
On a H->L transition of A, Y2 follows with standard logic propagation delay. This ensures the backlight is off immediately after loss of video
Y1 is simply an inverted version of A, with no delay

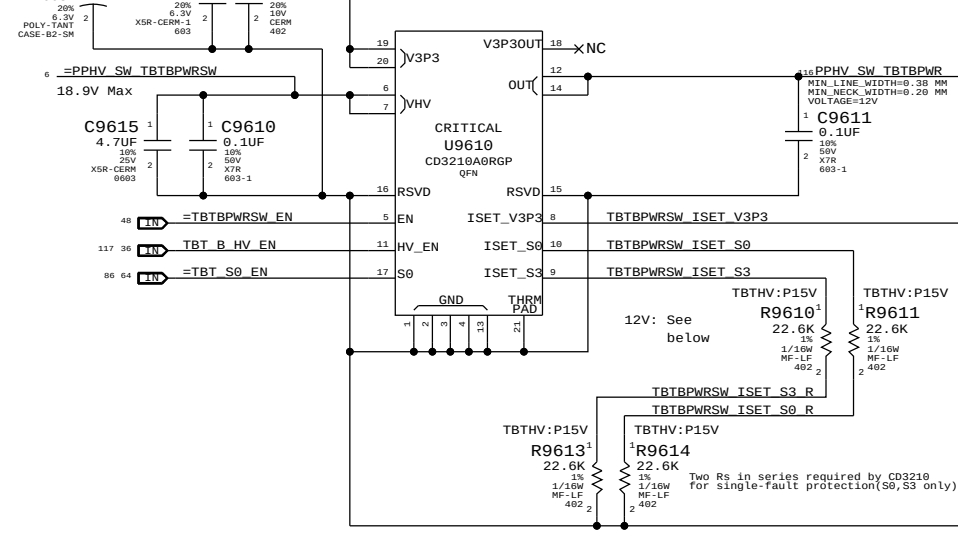
PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
311S0566	1	ONE BIT SELECTABLE DELAY	U9500	Y	

SYNC MASTER=D8 MLB		SYNC DATE=03/21/2012	
PAGE TITLE		Internal DP Support	
Apple Inc.		DRAWING NUMBER	051-9504
		REVISION	5.3.0
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:		PAGE	95 OF 143
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		SHEET	87 OF 117
II NOT TO REPRODUCE OR COPY IT			
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IV ALL RIGHTS RESERVED			

3.3V/HV Power MUX

V3P3 must be S4 to support wake from Thunderbolt devices.

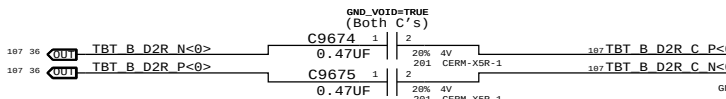
	Nominal	Min	Max
IV3P3	1100mA	1030mA	1200mA
IHV50	890mA	830mA	930mA (assumes 15V, 12W minimum)
IHV53	890mA	830mA	930mA (assumes 3S, 9-12.6V, 7.5-11.7W)



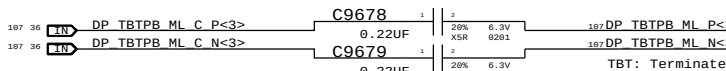
For 12V systems:

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
114S0338	2	RES,MTL FILM,1/16W,17.8K,1,0402,SMD,LF	R9610,R9613		TBTHV:P12V
114S0338	2	RES,MTL FILM,1/16W,17.8K,1,0402,SMD,LF	R9611,R9614		TBTHV:P12V

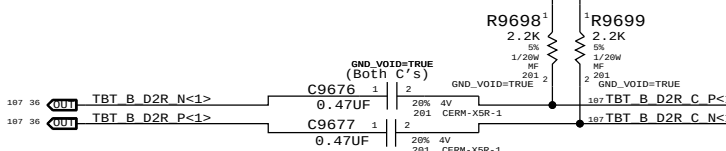
	Nominal	Min	Max
IHV50/S3	1120mA	1090mA	1170mA (12W minimum)



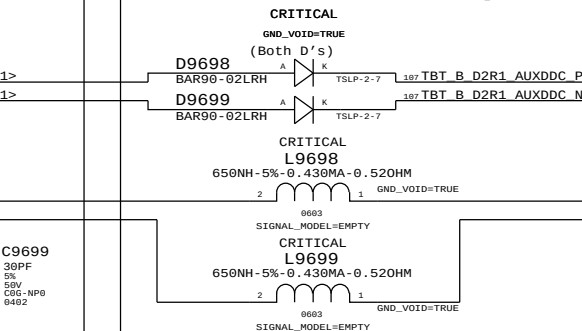
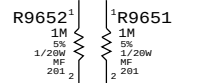
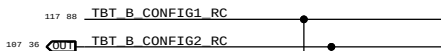
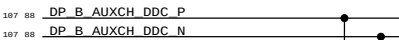
NOTE: Polarity Swapped for Layout!



TBT: Terminated

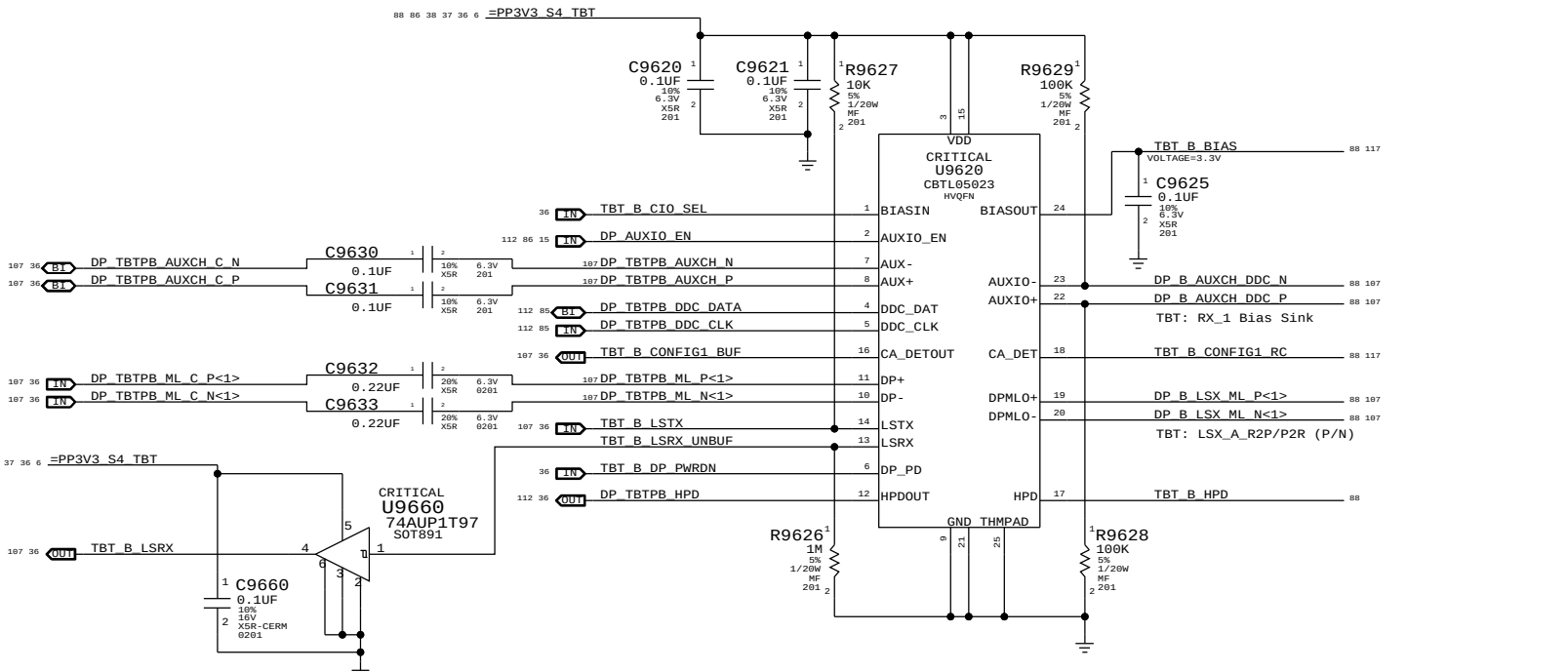


NOTE: Polarity Swapped for Layout!

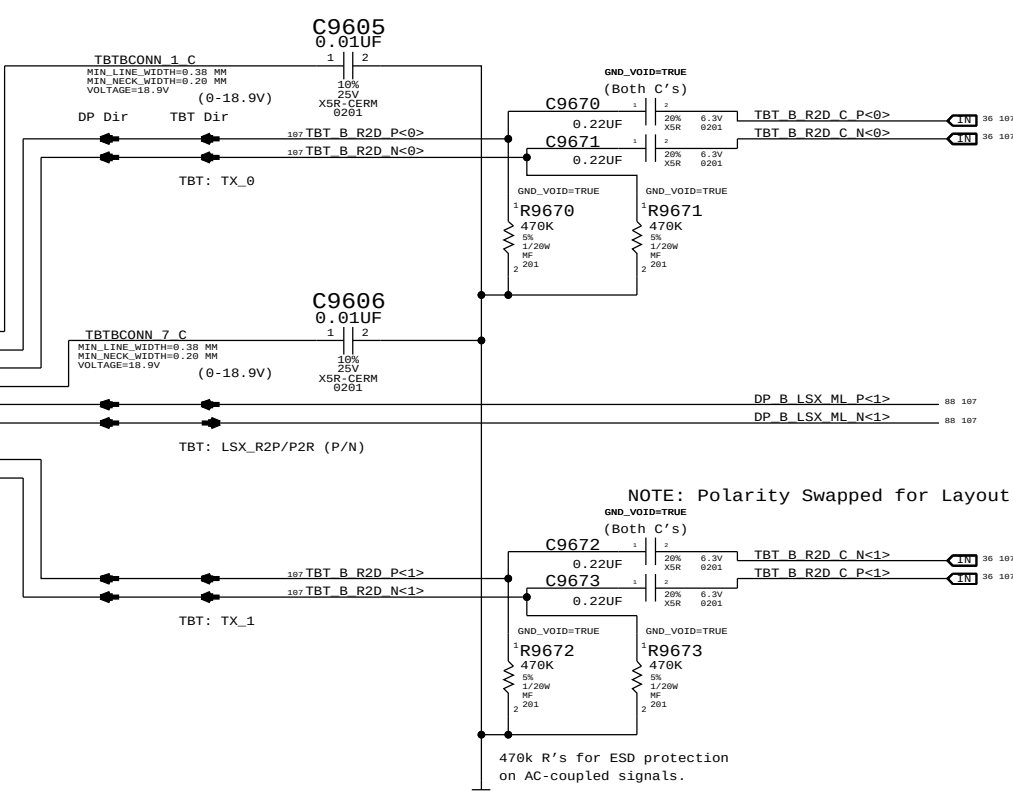
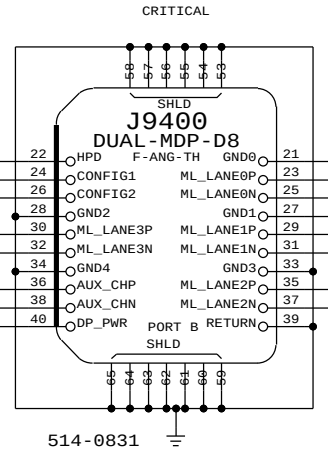
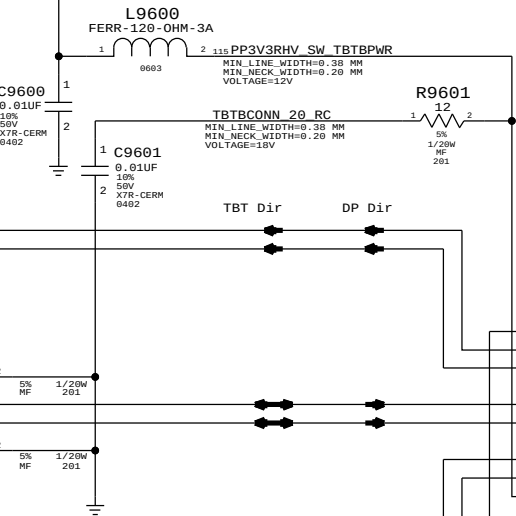


DP Source must pull down HPD input with greater than or equal to 100K (DPv1.1a).

Sink HPD range:
High: 2.0 - 5.0V
Low: 0 - 0.8V

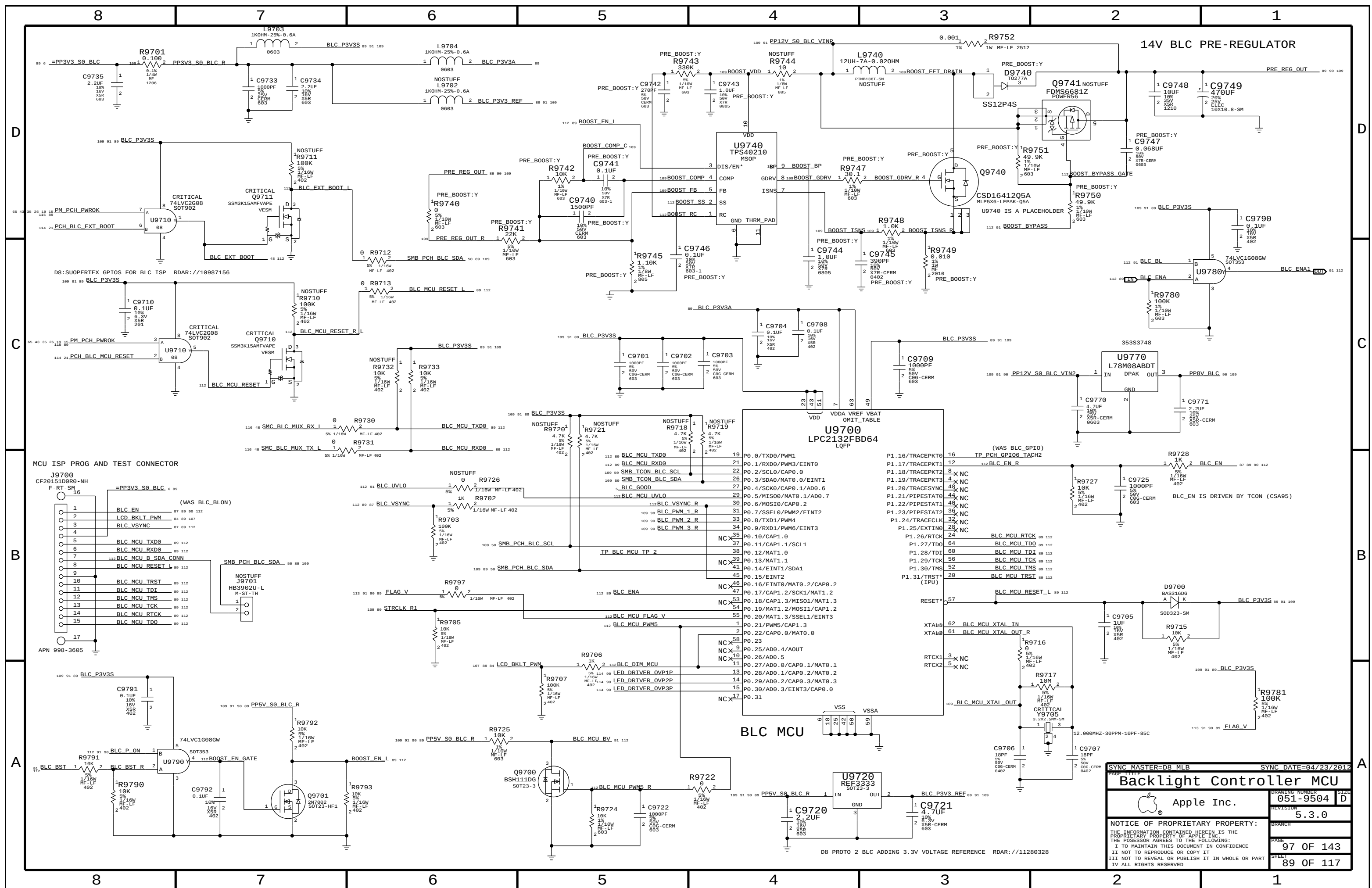


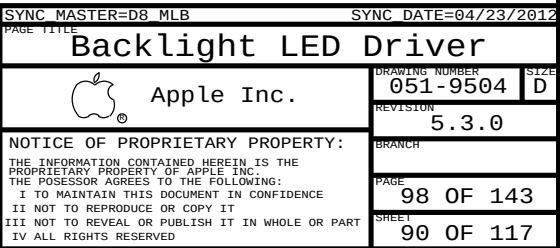
Thunderbolt Connector B

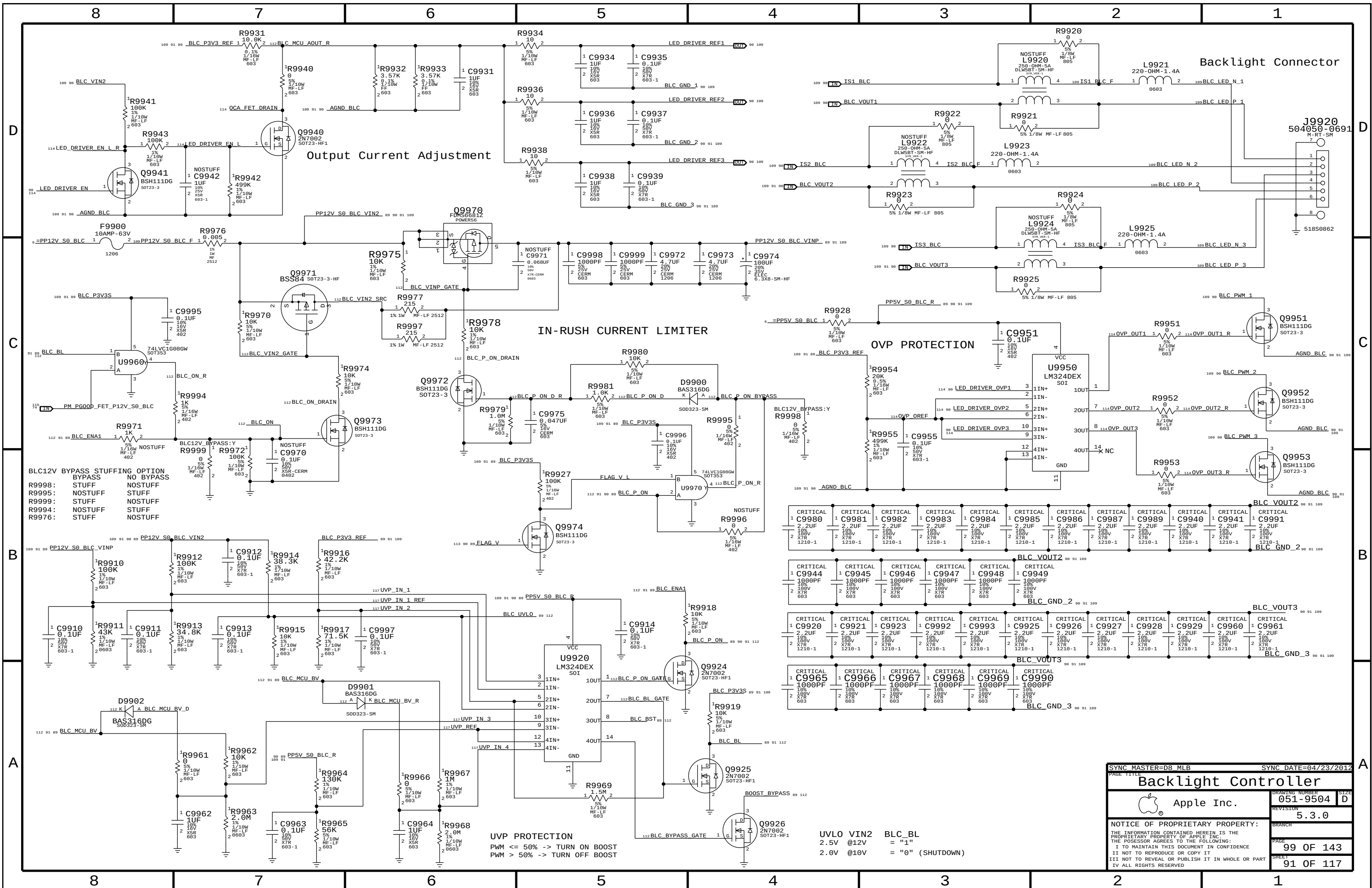


NOTE: Polarity Swapped for Layout!

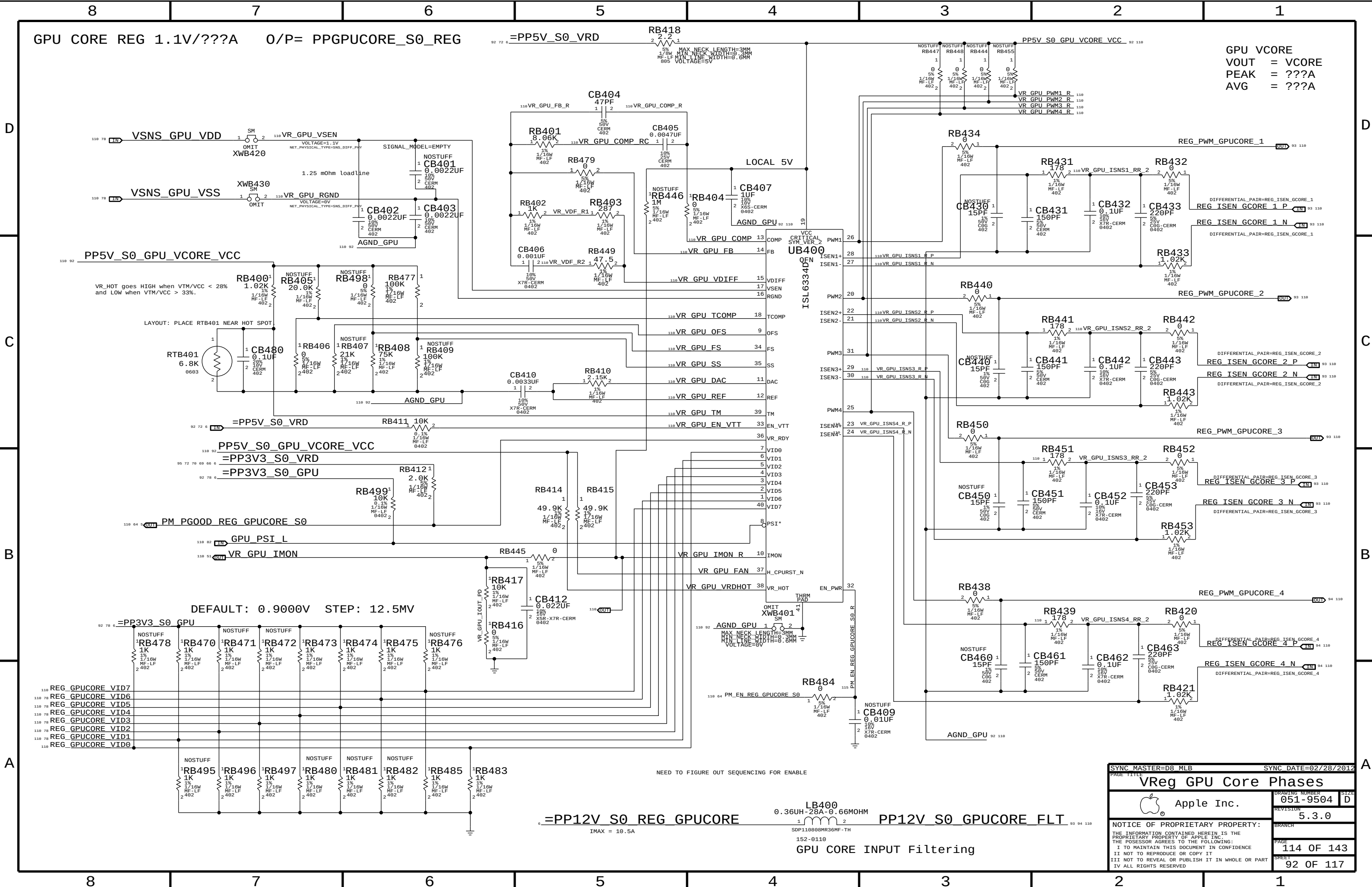
SYNC MASTER=D8 AARON	SYNC DATE=03/13/2012
Thunderbolt Connector B	
Apple Inc.	DRAWING NUMBER 051-9504
REVISION 5.3.0	SIZE D
NOTICE OF PROPRIETARY PROPERTY:	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:	
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE	
II NOT TO REPRODUCE OR COPY IT	
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART	
IV ALL RIGHTS RESERVED	
PAGE 96 OF 143	SHEET 88 OF 117





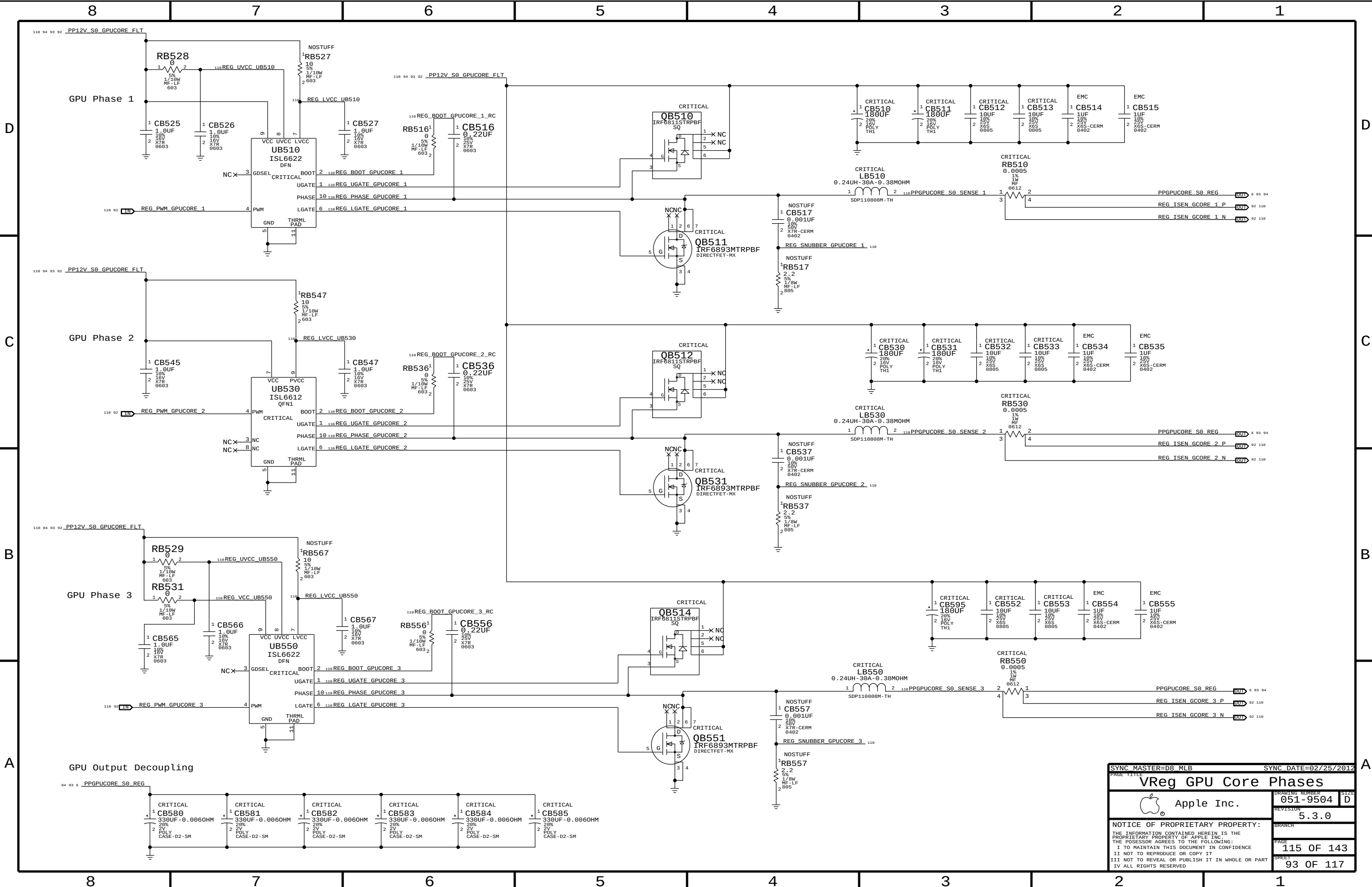


SYNC MASTER=D8 MLB		SYNC DATE=04/23/2012	
PAGE TITLE		Backlight Controller	
Apple Inc.		DRAWING NUMBER	051-9504
		REVISION	5.3.0
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:		PAGE	99 OF 143
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		SHEET	91 OF 117
II NOT TO REPRODUCE OR COPY IT			
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IV ALL RIGHTS RESERVED			



GPU VCORE
VOUT = VCORE
PEAK = ???A
AVG = ???A

SYNC MASTER=D8 MLB		SYNC DATE=02/28/2012	
PAGE TITLE		DRAWING NUMBER	
VReg GPU Core Phases		051-9504	
Apple Inc.		REVISION	
		5.3.0	
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:		PAGE	
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		114 OF 143	
II NOT TO REPRODUCE OR COPY IT		SHEET	
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART		92 OF 117	
IV ALL RIGHTS RESERVED			



SYNC MASTER=D8_MLB

SYNC DATE=02/25/2012

VReg GPU Core Phases

 Apple Inc.

DRAWING NUMBER

051-9504

SIZE

D

REVISION

5.3.0

NOTICE OF PROPRIETARY PROPERTY:

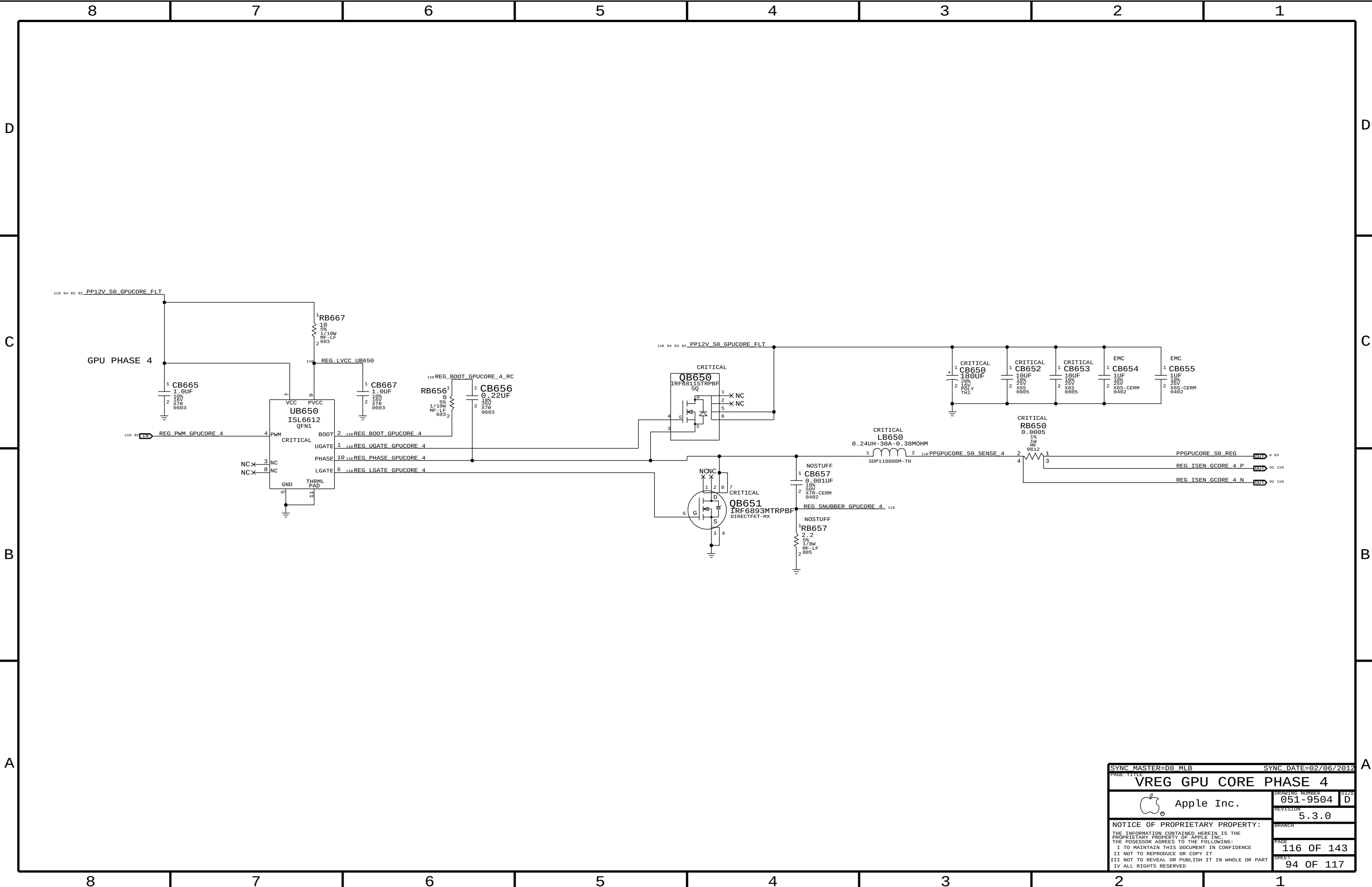
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
IV ALL RIGHTS RESERVED

PAGE

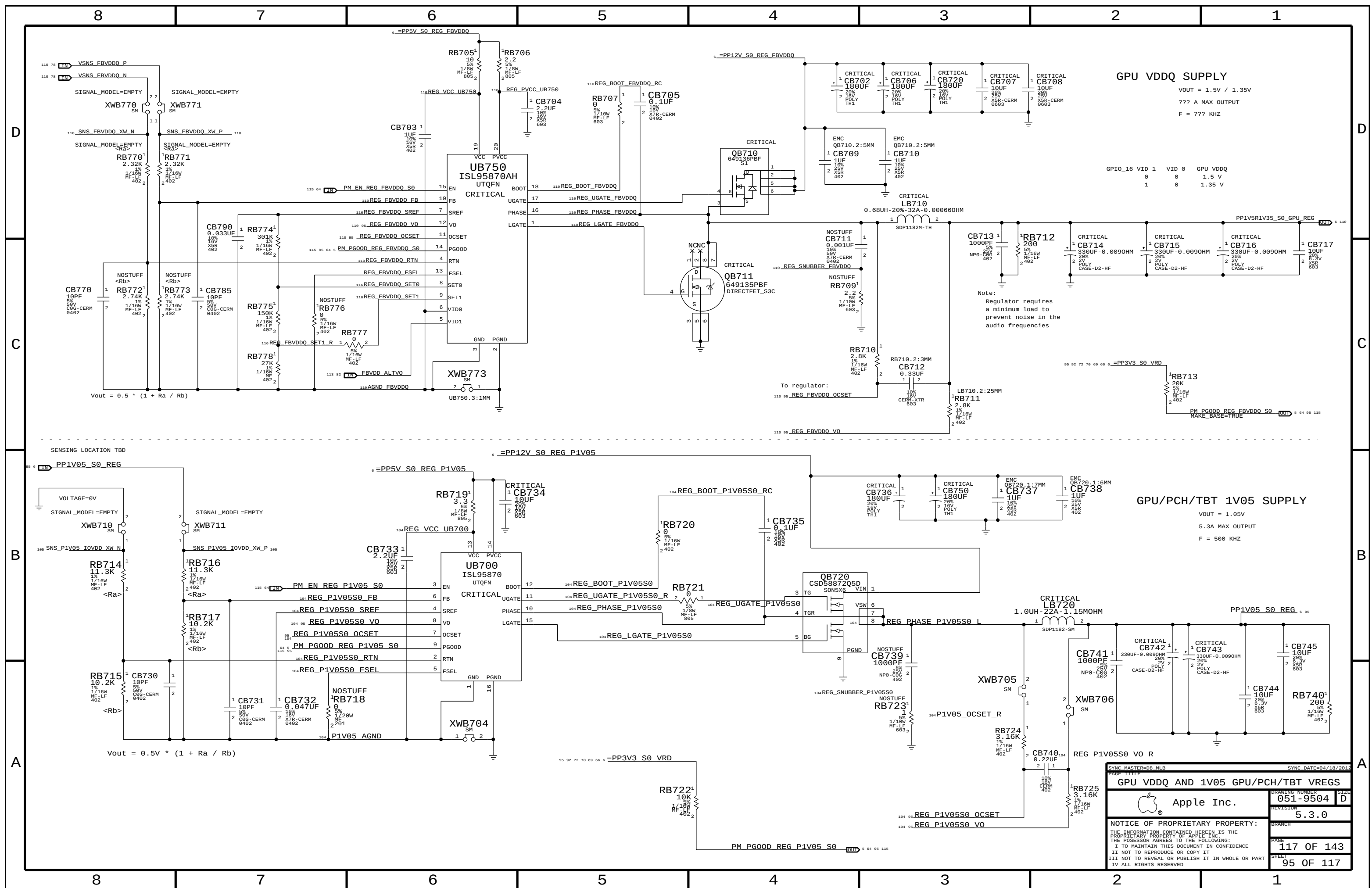
115 OF 143

SHEET

93 OF 117



SYNC MASTER=D8_MLB		SYNC DATE=02/06/2012	
PAGE TITLE		VREG GPU CORE PHASE 4	
DRAWING NUMBER		051-9504	SIZE D
REVISION		5.3.0	BRANCH
PAGE		116 OF 143	SHEET
SHEET		94 OF 117	
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED			



D8 BOARD SPECIFIC PHYSICAL AND SPACING CONSTRAINTS

BOARD LAYERS	BOARD AREAS	BOARD UNITS (MIL or MM)	ALLEGRO VERSION
TOP, ISL2, ISL3, ISL4, ISL5, ISL6, ISL7, ISL8, ISL9, ISL10, ISL11, BOTTOM	NO_TYPE, BGA, BGA_TBT	MM	16.2

General Physical Rule Definitions

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DEFAULT	*	Y	0.1 MM	=50_OHM_SE	10 MM	0 MM	0 MM
STANDARD	*	Y	=DEFAULT	=DEFAULT	10 MM	=DEFAULT	=DEFAULT

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
34_OHM_SE	*	Y	0.185 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
34_OHM_SE	ISL5, ISL8	Y	0.205 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
34_OHM_SE	TOP, BOTTOM	Y	0.220 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
39_OHM_SE	*	Y	0.150 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
39_OHM_SE	ISL5, ISL8	Y	0.165 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
39_OHM_SE	TOP, BOTTOM	Y	0.175 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
42_OHM_SE	*	Y	0.130 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
42_OHM_SE	ISL5, ISL8	Y	0.145 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
42_OHM_SE	TOP, BOTTOM	Y	0.155 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
45_OHM_SE	*	Y	0.115 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
45_OHM_SE	ISL5, ISL8	Y	0.126 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
45_OHM_SE	TOP, BOTTOM	Y	0.135 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
50_OHM_SE	*	Y	0.090 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
50_OHM_SE	ISL5, ISL8	Y	0.100 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
50_OHM_SE	TOP,BOTTOM	Y	0.105 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
55_OHM_SE	*	Y	0.075 MM	0.075 MM	=STANDARD	=STANDARD	=STANDARD
55_OHM_SE	ISL5, ISL8	Y	0.080 MM	0.080 MM	=STANDARD	=STANDARD	=STANDARD
55_OHM_SE	TOP, BOTTOM	Y	0.085 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
68_OHM_DIFF	*	Y	0.171 MM	0.085 MM	=STANDARD	0.130 MM	0.1 MM
68_OHM_DIFF	TOP,BOTTOM	Y	0.185 MM	0.085 MM	=STANDARD	0.150 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
80_OHM_DIFF	*	Y	0.136 MM	0.085 MM	=STANDARD	0.190 MM	0.1 MM
80_OHM_DIFF	TOP,BOTTOM	Y	0.141 MM	0.085 MM	=STANDARD	0.185 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
85_OHM_DIFF	*	Y	0.121 MM	0.085 MM	=STANDARD	0.190 MM	0.1 MM
85_OHM_DIFF	TOP,BOTTOM	Y	0.125 MM	0.085 MM	=STANDARD	0.190 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
90_OHM_DIFF	*	Y	0.109 MM	0.085 MM	=STANDARD	0.200 MM	0.1 MM
90_OHM_DIFF	TOP,BOTTOM	Y	0.111 MM	0.085 MM	=STANDARD	0.200 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
100_OHM_DIFF	*	Y	0.086 MM	0.085 MM	=STANDARD	0.200 MM	0.1 MM
100_OHM_DIFF	TOP,BOTTOM	Y	0.090 MM	0.085 MM	=STANDARD	0.230 MM	0.1 MM

Compensation Physical Rule Definition

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
COMP_SE	*	Y	0.305 MM	0.105 MM	3 MM	=STANDARD	=STANDARD

NOTE: line width based on 12 mil recommendation
NOTE: neck width based on 4 mil recommendation

General Spacing Definitions

Default

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DEFAULT	*	0.1 MM	?
STANDARD	*	=DEFAULT	?

Fixed and Dielectric

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
1:1_SPACING	*	0.1 MM	?
1X_DIELECTRIC	TOP, BOTTOM	0.071 MM	?
1X_DIELECTRIC	ISL3, ISL10	0.101 MM	?
1X_DIELECTRIC	*	0.076 MM	?

Board Stack-up

FINISHED BOARD THICKNESS: 1.94 MM

Layer	Material	Weight	Thickness
Top	Signal	0.5 oz (Cu plated)	
	Prepreg		0.071 MM
	Plane	1 oz	
2	Core		0.101 MM
	Signal	0.5 oz	
	Prepreg		0.115 MM
4	Plane	1 oz	
	Core		0.076 MM
	Signal	0.5 oz	
5	Prepreg		0.380 MM
	Plane	1 oz	
	Core		0.076 MM
7	Plane	1 oz	
	Prepreg		0.380 MM
	Signal	0.5 oz	
8	Core		0.076 MM
	Plane	1 oz	
	Prepreg		0.115 MM
10	Signal	0.5 oz	
	Core		0.101 MM
	Plane	1 oz	
11	Prepreg		0.071 MM
	Signal	0.5 oz (Cu plated)	
	Btm		

BGA

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
BGA_P1MM	*	=STANDARD	?

Power and Common

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GND_ISO	*	=STANDARD	8000
GND_P2MM	*	=2:1_SPACING	1000
PWR_P2MM	*	=2:1_SPACING	1100

GENERIC

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GENERIC_ISO	*	=1:1_SPACING	?
PM_ISO	*	=1:1_SPACING	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
PM	*	*	PM_ISO
PM	GND	*	DEFAULT

BGA Area Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
*	*	BGA	BGA_P1MM

SYNCH MASTER=D8 KOSECOFF		SYNCH DATE=03/19/2012	
PAGE TITLE			
D8 RULE DEFINITIONS			
		DRAWING NUMBER	
Apple Inc.		051-9504	
		SIZE	
		D	
		REVISION	
		5.3.0	
NOTICE OF PROPRIETARY PROPERTY:			
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:			
I. TO MAINTAIN THIS DOCUMENT IN CONFIDENCE			
II. NOT TO REPRODUCE OR COPY IT			
III. NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IV. ALL RIGHTS RESERVED			
		BRANCH	
		PAGE	
		120 OF 143	
		SHEET	
		96 OF 117	

DDR3

DDR3-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DDR_34S	*	=34_OHM_SE	=34_OHM_SE	=34_OHM_SE	=34_OHM_SE	=STANDARD	=STANDARD
DDR_34S	BOTTOM	=34_OHM_SE	=34_OHM_SE	=34_OHM_SE	2.0 MM	=STANDARD	=STANDARD
DDR_39S	*	=39_OHM_SE	=39_OHM_SE	=39_OHM_SE	=39_OHM_SE	=STANDARD	=STANDARD
DDR_42S	*	=42_OHM_SE	=42_OHM_SE	=42_OHM_SE	=42_OHM_SE	=STANDARD	=STANDARD
DDR_42S	BOTTOM	=42_OHM_SE	=55_OHM_SE	=55_OHM_SE	2.0 MM	=STANDARD	=STANDARD
DDR_42S	ISL5, ISL8	=42_OHM_SE	=55_OHM_SE	=55_OHM_SE	2.0 MM	=STANDARD	=STANDARD
DDR_42S_D	*	=42_OHM_SE	=42_OHM_SE	=42_OHM_SE	=42_OHM_SE	0.1016 MM	0.1016 MM
DDR_42S_D	BOTTOM	=42_OHM_SE	=55_OHM_SE	=55_OHM_SE	2.0 MM	0.1016 MM	0.1016 MM
DDR_42S_D	ISL5, ISL8	=42_OHM_SE	=55_OHM_SE	=55_OHM_SE	2.0 MM	0.1016 MM	0.1016 MM
DDR_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
DDR_68D	*	=68_OHM_DIFF	=68_OHM_DIFF	=68_OHM_DIFF	=68_OHM_DIFF	=68_OHM_DIFF	=68_OHM_DIFF

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
POWER_DDR_P4MM	*	Y	0.400 MM	0.100 MM	3.0 MM	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
POWER_DDR_PHY	*	POWER_DDR_P4MM
DDR_CLK_PHY	*	DDR_68D
DDR_CTRL_PHY	*	DDR_39S
DDR_CMD_PHY	*	DDR_34S
DDR_DQ_PHY	*	DDR_42S
DDR_DQS_PHY	*	DDR_42S_D

DDR3-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DDR_CLK_ISO	TOP, BOTTOM	=5.5X_DIELECTRIC	?
DDR_CLK_ISO	ISL3, ISL10	=4.9X_DIELECTRIC	?
DDR_CLK_ISO	*	=6.5X_DIELECTRIC	?
DDR_CTRL_ISO	TOP, BOTTOM	=4.5X_DIELECTRIC	?
DDR_CTRL_ISO	ISL3, ISL10	=4.0X_DIELECTRIC	?
DDR_CTRL_ISO	*	=5.3X_DIELECTRIC	?
DDR_CTRL2CTRL	TOP, BOTTOM	=3.0X_DIELECTRIC	?
DDR_CTRL2CTRL	ISL3, ISL10	=2.6X_DIELECTRIC	?
DDR_CTRL2CTRL	*	=3.5X_DIELECTRIC	?
DDR_CMD_ISO	TOP, BOTTOM	=4.5X_DIELECTRIC	?
DDR_CMD_ISO	ISL3, ISL10	=4.0X_DIELECTRIC	?
DDR_CMD_ISO	*	=5.3X_DIELECTRIC	?
DDR_CMD2CMD	TOP, BOTTOM	=2.3X_DIELECTRIC	?
DDR_CMD2CMD	ISL3, ISL10	=2.0X_DIELECTRIC	?
DDR_CMD2CMD	*	=2.7X_DIELECTRIC	?
DDR_DATA_ISO	TOP, BOTTOM	=4.5X_DIELECTRIC	?
DDR_DATA_ISO	ISL3, ISL10	=4.0X_DIELECTRIC	?
DDR_DATA_ISO	*	=5.3X_DIELECTRIC	?
DDR_DQ2DQ	TOP, BOTTOM	=3.2X_DIELECTRIC	900
DDR_DQ2DQ	ISL3, ISL10	=2.8X_DIELECTRIC	900
DDR_DQ2DQ	*	=3.8X_DIELECTRIC	900
DDR_DQ2DQS	TOP, BOTTOM	=3.7X_DIELECTRIC	?
DDR_DQ2DQS	ISL3, ISL10	=3.3X_DIELECTRIC	?
DDR_DQ2DQS	*	=4.4X_DIELECTRIC	?
DDR_BL2BL	TOP, BOTTOM	=4.5X_DIELECTRIC	?
DDR_BL2BL	ISL3, ISL10	=4.0X_DIELECTRIC	?
DDR_BL2BL	*	=5.3X_DIELECTRIC	?
DDR_CH2CH	TOP, BOTTOM	=9.1X_DIELECTRIC	?
DDR_CH2CH	ISL3, ISL10	=8.2X_DIELECTRIC	?
DDR_CH2CH	*	=10.9X_DIELECTRIC	?
CMD2DATA_ISO	TOP, BOTTOM	=7X_DIELECTRIC	?
CMD2DATA_ISO	ISL3, ISL10	=5X_DIELECTRIC	?
CMD2DATA_ISO	*	=5X_DIELECTRIC	?

DDR3 Power-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
POWER_DDR_ISO	*	=4.3X_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
POWER_DDR	*	*	POWER_DDR_ISO

Constraints

Clocks: CK[3:0], CK#[3:0]

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
DDR_CLK	*	*	DDR_CLK_ISO

Control: CS#[3:0], CKE[3:0], ODT[3:0]

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
DDR_CTRL	*	*	DDR_CTRL_ISO
DDR_CTRL	DDR_CTRL	*	DDR_CTRL2CTRL

Command: MA[15:0], RAS#, CAS#, WE# BS[2:0]

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
DDR_CMD	*	*	DDR_CMD_ISO
DDR_CMD	DDR_CMD	*	DDR_CMD2CMD
DDR_CMD	DDR_A_DQ_BYTE*	*	CMD2DATA_ISO
DDR_CMD	DDR_B_DQ_BYTE*	*	CMD2DATA_ISO
DDR_CMD	DDR_B_DQS*	*	CMD2DATA_ISO

Data: DQS[7:0], DQS#[7:0], DQ[63:0]

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
DDR_A_DQ_BYTE*	*	*	DDR_DATA_ISO
DDR_A_DQS*	*	*	DDR_DATA_ISO
DDR_B_DQ_BYTE*	*	*	DDR_DATA_ISO
DDR_B_DQS*	*	*	DDR_DATA_ISO
DDR_*_DQ_BYTE*	=SAME	*	DDR_DQ2DQ
DDR_A_DQ_BYTE*	DDR_A_DQS*	*	DDR_DQ2DQS
DDR_A_DQ_BYTE*	DDR_A_DQ_BYTE*	*	DDR_BL2BL
DDR_B_DQ_BYTE*	DDR_B_DQS*	*	DDR_DQ2DQS
DDR_B_DQ_BYTE*	DDR_B_DQ_BYTE*	*	DDR_BL2BL
DDR_A_*	DDR_B_*	*	DDR_CH2CH

Note (1):

Deliberately set DQ to DQS spacing to 3:1 to avoid adding complexity to constraints, even though it can be less. Only one rule per channel is needed by trading off a little space.

Note (2):

Intel suggests 25 mil (0.65 mm) spacing for via to channel, and via to pad to two different channels. DDR3 draws about 20 mA per trace with edge rates in the 100s of ps. The main coupling mechanism is capacitive. A 0.65 mm spacing is used for power nets, which draw far more current (inductive coupling however). These rules are far too conservative. To meet these rules, the spacing must be applied to the net.

Note (3):

In order for the constraints DDR_*_DQ_BYTE* to =SAME to work out over DDR_{A,B}_DQ_BYTE* to DDR_{A,B}_DQ_BYTE* so that the small intra-bytelane spacing is used, the spacing rule DDR_DQ2DQ must have a weight greater than DDR_BL2BL.

DDR3

Electrical Constraint Set	Physical	Spacing
Channel A		
1E97D DDR A CLK0	DDR_CLK_PHY	DDR_CLK
1E97D DDR A CLK0	DDR_CLK_PHY	DDR_CLK
1E97D DDR A CLK1	DDR_CLK_PHY	DDR_CLK
1E97D DDR A CLK1	DDR_CLK_PHY	DDR_CLK
1E97D DDR A CTRL0	DDR_CTRL_PHY	DDR_CTRL
1E97D DDR A CTRL0	DDR_CTRL_PHY	DDR_CTRL
1E97D DDR A CTRL0	DDR_CTRL_PHY	DDR_CTRL
1E97D DDR A CTRL1	DDR_CTRL_PHY	DDR_CTRL
1E97D DDR A CTRL1	DDR_CTRL_PHY	DDR_CTRL
1E97D DDR A CTRL1	DDR_CTRL_PHY	DDR_CTRL
1E97D DDR A CMD	DDR_CMD_PHY	DDR_CMD
1E97D DDR A CMD	DDR_CMD_PHY	DDR_CMD
1E97D DDR A CMD	DDR_CMD_PHY	DDR_CMD
1E97D DDR A CMD	DDR_CMD_PHY	DDR_CMD
1E97D DDR A CMD	DDR_CMD_PHY	DDR_CMD
1E97D DDR A DQ BYTE0	DDR_DQ_PHY	DDR_A_DQ_BYTE0
1E97D DDR A DQ BYTE1	DDR_DQ_PHY	DDR_A_DQ_BYTE1
1E97D DDR A DQ BYTE2	DDR_DQ_PHY	DDR_A_DQ_BYTE2
1E97D DDR A DQ BYTE3	DDR_DQ_PHY	DDR_A_DQ_BYTE3
1E97D DDR A DQ BYTE4	DDR_DQ_PHY	DDR_A_DQ_BYTE4
1E97D DDR A DQ BYTE5	DDR_DQ_PHY	DDR_A_DQ_BYTE5
1E97D DDR A DQ BYTE6	DDR_DQ_PHY	DDR_A_DQ_BYTE6
1E97D DDR A DQ BYTE7	DDR_DQ_PHY	DDR_A_DQ_BYTE7
1E97D DDR A DQS0	DDR_DQS_PHY	DDR_A_DQS0
1E97D DDR A DQS0	DDR_DQS_PHY	DDR_A_DQS0
1E97D DDR A DQS1	DDR_DQS_PHY	DDR_A_DQS1
1E97D DDR A DQS1	DDR_DQS_PHY	DDR_A_DQS1
1E97D DDR A DQS2	DDR_DQS_PHY	DDR_A_DQS2
1E97D DDR A DQS2	DDR_DQS_PHY	DDR_A_DQS2
1E97D DDR A DQS3	DDR_DQS_PHY	DDR_A_DQS3
1E97D DDR A DQS3	DDR_DQS_PHY	DDR_A_DQS3
1E97D DDR A DQS4	DDR_DQS_PHY	DDR_A_DQS4
1E97D DDR A DQS4	DDR_DQS_PHY	DDR_A_DQS4
1E97D DDR A DQS5	DDR_DQS_PHY	DDR_A_DQS5
1E97D DDR A DQS5	DDR_DQS_PHY	DDR_A_DQS5
1E97D DDR A DQS6	DDR_DQS_PHY	DDR_A_DQS6
1E97D DDR A DQS6	DDR_DQS_PHY	DDR_A_DQS6
1E97D DDR A DQS7	DDR_DQS_PHY	DDR_A_DQS7
1E97D DDR A DQS7	DDR_DQS_PHY	DDR_A_DQS7
Channel B		
1E97D DDR B CLK0	DDR_CLK_PHY	DDR_CLK
1E97D DDR B CLK0	DDR_CLK_PHY	DDR_CLK
1E97D DDR B CLK1	DDR_CLK_PHY	DDR_CLK
1E97D DDR B CLK1	DDR_CLK_PHY	DDR_CLK
1E97D DDR B CTRL0	DDR_CTRL_PHY	DDR_CTRL
1E97D DDR B CTRL0	DDR_CTRL_PHY	DDR_CTRL
1E97D DDR B CTRL0	DDR_CTRL_PHY	DDR_CTRL
1E97D DDR B CTRL1	DDR_CTRL_PHY	DDR_CTRL
1E97D DDR B CTRL1	DDR_CTRL_PHY	DDR_CTRL
1E97D DDR B CTRL1	DDR_CTRL_PHY	DDR_CTRL
1E97D DDR B CMD	DDR_CMD_PHY	DDR_CMD
1E97D DDR B CMD	DDR_CMD_PHY	DDR_CMD
1E97D DDR B CMD	DDR_CMD_PHY	DDR_CMD
1E97D DDR B CMD	DDR_CMD_PHY	DDR_CMD
1E97D DDR B CMD	DDR_CMD_PHY	DDR_CMD
1E97D DDR B DQ BYTE0	DDR_DQ_PHY	DDR_B_DQ_BYTE0
1E97D DDR B DQ BYTE1	DDR_DQ_PHY	DDR_B_DQ_BYTE1
1E97D DDR B DQ BYTE2	DDR_DQ_PHY	DDR_B_DQ_BYTE2
1E97D DDR B DQ BYTE3	DDR_DQ_PHY	DDR_B_DQ_BYTE3
1E97D DDR B DQ BYTE4	DDR_DQ_PHY	DDR_B_DQ_BYTE4
1E97D DDR B DQ BYTE5	DDR_DQ_PHY	DDR_B_DQ_BYTE5
1E97D DDR B DQ BYTE6	DDR_DQ_PHY	DDR_B_DQ_BYTE6
1E97D DDR B DQ BYTE7	DDR_DQ_PHY	DDR_B_DQ_BYTE7
1E97D DDR B DQS0	DDR_DQS_PHY	DDR_B_DQS0
1E97D DDR B DQS0	DDR_DQS_PHY	DDR_B_DQS0
1E97D DDR B DQS1	DDR_DQS_PHY	DDR_B_DQS1
1E97D DDR B DQS1	DDR_DQS_PHY	DDR_B_DQS1
1E97D DDR B DQS2	DDR_DQS_PHY	DDR_B_DQS2
1E97D DDR B DQS2	DDR_DQS_PHY	DDR_B_DQS2
1E97D DDR B DQS3	DDR_DQS_PHY	DDR_B_DQS3
1E97D DDR B DQS3	DDR_DQS_PHY	DDR_B_DQS3
1E97D DDR B DQS4	DDR_DQS_PHY	DDR_B_DQS4
1E97D DDR B DQS4	DDR_DQS_PHY	DDR_B_DQS4
1E97D DDR B DQS5	DDR_DQS_PHY	DDR_B_DQS5
1E97D DDR B DQS5	DDR_DQS_PHY	DDR_B_DQS5
1E97D DDR B DQS6	DDR_DQS_PHY	DDR_B_DQS6
1E97D DDR B DQS6	DDR_DQS_PHY	DDR_B_DQS6
1E97D DDR B DQS7	DDR_DQS_PHY	DDR_B_DQS7
1E97D DDR B DQS7	DDR_DQS_PHY	DDR_B_DQS7
Reset	DDR_S0S	CPU
1E97D		
MEM A CLK P<1..0>		
MEM A CLK N<1..0>		
MEM A CLK P<3..2>		
MEM A CLK N<3..2>		
MEM A CKE<1..0>		
MEM A CS L<1..0>		
MEM A ODT<1..0>		
MEM A CKE<3..2>		
MEM A CS L<3..2>		
MEM A ODT<3..2>		
MEM A A<15..0>		
MEM A BA<2..0>		
MEM A RAS L		
MEM A CAS L		
MEM A WE L		
MEM A DQ<7..0>		
MEM A DQ<15..8>		
MEM A DQ<23..16>		
MEM A DQ<31..24>		
MEM A DQ<39..32>		
MEM A DQ<47..40>		
MEM A DQ<55..48>		
MEM A DQ<63..56>		
MEM A DQS P<0>		
MEM A DQS N<0>		
MEM A DQS P<1>		
MEM A DQS N<1>		
MEM A DQS P<2>		
MEM A DQS N<2>		
MEM A DQS P<3>		
MEM A DQS N<3>		
MEM A DQS P<4>		
MEM A DQS N<4>		
MEM A DQS P<5>		
MEM A DQS N<5>		
MEM A DQS P<6>		
MEM A DQS N<6>		
MEM A DQS P<7>		
MEM A DQS N<7>		
MEM B CLK P<1..0>		
MEM B CLK N<1..0>		
MEM B CLK P<3..2>		
MEM B CLK N<3..2>		
MEM B CKE<1..0>		
MEM B CS L<1..0>		
MEM B ODT<1..0>		
MEM B CKE<3..2>		
MEM B CS L<3..2>		
MEM B ODT<3..2>		
MEM B A<15..0>		
MEM B BA<2..0>		
MEM B RAS L		
MEM B CAS L		
MEM B WE L		
MEM B DQ<7..0>		
MEM B DQ<15..8>		
MEM B DQ<23..16>		
MEM B DQ<31..24>		
MEM B DQ<39..32>		
MEM B DQ<47..40>		
MEM B DQ<55..48>		
MEM B DQ<63..56>		
MEM B DQS P<0>		
MEM B DQS N<0>		
MEM B DQS P<1>		
MEM B DQS N<1>		
MEM B DQS P<2>		
MEM B DQS N<2>		
MEM B DQS P<3>		
MEM B DQS N<3>		
MEM B DQS P<4>		
MEM B DQS N<4>		
MEM B DQS P<5>		
MEM B DQS N<5>		
MEM B DQS P<6>		
MEM B DQS N<6>		
MEM B DQS P<7>		
MEM B DQS N<7>		

SYNC MASTER=D8 KOSECOFF		SYNC DATE=03/19/2012	
PAGE TITLE		DDR3 Constraints	
DRAWING NUMBER		051-9504	SIZE D
REVISION		5.3.0	BRANCH
NOTICE OF PROPRIETARY PROPERTY:		PAGE	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:		121 OF 143	
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		SHEET	
II NOT TO REPRODUCE OR COPY IT		97 OF 117	
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IV ALL RIGHTS RESERVED			

PCI EXPRESS

PCIe-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCIE_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
PCIE_80D	*	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF
PCIE_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
PCIE_PHY	*	PCIE_80D
CLK_PCIE_PHY	*	PCIE_90D
COMP_PCIE_PHY	*	COMP_SE

PCIe-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PCIE_ISO	TOP, BOTTOM	=5X_DIELECTRIC	?
PCIE_ISO	*	=4X_DIELECTRIC	?
CLK_PCIE_ISO	*	=5:1_SPACING	?
COMP_PCIE_ISO	*	=4:1_SPACING	?

Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
PCIE	*	*	PCIE_ISO
CLK_PCIE	*	*	CLK_PCIE_ISO
COMP_PCIE	*	*	COMP_PCIE_ISO

PCIe (PCH)

Electrical Constraint Set		Physical	Spacing	
x1 AirPort				
H599	PCIE_GEN2_R2D_CONN_AP	PCIE_PHY	PCIE	PCIE AP R2D P NO TEST TRUE 35
H600	PCIE_GEN2_R2D_CONN_AP	PCIE_PHY	PCIE	PCIE AP R2D N NO TEST TRUE 35
H601		PCIE_PHY	PCIE	PCIE AP R2D C P NO TEST TRUE 18 35
H602		PCIE_PHY	PCIE	PCIE AP R2D C N NO TEST TRUE 18 35
H603	PCIE_GEN2_D2R_CONN_AP	PCIE_PHY	PCIE	PCIE AP D2R P NO TEST TRUE 18 35
H604	PCIE_GEN2_D2R_CONN_AP	PCIE_PHY	PCIE	PCIE AP D2R N NO TEST TRUE 18 35
H605	PCIE_REF_CLK_CONN	CLK PCIE_PHY	CLK PCIE	PCIE CLK100M_AP P 18 35
H606	PCIE_REF_CLK_CONN	CLK PCIE_PHY	CLK PCIE	PCIE CLK100M_AP N 18 35
x1 Caesar IV				
H607	PCIE_GEN2_R2D	PCIE_PHY	PCIE	PCIE ENET_R2D P NO TEST TRUE 39
H608	PCIE_GEN2_R2D	PCIE_PHY	PCIE	PCIE ENET_R2D N NO TEST TRUE 39
H609		PCIE_PHY	PCIE	PCIE ENET_R2D C P NO TEST TRUE 18 39
H610		PCIE_PHY	PCIE	PCIE ENET_R2D C N NO TEST TRUE 18 39
H611	PCIE_GEN2_D2R	PCIE_PHY	PCIE	PCIE ENET_D2R P NO TEST TRUE 18 39
H612	PCIE_GEN2_D2R	PCIE_PHY	PCIE	PCIE ENET_D2R N NO TEST TRUE 18 39
H613		PCIE_PHY	PCIE	PCIE ENET_D2R C P NO TEST TRUE 39
H614		PCIE_PHY	PCIE	PCIE ENET_D2R C N NO TEST TRUE 39
H615	PCIE_REF_CLK	CLK PCIE_PHY	CLK PCIE	PCIE CLK100M_ENET P 18 39
H616	PCIE_REF_CLK	CLK PCIE_PHY	CLK PCIE	PCIE CLK100M_ENET N 18 39

PCIE (PCH - TBT)

Electrical Constraint Set		Physical	Spacing	
x4 Thunderbolt				
ES79	PCIE_GEN2_R2D_PTNV	PCIE_PHY	PCIE	PCIE TBT R2D P<2...0> NO TEST-TRUE 36
ES79	PCIE_GEN2_R2D_PTNV	PCIE_PHY	PCIE	PCIE TBT R2D N<2...0> NO TEST-TRUE 36
ES80	PCIE_GEN2_R2D	PCIE_PHY	PCIE	PCIE TBT R2D P<3> NO TEST-TRUE 36
ES80	PCIE_GEN2_R2D	PCIE_PHY	PCIE	PCIE TBT R2D N<3> NO TEST-TRUE 36
ES82		PCIE_PHY	PCIE	PCIE TBT R2D C P<3...0> NO TEST-TRUE 18 36
ES82		PCIE_PHY	PCIE	PCIE TBT R2D C N<3...0> NO TEST-TRUE 18 36
ES90		PCIE_PHY	PCIE	PCIE TBT D2R P<3...0> NO TEST-TRUE 18 36
ES90		PCIE_PHY	PCIE	PCIE TBT D2R N<3...0> NO TEST-TRUE 18 36
ES92	PCIE_GEN2_D2R_PTNV	PCIE_PHY	PCIE	PCIE TBT D2R C P<1...0> NO TEST-TRUE 36
ES92	PCIE_GEN2_D2R_PTNV	PCIE_PHY	PCIE	PCIE TBT D2R C N<1...0> NO TEST-TRUE 36
ES92	PCIE_GEN2_D2R	PCIE_PHY	PCIE	PCIE TBT D2R C P<2> NO TEST-TRUE 36
ES92	PCIE_GEN2_D2R	PCIE_PHY	PCIE	PCIE TBT D2R C N<2> NO TEST-TRUE 36
ES92	PCIE_GEN2_D2R_PTNV	PCIE_PHY	PCIE	PCIE TBT D2R C P<3> NO TEST-TRUE 36
ES92	PCIE_GEN2_D2R_PTNV	PCIE_PHY	PCIE	PCIE TBT D2R C N<3> NO TEST-TRUE 36
ES90	PCIE_REF_CLK	CLK PCIE_PHY	CLK PCIE	PCIE CLK100M TBT P 18 36
ES90	PCIE_REF_CLK	CLK PCIE_PHY	CLK PCIE	PCIE CLK100M TBT N 18 36

PCIe (CPU)

Electrical Constraint Set	Physical	Spacing	
x16 Graphics			
E600 PCIE_GEN3_R2D	PCIE_PHY	PCIE	PEG R2D P<15> NO_TEST=TRUE 10 75
E601 PCIE_GEN3_R2D	PCIE_PHY	PCIE	PEG R2D N<15> NO_TEST=TRUE 10 75
E602	PCIE_PHY	PCIE	PEG R2D C P<15> NO_TEST=TRUE 75
E603	PCIE_PHY	PCIE	PEG R2D C N<15> NO_TEST=TRUE 75
E604 PCIE_GEN3_D2R	PCIE_PHY	PCIE	PEG D2R P<15> NO_TEST=TRUE 75
E605 PCIE_GEN3_D2R	PCIE_PHY	PCIE	PEG D2R N<15> NO_TEST=TRUE 75
E606	PCIE_PHY	PCIE	PEG D2R C P<15> NO_TEST=TRUE 10 75
E607	PCIE_PHY	PCIE	PEG D2R C N<15> NO_TEST=TRUE 10 75
E608 PCIE_GEN3_R2D	PCIE_PHY	PCIE	PEG R2D P<14> NO_TEST=TRUE 10 75
E609 PCIE_GEN3_R2D	PCIE_PHY	PCIE	PEG R2D N<14> NO_TEST=TRUE 10 75
E610	PCIE_PHY	PCIE	PEG R2D C P<14> NO_TEST=TRUE 75
E611	PCIE_PHY	PCIE	PEG R2D C N<14> NO_TEST=TRUE 75
E612 PCIE_GEN3_D2R	PCIE_PHY	PCIE	PEG D2R P<14> NO_TEST=TRUE 75
E613 PCIE_GEN3_D2R	PCIE_PHY	PCIE	PEG D2R N<14> NO_TEST=TRUE 75
E614	PCIE_PHY	PCIE	PEG D2R C P<14> NO_TEST=TRUE 10 75
E615	PCIE_PHY	PCIE	PEG D2R C N<14> NO_TEST=TRUE 10 75
E616 PCIE_GEN3_R2D	PCIE_PHY	PCIE	PEG R2D P<13> NO_TEST=TRUE 10 75
E617 PCIE_GEN3_R2D	PCIE_PHY	PCIE	PEG R2D N<13> NO_TEST=TRUE 10 75
E618	PCIE_PHY	PCIE	PEG R2D C P<13> NO_TEST=TRUE 75
E619	PCIE_PHY	PCIE	PEG R2D C N<13> NO_TEST=TRUE 75
E620 PCIE_GEN3_D2R_P1NV	PCIE_PHY	PCIE	PEG D2R P<13> NO_TEST=TRUE 75
E621 PCIE_GEN3_D2R_P1NV	PCIE_PHY	PCIE	PEG D2R N<13> NO_TEST=TRUE 75
E622	PCIE_PHY	PCIE	PEG D2R C P<13> NO_TEST=TRUE 10 75
E623	PCIE_PHY	PCIE	PEG D2R C N<13> NO_TEST=TRUE 10 75
E624 PCIE_GEN3_R2D	PCIE_PHY	PCIE	PEG R2D P<12> NO_TEST=TRUE 10 75
E625 PCIE_GEN3_R2D	PCIE_PHY	PCIE	PEG R2D N<12> NO_TEST=TRUE 10 75
E626	PCIE_PHY	PCIE	PEG R2D C P<12> NO_TEST=TRUE 75
E627	PCIE_PHY	PCIE	PEG R2D C N<12> NO_TEST=TRUE 75
E628 PCIE_GEN3_D2R	PCIE_PHY	PCIE	PEG D2R P<12> NO_TEST=TRUE 75
E629 PCIE_GEN3_D2R	PCIE_PHY	PCIE	PEG D2R N<12> NO_TEST=TRUE 75
E630	PCIE_PHY	PCIE	PEG D2R C P<12> NO_TEST=TRUE 10 75
E631	PCIE_PHY	PCIE	PEG D2R C N<12> NO_TEST=TRUE 10 75
E632 PCIE_GEN3_R2D	PCIE_PHY	PCIE	PEG R2D P<11> NO_TEST=TRUE 10 75
E633 PCIE_GEN3_R2D	PCIE_PHY	PCIE	PEG R2D N<11> NO_TEST=TRUE 10 75
E634	PCIE_PHY	PCIE	PEG R2D C P<11> NO_TEST=TRUE 75
E635	PCIE_PHY	PCIE	PEG R2D C N<11> NO_TEST=TRUE 75
E636 PCIE_GEN3_D2R	PCIE_PHY	PCIE	PEG D2R P<11> NO_TEST=TRUE 75
E637 PCIE_GEN3_D2R	PCIE_PHY	PCIE	PEG D2R N<11> NO_TEST=TRUE 75
E638	PCIE_PHY	PCIE	PEG D2R C P<11> NO_TEST=TRUE 10 75
E639	PCIE_PHY	PCIE	PEG D2R C N<11> NO_TEST=TRUE 10 75
E640 PCIE_GEN3_R2D_P1NV	PCIE_PHY	PCIE	PEG R2D P<10> NO_TEST=TRUE 10 75
E641 PCIE_GEN3_R2D_P1NV	PCIE_PHY	PCIE	PEG R2D N<10> NO_TEST=TRUE 10 75
E642	PCIE_PHY	PCIE	PEG R2D C P<10> NO_TEST=TRUE 75
E643	PCIE_PHY	PCIE	PEG R2D C N<10> NO_TEST=TRUE 75
E644 PCIE_GEN3_D2R	PCIE_PHY	PCIE	PEG D2R P<10> NO_TEST=TRUE 75
E645 PCIE_GEN3_D2R	PCIE_PHY	PCIE	PEG D2R N<10> NO_TEST=TRUE 75
E646	PCIE_PHY	PCIE	PEG D2R C P<10> NO_TEST=TRUE 10 75
E647	PCIE_PHY	PCIE	PEG D2R C N<10> NO_TEST=TRUE 10 75
E648 PCIE_GEN3_R2D	PCIE_PHY	PCIE	PEG R2D P<9> NO_TEST=TRUE 10 75
E649 PCIE_GEN3_R2D	PCIE_PHY	PCIE	PEG R2D N<9> NO_TEST=TRUE 10 75
E650	PCIE_PHY	PCIE	PEG R2D C P<9> NO_TEST=TRUE 75
E651	PCIE_PHY	PCIE	PEG R2D C N<9> NO_TEST=TRUE 75
E652 PCIE_GEN3_D2R	PCIE_PHY	PCIE	PEG D2R P<9> NO_TEST=TRUE 75
E653 PCIE_GEN3_D2R	PCIE_PHY	PCIE	PEG D2R N<9> NO_TEST=TRUE 75
E654	PCIE_PHY	PCIE	PEG D2R C P<9> NO_TEST=TRUE 10 75
E655	PCIE_PHY	PCIE	PEG D2R C N<9> NO_TEST=TRUE 10 75
E656 PCIE_GEN3_R2D_P1NV	PCIE_PHY	PCIE	PEG R2D P<8> NO_TEST=TRUE 10 75
E657 PCIE_GEN3_R2D_P1NV	PCIE_PHY	PCIE	PEG R2D N<8> NO_TEST=TRUE 10 75
E658	PCIE_PHY	PCIE	PEG R2D C P<8> NO_TEST=TRUE 75
E659	PCIE_PHY	PCIE	PEG R2D C N<8> NO_TEST=TRUE 75
E660 PCIE_GEN3_D2R_P1NV	PCIE_PHY	PCIE	PEG D2R P<8> NO_TEST=TRUE 75
E661 PCIE_GEN3_D2R_P1NV	PCIE_PHY	PCIE	PEG D2R N<8> NO_TEST=TRUE 75
E662	PCIE_PHY	PCIE	PEG D2R C P<8> NO_TEST=TRUE 10 75
E663	PCIE_PHY	PCIE	PEG D2R C N<8> NO_TEST=TRUE 10 75

PCIe (CPU)

[illegible]

SYNC MASTER=D8 AARON		SYNC DATE=03/13/2012	
PAGE TITLE			
CPU PCIe Constraints			
 Apple Inc.	DRAWING NUMBER	SIZE	
	051-9504	D	
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE I NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED	REVISION	5.3.0	
	BRANCH		
	PAGE	122 OF 143	
SHEET	98 OF 117		

DMI

DMI-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DMI_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
DMI_PHY	*	DMI_85D

CPU Misc / FDI

CPU-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CPU_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
COMP_FDI_SE	*	Y	0.3 MM	0.25 MM	3 MM	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
CPU_PHY	*	CPU_50S
COMP_FDI_PHY	*	COMP_FDI_SE

CPU-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CPU_ISO	*	=3:1_SPACING	?
COMP_FDI_ISO	*	=4:1_SPACING	?

FDI Compensation Rules (mils) (Maho Bay PDG, Intel Doc# 473718)

Table	Trace	Design	Iso	Design	Comments
6-4	10	11.81	-	15.75	Using PCIe guidelines

Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CPU	*	*	CPU_ISO
COMP_FDI	*	*	COMP_FDI_ISO

XDP

XDP-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
XDP_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
XDP_PHY	*	XDP_50S

XDP-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
XDP_ISO	*	=2:1_SPACING	?
XDP_CLK_ISO	*	=4:1_SPACING	?

Desktop Debug Design Guide (Intel Doc# 430883)

Section	Imp	Design	Iso	Design	Comments
1.5	45-65	50	-	15.75	Isolation is for JTAG clocks. All signals default are 50 Ohm SE.

Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
XDP	*	*	XDP_ISO
CLK_JTAG	*	*	XDP_CLK_ISO
OBS_STROBE	*	*	XDP_CLK_ISO

DMI

Electrical Constraint Set	Physical	Spacing	
DMI			
DMI N2S	DMI PHY	PCIE	DMI N2S P<3..0> NO TEST=TRUE 10 19
DMI N2S	DMI PHY	PCIE	DMI N2S N<3..0> NO TEST=TRUE 10 19
DMI S2N	DMI PHY	PCIE	DMI S2N P<3..0> NO TEST=TRUE 10 19
DMI S2N	DMI PHY	PCIE	DMI S2N N<3..0> NO TEST=TRUE 10 19
PCIE REF CLK	CLK PCIE PHY	CLK PCIE	DMI CLK100M CPU P 11 18
PCIE REF CLK	CLK PCIE PHY	CLK PCIE	DMI CLK100M CPU N 11 18
DMI Compensation			
COMP PCIE PHY	COMP PCIE PHY	COMP PCIE	PCH_DMI_COMP 19
COMP PCIE PHY	COMP PCIE PHY	COMP PCIE	PCH_DMI2RBIAS 19

CPU Misc.

Electrical Constraint Set	Physical	Spacing	
GPU Misc.			
PR000	GPU_PHY	GPU	GPU_SKTOCC_L 11 64
PR001	GPU_PHY	GPU	GPU_PROC_SEL 11 10
PR002	GPU_PHY	GPU	GPU_CATERR_L 11 48
PR003	GPU_PHY	GPU	GPU_PECI 11 21 47 48
PR004	GPU_PHY	GPU	GPU_PROCHOT_L 11 47 48 60
PR005	GPU_PHY	GPU	GPU_PROCHOT_R_L 11
PR006	GPU_PHY	GPU	GPU_THRMTRIP_L 11 48
PR007	GPU_PHY	GPU	GPU_RESET_L 11 26
PR008	GPU_PHY	GPU	PLT_RESET_LS1V05_L 11
PR009	GPU_PHY	GPU	PM_SYNC 11 10
PR010	GPU_PHY	GPU	GPU_PWRGD 11 21 25 28
PR011	GPU_PHY	GPU	PM_MEM_PWRGD 11 19 28
PR012	GPU_PHY	GPU	PM_MEM_PWRGD_R 11
PR013	GPU_PHY	GPU	GPU_MEM_RESET_L 11 28

FDI

Electrical Constraint Set	Physical	Spacing	
FDI Compensation			
FE200	COMP_FDI_PHY	COMP_FDI	CPU_FDI_COMP10

XDP

Electrical Constraint Set	Physical	Spacing	
XDP_MISC.			
FEWD XDP_BPM_L	XDP_PHY	XDP	XDP BPM L<7..0> 11 25
FEWD	XDP_PHY	XDP	CPU_CFG<17..16> 10 25
FEWD	XDP_PHY	XDP	CPU_CFG<11..0> 10 15 25
FEWD	XDP_PHY	XDP	XDP_DBRESET_L 11 25
FEWD	XDP_PHY	XDP	XDP_CPU_PWRGD 25
FEWD	XDP_PHY	XDP	XDP_CPU_PWRBTN_L 25
FEWD	XDP_PHY	XDP	XDP_CPU_CFG<0> 25
FEWD	XDP_PHY	XDP	XDP_VR_READY 25
FEWD	XDP_PHY	XDP	XDP_CPU_PLTRST_L 25 26
FEWD	XDP_PHY	XDP	XDP_PCH_PWRGD 25
FEWD	XDP_PHY	XDP	XDP_PCH_PWRBTN_L 25
FEWD	XDP_PHY	XDP	XDP_PCH_PLTRST_L 25 26
FEWD	CLK_PCIE_PHY	CLK_PCIE	ITPCPU_CLK100M_P 11 15
	CLK_PCIE_PHY	CLK_PCIE	ITPCPU_CLK100M_N 11 15
FEWD TTP_CLK_CONN	CLK_PCIE_PHY	CLK_PCIE	ITPXDPCLK100M_P 10 15 25
FEWD TTP_CLK_CONN	CLK_PCIE_PHY	CLK_PCIE	ITPXDPCLK100M_N 10 15 25
FEWD	CLK_PCIE_PHY	CLK_PCIE	XDP_CPU_CLK100M_P 25
FEWD	CLK_PCIE_PHY	CLK_PCIE	XDP_CPU_CLK100M_N 25
CPU_JTAG			
FEWD	XDP_PHY	CLK_JTAG	XDP_CPU_TCK 11 25
FEWD	XDP_PHY	XDP	XDP_CPU_TMS 11 25
FEWD	XDP_PHY	XDP	XDP_CPU_TDI 11 25
FEWD	XDP_PHY	XDP	XDP_CPU_TDO 11 25
FEWD	XDP_PHY	XDP	XDP_CPU_TRST_L 11 25
FEWD	XDP_PHY	XDP	XDP_CPU_PRDY_L 11 25
FEWD	XDP_PHY	XDP	XDP_CPU_PREQ_L 11 25
PCH_JTAG			
FEWD	XDP_PHY	CLK_JTAG	XDP_PCH_TCK 10 25
FEWD	XDP_PHY	XDP	XDP_PCH_TMS 10 25
FEWD	XDP_PHY	XDP	XDP_PCH_TDI 10 25
FEWD	XDP_PHY	XDP	XDP_PCH_TDO 10 25

Chipset Test Interface

Electrical Constraint Set	Physical	Spacing	
OBS PCH Side			
H197	XDP_PCH_OBS	XDP_PHY	XDP USB_EXT_A OC_R_L
H197	XDP_PCH_OBS	XDP_PHY	XDP USB_EXT_B OC_R_L
H200	XDP_PCH_OBS	XDP_PHY	XDP USB_EXT_C OC_R_L
H200	XDP_PCH_OBS	XDP_PHY	XDP USB_EXT_D OC_R_L
H200	XDP_PCH_OBS	XDP_PHY	XDP USB_EXT_B OC_EHCI_R_L
H200	XDP_PCH_OBS	XDP_PHY	XDP USB_EXT_D OC_EHCI_R_L
H200	XDP_PCH_OBS	XDP_PHY	AP_PWR_EN
H207	XDP_PCH_OBS	XDP_PHY	SDCONN_STATE_CHANGE_R
H207	XDP_PCH_OBS	XDP_PHY	TBT_CIO_PLUG_EVENT_R
H210	XDP_PCH_OBS	XDP_PHY	ISOLATE_CPU_MEM_R_L
H210	XDP_PCH_OBS	XDP_PHY	GPU_GOOD_R
H210	XDP_PCH_OBS	XDP_PHY	DP_AUXCH_ISOL_R
H210	XDP_PCH_OBS	XDP_PHY	SATARDRV_R_EN
H210	XDP_PCH_OBS	XDP_PHY	DP_TBT_SEL_R
H210	XDP_PCH_OBS	XDP_PHY	JTAG_TBT_TCK_R
H210	XDP_PCH_OBS	XDP_PHY	AUD_IPHS_SWITCH_EN_PCH_R
H210	XDP_PCH_OBS	XDP_PHY	ENET_LOW_PWR_PCH_R
H210	XDP_PCH_OBS_UNUSED	XDP_PHY	OBS_STROBE XDP_PIN03
OBS XDP Side			
H200		XDP_PHY	XDP_DA0_USB_EXT_A_OC_L
H207		XDP_PHY	XDP_DA1_USB_EXT_B_OC_L
H207		XDP_PHY	XDP_DA2_USB_EXT_C_OC_L
H207		XDP_PHY	XDP_DA3_USB_EXT_D_OC_L
H207		XDP_PHY	XDP_DB0_USB_EXT_B_OC_EHCI_L
H207		XDP_PHY	XDP_DB1_USB_EXT_D_OC_EHCI_L
H207		XDP_PHY	XDP_DB2_AP_PWR_EN
H207		XDP_PHY	XDP_DB3_SDCONN_STATE_CHANGE
H207		XDP_PHY	OBS_STROBE XDP_FC1_TBT_CIO_PLUG_EVENT
H210		XDP_PHY	XDP_DC0_ISOLATE_CPU_MEM_L
H210		XDP_PHY	XDP_DC1_GPU_GOOD
H220		XDP_PHY	XDP_DC2_DP_AUXCH_ISOL
H220		XDP_PHY	XDP_DC3_SATARDRV_R_EN
H220		XDP_PHY	XDP_DD0_DP_TBT_SEL
H220		XDP_PHY	XDP_DD1_JTAG_TBT_TCK
H220		XDP_PHY	XDP_DD2_AUD_IPHS_SWITCH_EN_PCH
H220		XDP_PHY	XDP_DD3_ENET_LOW_PWR_PCH
H220		XDP_PHY	OBS_STROBE XDP_FC0_PCH_GPIO15
Standard usage branch off from OBS			
H220	XDP_PHY	XDP	USB_EXT_A_OC_L
H220	XDP_PHY	XDP	USB_EXT_B_OC_L
H220	XDP_PHY	XDP	USB_EXT_C_OC_L
H220	XDP_PHY	XDP	USB_EXT_D_OC_L
H220	XDP_PHY	XDP	USB_EXT_B_OC_EHCI_L
H220	XDP_PHY	XDP	USB_EXT_D_OC_EHCI_L
H220	XDP_PHY	XDP	AP_PWR_EN
H220	XDP_PHY	XDP	SDCONN_STATE_CHANGE
H220	XDP_PHY	OBS_STROBE	TBT_CIO_PLUG_EVENT
H220	XDP_PHY	XDP	ISOLATE_CPU_MEM_L
H220	XDP_PHY	XDP	GPU_GOOD
H220	XDP_PHY	XDP	DP_AUXCH_ISOL
H220	XDP_PHY	XDP	SATARDRV_R_EN
H220	XDP_PHY	XDP	DP_TBT_SEL
H220	XDP_PHY	XDP	JTAG_TBT_TCK
H220	XDP_PHY	XDP	JTAG_TBT_TCK_ISOL
H220	XDP_PHY	XDP	AUD_IPHS_SWITCH_EN_PCH
H220	XDP_PHY	XDP	ENET_LOW_PWR_PCH
H220	XDP_PHY	XDP	DP_AUXCH_ISOL_EN

SYNCH MASTER=D8 ROSITA		SYNCH DATE=03/23/2012	
PAGE TITLE			
CPU MISC/DMI/FDI/XDP Constraints			
 Apple Inc.		DRAWING NUMBER 051-9504	SIZE D
		REVISION 5.3.0	
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		PAGE 123 OF 143	
		SHEET 99 OF 117	

SATA

SATA-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SATA_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
SATA_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
SATA_PHY	*	SATA_90D
COMP_SATA_PHY	*	SATA_50S

SATA-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SATA_ISO	*	=7.2X_DIELECTRIC	?
COMP_SATA_ISO	*	=5.4X_DIELECTRIC	?

Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
SATA	*	*	SATA_ISO
COMP_SATA	*	*	COMP_SATA_ISO

SATA Min Spacing Rules (mils) (Maho Bay PDG, Intel Doc# 473718)

Section	Imp	Design	Iso	Design	Comments
15.2.1	90	95	20	23.62	SATA Gen2, SATA Gen3

SATA Compensation Rules (mils)

Table	Imp	Design	Iso	Design	Comments
15-3	50	50	15	15.75	SATA Gen2, SATA Gen3

SATA

Electrical Constraint Set	Physical	Spacing	
PCH SATA Port 0 (HDD)			
ED01 SATA_R2D	SATA_PHY	SATA	SATA_HDD_R2D_P 44
ED02 SATA_R2D	SATA_PHY	SATA	SATA_HDD_R2D_N 44
ED03	SATA_PHY	SATA	SATA_HDD_R2D_C_P 18 44
ED04	SATA_PHY	SATA	SATA_HDD_R2D_C_N 18 44
ED05 SATA_D2R	SATA_PHY	SATA	SATA_HDD_D2R_P 18 44
ED06 SATA_D2R	SATA_PHY	SATA	SATA_HDD_D2R_N 18 44
ED07	SATA_PHY	SATA	SATA_HDD_D2R_C_P 44
ED08	SATA_PHY	SATA	SATA_HDD_D2R_C_N 44
PCH SATA Port 1 (SSD)			
ED09 SATA_R2D_MUX_SSD	SATA_PHY	SATA	SATA_SSD_R2D_P 18 44
ED10 SATA_R2D_MUX_SSD	SATA_PHY	SATA	SATA_SSD_R2D_N 18 44
ED11 SATA_D2R_MUX_SSD	SATA_PHY	SATA	SATA_SSD_D2R_P 18 44
ED12 SATA_D2R_MUX_SSD	SATA_PHY	SATA	SATA_SSD_D2R_N 18 44
PCH SATA Compensation			
ED13A	COMP_SATA_PHY	COMP_SATA	PCH_SATAICOMP 18
ED13B	COMP_SATA_PHY	COMP_SATA	PCH_SATA3COMP 18
ED13C	COMP_SATA_PHY	COMP_SATA	PCH_SATA3RBIAS 18

Unused

Electrical Constraint Set	Physical	Spacing

PCH

PCH-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCH_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
CLK_PCH_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

PCH-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CLK_PCH_ISO	*	=6.5X_DIELECTRIC	?	CLK_PCH	*	*	CLK_PCH_ISO
COMP_PCH_ISO	*	=2:1_SPACING	?	COMP_PCH	*	*	COMP_PCH_ISO
PCH_ISO	*	=3:1_SPACING	?	PCH	*	*	PCH_ISO

PCI

PCI-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCI_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
CLK_PCI_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

PCI-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_PCI_ISO	*	=3.6X_DIELECTRIC	?

LPC

LPC-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
LPC_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
CLK_LPC_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

LPC-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
LPC_ISO	*	=1.5:1_SPACING	?	LPC	*	*	LPC_ISO
CLK_LPC_ISO	*	=3.6X_DIELECTRIC	?	CLK_LPC	*	*	CLK_LPC_ISO

HDA

HDA-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
HDA_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

HDA-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
HDA_ISO	*	=2X_DIELECTRIC	?	HDA	*	*	HDA_ISO

Crystal

Crystal-specific Physical Rules

[illegible]

Crystal-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
XTAL_ISO	*	=4X_DIELECTRIC	?	XTAL	*	*	XTAL_ISO

SPI

SPI-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SPI_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

SPI-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
SPI_ISO	*	=2X_DIELECTRIC	?	SPI	*	*	SPI_ISO

PCI

Electrical Constraint Set	Physical	Spacing
PCI Clock		
100M	CLK_PCT_50S	CLK_PCT
100M	CLK_PCT_50S	CLK_PCT

LPC

Electrical Constraint Set	Physical	Spacing	
LPC			
	LPC_50S	LPC	LPC_AD<3...0> 18 47
	LPC_50S	LPC	LPC_R_AD<3...0> 18
	LPC_50S	LPC	LPC_FRAME_L 18 47
	LPC_50S	LPC	LFRAME_L 18
LPC Clocks			
	CLK_LPC_50S	CLK_LPC	LPC_CLK33M_LPCPLUS 26 49
	CLK_LPC_50S	CLK_LPC	LPC_CLK33M_LPCPLUS_R 26 26
	CLK_LPC_50S	CLK_LPC	LPC_CLK33M_SMC 26 47
	CLK_LPC_50S	CLK_LPC	LPC_CLK33M_SMC_R 26 26

PCH Clocks

Electrical Constraint Set	Physical	Spacing	
PCH Reference Clock			
 CLK_PCH_50S	CLK_PCH		SYSCLK_CLK25M_SB 26
 CLK_PCH_50S	CLK_PCH		PCH_CLK25M_XTALIN 18 26
PCH Ref Clock Comp			
 PCH_50S	COMP_PCH		PCH_XCLK_RCMP 18
PCH RTC 32K			
 CLK_XTAL	XTAL		PCH_CLK32K_RTCX1 18 26
 CLK_XTAL	XTAL		PCH_CLK32K_RTCX2 18 26
 CLK_XTAL	XTAL		PCH_CLK32K_RTCX2_R 26
SMC 32K			
 CLK_PCH_50S	CLK_PCH		PM_CLK32K_SUSCLK_R 19 48
 CLK_PCH_50S	CLK_PCH		SMC_CLK32K 47 48

25 MHz Reference Clocks

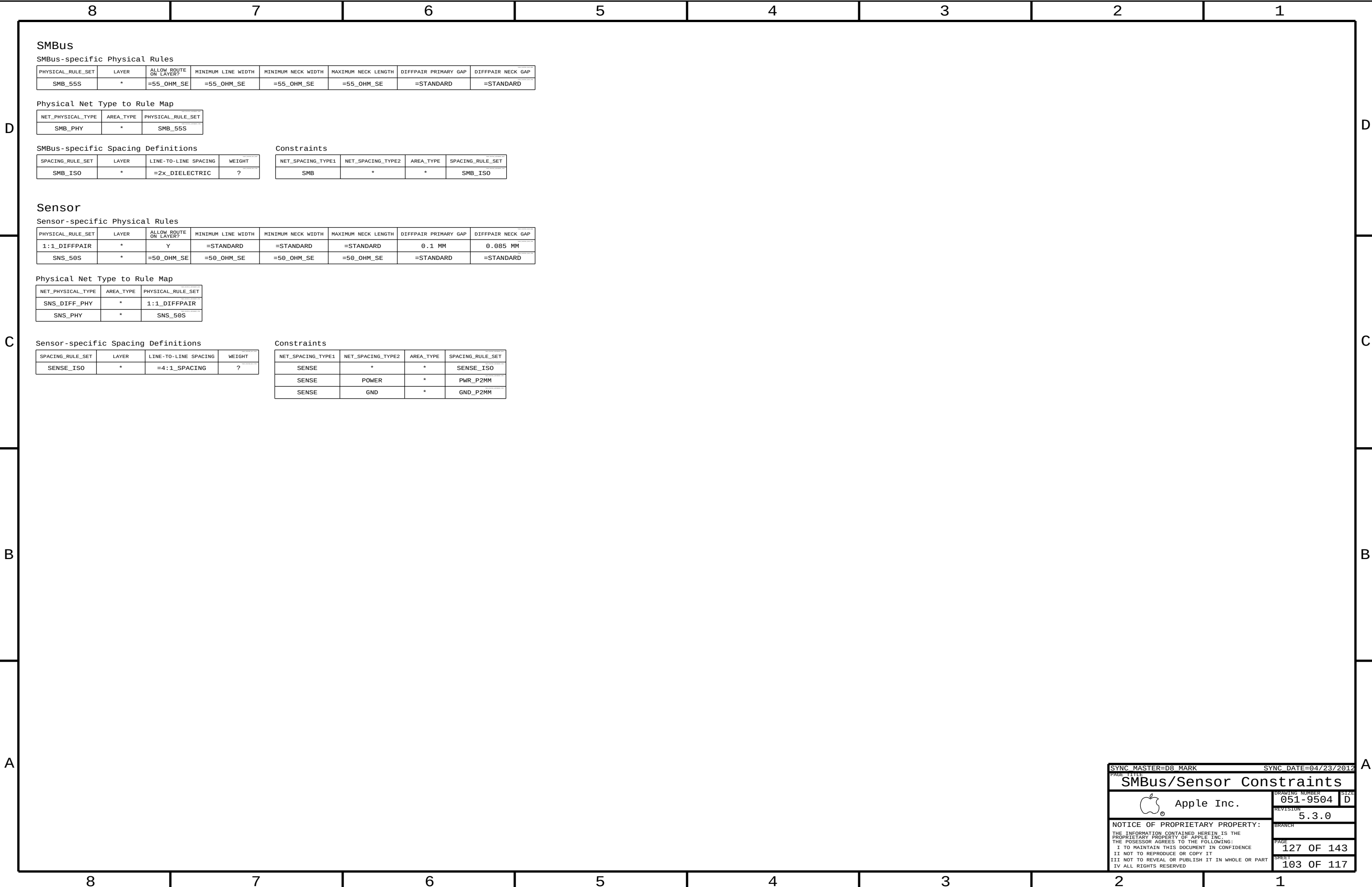
Electrical Constraint Set	Physical	Spacing	
25M Reference Crystal			
	CLK_XTAL	XTAL	SYSCLK_CLK25M_X1 26
	CLK_XTAL	XTAL	SYSCLK_CLK25M_X2 26
	CLK_XTAL	XTAL	SYSCLK_CLK25M_X2_R 26
25M Reference Clocks			
	CLK_PCH_50S	CLK_PCH	SYSCLK_CLK25M_ENET 26 39
	CLK_PCH_50S	CLK_PCH	SYSCLK_CLK25M_ENET_R 26
	CLK_PCH_50S	CLK_PCH	SYSCLK_CLK25M_TBT 26 36
	CLK_PCH_50S	CLK_PCH	SYSCLK_CLK25M_TBT_R 36

HDA

Electrical Constraint Set	Physical	Spacing	
HDA			
H80P	HDA_56S	HDA	HDA_BIT_CLK 18 56
H80P	HDA_56S	HDA	HDA_BIT_CLK_R 18
H80P	HDA_56S	HDA	HDA_RST_L 18 56
H80P	HDA_56S	HDA	HDA_RST_R_L 18
H80P	HDA_56S	HDA	HDA_SDOUT 18 56
H80P	HDA_56S	HDA	HDA_SDOUT_R 15 18
H80P	HDA_56S	HDA	HDA_SYNC 15 18 56
H80P	HDA_56S	HDA	HDA_SYNC_R 18
H80P	HDA_56S	HDA	HDA_SDI0 18 56
H80P	HDA_56S	HDA	AUD_SDI_R 56
H80P	HDA_56S	HDA	SPI_DESCRIPTOR_OVERRIDE_R 15
SPDIF			
H80P		HDA	AUD_SPDIF_CHIP 56
H80P		HDA	AUD_SPDIF_OUT 56 60

SPI Bootrom

Electrical Constraint Set	Physical	Spacing	
SPI ROM			
H09D	SPI 56S	SPT	SPI CLK R 18 49
H09D	SPI 56S	SPT	SPI CLK 49
H09D	SPI 56S	SPT	SPI ALT CLK 49
H09D	SPI 56S	SPT	SPI SMC CLK 47 48
H09D	SPI 56S	SPT	SPI MLB CLK 48 49
H09D	SPI 56S	SPT	SPI CS0 R L 18 49
H09D	SPI 56S	SPT	SPI CS0 L 49
H09D	SPI 56S	SPT	SPI ALT CS L 49
H09D	SPI 56S	SPT	SPI SMC CS L 47 48
H09D	SPI 56S	SPT	SPI MLB CS L 48 49
H09D	SPI 56S	SPT	SPI MOSI R 18 49
H09D	SPI 56S	SPT	SPI MOSI 49
H09D	SPI 56S	SPT	SPI ALT MOSI 49
H09D	SPI 56S	SPT	SPI SMC MOSI 47 48
H09D	SPI 56S	SPT	SPI MLB MOSI 48 49
H09D	SPI 56S	SPT	SPI MISO 18 49
H09D	SPI 56S	SPT	SPI ALT MISO 49
H09D	SPI 56S	SPT	SPI SMC MISO 47 48
H09D	SPI 56S	SPT	SPI MLB MISO 48 49
H09D	SPI 56S	SPT	SPIROM USE MLB 21 49



D

D

C

C

B

B

A

A

SYNC_MASTER=D8_MARK

SYNC_DATE=04/23/2012

PAGE_TITLE

SMBus/Sensor Constraints

 Apple Inc.

THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
IV ALL RIGHTS RESERVED

DRAWING_NUMBER051-9504

REVISION5.3.0

BRANCH

PAGE127 OF 143

SHEET103 OF 117

DC - DC

Power-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
GND_P3MM	*	Y	0.300 MM	0.200 MM	3.0 MM	=STANDARD	=STANDARD
GND_P5MM	*	Y	0.500 MM	0.200 MM	3.0 MM	=STANDARD	=STANDARD
POWER_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
POWER_P3MM	*	Y	0.300 MM	0.200 MM	3.0 MM	=STANDARD	=STANDARD
POWER_P6MM	*	Y	0.600 MM	0.200 MM	3.0 MM	=STANDARD	=STANDARD
POWER_P5MM	*	Y	0.500 MM	0.200 MM	3.0 MM	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
VR_CTL_PHY	*	POWER_P5MM
VR_CTL_PHY	BGA	STANDARD
VR_VID_PHY	*	POWER_50S
POWER_PHY	*	POWER_P6MM
GND_PHY	*	GND_P5MM

Constraints

Power and Common

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
POWER	*	*	POWER_ISO
GND	*	*	GND_ISO

Power-specific Spacing Definitions

Power and Common

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
POWER_ISO	*	=STANDARD	?

DC-DC Baddies

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SWNODE_TSO	*	=8:1_SPACING	1000
SWNODE_SW2SW	*	=1:1_SPACING	?
SWNODE_SW2PWR	*	=2:1_SPACING	?
SWNODE_SW2GND	*	=2:1_SPACING	?
SWNODE_LG2SW	*	=3:1_SPACING	?

DC-DC Control

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
VR_CTL_ISO	*	=3:1_SPACING	?
VR_VID_ISO	*	=4X_DIELECTRIC	?

CPU VccSA

Electrical Constraint Set	Physical	Spacing	Voltage	D1D2	NO_TEST	
Input Bus						
I24	POWER_PHY	POWER	5V			REG_VCC_U7500 70
I25	POWER_PHY	POWER	5V			REG_PVCC_U7500 70
Local Ground						
I26	GND_PHY	GND	0V			AGND_VCCSAS0 70
VCCIO						
I27	POWER_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_VCCSAS0 70
I28	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_B00T_VCCSAS0 70
I29	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_B00T_VCCSAS0_RC 70
I30	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_VCCSAS0 70
I31	VR_CTL_PHY	VR_LGATE	12V	TRUE		REG_LGATE_VCCSAS0 70
I32	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_SNUBBER_VCCSAS0 70
I33	VR_CTL_PHY	VR_CTL				REG_VCCSAS0_OCSET 70
I34	VR_CTL_PHY	VR_CTL				REG_VCCSAS0_VO 70
I35		SENSE				SNS_CPU_VCCSA 13 70
I36 VSNS_CPU_VCCSA	SNS_DIFF_PHY	SENSE				SNS_VCCSAS0_XW_P 70
I37 VSNS_CPU_VCCSA	SNS_DIFF_PHY	SENSE				SNS_VCCSAS0_XW_N 70
I38		SENSE				REG_VCCSAS0_FB 70
I39		SENSE				REG_VCCSAS0_RTN 70
I40	VR_CTL_PHY	VR_CTL				REG_VCCSAS0_SREF 70
I41	VR_CTL_PHY	VR_CTL				REG_VCCSAS0_FSEL 70
Output Bus						
I42	POWER_PHY	POWER	0_925V			PPVCCSA_S0 6

```
CPU VccIO PCH 1.05V S0
```

Electrical Constraint Set	Physical	Spacing	Voltage	DIDT	NO_TEST	
Input Bus						
 P0	POWER_PHY	POWER	5V			REG_VCC_U7400 69
 P2	POWER_PHY	POWER	5V			REG_PVCC_U7400 69
Local Ground						
 A0	GND_PHY	GND	0V			AGND_CPU_P1V05S0 69
1.05V S0						
 P5	POWER_PHY	VB_SWITCH	12V	TRUE		REG_CPU_PHASE_P1V05S0 69
 P7	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_CPU_BOOT_P1V05S0 69
 P0	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_CPU_BOOT_P1V05S0_RC 69
 P0	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_CPU_UGATE_P1V05S0 69
 P0	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_CPU_UGATE_P1V05S0_R 69
 P0E0	VR_CTL_PHY	VR_LGATE	12V	TRUE		REG_CPU_LGATE_P1V05S0 69
 P0E2	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_CPU_SNUBBER_P1V05S0 69
 P0S0	VR_CTL_PHY	VR_CTL				REG_CPU_P1V05S0_OCSET 69
 P0S0	VR_CTL_PHY	VR_CTL				REG_CPU_P1V05S0_VD 69
 P0S0	VSNS_CPU_VCCIO	SENSE				SNS_CPU_VCCIO_P 13 69
 P0S0	VSNS_CPU_VCCIO	SENSE				SNS_CPU_VCCIO_N 13 69
 P0E0	SNS_DIEF_PHY	SENSE				SNS_CPU_P1V05S0_XW_P 69
 P0E0	SNS_DIEF_PHY	SENSE				SNS_CPU_P1V05S0_XW_N 69
 P0S0		SENSE				REG_CPU_P1V05S0_FB 69
 P0E0		SENSE				REG_CPU_P1V05S0_RTN 69
 P0E0	VR_CTL_PHY	VR_CTL				REG_CPU_P1V05S0_SREF 69
 P0E0	VR_CTL_PHY	VR_CTL				REG_CPU_P1V05S0_FSEL 69
Output Bus						
 P0P0	POWER_PHY	POWER	1.05V			PP1V05_S0_CPU 6

PCH/GPU/TBT 1.05V S0

Electrical Constraint Set	Physical	Spacing	Voltage	DIDT	NO_TEST	
Input Bus						
REG	POWER_PHY	POWER	5V			REG VCC UB700
Local Ground						
LGND	GND_PHY	GND	0V			P1V05 AGND
1.05V S0						
PGND	POWER_PHY	VR_SWITCH	12V	TRUE		REG PHASE P1V05S0
REG	POWER_PHY	VR_SWITCH	12V	TRUE		REG PHASE P1V05S0_L
VR_CTL_PHY	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG BOOT P1V05S0
VR_CTL_PHY	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG BOOT P1V05S0_RC
VR_CTL_PHY	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG UGATE P1V05S0
VR_CTL_PHY	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG UGATE P1V05S0_R
VR_CTL_PHY	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG LGATE P1V05S0
VR_CTL_PHY	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG SNUBBER P1V05S0
VR_CTL_PHY	VR_CTL_PHY	VR_CTL				REG P1V05S0_OCSET
VR_CTL_PHY	VR_CTL_PHY	VR_CTL				REG P1V05S0_VO
POWER_PHY	POWER_PHY	VR_SWITCH	12V			REG P1V05S0_VO_R
POWER_PHY	POWER_PHY	VR_SWITCH	12V			P1V05_OCSET_R
		SENSE				REG P1V05S0_FB
		SENSE				REG P1V05S0_RTN
VR_CTL_PHY	VR_CTL_PHY	VR_CTL				REG P1V05S0_SREF
VR_CTL_PHY	VR_CTL_PHY	VR_CTL				REG P1V05S0_FSEL
Output Bus						
PP1V05	POWER_PHY	POWER	1.05V			PP1V05_S0_PCH
PP1V05	POWER_PHY	POWER	1.05V			PP1V05_S0
PP1V05	POWER_PHY	POWER	1.05V			PP1V05_TBTL
PP1V05	POWER_PHY	POWER	1.05V			PP1V05_TRTCTIO

SYNC MASTER=D8 MARK		SYNC DATE=02/10/2012	
PAGE TITLE			
VReg Constraints			
	Apple Inc.		DRAWING NUMBER 051-9504
			SIZE D
		REVISION 5.3.0	
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE I NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		BRANCH PAGE 128 OF 143	
		SHEET 104 OF 117	

8	7	6	5	4	3	2	1																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																													
CPU Core Phases				CPU AXG Phase and Core Controller																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
<table><tr><th>Electrical Constraint Set</th><th>Physical</th><th>Spacing</th><th>Voltage</th><th>DIDT</th><th>NO_TEST</th><th></th></tr><tr><td colspan="7">Input Bus</td></tr><tr><td>1E880</td><td>POWER_PHY</td><td>POWER</td><td>12V</td><td></td><td></td><td>PP12V_S0_CPUCORE FLT</td></tr><tr><td>1E880</td><td>POWER_PHY</td><td>POWER</td><td>5V</td><td></td><td></td><td>REG_VCC_U7100</td></tr><tr><td colspan="7">Local Ground</td></tr><tr><td>1E880</td><td>GND_PHY</td><td>GND</td><td>0V</td><td></td><td></td><td>AGND_CPU</td></tr><tr><td colspan="7">Phase 1</td></tr><tr><td>1E880</td><td>POWER_PHY</td><td>POWER</td><td>12V</td><td></td><td></td><td>REG_LVCC_U7210</td></tr><tr><td>1E880</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_PWM_CPUCORE_1</td></tr><tr><td>1E880</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_PWM_CPUCORE_1 R</td></tr><tr><td>1E880</td><td>POWER_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td><td>REG PHASE CPUCORE_1</td></tr><tr><td>1E880</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td><td>REG_BOOT CPUCORE_1</td></tr><tr><td>1E880</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td>TRUE</td><td>REG_BOOT CPUCORE_1 RC</td></tr><tr><td>1E880</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td><td>REG_UGATE CPUCORE_1</td></tr><tr><td>1E880</td><td>VR_CTL_PHY</td><td>VR_LGATE</td><td>12V</td><td>TRUE</td><td></td><td>REG_LGATE CPUCORE_1</td></tr><tr><td>1E880</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td>TRUE</td><td>REG_SNUBBER CPUCORE_1</td></tr><tr><td>1E880</td><td>POWER_PHY</td><td>POWER</td><td>1.1V</td><td></td><td></td><td>PPCPUCORE_S0 SENSE_1</td></tr><tr><td>1E880</td><td>TSNS_CPU CORE</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td></td><td>REG ISENCORE_1 P</td></tr><tr><td>1E880</td><td>TSNS_CPU CORE</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td></td><td>REG ISENCORE_1 N</td></tr><tr><td>1E880</td><td></td><td>SENSE</td><td></td><td></td><td></td><td>REG ISENCORE_1 NR</td></tr><tr><td colspan="7">Phase 2</td></tr><tr><td>1E880</td><td>POWER_PHY</td><td>POWER</td><td>12V</td><td></td><td></td><td>REG_LVCC_U7230</td></tr><tr><td>1E880</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_PWM_CPUCORE_2</td></tr><tr><td>1E880</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_PWM_CPUCORE_2 R</td></tr><tr><td>1E880</td><td>POWER_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td><td>REG PHASE CPUCORE_2</td></tr><tr><td>1E880</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td><td>REG_BOOT CPUCORE_2</td></tr><tr><td>1E880</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td>TRUE</td><td>REG_BOOT CPUCORE_2 RC</td></tr><tr><td>1E880</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td><td>REG_UGATE CPUCORE_2</td></tr><tr><td>1E880</td><td>VR_CTL_PHY</td><td>VR_LGATE</td><td>12V</td><td>TRUE</td><td></td><td>REG_LGATE CPUCORE_2</td></tr><tr><td>1E880</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td>TRUE</td><td>REG_SNUBBER CPUCORE_2</td></tr><tr><td>1E880</td><td>POWER_PHY</td><td>POWER</td><td>1.1V</td><td></td><td></td><td>PPCPUCORE_S0 SENSE_2</td></tr><tr><td>1E880</td><td>TSNS_CPU CORE</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td></td><td>REG ISENCORE_2 P</td></tr><tr><td>1E880</td><td>TSNS_CPU CORE</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td></td><td>REG ISENCORE_2 N</td></tr><tr><td>1E880</td><td></td><td>SENSE</td><td></td><td></td><td></td><td>REG ISENCORE_2 NR</td></tr><tr><td colspan="7">Phase 3</td></tr><tr><td>1E880</td><td>POWER_PHY</td><td>POWER</td><td>12V</td><td></td><td></td><td>REG_LVCC_U7250</td></tr><tr><td>1E880</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_PWM_CPUCORE_3</td></tr><tr><td>1E880</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_PWM_CPUCORE_3 R</td></tr><tr><td>1E880</td><td>POWER_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td><td>REG PHASE CPUCORE_3</td></tr><tr><td>1E880</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td><td>REG_BOOT CPUCORE_3</td></tr><tr><td>1E880</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td>TRUE</td><td>REG_BOOT CPUCORE_3 RC</td></tr><tr><td>1E880</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td><td>REG_UGATE CPUCORE_3</td></tr><tr><td>1E880</td><td>VR_CTL_PHY</td><td>VR_LGATE</td><td>12V</td><td>TRUE</td><td></td><td>REG_LGATE CPUCORE_3</td></tr><tr><td>1E880</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td>TRUE</td><td>REG_SNUBBER CPUCORE_3</td></tr><tr><td>1E880</td><td>POWER_PHY</td><td>POWER</td><td>1.1V</td><td></td><td></td><td>PPCPUCORE_S0 SENSE_3</td></tr><tr><td>1E880</td><td>TSNS_CPU CORE</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td></td><td>REG ISENCORE_3 P</td></tr><tr><td>1E880</td><td>TSNS_CPU CORE</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td></td><td>REG ISENCORE_3 N</td></tr><tr><td>1E880</td><td></td><td>SENSE</td><td></td><td></td><td></td><td>REG ISENCORE_3 NR</td></tr><tr><td colspan="7">Phase 4</td></tr><tr><td>1E880</td><td>POWER_PHY</td><td>POWER</td><td>12V</td><td></td><td></td><td>REG_LVCC_U7310</td></tr><tr><td>1E880</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_PWM_CPUCORE_4</td></tr><tr><td>1E880</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_PWM_CPUCORE_4 R</td></tr><tr><td>1E880</td><td>POWER_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td><td>REG PHASE CPUCORE_4</td></tr><tr><td>1E880</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td><td>REG_BOOT CPUCORE_4</td></tr><tr><td>1E880</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td>TRUE</td><td>REG_BOOT CPUCORE_4 RC</td></tr><tr><td>1E880</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td><td>REG_UGATE CPUCORE_4</td></tr><tr><td>1E880</td><td>VR_CTL_PHY</td><td>VR_LGATE</td><td>12V</td><td>TRUE</td><td></td><td>REG_LGATE CPUCORE_4</td></tr><tr><td>1E880</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td>TRUE</td><td>REG_SNUBBER CPUCORE_4</td></tr><tr><td>1E880</td><td>POWER_PHY</td><td>POWER</td><td>1.1V</td><td></td><td></td><td>PPCPUCORE_S0 SENSE_4</td></tr><tr><td>1E880</td><td>TSNS_CPU CORE</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td></td><td>REG ISENCORE_4 P</td></tr><tr><td>1E880</td><td>TSNS_CPU CORE</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td></td><td>REG ISENCORE_4 N</td></tr><tr><td>1E880</td><td></td><td>SENSE</td><td></td><td></td><td></td><td>REG ISENCORE_4 NR</td></tr><tr><td>1E880</td><td>TSNS_CPU CORE</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td></td><td>SNS CORE_XW_P</td></tr><tr><td>1E880</td><td>TSNS_CPU CORE</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td></td><td>SNS CORE_XW_N</td></tr><tr><td>1E880</td><td>TSNS_CPU CORE</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td></td><td>SNS CORE_R_P</td></tr><tr><td>1E880</td><td>TSNS_CPU CORE</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td></td><td>SNS CORE_R_N</td></tr><tr><td>1E880</td><td>TSNS_CPU CORE</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td></td><td>SNS CPU_VAXG_P</td></tr><tr><td>1E880</td><td>TSNS_CPU CORE</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td></td><td>SNS CPU_VAXG_N</td></tr><tr><td>1E880</td><td>TSNS_CPU CORE</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td></td><td>SNS CPU_VCORE_P</td></tr><tr><td>1E880</td><td>TSNS_CPU CORE</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td></td><td>SNS CPU_VCORE_N</td></tr><tr><td>1E880</td><td>TSNS_CPU CORE</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td></td><td>SNS_P1V05_IOVDD_XW_P</td></tr><tr><td>1E880</td><td>TSNS_CPU CORE</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td></td><td>SNS_P1V05_IOVDD_XW_N</td></tr></table>				Electrical Constraint Set	Physical	Spacing	Voltage	DIDT	NO_TEST		Input Bus							1E880	POWER_PHY	POWER	12V			PP12V_S0_CPUCORE FLT	1E880	POWER_PHY	POWER	5V			REG_VCC_U7100	Local Ground							1E880	GND_PHY	GND	0V			AGND_CPU	Phase 1							1E880	POWER_PHY	POWER	12V			REG_LVCC_U7210	1E880	VR_CTL_PHY	VR_CTL				REG_PWM_CPUCORE_1	1E880	VR_CTL_PHY	VR_CTL				REG_PWM_CPUCORE_1 R	1E880	POWER_PHY	VR_SWITCH	12V	TRUE		REG PHASE CPUCORE_1	1E880	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_BOOT CPUCORE_1	1E880	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT CPUCORE_1 RC	1E880	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE CPUCORE_1	1E880	VR_CTL_PHY	VR_LGATE	12V	TRUE		REG_LGATE CPUCORE_1	1E880	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_SNUBBER CPUCORE_1	1E880	POWER_PHY	POWER	1.1V			PPCPUCORE_S0 SENSE_1	1E880	TSNS_CPU CORE	SNS_DIFF_PHY	SENSE			REG ISENCORE_1 P	1E880	TSNS_CPU CORE	SNS_DIFF_PHY	SENSE			REG ISENCORE_1 N	1E880		SENSE				REG ISENCORE_1 NR	Phase 2							1E880	POWER_PHY	POWER	12V			REG_LVCC_U7230	1E880	VR_CTL_PHY	VR_CTL				REG_PWM_CPUCORE_2	1E880	VR_CTL_PHY	VR_CTL				REG_PWM_CPUCORE_2 R	1E880	POWER_PHY	VR_SWITCH	12V	TRUE		REG PHASE CPUCORE_2	1E880	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_BOOT CPUCORE_2	1E880	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT CPUCORE_2 RC	1E880	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE CPUCORE_2	1E880	VR_CTL_PHY	VR_LGATE	12V	TRUE		REG_LGATE CPUCORE_2	1E880	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_SNUBBER CPUCORE_2	1E880	POWER_PHY	POWER	1.1V			PPCPUCORE_S0 SENSE_2	1E880	TSNS_CPU CORE	SNS_DIFF_PHY	SENSE			REG ISENCORE_2 P	1E880	TSNS_CPU CORE	SNS_DIFF_PHY	SENSE			REG ISENCORE_2 N	1E880		SENSE				REG ISENCORE_2 NR	Phase 3							1E880	POWER_PHY	POWER	12V			REG_LVCC_U7250	1E880	VR_CTL_PHY	VR_CTL				REG_PWM_CPUCORE_3	1E880	VR_CTL_PHY	VR_CTL				REG_PWM_CPUCORE_3 R	1E880	POWER_PHY	VR_SWITCH	12V	TRUE		REG PHASE CPUCORE_3	1E880	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_BOOT CPUCORE_3	1E880	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT CPUCORE_3 RC	1E880	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE CPUCORE_3	1E880	VR_CTL_PHY	VR_LGATE	12V	TRUE		REG_LGATE CPUCORE_3	1E880	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_SNUBBER CPUCORE_3	1E880	POWER_PHY	POWER	1.1V			PPCPUCORE_S0 SENSE_3	1E880	TSNS_CPU CORE	SNS_DIFF_PHY	SENSE			REG ISENCORE_3 P	1E880	TSNS_CPU CORE	SNS_DIFF_PHY	SENSE			REG ISENCORE_3 N	1E880		SENSE				REG ISENCORE_3 NR	Phase 4							1E880	POWER_PHY	POWER	12V			REG_LVCC_U7310	1E880	VR_CTL_PHY	VR_CTL				REG_PWM_CPUCORE_4	1E880	VR_CTL_PHY	VR_CTL				REG_PWM_CPUCORE_4 R	1E880	POWER_PHY	VR_SWITCH	12V	TRUE		REG PHASE CPUCORE_4	1E880	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_BOOT CPUCORE_4	1E880	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT CPUCORE_4 RC	1E880	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE CPUCORE_4	1E880	VR_CTL_PHY	VR_LGATE	12V	TRUE		REG_LGATE CPUCORE_4	1E880	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_SNUBBER CPUCORE_4	1E880	POWER_PHY	POWER	1.1V			PPCPUCORE_S0 SENSE_4	1E880	TSNS_CPU CORE	SNS_DIFF_PHY	SENSE			REG ISENCORE_4 P	1E880	TSNS_CPU CORE	SNS_DIFF_PHY	SENSE			REG ISENCORE_4 N	1E880		SENSE				REG ISENCORE_4 NR	1E880	TSNS_CPU CORE	SNS_DIFF_PHY	SENSE			SNS CORE_XW_P	1E880	TSNS_CPU CORE	SNS_DIFF_PHY	SENSE			SNS CORE_XW_N	1E880	TSNS_CPU CORE	SNS_DIFF_PHY	SENSE			SNS CORE_R_P	1E880	TSNS_CPU CORE	SNS_DIFF_PHY	SENSE			SNS CORE_R_N	1E880	TSNS_CPU CORE	SNS_DIFF_PHY	SENSE			SNS CPU_VAXG_P	1E880	TSNS_CPU CORE	SNS_DIFF_PHY	SENSE			SNS CPU_VAXG_N	1E880	TSNS_CPU CORE	SNS_DIFF_PHY	SENSE			SNS CPU_VCORE_P	1E880	TSNS_CPU CORE	SNS_DIFF_PHY	SENSE			SNS CPU_VCORE_N	1E880	TSNS_CPU CORE	SNS_DIFF_PHY	SENSE			SNS_P1V05_IOVDD_XW_P	1E880	TSNS_CPU CORE	SNS_DIFF_PHY	SENSE			SNS_P1V05_IOVDD_XW_N	<table><tr><th>Electrical Constraint Set</th><th>Physical</th><th>Spacing</th><th>Voltage</th><th>DIDT</th><th>NO_TEST</th><th></th></tr><tr><td colspan="7">AXG</td></tr><tr><td>1E880</td><td>POWER_PHY</td><td>POWER</td><td>12V</td><td></td><td></td><td>REG_LVCC_U7330</td></tr><tr><td>1E880</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_PWM_CPUAXG</td></tr><tr><td>1E880</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_PWM_CPUAXG_R</td></tr><tr><td>1E880</td><td>POWER_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td><td>REG PHASE CPUAXG</td></tr><tr><td>1E880</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td><td>REG_BOOT CPUAXG</td></tr><tr><td>1E880</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td>TRUE</td><td>REG_BOOT CPUAXG_RC</td></tr><tr><td>1E880</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td><td>REG_UGATE CPUAXG</td></tr><tr><td>1E880</td><td>VR_CTL_PHY</td><td>VR_LGATE</td><td>12V</td><td>TRUE</td><td></td><td>REG_LGATE CPUAXG</td></tr><tr><td>1E880</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td>TRUE</td><td>REG_SNUBBER CPUAXG</td></tr><tr><td>1E880</td><td>POWER_PHY</td><td>POWER</td><td>1.1V</td><td></td><td></td><td>PPCPUAXG_S0 SENSE</td></tr><tr><td>1E880</td><td>TSNS_CPU AXG</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td></td><td>REG ISENAXG_P</td></tr><tr><td>1E880</td><td>TSNS_CPU AXG</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td></td><td>REG ISENAXG_N</td></tr><tr><td>1E880</td><td></td><td>SENSE</td><td></td><td></td><td></td><td>REG ISENAXG_PR</td></tr><tr><td>1E880</td><td></td><td>SENSE</td><td></td><td></td><td></td><td>REG ISENAXG_NR</td></tr><tr><td>1E880</td><td>TSNS_CPU AXG</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td></td><td>SNS_AXG_R_P</td></tr><tr><td>1E880</td><td>TSNS_CPU AXG</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td></td><td>SNS_AXG_R_N</td></tr><tr><td>1E880</td><td>TSNS_CPU AXG</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td></td><td>SNS_AXG_XW_P</td></tr><tr><td>1E880</td><td>TSNS_CPU AXG</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td></td><td>SNS_AXG_XW_N</td></tr><tr><td colspan="7">ISL6364</td></tr><tr><td>1E880</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_CPUCORE_COMP</td></tr><tr><td>1E880</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>CPUCORE_COMP_RC</td></tr><tr><td>1E880</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_CPUCORE_FB</td></tr><tr><td>1E880</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>CPUCORE_FB_RC</td></tr><tr><td>1E880</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>CPUCORE_FB_R_1</td></tr><tr><td>1E880</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>CPUCORE_FB_R_2</td></tr><tr><td>1E880</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>CPUCORE_PSICOMP_RC</td></tr><tr><td>1E880</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_CPUCORE_PSICOMP</td></tr><tr><td>1E880</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_CPUCORE_HFCOMP</td></tr><tr><td>1E880</td><td></td><td>SENSE</td><td></td><td></td><td></td><td>REG_CPUCORE_VSEN</td></tr><tr><td>1E880</td><td></td><td>SENSE</td><td></td><td></td><td></td><td>REG_CPUCORE_RGND</td></tr><tr><td>1E880</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_CPUCORE_IMON</td></tr><tr><td>1E880</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>CPUCORE_IMON_R</td></tr><tr><td>1E880</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_CPUCORE_TM</td></tr><tr><td>1E880</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_CPUCORE_SUTH</td></tr><tr><td>1E880</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_CPUCORE_NPSI</td></tr><tr><td>1E880</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_CPUCORE_FVID</td></tr><tr><td>1E880</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_CPUCORE_IAUTO</td></tr><tr><td>1E880</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_CPUCORE_SW_FREQ</td></tr><tr><td>1E880</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_CPUCORE_RAMPADJ</td></tr><tr><td>1E880</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_CPUCORE_EN_PWR</td></tr><tr><td>1E880</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>CPUCORE_EN_PWR_R</td></tr><tr><td>1E880</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_CPUCORE_RSET</td></tr><tr><td>1E880</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_CPUAXG_COMP</td></tr><tr><td>1E880</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>CPUAXG_COMP_RC</td></tr><tr><td>1E880</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_CPUAXG_FB</td></tr><tr><td>1E880</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>CPUAXG_FB_RC</td></tr><tr><td>1E880</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>CPUAXG_FB_R_1</td></tr><tr><td>1E880</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>CPUAXG_FB_R_2</td></tr><tr><td>1E880</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_CPUAXG_HFCOMP</td></tr><tr><td>1E880</td><td></td><td>SENSE</td><td></td><td></td><td></td><td>REG_CPUAXG_VSEN</td></tr><tr><td>1E880</td><td></td><td>SENSE</td><td></td><td></td><td></td><td>REG_CPUAXG_RGND</td></tr><tr><td>1E880</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_CPUAXG_IMON</td></tr><tr><td>1E880</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>CPUAXG_IMON_R</td></tr><tr><td>1E880</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_CPUAXG_TM</td></tr><tr><td>1E880</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_CPUAXG_TCOMP</td></tr><tr><td>1E880</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_CPUAXG_SW_FREQ</td></tr><tr><td>1E880</td><td>VR_VID_PHY</td><td>VR_VID</td><td></td><td></td><td></td><td>CPU_VIDSCLK</td></tr><tr><td>1E880</td><td>VR_VID_PHY</td><td>VR_VID</td><td></td><td></td><td></td><td>CPU_VIDSCLK_R</td></tr><tr><td>1E880</td><td>VR_VID_PHY</td><td>VR_VID</td><td></td><td></td><td></td><td>CPU_VIDALERT_L</td></tr><tr><td>1E880</td><td>VR_VID_PHY</td><td>VR_VID</td><td></td><td></td><td></td><td>CPU_VIDALERT_R_L</td></tr><tr><td>1E880</td><td>VR_VID_PHY</td><td>VR_VID</td><td></td><td></td><td></td><td>CPU_VIDSOUT</td></tr><tr><td>1E880</td><td>VR_VID_PHY</td><td>VR_VID</td><td></td><td></td><td></td><td>CPU_VIDSOUT_R</td></tr><tr><td colspan="7">Output Bus</td></tr><tr><td>1E880</td><td>POWER_PHY</td><td>POWER</td><td>1.1V</td><td></td><td></td><td>PPVCORE_S0_CPU</td></tr><tr><td>1E880</td><td>POWER_PHY</td><td>POWER</td><td>1.1V</td><td></td><td></td><td>PPVAXG_S0</td></tr></table>				Electrical Constraint Set	Physical	Spacing	Voltage	DIDT	NO_TEST		AXG							1E880	POWER_PHY	POWER	12V			REG_LVCC_U7330	1E880	VR_CTL_PHY	VR_CTL				REG_PWM_CPUAXG	1E880	VR_CTL_PHY	VR_CTL				REG_PWM_CPUAXG_R	1E880	POWER_PHY	VR_SWITCH	12V	TRUE		REG PHASE CPUAXG	1E880	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_BOOT CPUAXG	1E880	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT CPUAXG_RC	1E880	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE CPUAXG	1E880	VR_CTL_PHY	VR_LGATE	12V	TRUE		REG_LGATE CPUAXG	1E880	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_SNUBBER CPUAXG	1E880	POWER_PHY	POWER	1.1V			PPCPUAXG_S0 SENSE	1E880	TSNS_CPU AXG	SNS_DIFF_PHY	SENSE			REG ISENAXG_P	1E880	TSNS_CPU AXG	SNS_DIFF_PHY	SENSE			REG ISENAXG_N	1E880		SENSE				REG ISENAXG_PR	1E880		SENSE				REG ISENAXG_NR	1E880	TSNS_CPU AXG	SNS_DIFF_PHY	SENSE			SNS_AXG_R_P	1E880	TSNS_CPU AXG	SNS_DIFF_PHY	SENSE			SNS_AXG_R_N	1E880	TSNS_CPU AXG	SNS_DIFF_PHY	SENSE			SNS_AXG_XW_P	1E880	TSNS_CPU AXG	SNS_DIFF_PHY	SENSE			SNS_AXG_XW_N	ISL6364							1E880	VR_CTL_PHY	VR_CTL				REG_CPUCORE_COMP	1E880	VR_CTL_PHY	VR_CTL				CPUCORE_COMP_RC	1E880	VR_CTL_PHY	VR_CTL				REG_CPUCORE_FB	1E880	VR_CTL_PHY	VR_CTL				CPUCORE_FB_RC	1E880	VR_CTL_PHY	VR_CTL				CPUCORE_FB_R_1	1E880	VR_CTL_PHY	VR_CTL				CPUCORE_FB_R_2	1E880	VR_CTL_PHY	VR_CTL				CPUCORE_PSICOMP_RC	1E880	VR_CTL_PHY	VR_CTL				REG_CPUCORE_PSICOMP	1E880	VR_CTL_PHY	VR_CTL				REG_CPUCORE_HFCOMP	1E880		SENSE				REG_CPUCORE_VSEN	1E880		SENSE				REG_CPUCORE_RGND	1E880	VR_CTL_PHY	VR_CTL				REG_CPUCORE_IMON	1E880	VR_CTL_PHY	VR_CTL				CPUCORE_IMON_R	1E880	VR_CTL_PHY	VR_CTL				REG_CPUCORE_TM	1E880	VR_CTL_PHY	VR_CTL				REG_CPUCORE_SUTH	1E880	VR_CTL_PHY	VR_CTL				REG_CPUCORE_NPSI	1E880	VR_CTL_PHY	VR_CTL				REG_CPUCORE_FVID	1E880	VR_CTL_PHY	VR_CTL				REG_CPUCORE_IAUTO	1E880	VR_CTL_PHY	VR_CTL				REG_CPUCORE_SW_FREQ	1E880	VR_CTL_PHY	VR_CTL				REG_CPUCORE_RAMPADJ	1E880	VR_CTL_PHY	VR_CTL				REG_CPUCORE_EN_PWR	1E880	VR_CTL_PHY	VR_CTL				CPUCORE_EN_PWR_R	1E880	VR_CTL_PHY	VR_CTL				REG_CPUCORE_RSET	1E880	VR_CTL_PHY	VR_CTL				REG_CPUAXG_COMP	1E880	VR_CTL_PHY	VR_CTL				CPUAXG_COMP_RC	1E880	VR_CTL_PHY	VR_CTL				REG_CPUAXG_FB	1E880	VR_CTL_PHY	VR_CTL				CPUAXG_FB_RC	1E880	VR_CTL_PHY	VR_CTL				CPUAXG_FB_R_1	1E880	VR_CTL_PHY	VR_CTL				CPUAXG_FB_R_2	1E880	VR_CTL_PHY	VR_CTL				REG_CPUAXG_HFCOMP	1E880		SENSE				REG_CPUAXG_VSEN	1E880		SENSE				REG_CPUAXG_RGND	1E880	VR_CTL_PHY	VR_CTL				REG_CPUAXG_IMON	1E880	VR_CTL_PHY	VR_CTL				CPUAXG_IMON_R	1E880	VR_CTL_PHY	VR_CTL				REG_CPUAXG_TM	1E880	VR_CTL_PHY	VR_CTL				REG_CPUAXG_TCOMP	1E880	VR_CTL_PHY	VR_CTL				REG_CPUAXG_SW_FREQ	1E880	VR_VID_PHY	VR_VID				CPU_VIDSCLK	1E880	VR_VID_PHY	VR_VID				CPU_VIDSCLK_R	1E880	VR_VID_PHY	VR_VID				CPU_VIDALERT_L	1E880	VR_VID_PHY	VR_VID				CPU_VIDALERT_R_L	1E880	VR_VID_PHY	VR_VID				CPU_VIDSOUT	1E880	VR_VID_PHY	VR_VID				CPU_VIDSOUT_R	Output Bus							1E880	POWER_PHY	POWER	1.1V			PPVCORE_S0_CPU	1E880	POWER_PHY	POWER	1.1V			PPVAXG_S0
Electrical Constraint Set	Physical	Spacing	Voltage	DIDT	NO_TEST																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																															
Input Bus																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																				
1E880	POWER_PHY	POWER	12V			PP12V_S0_CPUCORE FLT																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	POWER_PHY	POWER	5V			REG_VCC_U7100																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
Local Ground																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																				
1E880	GND_PHY	GND	0V			AGND_CPU																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
Phase 1																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																				
1E880	POWER_PHY	POWER	12V			REG_LVCC_U7210																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	VR_CTL_PHY	VR_CTL				REG_PWM_CPUCORE_1																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	VR_CTL_PHY	VR_CTL				REG_PWM_CPUCORE_1 R																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	POWER_PHY	VR_SWITCH	12V	TRUE		REG PHASE CPUCORE_1																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_BOOT CPUCORE_1																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT CPUCORE_1 RC																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE CPUCORE_1																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	VR_CTL_PHY	VR_LGATE	12V	TRUE		REG_LGATE CPUCORE_1																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_SNUBBER CPUCORE_1																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	POWER_PHY	POWER	1.1V			PPCPUCORE_S0 SENSE_1																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	TSNS_CPU CORE	SNS_DIFF_PHY	SENSE			REG ISENCORE_1 P																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	TSNS_CPU CORE	SNS_DIFF_PHY	SENSE			REG ISENCORE_1 N																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880		SENSE				REG ISENCORE_1 NR																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
Phase 2																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																				
1E880	POWER_PHY	POWER	12V			REG_LVCC_U7230																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	VR_CTL_PHY	VR_CTL				REG_PWM_CPUCORE_2																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	VR_CTL_PHY	VR_CTL				REG_PWM_CPUCORE_2 R																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	POWER_PHY	VR_SWITCH	12V	TRUE		REG PHASE CPUCORE_2																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_BOOT CPUCORE_2																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT CPUCORE_2 RC																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE CPUCORE_2																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	VR_CTL_PHY	VR_LGATE	12V	TRUE		REG_LGATE CPUCORE_2																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_SNUBBER CPUCORE_2																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	POWER_PHY	POWER	1.1V			PPCPUCORE_S0 SENSE_2																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	TSNS_CPU CORE	SNS_DIFF_PHY	SENSE			REG ISENCORE_2 P																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	TSNS_CPU CORE	SNS_DIFF_PHY	SENSE			REG ISENCORE_2 N																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880		SENSE				REG ISENCORE_2 NR																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
Phase 3																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																				
1E880	POWER_PHY	POWER	12V			REG_LVCC_U7250																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	VR_CTL_PHY	VR_CTL				REG_PWM_CPUCORE_3																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	VR_CTL_PHY	VR_CTL				REG_PWM_CPUCORE_3 R																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	POWER_PHY	VR_SWITCH	12V	TRUE		REG PHASE CPUCORE_3																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_BOOT CPUCORE_3																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT CPUCORE_3 RC																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE CPUCORE_3																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	VR_CTL_PHY	VR_LGATE	12V	TRUE		REG_LGATE CPUCORE_3																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_SNUBBER CPUCORE_3																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	POWER_PHY	POWER	1.1V			PPCPUCORE_S0 SENSE_3																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	TSNS_CPU CORE	SNS_DIFF_PHY	SENSE			REG ISENCORE_3 P																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	TSNS_CPU CORE	SNS_DIFF_PHY	SENSE			REG ISENCORE_3 N																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880		SENSE				REG ISENCORE_3 NR																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
Phase 4																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																				
1E880	POWER_PHY	POWER	12V			REG_LVCC_U7310																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	VR_CTL_PHY	VR_CTL				REG_PWM_CPUCORE_4																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	VR_CTL_PHY	VR_CTL				REG_PWM_CPUCORE_4 R																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	POWER_PHY	VR_SWITCH	12V	TRUE		REG PHASE CPUCORE_4																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_BOOT CPUCORE_4																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT CPUCORE_4 RC																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE CPUCORE_4																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	VR_CTL_PHY	VR_LGATE	12V	TRUE		REG_LGATE CPUCORE_4																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_SNUBBER CPUCORE_4																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	POWER_PHY	POWER	1.1V			PPCPUCORE_S0 SENSE_4																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	TSNS_CPU CORE	SNS_DIFF_PHY	SENSE			REG ISENCORE_4 P																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	TSNS_CPU CORE	SNS_DIFF_PHY	SENSE			REG ISENCORE_4 N																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880		SENSE				REG ISENCORE_4 NR																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	TSNS_CPU CORE	SNS_DIFF_PHY	SENSE			SNS CORE_XW_P																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	TSNS_CPU CORE	SNS_DIFF_PHY	SENSE			SNS CORE_XW_N																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	TSNS_CPU CORE	SNS_DIFF_PHY	SENSE			SNS CORE_R_P																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	TSNS_CPU CORE	SNS_DIFF_PHY	SENSE			SNS CORE_R_N																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	TSNS_CPU CORE	SNS_DIFF_PHY	SENSE			SNS CPU_VAXG_P																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	TSNS_CPU CORE	SNS_DIFF_PHY	SENSE			SNS CPU_VAXG_N																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	TSNS_CPU CORE	SNS_DIFF_PHY	SENSE			SNS CPU_VCORE_P																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	TSNS_CPU CORE	SNS_DIFF_PHY	SENSE			SNS CPU_VCORE_N																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	TSNS_CPU CORE	SNS_DIFF_PHY	SENSE			SNS_P1V05_IOVDD_XW_P																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	TSNS_CPU CORE	SNS_DIFF_PHY	SENSE			SNS_P1V05_IOVDD_XW_N																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
Electrical Constraint Set	Physical	Spacing	Voltage	DIDT	NO_TEST																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																															
AXG																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																				
1E880	POWER_PHY	POWER	12V			REG_LVCC_U7330																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	VR_CTL_PHY	VR_CTL				REG_PWM_CPUAXG																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	VR_CTL_PHY	VR_CTL				REG_PWM_CPUAXG_R																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	POWER_PHY	VR_SWITCH	12V	TRUE		REG PHASE CPUAXG																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_BOOT CPUAXG																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT CPUAXG_RC																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE CPUAXG																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	VR_CTL_PHY	VR_LGATE	12V	TRUE		REG_LGATE CPUAXG																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_SNUBBER CPUAXG																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	POWER_PHY	POWER	1.1V			PPCPUAXG_S0 SENSE																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	TSNS_CPU AXG	SNS_DIFF_PHY	SENSE			REG ISENAXG_P																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	TSNS_CPU AXG	SNS_DIFF_PHY	SENSE			REG ISENAXG_N																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880		SENSE				REG ISENAXG_PR																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880		SENSE				REG ISENAXG_NR																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	TSNS_CPU AXG	SNS_DIFF_PHY	SENSE			SNS_AXG_R_P																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	TSNS_CPU AXG	SNS_DIFF_PHY	SENSE			SNS_AXG_R_N																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	TSNS_CPU AXG	SNS_DIFF_PHY	SENSE			SNS_AXG_XW_P																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	TSNS_CPU AXG	SNS_DIFF_PHY	SENSE			SNS_AXG_XW_N																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
ISL6364																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																				
1E880	VR_CTL_PHY	VR_CTL				REG_CPUCORE_COMP																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	VR_CTL_PHY	VR_CTL				CPUCORE_COMP_RC																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	VR_CTL_PHY	VR_CTL				REG_CPUCORE_FB																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	VR_CTL_PHY	VR_CTL				CPUCORE_FB_RC																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	VR_CTL_PHY	VR_CTL				CPUCORE_FB_R_1																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	VR_CTL_PHY	VR_CTL				CPUCORE_FB_R_2																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	VR_CTL_PHY	VR_CTL				CPUCORE_PSICOMP_RC																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	VR_CTL_PHY	VR_CTL				REG_CPUCORE_PSICOMP																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	VR_CTL_PHY	VR_CTL				REG_CPUCORE_HFCOMP																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880		SENSE				REG_CPUCORE_VSEN																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880		SENSE				REG_CPUCORE_RGND																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	VR_CTL_PHY	VR_CTL				REG_CPUCORE_IMON																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	VR_CTL_PHY	VR_CTL				CPUCORE_IMON_R																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	VR_CTL_PHY	VR_CTL				REG_CPUCORE_TM																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	VR_CTL_PHY	VR_CTL				REG_CPUCORE_SUTH																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	VR_CTL_PHY	VR_CTL				REG_CPUCORE_NPSI																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	VR_CTL_PHY	VR_CTL				REG_CPUCORE_FVID																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	VR_CTL_PHY	VR_CTL				REG_CPUCORE_IAUTO																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	VR_CTL_PHY	VR_CTL				REG_CPUCORE_SW_FREQ																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	VR_CTL_PHY	VR_CTL				REG_CPUCORE_RAMPADJ																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	VR_CTL_PHY	VR_CTL				REG_CPUCORE_EN_PWR																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	VR_CTL_PHY	VR_CTL				CPUCORE_EN_PWR_R																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	VR_CTL_PHY	VR_CTL				REG_CPUCORE_RSET																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	VR_CTL_PHY	VR_CTL				REG_CPUAXG_COMP																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	VR_CTL_PHY	VR_CTL				CPUAXG_COMP_RC																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	VR_CTL_PHY	VR_CTL				REG_CPUAXG_FB																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	VR_CTL_PHY	VR_CTL				CPUAXG_FB_RC																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	VR_CTL_PHY	VR_CTL				CPUAXG_FB_R_1																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	VR_CTL_PHY	VR_CTL				CPUAXG_FB_R_2																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	VR_CTL_PHY	VR_CTL				REG_CPUAXG_HFCOMP																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880		SENSE				REG_CPUAXG_VSEN																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880		SENSE				REG_CPUAXG_RGND																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	VR_CTL_PHY	VR_CTL				REG_CPUAXG_IMON																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	VR_CTL_PHY	VR_CTL				CPUAXG_IMON_R																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	VR_CTL_PHY	VR_CTL				REG_CPUAXG_TM																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	VR_CTL_PHY	VR_CTL				REG_CPUAXG_TCOMP																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	VR_CTL_PHY	VR_CTL				REG_CPUAXG_SW_FREQ																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	VR_VID_PHY	VR_VID				CPU_VIDSCLK																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	VR_VID_PHY	VR_VID				CPU_VIDSCLK_R																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	VR_VID_PHY	VR_VID				CPU_VIDALERT_L																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	VR_VID_PHY	VR_VID				CPU_VIDALERT_R_L																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	VR_VID_PHY	VR_VID				CPU_VIDSOUT																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	VR_VID_PHY	VR_VID				CPU_VIDSOUT_R																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
Output Bus																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																				
1E880	POWER_PHY	POWER	1.1V			PPVCORE_S0_CPU																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E880	POWER_PHY	POWER	1.1V			PPVAXG_S0																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
87 Apple Inc. NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																				
8	7	6	5	4	3	2	1																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																													

SYNC_MASTER=D8_MARK

SYNC_DATE=02/10/2012

CPU VReg Constraints

Apple Inc.

051-9504

5.3.0

129 OF 143

105 OF 117

NOTICE OF PROPRIETARY PROPERTY:
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
IV ALL RIGHTS RESERVED

GDDR5

GDDR5-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
GDDR_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD
GDDR_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	12 MM	=STANDARD	=STANDARD
GDDR_80D	*	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
GDDR_MA_PHY	*	GDDR_45S
GDDR_ADBI_PHY	*	GDDR_45S
GDDR_CTRL_PHY	*	GDDR_45S
GDDR_CLK_PHY	*	GDDR_80D
GDDR_DQ_PHY	*	GDDR_45S
GDDR_EDC_PHY	*	GDDR_45S
GDDR_DBI_PHY	*	GDDR_45S
GDDR_WCK_PHY	*	GDDR_80D

Main Segment Min Spacing Rules for 4.5 Gbps or Less (AMD Doc# 49919)

[illegible]

GDDR5-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GDDR_ISO	*	=3x_DIELECTRIC	?
GDDR_ISO	TOP, BOTTOM	=5x_DIELECTRIC	?
GDDR_MA2MA	*	=2x_DIELECTRIC	?
GDDR_MA2MA	TOP, BOTTOM	=3x_DIELECTRIC	?
GDDR_ADBI2ADBI	*	=2x_DIELECTRIC	?
GDDR_ADBI2ADBI	TOP, BOTTOM	=2x_DIELECTRIC	?
GDDR_CTRL2CTRL	*	=2x_DIELECTRIC	?
GDDR_CTRL2CTRL	TOP, BOTTOM	=2x_DIELECTRIC	?
GDDR_CLK2CLK	*	=3x_DIELECTRIC	?
GDDR_CLK2CLK	TOP, BOTTOM	=5x_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GDDR_DQ2DQ	*	=3x_DIELECTRIC	?
GDDR_DQ2DQ	TOP, BOTTOM	=3x_DIELECTRIC	?
GDDR_EDC_ISO	*	=3X_DIELECTRIC	?
GDDR_EDC_ISO	TOP, BOTTOM	=5X_DIELECTRIC	?
GDDR_EDC2EDC	*	=3X_DIELECTRIC	?
GDDR_EDC2EDC	TOP, BOTTOM	=5X_DIELECTRIC	?
GDDR_DBI2DBI	*	=3x_DIELECTRIC	?
GDDR_DBI2DBI	TOP, BOTTOM	=3x_DIELECTRIC	?
GDDR_WCK2WCK	*	=3X_DIELECTRIC	?
GDDR_WCK2WCK	TOP, BOTTOM	=5x_DIELECTRIC	?

Constraints (x in $\{A, B\}$, y in $\{0, 1\}$)

Memory Address: M_{Axy}[8:0]

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GDDR_*_*_MA	*	*	GDDR_ISO
GDDR_*_*_MA	=SAME	*	GDDR_MA2MA

Address Dynamic Bus Inversion: ADBIx

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GDDR_*_*_ADBI	*	*	GDDR_ISO
GDDR_*_*_ADBI	=SAME	*	GDDR_ADBI2ADBI

Control: Reset, CKExy, CSxy, WExy, RASxy, CASxy

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GDDR_CTRL	*	*	GDDR_ISO
GDDR_*_*_CTRL	*	*	GDDR_ISO
GDDR_*_*_*_CTRL	=SAME	*	GDDR_CTRL2CTRL

Clock: CKxy

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GDDR_*_*_CLK	*	*	GDDR_ISO
GDDR_*_*_CLK	=SAME	*	GDDR_CLK2CLK

GPU

GPU-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CLK_GPU_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

GPU-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_GPU_ISO	*	=4:1_SPACING	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CLK_GPU	*	*	CLK_GPU_ISO

GDDR5 Frame Buffer A

Electrical Constraint Set		Physical	Spacing		
Memory Address					
MEM0	GDDR A0 MA	GDDR NA PHY	GDDR A 0 MA	FB A0 A<8..0>	NO TEST=TRUE 77 79
MEM1	GDDR A1 MA	GDDR NA PHY	GDDR A 1 MA	FB A1 A<8..0>	NO TEST=TRUE 77 79
Address Dynamic Bus Inv					
FE00	GDDR A0 ADBT	GDDR ADBT PHY	GDDR A 0 ADBT	FB A0 ABT L	NO TEST=TRUE 77 79
FE01	GDDR A1 ADBT	GDDR ADBT PHY	GDDR A 1 ADBT	FB A1 ABT L	NO TEST=TRUE 77 79
Control					
FE04	GDDR A0 CKE	GDDR CTRL PHY	GDDR A 0 CTRL	FB A0 CKE L	NO TEST=TRUE 77 79
FE05	GDDR A0 CTRL	GDDR CTRL PHY	GDDR A 0 CTRL	FB A0 CS L	NO TEST=TRUE 77 79
FE06	GDDR A0 CTRL	GDDR CTRL PHY	GDDR A 0 CTRL	FB A0 WE L	NO TEST=TRUE 77 79
FE07	GDDR A0 CTRL	GDDR CTRL PHY	GDDR A 0 CTRL	FB A0 CAS L	NO TEST=TRUE 77 79
FE08	GDDR A0 CTRL	GDDR CTRL PHY	GDDR A 0 CTRL	FB A0 RAS L	NO TEST=TRUE 77 79
FE09	GDDR A1 CKE	GDDR CTRL PHY	GDDR A 1 CTRL	FB A1 CKE L	NO TEST=TRUE 77 79
FE0A	GDDR A1 CTRL	GDDR CTRL PHY	GDDR A 1 CTRL	FB A1 CS L	NO TEST=TRUE 77 79
FE0B	GDDR A1 CTRL	GDDR CTRL PHY	GDDR A 1 CTRL	FB A1 WE L	NO TEST=TRUE 77 79
FE0C	GDDR A1 CTRL	GDDR CTRL PHY	GDDR A 1 CTRL	FB A1 CAS L	NO TEST=TRUE 77 79
FE0D	GDDR A1 CTRL	GDDR CTRL PHY	GDDR A 1 CTRL	FB A1 RAS L	NO TEST=TRUE 77 79
Clock					
FE0E	GDDR A0 CLK	GDDR CLK PHY	GDDR A 0 CLK	FB A0 CLK P	NO TEST=TRUE 77 79
FE0F	GDDR A0 CLK	GDDR CLK PHY	GDDR A 0 CLK	FB A0 CLK N	NO TEST=TRUE 77 79
FE10	GDDR A1 CLK	GDDR CLK PHY	GDDR A 1 CLK	FB A1 CLK P	NO TEST=TRUE 77 79
FE11	GDDR A1 CLK	GDDR CLK PHY	GDDR A 1 CLK	FB A1 CLK N	NO TEST=TRUE 77 79
Data					
FE12	GDDR A0 DQ BYTE0	GDDR DQ PHY	GDDR A 0 DQ	FB A0 DQ<7..0>	NO TEST=TRUE 77 79
FE13	GDDR A0 DQ BYTE1	GDDR DQ PHY	GDDR A 0 DQ	FB A0 DQ<15..8>	NO TEST=TRUE 77 79
FE14	GDDR A0 DQ BYTE2	GDDR DQ PHY	GDDR A 0 DQ	FB A0 DQ<23..16>	NO TEST=TRUE 77 79
FE15	GDDR A0 DQ BYTE3	GDDR DQ PHY	GDDR A 0 DQ	FB A0 DQ<31..24>	NO TEST=TRUE 77 79
FE16	GDDR A1 DQ BYTE0	GDDR DQ PHY	GDDR A 1 DQ	FB A1 DQ<7..0>	NO TEST=TRUE 77 79
FE17	GDDR A1 DQ BYTE1	GDDR DQ PHY	GDDR A 1 DQ	FB A1 DQ<15..8>	NO TEST=TRUE 77 79
FE18	GDDR A1 DQ BYTE2	GDDR DQ PHY	GDDR A 1 DQ	FB A1 DQ<23..16>	NO TEST=TRUE 77 79
FE19	GDDR A1 DQ BYTE3	GDDR DQ PHY	GDDR A 1 DQ	FB A1 DQ<31..24>	NO TEST=TRUE 77 79
Error Detection					
FE20	GDDR A0 EDC0	GDDR EDC PHY	GDDR A 0 EDC	FB A0 EDC<0>	NO TEST=TRUE 77 79
FE21	GDDR A0 EDC1	GDDR EDC PHY	GDDR A 0 EDC	FB A0 EDC<1>	NO TEST=TRUE 77 79
FE22	GDDR A0 EDC2	GDDR EDC PHY	GDDR A 0 EDC	FB A0 EDC<2>	NO TEST=TRUE 77 79
FE23	GDDR A0 EDC3	GDDR EDC PHY	GDDR A 0 EDC	FB A0 EDC<3>	NO TEST=TRUE 77 79
FE24	GDDR A1 EDC0	GDDR EDC PHY	GDDR A 1 EDC	FB A1 EDC<0>	NO TEST=TRUE 77 79
FE25	GDDR A1 EDC1	GDDR EDC PHY	GDDR A 1 EDC	FB A1 EDC<1>	NO TEST=TRUE 77 79
FE26	GDDR A1 EDC2	GDDR EDC PHY	GDDR A 1 EDC	FB A1 EDC<2>	NO TEST=TRUE 77 79
FE27	GDDR A1 EDC3	GDDR EDC PHY	GDDR A 1 EDC	FB A1 EDC<3>	NO TEST=TRUE 77 79
Data Dynamic Bus Inv					
FE28	GDDR A0 DBT0	GDDR DBT PHY	GDDR A 0 DBT	FB A0 DBT L<0>	NO TEST=TRUE 77 79
FE29	GDDR A0 DBT1	GDDR DBT PHY	GDDR A 0 DBT	FB A0 DBT L<1>	NO TEST=TRUE 77 79
FE2A	GDDR A0 DBT2	GDDR DBT PHY	GDDR A 0 DBT	FB A0 DBT L<2>	NO TEST=TRUE 77 79
FE2B	GDDR A0 DBT3	GDDR DBT PHY	GDDR A 0 DBT	FB A0 DBT L<3>	NO TEST=TRUE 77 79
FE2C	GDDR A1 DBT0	GDDR DBT PHY	GDDR A 1 DBT	FB A1 DBT L<0>	NO TEST=TRUE 77 79
FE2D	GDDR A1 DBT1	GDDR DBT PHY	GDDR A 1 DBT	FB A1 DBT L<1>	NO TEST=TRUE 77 79
FE2E	GDDR A1 DBT2	GDDR DBT PHY	GDDR A 1 DBT	FB A1 DBT L<2>	NO TEST=TRUE 77 79
FE2F	GDDR A1 DBT3	GDDR DBT PHY	GDDR A 1 DBT	FB A1 DBT L<3>	NO TEST=TRUE 77 79
Forwarded Clock					
FE30	GDDR A0 WCK0	GDDR WCK PHY	GDDR A 0 WCK	FB A0 WCLK P<0>	NO TEST=TRUE 77 79
FE31	GDDR A0 WCK0	GDDR WCK PHY	GDDR A 0 WCK	FB A0 WCLK N<0>	NO TEST=TRUE 77 79
FE32	GDDR A0 WCK1	GDDR WCK PHY	GDDR A 0 WCK	FB A0 WCLK P<1>	NO TEST=TRUE 77 79
FE33	GDDR A0 WCK1	GDDR WCK PHY	GDDR A 0 WCK	FB A0 WCLK N<1>	NO TEST=TRUE 77 79
FE34	GDDR A1 WCK0	GDDR WCK PHY	GDDR A 1 WCK	FB A1 WCLK P<0>	NO TEST=TRUE 77 79
FE35	GDDR A1 WCK0	GDDR WCK PHY	GDDR A 1 WCK	FB A1 WCLK N<0>	NO TEST=TRUE 77 79
FE36	GDDR A1 WCK1	GDDR WCK PHY	GDDR A 1 WCK	FB A1 WCLK P<1>	NO TEST=TRUE 77

GPU

Electrical Constraint Set	Physical	Spacing	
Clocks			
PEX0	CLK_GPU_55S	CLK_GPU	PEX_TSTCLK_0_PL 75
PEX0	CLK_GPU_55S	CLK_GPU	PEX_TSTCLK_0_NG 75
PEX0	CLK_PCTE_PHY	CLK_PCTE	GPU_TESTMODE 81
PEX0	CLK_PCTE_PHY	CLK_PCTE	GPU_PEX_TERM 75
PEX0	CLK_GPU_55S	CLK_GPU	FB_A0_CK_MID NO_TEST=TRUE 79
PEX0	CLK_GPU_55S	CLK_GPU	FB_A1_CK_MID NO_TEST=TRUE 78
PEX0	CLK_GPU_55S	CLK_GPU	FB_B0_CK_MID NO_TEST=TRUE 80
PEX0	CLK_GPU_55S	CLK_GPU	FB_B1_CK_MID NO_TEST=TRUE 80
PEX0	CLK_GPU_55S	CLK_GPU	FB_C0_CK_MID NO_TEST=TRUE 80
PEX0	CLK_GPU_55S	CLK_GPU	FB_C1_CK_MID NO_TEST=TRUE 80
PEX0	CLK_GPU_55S	CLK_GPU	FB_D0_CK_MID NO_TEST=TRUE 80
PEX0	CLK_GPU_55S	CLK_GPU	FB_D1_CK_MID NO_TEST=TRUE 80
PEX0	CLK_GPU_55S	CLK_GPU	GPU_JTAG_TCK 78 81
PEX0	CLK_GPU_55S	CLK_GPU	GPU_ROM_SCLK 81 82
PEX0	CLK_GPU_55S	CLK_GPU	GPU_ROM_SCLK_R 82
PEX0	CLK_GPU_55S	CLK_GPU	GPU_XTALOUT 81 82
PEX0	CLK_GPU_55S	CLK_GPU	GPU_OSC_27M_XTALIN 81 82
PEX0	CLK_GPU_55S	CLK_GPU	GPU_OSC_27M_XTALIN 81 82
SMB			
PEX0	SMB_PHY	SMB	GPU_SMB_CLK 81 82
PEX0	SMB_PHY	SMB	GPU_SMB_DAT 81 82
PEX0	SMB_PHY	SMB	GPU_SMB_CLK_R 50 82
PEX0	SMB_PHY	SMB	GPU_SMB_DAT_R 50 82
PCIe Compensation			
PEX0	PCIE_S0S	COMP_PCIE	FB_CAL_PD_VDDQ 77
PEX0	PCIE_S0S	COMP_PCIE	FB_CAL_PU_GND 77
PEX0	PCIE_S0S	COMP_PCIE	FB_CAL_TERM_GND 77

GDDR5 Frame Buffer B

Electrical Constraint Set		Physical	Spacing		
Memory Address					
H009	GDDR_B0_MA	GDDR_MA_PHY	GDDR_B_0_MA	FB_B0_A<8..0>	NO TEST-TRUE 77 80
H009	GDDR_B1_MA	GDDR_MA_PHY	GDDR_B_1_MA	FB_B1_A<8..0>	NO TEST-TRUE 77 80
Address Dynamic Bus Inv					
H009	GDDR_B0_ADBT	GDDR_ADBT_PHY	GDDR_B_0_ADBT	FB_B0_ABT_L	NO TEST-TRUE 77 80
H009	GDDR_B1_ADBT	GDDR_ADBT_PHY	GDDR_B_1_ADBT	FB_B1_ABT_L	NO TEST-TRUE 77 80
Control					
H009	GDDR_B0_CKE	GDDR_CTRL_PHY	GDDR_B_0_CTRL	FB_B0_CKE_L	NO TEST-TRUE 77 80
H009	GDDR_B0_CTRL	GDDR_CTRL_PHY	GDDR_B_0_CTRL	FB_B0_CS_L	NO TEST-TRUE 77 80
H009	GDDR_B0_CTRL	GDDR_CTRL_PHY	GDDR_B_0_CTRL	FB_B0_WE_L	NO TEST-TRUE 77 80
H009	GDDR_B0_CTRL	GDDR_CTRL_PHY	GDDR_B_0_CTRL	FB_B0_CAS_L	NO TEST-TRUE 77 80
H009	GDDR_B0_CTRL	GDDR_CTRL_PHY	GDDR_B_0_CTRL	FB_B0_RAS_L	NO TEST-TRUE 77 80
H009	GDDR_B1_CKE	GDDR_CTRL_PHY	GDDR_B_1_CTRL	FB_B1_CKE_L	NO TEST-TRUE 77 80
H009	GDDR_B1_CTRL	GDDR_CTRL_PHY	GDDR_B_1_CTRL	FB_B1_CS_L	NO TEST-TRUE 77 80
H009	GDDR_B1_CTRL	GDDR_CTRL_PHY	GDDR_B_1_CTRL	FB_B1_WE_L	NO TEST-TRUE 77 80
H009	GDDR_B1_CTRL	GDDR_CTRL_PHY	GDDR_B_1_CTRL	FB_B1_CAS_L	NO TEST-TRUE 77 80
H009	GDDR_B1_CTRL	GDDR_CTRL_PHY	GDDR_B_1_CTRL	FB_B1_RAS_L	NO TEST-TRUE 77 80
Clock					
H009	GDDR_B0_CLK	GDDR_CLK_PHY	GDDR_B_0_CLK	FB_B0_CLK_P	NO TEST-TRUE 77 80
H009	GDDR_B0_CLK	GDDR_CLK_PHY	GDDR_B_0_CLK	FB_B0_CLK_N	NO TEST-TRUE 77 80
H009	GDDR_B1_CLK	GDDR_CLK_PHY	GDDR_B_1_CLK	FB_B1_CLK_P	NO TEST-TRUE 77 80
H009	GDDR_B1_CLK	GDDR_CLK_PHY	GDDR_B_1_CLK	FB_B1_CLK_N	NO TEST-TRUE 77 80
Data					
H009	GDDR_B0_DQ_BYTE0	GDDR_DQ_PHY	GDDR_B_0_DQ	FB_B0_DQ<7..0>	NO TEST-TRUE 77 80
H009	GDDR_B0_DQ_BYTE1	GDDR_DQ_PHY	GDDR_B_0_DQ	FB_B0_DQ<15..8>	NO TEST-TRUE 77 80
H009	GDDR_B0_DQ_BYTE2	GDDR_DQ_PHY	GDDR_B_0_DQ	FB_B0_DQ<23..16>	NO TEST-TRUE 77 80
H009	GDDR_B0_DQ_BYTE3	GDDR_DQ_PHY	GDDR_B_0_DQ	FB_B0_DQ<31..24>	NO TEST-TRUE 77 80
H009	GDDR_B1_DQ_BYTE0	GDDR_DQ_PHY	GDDR_B_1_DQ	FB_B1_DQ<7..0>	NO TEST-TRUE 77 80
H009	GDDR_B1_DQ_BYTE1	GDDR_DQ_PHY	GDDR_B_1_DQ	FB_B1_DQ<15..8>	NO TEST-TRUE 77 80
H009	GDDR_B1_DQ_BYTE2	GDDR_DQ_PHY	GDDR_B_1_DQ	FB_B1_DQ<23..16>	NO TEST-TRUE 77 80
H009	GDDR_B1_DQ_BYTE3	GDDR_DQ_PHY	GDDR_B_1_DQ	FB_B1_DQ<31..24>	NO TEST-TRUE 77 80
Error Detection					
H009	GDDR_B0_EDC0	GDDR_EDC_PHY	GDDR_B_0_EDC	FB_B0_EDC<0>	NO TEST-TRUE 77 80
H009	GDDR_B0_EDC1	GDDR_EDC_PHY	GDDR_B_0_EDC	FB_B0_EDC<1>	NO TEST-TRUE 77 80
H009	GDDR_B0_EDC2	GDDR_EDC_PHY	GDDR_B_0_EDC	FB_B0_EDC<2>	NO TEST-TRUE 77 80
H009	GDDR_B0_EDC3	GDDR_EDC_PHY	GDDR_B_0_EDC	FB_B0_EDC<3>	NO TEST-TRUE 77 80
H009	GDDR_B1_EDC0	GDDR_EDC_PHY	GDDR_B_1_EDC	FB_B1_EDC<0>	NO TEST-TRUE 77 80
H009	GDDR_B1_EDC1	GDDR_EDC_PHY	GDDR_B_1_EDC	FB_B1_EDC<1>	NO TEST-TRUE 77 80
H009	GDDR_B1_EDC2	GDDR_EDC_PHY	GDDR_B_1_EDC	FB_B1_EDC<2>	NO TEST-TRUE 77 80
H009	GDDR_B1_EDC3	GDDR_EDC_PHY	GDDR_B_1_EDC	FB_B1_EDC<3>	NO TEST-TRUE 77 80
Data Dynamic Bus Inv					
H009	GDDR_B0_DBT0	GDDR_DBT_PHY	GDDR_B_0_DBT	FB_B0_DBT_L<0>	NO TEST-TRUE 77 80
H009	GDDR_B0_DBT1	GDDR_DBT_PHY	GDDR_B_0_DBT	FB_B0_DBT_L<1>	NO TEST-TRUE 77 80
H009	GDDR_B0_DBT2	GDDR_DBT_PHY	GDDR_B_0_DBT	FB_B0_DBT_L<2>	NO TEST-TRUE 77 80
H009	GDDR_B0_DBT3	GDDR_DBT_PHY	GDDR_B_0_DBT	FB_B0_DBT_L<3>	NO TEST-TRUE 77 80
H009	GDDR_B1_DBT0	GDDR_DBT_PHY	GDDR_B_1_DBT	FB_B1_DBT_L<0>	NO TEST-TRUE 77 80
H009	GDDR_B1_DBT1	GDDR_DBT_PHY	GDDR_B_1_DBT	FB_B1_DBT_L<1>	NO TEST-TRUE 77 80
H009	GDDR_B1_DBT2	GDDR_DBT_PHY	GDDR_B_1_DBT	FB_B1_DBT_L<2>	NO TEST-TRUE 77 80
H009	GDDR_B1_DBT3	GDDR_DBT_PHY	GDDR_B_1_DBT	FB_B1_DBT_L<3>	NO TEST-TRUE 77 80
Forwarded Clock					
H009	GDDR_B0_WCLK	GDDR_WCK_PHY	GDDR_B_0_WCK	FB_B0_WCLK_P<0>	NO TEST-TRUE 77 80
H009	GDDR_B0_WCK0	GDDR_WCK_PHY	GDDR_B_0_WCK	FB_B0_WCLK_N<0>	NO TEST-TRUE 77 80
H009	GDDR_B0_WCK1	GDDR_WCK_PHY	GDDR_B_0_WCK	FB_B0_WCLK_P<1>	NO TEST-TRUE 77 80
H009	GDDR_B0_WCK1	GDDR_WCK_PHY	GDDR_B_0_WCK	FB_B0_WCLK_N<1>	NO TEST-TRUE 77 80
H009	GDDR_B1_WCK0	GDDR_WCK_PHY	GDDR_B_1_WCK	FB_B1_WCLK_P<0>	NO TEST-TRUE 77

Frame Buffer Reset

Electrical Constraint Set		Physical	Spacing	
Reset				
H445	GDNR_A0_RESET	GDNR_S05	GDNR_CTR1	FB_A0_RESET_L
H455	GDNR_A1_RESET	GDNR_S05	GDNR_CTR1	FB_A1_RESET_L
H460	GDNR_B0_RESET	GDNR_S05	GDNR_CTR1	FB_B0_RESET_L
H475	GDNR_B1_RESET	GDNR_S05	GDNR_CTR1	FB_B1_RESET_L
H485	GDNR_C0_RESET	GDNR_S05	GDNR_CTR1	FB_C0_RESET_L
H495	GDNR_C1_RESET	GDNR_S05	GDNR_CTR1	FB_C1_RESET_L
H705	GDNR_D0_RESET	GDNR_S05	GDNR_CTR1	FB_D0_RESET_L
H715	GDNR_D1_RESET	GDNR_S05	GDNR_CTR1	FB_D1_RESET_L

SYNCH MASTER=D8 AARON		SYNCH DATE=03/13/2012	
PAGE TITLE			
GDDR5/GPU Constraints			
 Apple Inc.		DRAWING NUMBER	051-9504
		SIZE	D
		REVISION	5.3.0
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE			
PROPRIETARY PROPERTY OF APPLE INC.			
THE POSSESSOR AGREES TO THE FOLLOWING:		PAGE	
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		132 OF 143	
II NOT TO REPRODUCE OR COPY IT			
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART		SHEET	
IV ALL RIGHTS RESERVED		108 OF 117	

D

C

B

A

Con

BLC High Voltage Output

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
BLC_HV	BLC_HV	*	BLC_CTL_ISO
BLC_HV	*	*	BLC_HV_ISO

BLC Baddies

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
BLC_PHASE	*	*	PHASE_ISO
BLC_PHASE	BLC_PHASE	*	PHASE_SW2SW
BLC_PHASE	POWER	*	PHASE_SW2PWR
BLC_PHASE	GND	*	PHASE_SW2GND

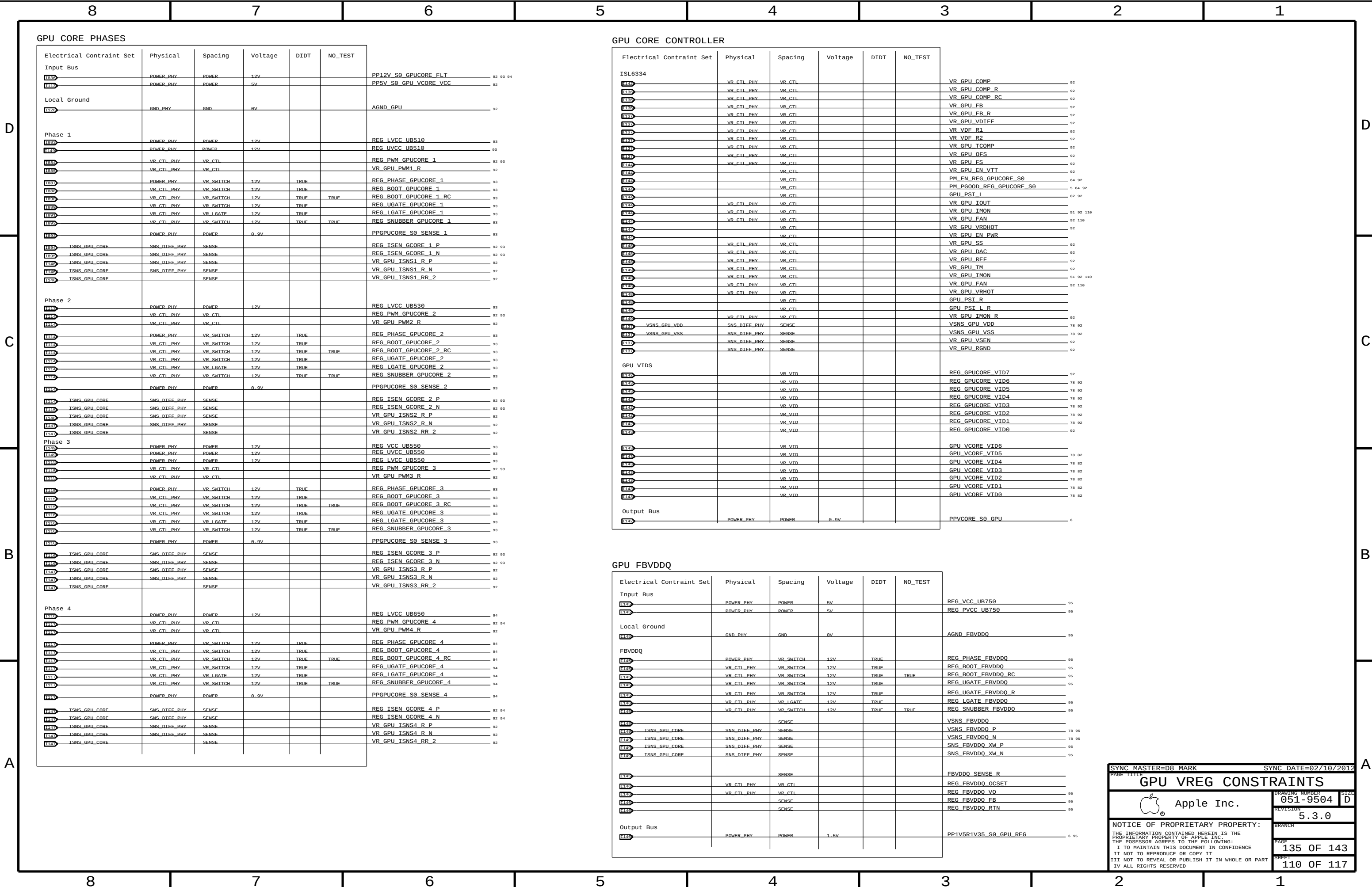
BLC Control

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
BLC_CTL	*	*	BLC_CTL_ISO

B

ADCB

Δ



Caesar IV (Ethernet/SD)

CIV-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
ENET_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
ENET_100D	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF
SD_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
ENET_COMP_PHY	*	ENET_50S
ENET_DIFF_PHY	*	ENET_100D
SD_PHY	*	SD_50S

CIV-specific Spacing Definitions
Ethernet

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
ENET_DIFF_ISO	*	=6:1_SPACING	?
ENET_DIFF2DIFF	*	=3:1_SPACING	?
ENET_TRANS_ISO	*	1.27 MM	?
COMP_ENET_ISO	*	=4:1_SPACING	?

Constraints
Ethernet

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
ENET_DIFF	*	*	ENET_DIFF_ISO
ENET_DIFF	ENET_DIFF	*	ENET_DIFF2DIFF
ENET_TRANS	*	*	ENET_TRANS_ISO
COMP_ENET	*	*	COMP_ENET_ISO
ENET_TRANS	ENET_TRANS	*	ENET_DIFF2DIFF

2 kV isolation

SD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SD_ISO	*	=3:1_SPACING	?

SD

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
SD	*	*	SD_ISO

VENI, VIDI, VICI!

Electrical Constraint Set	Physical	Spacing	
ENET_MDI_NORM	ENET_DIFF_PHY	ENET_DIFF	ENETCONN MDI P<3..0>
ENET_MDI_NORM	ENET_DIFF_PHY	ENET_DIFF	ENETCONN MDI N<3..0>
	ENET_DIFF_PHY	ENET_TRANS	ENETCONN MDI T P<3..0>
	ENET_DIFF_PHY	ENET_TRANS	ENETCONN MDI T N<3..0>
		ENET_TRANS	ENETCONN MCT3
		ENET_TRANS	ENETCONN MCT2
		ENET_TRANS	ENETCONN MCT1
		ENET_TRANS	ENETCONN MCT0
ENET_COMP_PHY	ENET_COMP_PHY	COMP_ENET	ENET_RDAC
SD_DATA	SD_PHY	SD	ENET_CR_DATA<7..0> NO_TEST=TRUE
	SD_PHY	SD	SDCONN_DATA<7..0>
	SD_PHY	SD	SDCONN_DATA R<7..0> NO_TEST=TRUE
SD_CMD	SD_PHY	SD	ENET_SD_CMD NO_TEST=TRUE
	SD_PHY	SD	SDCONN_CMD NO_TEST=TRUE
	SD_PHY	SD	SDCONN_CMD R NO_TEST=TRUE
SD_CLK	SD_PHY	SD	ENET_SD_CLK NO_TEST=TRUE
	SD_PHY	SD	SDCONN_CLK NO_TEST=TRUE
	SD_PHY	SD	SDCONN_CLK R NO_TEST=TRUE
	SD_PHY	SD	ENET_MEDIA_SENSE
	SD_PHY	SD	ENET_SD_DETECT L
	SD_PHY	SD	SDCONN_WP
CIV SPI	SPI_50S	SPI	ENET_SCLK
	SPI_50S	SPI	ENET_MISO
	SPI_50S	SPI	ENET_MOSI
	SPI_50S	SPI	ENET_CS_L

8	7	6	5	4	3	2	1
---	---	---	---	---	---	---	---

DCBA

AUTO-CONSTRAINTS PG 2

ENETCONN_*

Spacing	Netname
GENERIC ISO	ENETCONN_TCT

ENET_*

Physical	Spacing	Netname
GENERIC ISO	GENERIC ISO	ENET_ACT
GENERIC ISO	GENERIC ISO	ENET_ASF_GPIO
GENERIC ISO	GENERIC ISO	ENET_CLKREQ_L
PM	PM	ENET_CLKREQ_L_Q
PM	PM	ENET_CR_1V8_EN
PM	PM	ENET_CR_1V8_EN_R
PM	PM	ENET_CR_3V3_EN_L
PM	PM	ENET_CR_3V3_EN_L_R
PM	PM	ENET_CR_PWREN
XDP_PHY	XDP	ENET_LOW_PWR
PM	PM	ENET_PWR_EN_L
PM	PM	ENET_PWR_EN_L_R
PM	PM	ENET_RESET_L
PM	PM	ENET_SD_RESET_L
GENERIC ISO	GENERIC ISO	ENET_SR_DISABLE
GENERIC ISO	GENERIC ISO	ENET_SR_LX
PM	PM	ENET_WAKE_L

FAN_*

Spacing	Netname
GENERIC ISO	FAN_0_PWM_FET
GENERIC ISO	FAN_0_PWM_FILT
GENERIC ISO	FAN_0_TACH_FET
GENERIC ISO	FAN_0_TACH_FILT

FBVDD_*

Spacing	Netname
GENERIC ISO	FBVDD_ALTVO

FB_*

Spacing	Netname
GENERIC ISO	FB_A0_VREFC
GENERIC ISO	FB_A0_VREFD
GENERIC ISO	FB_A1_VREFC
GENERIC ISO	FB_A1_VREFD
GENERIC ISO	FB_B0_VREFC
GENERIC ISO	FB_B0_VREFD
GENERIC ISO	FB_B1_VREFC
GENERIC ISO	FB_B1_VREFD
GENERIC ISO	FB_VREF

FET_*

Physical	Spacing	Voltage	Netname
GENERIC ISO	GENERIC ISO		FET_EN_P12V_S0
GENERIC ISO	GENERIC ISO		FET_EN_P12V_S0_BLC
GENERIC ISO	GENERIC ISO		FET_EN_P12V_S0_BLC_R
GENERIC ISO	GENERIC ISO		FET_EN_P12V_S0_R
GENERIC ISO	GENERIC ISO		FET_EN_P12V_S5
GENERIC ISO	GENERIC ISO		FET_EN_P12V_S5_R
GENERIC ISO	GENERIC ISO		FET_EN_VDDQ_S0
GENERIC ISO	GENERIC ISO		FET_HDD_SLGSW
POWER_PHY	POWER	12V	FET_VCC_U7950
POWER_PHY	POWER	12V	FET_VCC_U7970
POWER_PHY	POWER	12V	FET_VCC_U7980

FLAG_*

Spacing	Netname
GENERIC ISO	FLAG_V

G3_*

Spacing	Netname
GENERIC ISO	G3_POWERON_L

GND_*

Spacing	Netname
SENSE	GND_SMC_AVSS

HDD_*

Spacing	Netname
GENERIC ISO	HDD_12V_S0_GATE
GENERIC ISO	HDD_00B_1V00_REF
PM	HDD_PWR_EN
GENERIC ISO	HDD_PWR_EN_L
GENERIC ISO	HDD_PWR_EN_R

I2C_*

Physical	Spacing	Netname
SMB_PHY	SMB	I2C_TCON_MAS_SCL
SMB_PHY	SMB	I2C_TCON_MAS_SDA

IFPD_*

Spacing	Netname
GENERIC ISO	IFPD_RSET

IFPEF_*

Spacing	Netname
GENERIC ISO	IFPEF_RSET

ISNSA_*

Electrical	Physical	Spacing	Netname
SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P12VG3H_N
SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P12VG3H_P
SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P12VS0_CPU_P1V05_N
SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P12VS0_CPU_P1V05_P
SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P12VS0_CPU_VCCSA_N
SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P12VS0_CPU_VCCSA_P
SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P12VS0_FBVDQ0_N
SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P12VS0_FBVDQ0_P
SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P12VS0_HDD_N
SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P12VS0_HDD_P
SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P12VS0_P1V05_N
SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P12VS0_P1V05_P
SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P1V05S0_PCH_N
SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P1V05S0_PCH_P
SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P1V5S0_CPU_MEM_N
SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P1V5S0_CPU_MEM_P
SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P3V3S0 SSD_N
SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P3V3S0 SSD_P
SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P3V3S4_AP_N
SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P3V3S4_AP_P
SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P5V50_HDD_N
SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P5V50_HDD_P
SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_PVDDQS3_DDR_N
SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_PVDDQS3_DDR_P

GPU_*

Physical	Spacing	Netname
GENERIC ISO	GENERIC ISO	GPU_ALT_VREF
GENERIC ISO	GENERIC ISO	GPU_BUFRSTN
GENERIC ISO	GENERIC ISO	GPU_IFPAB_PLLVDD
GENERIC ISO	GENERIC ISO	GPU_IFPA_IOVDD
GENERIC ISO	GENERIC ISO	GPU_IFPB_IOVDD
GENERIC ISO	GENERIC ISO	GPU_IFPC_IOVDD
XDP_PHY	XDP	GPU_JTAG_TDI
XDP_PHY	XDP	GPU_JTAG_TDO
XDP_PHY	XDP	GPU_JTAG_TMS
XDP_PHY	XDP	GPU_JTAG_TRST_L
GENERIC ISO	GENERIC ISO	GPU_MLS_STRAP0
GENERIC ISO	GENERIC ISO	GPU_MLS_STRAP1
GENERIC ISO	GENERIC ISO	GPU_MLS_STRAP2
GENERIC ISO	GENERIC ISO	GPU_MLS_STRAP3
GENERIC ISO	GENERIC ISO	GPU_MLS_STRAP4
GENERIC ISO	GENERIC ISO	GPU_RESET_L
SPI_50S	SPI	GPU_ROM_CS_L
SPI_50S	SPI	GPU_ROM_CS_L_R
SPI_50S	SPI	GPU_ROM_SI
SPI_50S	SPI	GPU_ROM_SI_R
SPI_50S	SPI	GPU_ROM_SO
SPI_50S	SPI	GPU_ROM_SO_R
SPI_50S	SPI	GPU_ROM_WP_L

AUTO-CONSTRAINTS GENERATED Fri 04/06/12 07:24:10 PM

PAGE TITLE
AUTO-CONSTRAINTS 2

DRAWING NUMBER
051-9504

REVISION
5.3.0

NOTICE OF PROPRIETARY PROPERTY:
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC.
THE POSSESSOR AGREES TO THE FOLLOWING:
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
IV ALL RIGHTS RESERVED

PAGE
139 OF 143

SHEET
113 OF 11

Diagram illustrating the AUTO-CONSTRAINTS PG 3, showing various constraints and their associated components across multiple categories (A, B, C, D).

Category D:

- ISNS_***: Constraints for ISNS components (Physical, Spacing, Netname).
- LPCPLUS_***: Constraints for LPCPLUS components (Spacing, Netname).
- LPC_***: Constraints for LPC components (Spacing, Netname).
- MEMVTT_***: Constraints for MEMVTT components (Spacing, Netname).
- MEM_***: Constraints for MEM components (Spacing, Netname).
- MOJO_***: Constraints for MOJO components (Physical, Spacing, Netname).
- OCA_***: Constraints for OCA components (Spacing, Netname).
- OVP_***: Constraints for OVP components (Spacing, Netname).
- P1V2_***: Constraints for P1V2 components (Spacing, Netname).
- P1V8_***: Constraints for P1V8 components (Spacing, Netname).
- P3V3AP_***: Constraints for P3V3AP components (Spacing, Netname).
- P3V3_***: Constraints for P3V3 components (Spacing, Netname).
- P3V42G3H_***: Constraints for P3V42G3H components (Spacing, Netname).
- P5V_***: Constraints for P5V components (Spacing, Netname).
- PCA9557D_***: Constraints for PCA9557D components (Spacing, Netname).
- PCH_***: Constraints for PCH components (Physical, Spacing, Netname).
- PCIE_***: Constraints for PCIE components (Spacing, Netname).
- PEG_***: Constraints for PEG components (Spacing, Netname).
- PG00D_***: Constraints for PG00D components (Spacing, Netname).
- PLT_***: Constraints for PLT components (Spacing, Netname).
- PM_***: Constraints for PM components (Spacing, Netname).

Category C:

- ISOLATE_***: Constraints for ISOLATE components (Spacing, Netname).

Category B:

- JTAG_***: Constraints for JTAG components (Physical, Spacing, Netname).
- LED_***: Constraints for LED components (Spacing, Netname).

Category A:

AUTO-CONSTRAINTS GENERATED Fri 04/06/12 07:24:10 PM

Metadata:

- PAGE TITLE: AUTO-CONSTRAINTS 3
- DRAWING NUMBER: 051-9504
- REVISION: 5.3.0
- NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED
- PAGE: 140 OF 143
- SHEET: 114 OF 117

8	7	6	5	4	3	2	1
---	---	---	---	---	---	---	---

D— 47 —DC

— 44

C644164

- 41

5 66

— 41

B26 4858

AUTO-CONSTRAINTS PG 6

SMC_*

Physical	Spacing	Netname
16	GENERIC ISO	SMC ROMBOOT 48 49
16	GENERIC ISO	SMC RUNTIME SCI L 15 21 47
16	GENERIC ISO	SMC RX L 47 48 49
16	PM	SMC S4 WAKESRC_EN 35 47 48
16	GENERIC ISO	SMC TCK 47 48 49
16	GENERIC ISO	SMC TDI 47 48 49
17	GENERIC ISO	SMC TDO 47 48 49
16	PM	SMC THRMTRIP 47 48
16	GENERIC ISO	SMC TMS 47 48 49
16	GENERIC ISO	SMC TO BLC_RX_L 48
16	GENERIC ISO	SMC TO BLC_TX_L 48
16	GENERIC ISO	SMC TX L 47 48 49
16	PM	SMC WAKE_L 47
16	GENERIC ISO	SMC WAKE_SCI_L 15 21 47
16	CLK XTAL	SMC XTAL 47 48
16	CLK XTAL	SMC XTAL_R 48

SML_*

Physical	Spacing	Netname
16	SMB_PHY	SMB 18 50
16	SMB_PHY	SMB 18 50
16	SMB_PHY	SMB 18 50
16	SMB_PHY	SMB 18 50

SPI_*

Spacing	Netname
16	GENERIC ISO SPI_DESCRIPTOR_OVERRIDE_L 15 47

SSD_*

Spacing	Netname
16	GENERIC ISO SSD_CLKREQ_L 15 18

SYS_*

Spacing	Netname
16	PM SYS_PWROK_R 65

TBT_*

Physical	Spacing	Voltage	Netname
12	GENERIC ISO	3.3V	TBT A BIAS 66
12	TBT_GEN_55S	TBT_GEN	TBT A_CONFIG1_RC 66
12	PM		TBT A_HV_EN 36 66
12	GENERIC ISO	3.3V	TBT B BIAS 68
12	TBT_GEN_55S	TBT_GEN	TBT B_CONFIG1_RC 68
12	PM		TBT B_HV_EN 36 68
12	GENERIC ISO		TBT_CLKREQ_ISOL_L 38
12	GENERIC ISO		TBT_CLKREQ_L 15 38
12	GENERIC ISO		TBT_DDC_XBAR_EN_L 36 85
12	GENERIC ISO		TBT_EN_CIO_PWR 38
12	PM		TBT_EN_CIO_PWR_L 36 38
12	GENERIC ISO		TBT_EN_LC_ISOL 38
12	GENERIC ISO		TBT_EN_LC_PWR 36 38
12	GENERIC ISO		TBT_PCH_CLKREQ_L 15 21
12	PM		TBT_PWR_EN 15 26 36
12	PM		TBT_PWR_EN_PCH 18 26
12	PM		TBT_PWR_ON_POC_RST_L 36 38
12	PM		TBT_PWR_REQ_L 15 20 36
12	PM		TBT_S0_EN 64
12	PM		TBT_SW_RESET_L 21 38

TSNS_*

Electrical	Physical	Spacing	Netname
12	SNS_TEMP	SNS_DIFF_PHY	SENSE TSNS_1_1_N 53
12	SNS_TEMP	SNS_DIFF_PHY	SENSE TSNS_1_1_P 53
12	SNS_TEMP	SNS_DIFF_PHY	SENSE TSNS_1_2_N 53
12	SNS_TEMP	SNS_DIFF_PHY	SENSE TSNS_1_2_P 53
12	SNS_TEMP	SNS_DIFF_PHY	SENSE TSNS_1_3_N 53
12	SNS_TEMP	SNS_DIFF_PHY	SENSE TSNS_1_3_P 53
12	SNS_TEMP	SNS_DIFF_PHY	SENSE TSNS_2_1_N 53
12	SNS_TEMP	SNS_DIFF_PHY	SENSE TSNS_2_1_P 53
12	SNS_TEMP	SNS_DIFF_PHY	SENSE TSNS_2_2_N 53
12	SNS_TEMP	SNS_DIFF_PHY	SENSE TSNS_2_2_P 53
12	SNS_TEMP	SNS_DIFF_PHY	SENSE TSNS_2_3_N 53
12	SNS_TEMP	SNS_DIFF_PHY	SENSE TSNS_2_3_P 53
12	SNS_TEMP	SNS_DIFF_PHY	SENSE TSNS_2_4_N 53
12	SNS_TEMP	SNS_DIFF_PHY	SENSE TSNS_2_4_P 53
12	SNS_TEMP	SNS_DIFF_PHY	SENSE TSNS_2_5_N 53
12	SNS_TEMP	SNS_DIFF_PHY	SENSE TSNS_2_5_P 53
12	SNS_TEMP	SNS_DIFF_PHY	SENSE TSNS_2_6_N 53
12	SNS_TEMP	SNS_DIFF_PHY	SENSE TSNS_2_6_P 53
12	SNS_TEMP	SNS_DIFF_PHY	SENSE TSNS_2_7_N 53
12	SNS_TEMP	SNS_DIFF_PHY	SENSE TSNS_2_7_P 53
12	SNS_TEMP	SNS_DIFF_PHY	SENSE TSNS_ACDC_N 6 53
12	SNS_TEMP	SNS_DIFF_PHY	SENSE TSNS_ACDC_P 6 53
12	SNS_TEMP	SNS_DIFF_PHY	SENSE TSNS_SKIN_N 53
12	SNS_TEMP	SNS_DIFF_PHY	SENSE TSNS_SKIN_P 53

UVP_*

Spacing	Netname
16	PM UVP_IN_1 91
16	PM UVP_IN_1_REF 91
16	PM UVP_IN_2 91
16	PM UVP_IN_3 91
16	PM UVP_IN_4 91
16	PM UVP_REF 91

VREFMRGN_*

Spacing	Netname
12	GENERIC ISO VREFMRGN_CA_SODIMMA_EN 34
12	GENERIC ISO VREFMRGN_CA_SODIMMB_EN 34

VTTCLAMP_*

Spacing	Netname
12	GENERIC ISO VTTCLAMP_EN 28
12	GENERIC ISO VTTCLAMP_L 28

WOL_*

Spacing	Netname
12	GENERIC ISO WOL_EN 15 21 48